

VISUAL THINKING *for* DESIGN

Colin Ware

*active vision, attention
visual queries, gist,
visual skills, color,
narrative, design*

Visual Thinking

For Design

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Visual Thinking For Design

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Contents

<i>Preface</i>	ix
VISUAL QUERIES	1
The Apparatus and Process of Seeing	5
The Act of Perception	8
Bottom-Up	10
Top-Down	12
Implications for Design	14
Nested Loops	17
Distributed Cognition	19
Conclusion	20
WHAT WE CAN EASILY SEE	23
The Machinery of Low-Level Feature Analysis	25
What and Where Pathways	26
Eye Movement Planning	26
What Stands Out = What We Can Bias for	27
Lessons for Design	33
Motion	36
Visual Search Strategies and Skills	37
The Detection Field	37
The Visual Search Process	39

Using Multiscale Structure to Design for Search	40
Conclusion	41

STRUCTURING TWO-DIMENSIONAL SPACE 43

2.5D Space	44
The Pattern-Processing Machinery	46
The Binding Problem: Features to Contours	46
The Generalized Contour	49
Texture Regions	50
Interference and Selective Tuning	51
Patterns, Channels, and Attention	52
Intermediate Patterns	53
Pattern Learning	54
Serial Processing	55
Visual Pattern Queries and the Apprehendable Chunk	55
Multi-chunk Queries	56
Spatial Layout	56
Horizontal and Vertical	57
Pattern for Design	58
Examples of Pattern Queries with Common Graphical Artifacts	60
Semantic Pattern Mappings	62

COLOR 65

The Color-Processing Machinery	66
Opponent Process Theory	68
Channel Properties	69
Principles for Design	75
Showing Detail	75
Color-Coding Information	77
Large and Small Areas	77
Emphasis and Highlighting	78
Color Sequences	80
Color on Shaded Surfaces	83
Semantics of Color	84
Conclusion	84

GETTING THE INFORMATION: VISUAL SPACE AND TIME 87

Depth Perception and Cue Theory	89
Stereoscopic Depth	94
Structure from Motion	95

2.5D DESIGN	95
How Much of the Third Dimension?	97
Affordances	99
The Where Pathway	100
Artificial Interactive Spaces	102
Space Traversal and Cognitive Costs	103
Conclusion	105

VISUAL OBJECTS, WORDS, AND MEANING 107

The Inferotemporal Cortex and the What Channel	108
Generalized Views from Patterns	109
Structured Objects	110
Gist and Scene Perception	112
Visual and Verbal Working Memory	114
Verbal Working Memory	115
Control of the Attention and the Cognitive Process	115
Long-term Memory	116
Priming	118
Getting into Visual Working Memory	118
Thinking in Action: Receiving a Cup of Coffee	120
Elaborations and Implications for Design	121
Make Objects Easy to Identify	121
Novelty	122
Images as Symbols	123
Meaning and Emotion	124
Imagery and Desire	125
Conclusion	126

VISUAL AND VERBAL NARRATIVE 129

Visual Thinking Versus Language-Based Thinking	130
Learned Symbols	131
Grammar and Logic	132
Comparing and Contrasting the Verbal and Written Modes	133
Linking Words and Images Through Diexis	135
PowerPoint Presentations and Pointing	136
Mirror Neurons: Copycat Cells	137
Visual Narrative: Capturing the Cognitive Thread	138
Q&A Patterns	139
Framing	139
FINSTs and Divided Attention	140
Shot transitions	141

Cartoons and Narrative Diagrams	142
Single-frame Narratives	144
Conclusion	145

CREATIVE META-SEEING 147

Mental Imagery	148
The Magic of the Scribble	152
Diagrams are Ideas Made Concrete	155
Requirements and Early Design	156
Visual Task Analysis	157
The Creative Design Loop	158
Cognitive Economics of Design Sketching	158
The Perceptual Critique	160
Meta-seeing with Design Prototypes	162
Visual Skill Development	163
Conclusion	164

THE DANCE OF MEANING 165

Review	166
Implications	172
Design to Support Pattern Finding	172
Optimizing the Cognitive Process	174
Learning and the Economics of Cognition	177
Attention and the Cognitive Thread	179
What's Next?	181

<i>Index</i>	183
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Preface

There has been a revolution in our understanding of human perception that goes under the name “active vision.” Active vision means that we should think about graphic designs as cognitive tools, enhancing and extending our brains. Although we can, to some extent, form mental images in our heads, we do much better when those images are out in the world, on paper or computer screen. Diagrams, maps, web pages, information graphics, visual instructions, and technical illustrations all help us to solve problems through a process of visual thinking. We are all cognitive cyborgs in this Internet age in the sense that we rely heavily on cognitive tools to amplify our mental abilities. Visual thinking tools are especially important because they harness the visual pattern finding part of the brain. Almost half the brain is devoted to the visual sense and the visual brain is exquisitely capable of interpreting graphical patterns, even scribbles, in many different ways. Often, to see a pattern is to understand the solution to a problem.

The active vision revolution is all about understanding perception as a dynamic process. Scientists used to think that we had rich images of the world in our heads built up from the information coming in through the eyes. Now we know that we only have the illusion of seeing the world in detail. In fact the brain grabs just those fragments that are needed to execute the current mental activity. The brain directs the eyes to move, tunes up parts of itself to receive the expected input, and extracts exactly what

is needed for our current thinking activity, whether that is reading a map, making a peanut butter and jelly sandwich, or looking at a poster. Our impression of a rich detailed world comes from the fact that we have the capability to extract anything we want at any moment through a movement of the eye that is literally faster than thought. This is automatic and so quick that we are unaware of doing it, giving us the illusion that we see stable detailed reality everywhere. The process of visual thinking is a kind of dance with the environment with some information stored internally and some externally and it is by understanding this dance that we can understand how graphic designs gain their meaning.

Active vision has profound implications for design and this is the subject of this book.

It is a book about how we think visually and what that understanding can tell us about how to design visual images. Understanding active vision tells us which colors and shapes will stand out clearly, how to organize space, and when we should use images instead of words to convey an idea.

Early on in the writing and image creation process I decided to “eat my own dog food” and apply active vision-based principles to the design of this book. One of these principles being that when text and images are related they should be placed in close proximity. This is not as easy as it sounds. It turns out that there is a reason why there are labeled figure legends in academic publishing (e.g. Figure 1, Figure 2, etc.). It makes the job of the compositor much easier. A compositor is a person whose specialty is to pack images and words on the page *without reading the text*. This leads to the labeled figure and the parenthetical phrase often found in academic publishing, “see Figure X”. This formula means that Figure X need not be on the same page as the accompanying text. It is a bad idea from the design perspective and a good idea from the perspective of the publisher. I decided to integrate text and words and avoid the use of “see Figure X” and the result was a difficult process and some conflict with a modern publishing house that does not, for example, invite authors to design meetings, even when the book is about design. The result is something of a design compromise but I am grateful to the individuals at Elsevier who helped me with what has been a challenging exercise.

There are many people who have helped. Diane Cerra with Elsevier was patient with the difficult demands I made and full of helpful advice when I needed it. Denise Penrose guided me through the later stages and came up with the compromise solution that is realized in these pages. Dennis Schaefer and Alisa Andreola helped with the design. Mary James and Paul Gottehrer provided cheerful and efficient support through the detailed

editing the production process. My wife, Dianne Ramey, read the whole thing twice and fixed a very great number of grammar and punctuation errors. I am very grateful to Paul Catanese of the New Media Department at San Francisco State University and David Laidlaw of the Computer Graphics Group at Brown University who provided content reviews and told me what was clear and what was not. I did major revisions to Chapters 3 and 8 as a result of their input.

This book is an introduction to what the burgeoning science of perception can tell us about visual design. It is intended for anyone who does design in a visual medium and it should be of special interest to anyone who does graphic design for the internet or who designs information graphics of one sort or another. Design can take ideas from anywhere, from art and culture as well as particular design genres. Science can enrich the mix.

Colin Ware
January 2008

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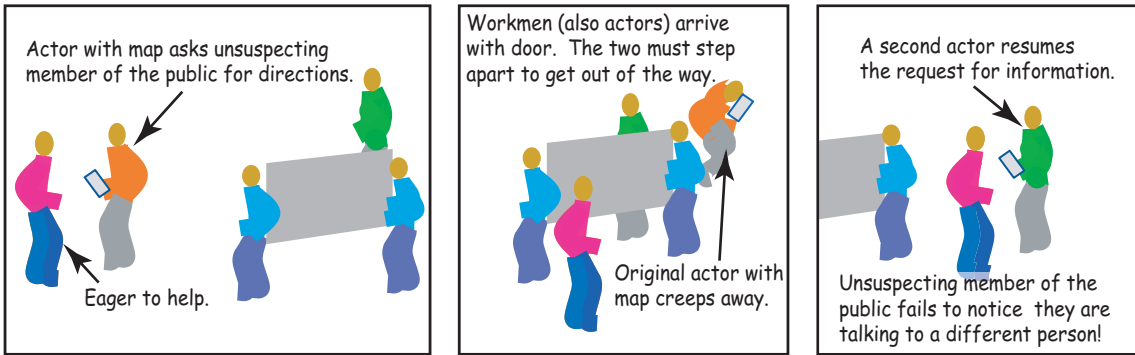
Chapter 1

Visual Queries

When we are awake, with our eyes open, we have the impression that we see the world vividly, completely, and in detail. But this impression is dead wrong. As scientists have devised increasingly elaborate tests to find out what is stored in the brain about the state of the visual world at any instant, the same answer has come back again and again—at any given instant, we apprehend only a tiny amount of the information in our surroundings, but it is usually just the right information to carry us through the task of the moment.

We cannot even remember new faces unless we are specifically paying attention. Consider the following remarkable “real world” experiment carried out by psychologists Daniel Simons and Daniel Levin.* A trained actor approached an unsuspecting member of the public, map in hand and in a crowded place with lots of pedestrian traffic, and began to ask for directions. Then, by means of a clever maneuver involving two workmen and a door, a second actor replaced the first in the middle of the conversation.

*Daniel J. Simons and Daniel T. Levin. 1998. Failure to detect changes to people during a real world interaction. *Psychonomic Bulletin and Review*. 5: 644–669.



The second actor could have different clothing and different hair color, yet more than 50 percent of the time the unsuspecting participants failed to notice the substitution. Incredibly, people even failed to notice a change in gender! In some of the experiments, a male actor started the dialogue and a female actor was substituted under the cover of the two workmen with the door, but still most people failed to spot the switch.

What is going on here? On the one hand, we have a subjective impression of being aware of everything, on the other hand, it seems, we see very little. How can this extraordinary finding be reconciled with our vivid impression that we see the whole visual environment? The solution, as psychologist Kevin O'Regan* puts it, is that "The world is its own memory." We see very little at any given instant, but we can sample any part of our visual environment so rapidly with swift eye movement, that we think we have all of it at once in our consciousness experience. We get what we need, when we need it. The reason why the unwitting participants in Simons and Levin's experiment failed to notice the changeover was that they were doing their best to concentrate on the map, and although they had undoubtedly glanced at the face of the person holding it, that information was not critical and was not retained. We have very little attentional capacity, and information unrelated to our current task is quickly replaced with something we need right now.

There is a very general lesson here about seeing and cognition. The brain, like all biological systems, has become optimized over millennia of evolution. Brains have a very high level of energy consumption and must be kept as small as possible, or our heads would topple us over. Keeping a copy of the world in our brains would be a huge waste of cognitive resources and completely unnecessary. It is much more efficient to have rapid access to the actual world—to see only what we attend to and only attend to what we need—for the task at hand.

*Kevin O'Regan's essay on the nature of the consciousness illusion brings into clear focus the fact that there is a major problem to be solved, how do we get a subjective impression of perceiving a detailed world, while all available evidence shows that we pick up very little information. It also points to the solution—just in time processing. J.K. O'Regan, 1992. Solving the "real" mysteries of visual perception: The world as an outside memory. *Canadian Journal of Psychology*. 46: 461–488.

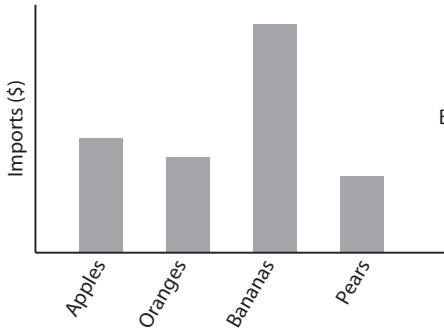
The one-tenth of a second or so that it takes to make an eye movement is such a short time in terms of the brain's neuron-based processing clock that it seems instantaneous. Our illusory impression that we are constantly aware of everything happens because our brains arrange for eye movements to occur and the particularly relevant information to be picked up just as we turn our attention to something we need. We do not have the whole visual world in conscious awareness. In truth, we have very little, but we can get anything we need through mechanisms that are rapid and unconscious. We are unaware that time has passed and cognitive effort has been expended. Exactly how we get the task-relevant information and construct meaning from it is a central focus of this book.

The understanding that we only sample the visual world on a kind of need-to-know basis leads to a profoundly different model of perception, one that has only emerged over the last decade or so as psychologists and neurophysiologists have devised new techniques to probe the brain.

According to this new view, visual thinking is a process that has the allocation of attention as its very essence. Attention, however, is multifaceted. Making an eye movement is an act of attending. The image on the retina is analyzed by further attention-related processes that tune our pattern-finding mechanisms to pull out the pattern most likely to help with whatever we are doing. At a cognitive level, we allocate scarce “working memory” resources to briefly retain in focal attention only to those pieces of information most likely to be useful. Seeing is all about attention. This new understanding leads to a revision of our thinking about the nature of visual consciousness. It is more accurate to say that we are conscious of the *field of information* to which we have *rapid access* rather than that we are immediately conscious of the world.

This new understanding also allows us to think about graphic design issues from a new and powerful perspective. We can now begin to develop a science of graphic design based on a scientific understanding of visual attention and pattern perception. To the extent to which it is possible to set out the message of this book in a single statement, the message is this: *Visual thinking consists of a series of acts of attention, driving eye movements and tuning our pattern-finding circuits.* These acts of attention are called *visual queries*, and understanding how visual queries work can make us better designers. When we interact with an information display, such as a map, diagram, chart, graph, or a poster on the wall, we are usually trying to solve some kind of cognitive problem. In the case of the map, it may be how to get from one location to another. In the case of the graph, it may be to determine the trend; for example, is the population

increasing or decreasing over time? What is the shape of the trend? The answers to our questions can be obtained by a series of searches for particular patterns—visual queries.



To find out which kind of fruit import is the largest by dollar value we make visual queries to find the tallest bar, then find and read the label beneath.



To find a fast route, we first make visual queries to find the starting and ending cities, then we make queries to find a connected red line, indicative of fast roads, between those points.

At this point, you may be considering an obvious objection. What about the occasions when we are not intensely involved in some particular task? Surely we are not continually constructing visual queries when we are sitting in conversation with someone, or strolling along a sidewalk, or listening to music. There are two answers to this. The first is that, indeed, we are not always thinking visually with reference to the external environment; for example, we might be musing about the verbal content of a conversation we had over the telephone. The second is we are mostly unaware of just how structured and directed our seeing processes are. Even when we are in face-to-face conversation with someone, we constantly monitor facial expressions, the gestures and gaze direction of that person, to pick up cues that supplement verbal information. If we walk on a path along the sidewalk of a city, we constantly monitor for obstacles and choose a path to take into account the other pedestrians. Our eyes make anticipatory movements to bumps and stones that may trip us, and our brains detect anything that may be on a trajectory to cross our path, triggering an eye movement to monitor it. Seeing while walking is, except on the smoothest and most empty road, a highly structured process.

To flesh out this model of visual thinking, we need to introduce key elements of the apparatus of vision and how each element functions.

THE APPARATUS AND PROCESS OF SEEING

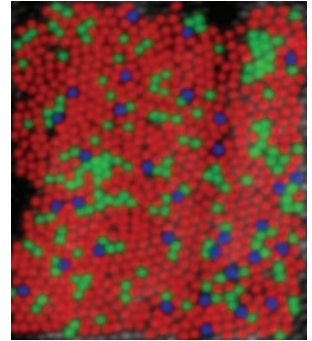
The eyes are something like little digital cameras. They contain lenses that focus an image on the eyeball. Many find the fact that the image is upside-down at the back of the eye to be a problem. But the brain is a computer, albeit quite unlike a digital silicon-based one, and it is as easy for the brain to compute with an upside-down image as a right-side-up image.

Just as a digital camera has an array of light-sensitive elements recording three different color values, so the eye also has an array of light-sensitive cones recording three different colors (leaving aside rods*). The analogy goes still further. Just as digital cameras compress image data for more compact transmission and storage, so several layers of cells in the retina extract what is most useful. As a result, information can be transmitted from the 100 million receptors in the eye to the brain by means of only 1 million fibers in the optic nerve.

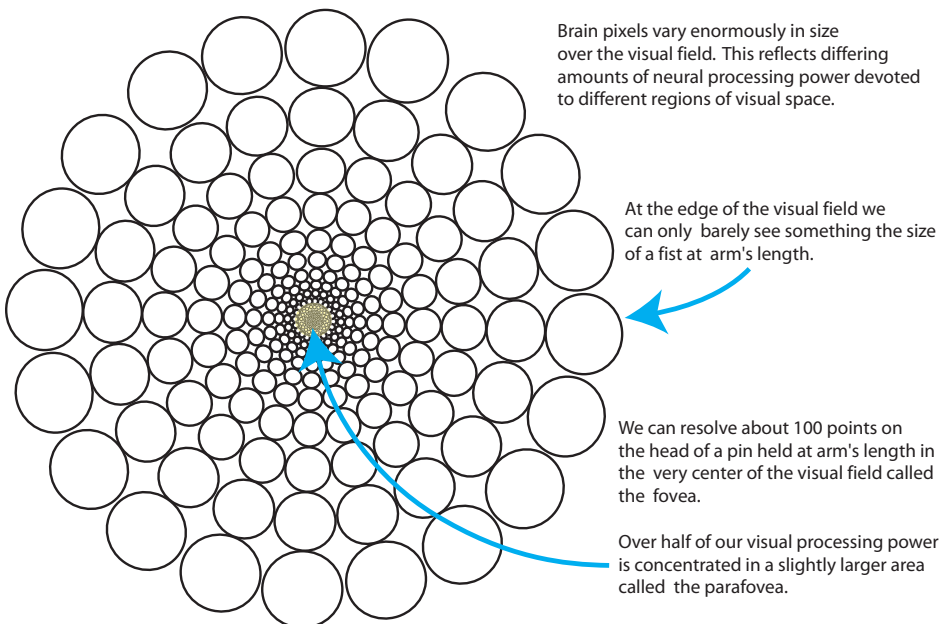
There is, however, a profound difference between the signal sent from the eye to the back of the brain for early-stage processing and the signal sent to a memory chip from the pixel array of a digital camera. *Brain pixels* are concentrated in a central region called the *fovea*, whereas camera pixels are arranged in a uniform grid. Also, brain pixels function as little image-processing computers, not just passive recorders.

Visual detail can only be seen via the fovea, at the very center of the visual field. Our vision is so good in this region that each eye can resolve about 100

*The human eye actually contains four different receptor types, three cone types and rods. However, because rods function mainly only at low light levels, in our modern, brightly lit world we can for all practical purposes treat the eye as a three-receptor system. It is because of this that we need only three different wavelength receptors in digital cameras.

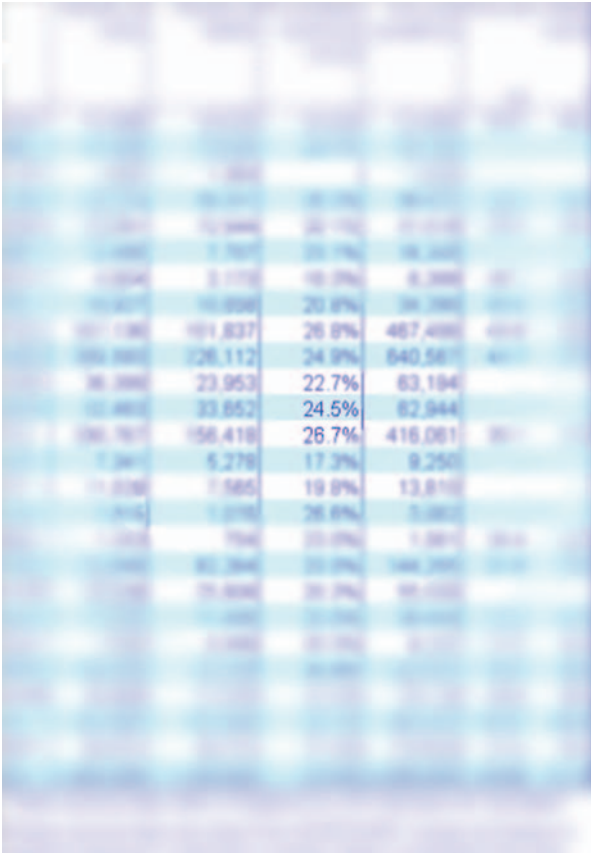


At the back of the eye is a mosaic of photo-receptors. Each responds to the amount of light falling on it. This example shows the central foveal region where three different cone receptors register different colors of light.

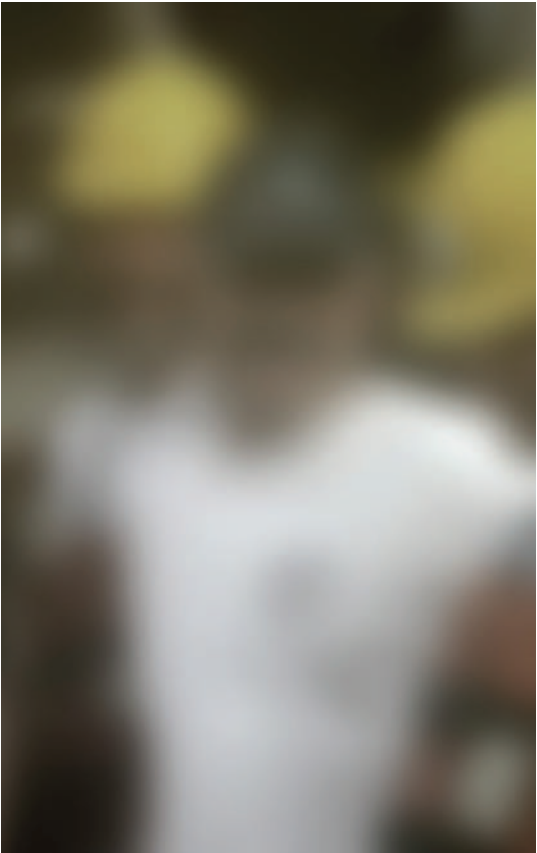


points on the head of a pin held at arm’s length, but the region is only about the size of our thumbnail held at arm’s length. At the edge of the visual field, vision is terrible; we can just resolve something the size of a human head. For example, we may be vaguely aware that someone is standing next to us, but unless we have already glanced in that direction we will not know who it is.

The non-uniformity of the visual processing power is such that half our visual brain power is directed to processing less than 5 percent of the visual world. This is why we have to move our eyes; it is the only way we can get all that brain power directed where it will be most useful. Non-uniformity is also one of the key pieces of evidence showing that we do not comprehend the world all at once. We cannot possibly grasp it all at once since our nervous systems only process details in a tiny location at any one instant.



We only process details in the center of the visual field. We pick up information by directing our foveas using rapid eye movements.



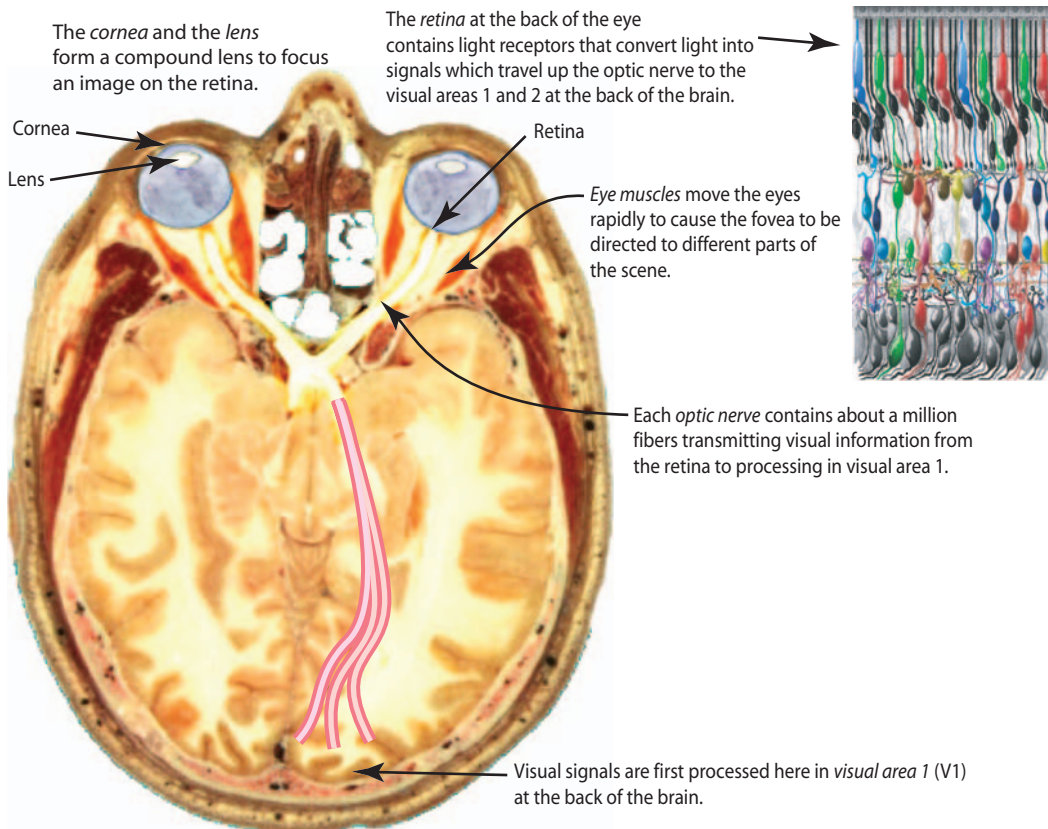
At the edge of the visual field, we can barely see that someone is standing next to us.

The term brain pixel was introduced earlier by way of contrast with digital camera pixels. Brain pixels provide a kind of distorted neural map covering the whole visual field. There are a great many tiny ones processing information from central regions where we direct our gaze and only a few very large ones processing information at the edge of the visual field. Because of this, we cannot see much out of the corners of our eyes.

Strong eye muscles attached to each eyeball rotate it rapidly so that different parts of the visual world become imaged on the central high-resolution fovea. The muscles accelerate the eyeball to an angular velocity up to 900 degrees per second, then stop it, all in less than one-tenth of a second. This movement is called a saccade, and during a saccadic eye movement, vision is suppressed. The eyes move in a series of jerks pointing the fovea at interesting and useful locations, pausing briefly at each, before flicking to the next point of interest. Controlling these eye movements is a key part of the skill of seeing.

We do not see the world as jerky, nor for the most part are we aware of moving our eyes, and this adds yet more evidence that we do not perceive what is directly available through our visual sense.

This illustration is based on a slice through the head of the visible man at the level of the eyeballs. Its color has been altered to show the eye muscles and the eyeballs more clearly.



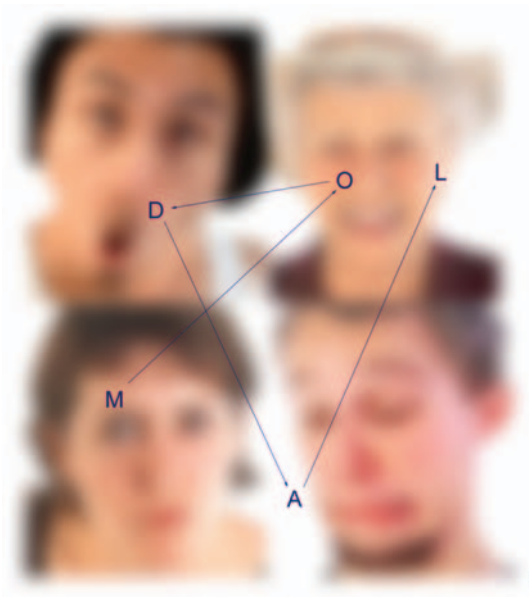
THE ACT OF PERCEPTION

The visual field has a big hole in it. Cover your left eye and look at the X. Move the page nearer and farther away, being sure to keep the X and the B horizontally aligned.

X

B

At some point the B should disappear. This is because the image of the B is falling on the blind spot, a region of the retina where there are no receptors at the point where the optic nerve and blood vessels enter the eye. We are unaware that we have this hole in our visual field. The brain does not know that it has a blind spot, just as it does not know how little of the world we see at each moment. This is more evidence that seeing is not at all the passive registration of information. Instead it is active and constructive.



Broadly speaking, the act of perception is determined by two kinds of processes: *bottom-up*, driven by the visual information in the pattern of light falling on the retina, and *top-down*, driven by the demands of attention, which in turn are determined by the needs of the tasks. The picture shown above is designed to demonstrate how top-down attention can influence what you see and how.

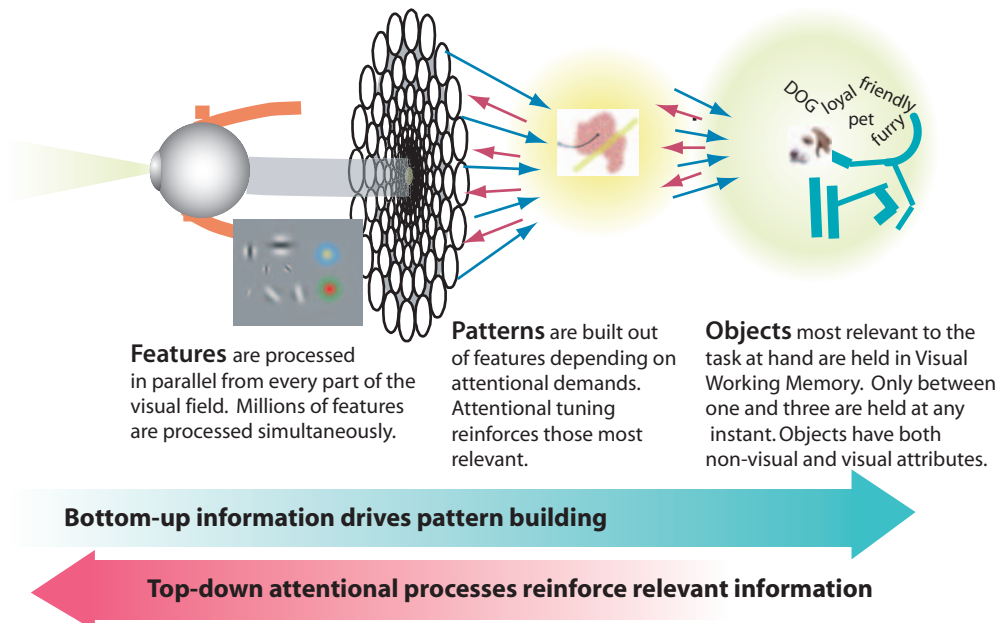
First look at the letters and lines. Start with the M and follow the sequence of lines and letters to see what word is spelled. You will find yourself making

a series of eye movements focusing your visual attention on the small area of each letter in turn. You will, of course, notice the faces in the background but as you perform the task they will recede from your consciousness.

Next look at the faces and try to interpret their expressions. You will find yourself focusing in turn on each of the faces and its specific features, such as the mouth or eyes, but also as you do this the letters and lines will recede from your consciousness. Thus, what you see depends on both the information in the pattern on the page as it is processed bottom-up through the various neural processing stages, and on the top-down effects of attention that determines both where you look and what you pull out from the patterns on the page.

There are actually two waves of neural activity that occur when our eye alights on a point of interest. An information-driven wave passes information first to the back of the brain along the optic nerve, then sweeps forward to the forebrain, and an attention-driven wave originates in the attention control centers of the forebrain and sweeps back, enhancing the most relevant information and suppressing less relevant information.

The neural machinery of the visual system is modular in the sense that distinct regions of the brain perform specific kinds of computations before passing the processed information on to some other region. The visual system has at least two dozen distinct processing modules, each performing some different computational task, but for the purposes of this overview we



will simplify to a three-stage model. The processing modules are organized in a hierarchy, with information being transferred both up and down from low-level brain-pixel processors to pattern and object processors. We shall consider it first from a bottom-up perspective, and then from a top-down perspective.

BOTTOM-UP

In the bottom-up view, information is successively selected and filtered so that meaningless low-level features in the first stage form into patterns in the second stage, and meaningful objects in the third stage.

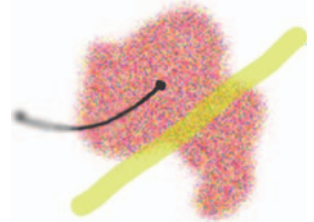
The main *feature processing* stage occurs after information arrives in the V1 cortex, having traveled up the optic nerve. There are more neurons devoted to this stage than any other. Perhaps as many as five billion neurons form a massively parallel processing machine simultaneously operating on information from only one million fibers in the optic nerve. Feature detection is done by several different kinds of brain pixel processors that are arranged in a distorted map of visual space. Some pull out little bits of size and orientation information, so that every part of the visual field is simultaneously processed for the existence of oriented edges or contours. Others compute red-green differences and yellow-blue differences, and still others process the elements of motion and the elements of stereoscopic depth. The brain has sufficient neurons in this stage to process every part of the visual field simultaneously for each kind of feature information. In later chapters, we will discuss how understanding features processing can help us design symbols that stand out distinctly.

At the intermediate level of the visual processing hierarchy, feature information is used to construct increasingly complex *patterns*. Visual space is divided up into regions of common texture and color. Long chains of features become connected to form continuous contours. Understanding how this occurs is critical for design because this is the level at which space becomes organized and different elements become linked or segregated. Some of the design principles that emerge at this level have been understood for over seventy years through the work of Gestalt psychology (*gestalt* means form or configuration in German). But there is also much that we have learned in the intervening years through the advent of modern neuroscience that refines and deepens our understanding.

At the top level of the hierarchy, information that has been processed from millions upon millions of simple features has been reduced and distilled through the pattern-processing stage to a small number of visual *objects*. The system that holds about three objects in attention at one



Some neurons that process elementary features respond to little packets of orientation and size information. Others respond best to redness, yellowness, greenness, and blueness. Still others respond to different directions of motion.



At the intermediate pattern finding stage of the visual system, patterns are formed out of elementary features. A string of features may form the boundary of a region having a particular color. The result is that the visual field is segmented into patterns.

time is called *visual working memory*. The small capacity of visual working memory is the reason why, in the experiment described at the start of this chapter, people failed to recognize that they were speaking to a different person. The information about the person they were talking to became displaced from their visual (and verbal) working memories by more immediate task-relevant data.

Although something labeled “dog” might be one of the objects we hold in our visual working memory, there is nothing like a picture of a dog in the head; rather we have a few visual details of the dog that have been recently fixated. These visual details are linked to various kinds of information that we already know about dogs through a network of association, and therein lies the power of the system. Concepts that dogs are loyal, pets, furry, and friendly may become activated and ready for use. In addition, various possibilities for action may become activated, leading to a heightened state of readiness. Actions such as petting the dog or avoiding the dog (depending on our concepts) become primed for activation. Of course if it is our own dog, “Millie,” a much richer set of associations become activated and the possibilities for action more varied. This momentary binding together of visual information with nonvisual concepts and action priming is central to what it means to perceive something.

The reason why we can make do with only three or four objects extracted from the blooming buzzing confusion of the world is that these few objects are made up of exactly what we need to help us perform the task of the moment. Each is a temporary nexus of meaning and action. Sometimes nexus objects are held in mind for a second or two; sometimes they only last for a tenth of a second. The greatly limited capacity of visual working memory is a major bottleneck in cognition, and it is the reason why we must often rely on external visual aids in the process of visual thinking.

It is tempting to think of visual working memory as the place real visual thinking occurs, but this is a mistake. One reason it is easy to think this way is that this is the way computers work. In a digital computer, all complex operations on data occur in the central processing unit. Everything else is about loading data, getting it lined up so that it is ready to be processed just when it is needed, and sending it back out again. The brain is not like this. There is actually far more processing going on in the lower-level feature processing and pattern-finding systems of the brain than in the visual working memory. It is much more accurate to think of visual thinking as a multicomponent cognitive system. Each part does something that is relatively simple. For example, the intermediate pattern processors detect and pass on information about a particular red patch of color that



When we see something, such as a dog, we do not simply form an image of that dog in our heads. Instead, the few features that we have directly fixated are bound together with the knowledge we have about dogs in general and this particular dog. Possible behaviors of the dog and actions we might take in response to it are also activated.

happens to be imaged on a particular part of the retina. An instant later this red patch may come to be labeled as “poppies.”

In many ways, the real power of *visual* thinking rests in pattern finding. Often to see a pattern is to find a solution to a problem. Seeing the path to the door tells us how to get out of the room, and that path is essentially a kind of visual pattern. Similarly, seeing the relative sizes of segments in a pie chart tells us which company has the greatest market shares.

Responses to visual patterns can be thought of as another type of pattern. (To make this point we briefly extend the use of the word “pattern” beyond its restricted sense as something done at a middle stage in visual processing.) For most mundane tasks we do not think through our actions from first principles. Instead, a response pattern like walking towards the door is triggered from a desire to leave the room. Indeed it is possible to think of intelligence in general as a collaboration of pattern-finding processors.* A way of responding to a pattern is also a pattern, and usually one we have executed many times before. A very common pattern of seeing and responding is the movement of a mouse cursor to the corner of a computer interface window, together with a mouse click to close that window. Response patterns are the essence of the skills that bind perception to action. But they have their negative side, too. They also cause us to ignore the great majority of the information that is available in the world so that we often miss things that are important.

*This view of intelligence as a kind of hierarchy of pattern finding systems has been elaborated in *On Intelligence* by Jeff Hawkins and Sandra Blakeslee (Times Books, New York, 2004).

TOP-DOWN

So far we have been focusing on vision as a bottom-up process:

retinal image → features → patterns → objects

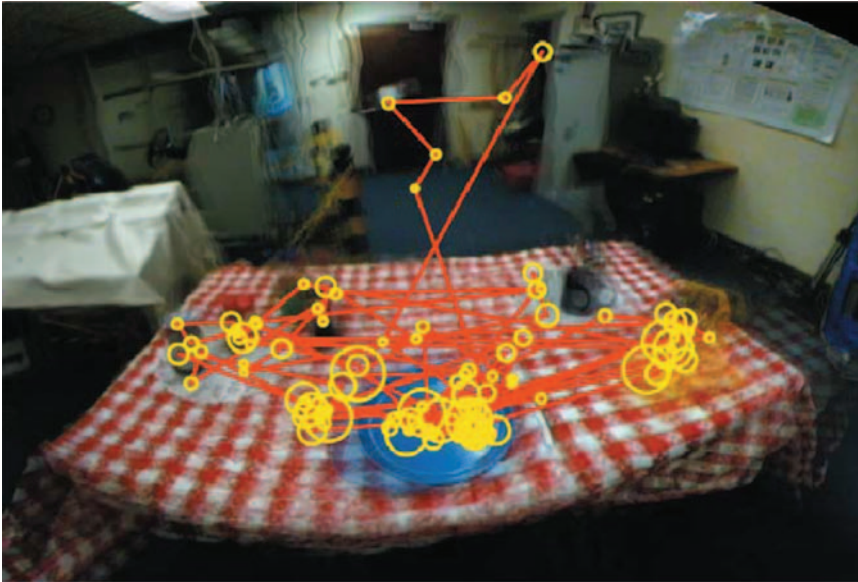
but every stage in this sequence contains corresponding top-down processes. In fact, there are more neurons sending signals back down the hierarchy than sending signals up the hierarchy.

We use the word *attention* to describe top-down processes. Top-down processes are driven by the need to accomplish some goal. This might be an action, such as reaching out and grasping a teacup or exiting a room. It might be a cognitive goal, such as understanding an idea expressed in a diagram. There is a constant linking and re-linking of different visual information with different kinds of nonvisual information. There is also a constant *priming* of action plans (so that if we have to act, we are ready) and action plans that are being executed. This linking and re-linking is the essence of high-level attention, but it also has implications for other lower-level processes.

At the low level of feature and elementary pattern analysis, top-down attention causes a bias in favor of the signals we are looking for. If we are looking for red spots then the red spot detectors will signal louder. If we are looking for slanted lines then slanted line feature detectors will have their signal enhanced. This biasing in favor of what we are seeking or anticipating occurs at every stage of processing. What we end up actually perceiving is the result of information about the world strongly biased according to what we are attempting to accomplish.

Perhaps the most important attentional process is the sequencing of eye movements. Psychologist Mary Hayhoe and computer scientist Dana Ballard collaborated in using a new technology that tracked individuals' eye movements while they were able to move freely.* This allowed them to study natural eye movements “in the wild” instead of the traditional laboratory setup with their heads rigidly fixed in a special apparatus.

*Mary Hayhoe and Dana Ballard. 2005. Eye movements in natural behavior. *Trends in Cognitive Science*. 9(4): 188–194.



The sequence of eye movements made by someone making a peanut butter and jelly sandwich. The yellow circles show the eye fixations. (Image courtesy of Mary M. Hayhoe.)

They had people carry out everyday tasks, such as making a peanut butter and jelly sandwich, and discovered a variety of eye movement patterns. Typically, people exhibited bursts of rapid eye movements when they first encountered the tools and ingredients laid out in front of them. This presumably allowed them to get a feel for the overall layout of the workspace. Each of these initial fixations was brief, usually one-tenth of a second or less. Once people got to work, they would make much longer fixations so that they could, for example, spread the peanut butter on the bread. Generally,

there was great economy in that objects were rarely looked at unnecessarily; instead, they were fixated using a “just-in-time strategy.” When people were performing some action, such as placing a lid on a jar, they did not look at what their hands were doing but looked ahead to the jar lid while one hand moved to grasp it. Once the lid was in hand, they looked ahead to fixate the top of the jar enabling the next movement of the lid. There were occasional longer-term look-aheads, where people would glance at something they might need to use sometime in the next minute or two. The overall impression we get from this research is of a remarkably efficient, skilled visual process with perception and action closely linked—the dominant principle being that we only get the information we need, when we need it.

How do we decide where to move our eyes in a visual search task? If our brains have not processed the scene, how do we know where to look? But if we already know what is there, why do we need to look? It’s a classic chicken and egg problem. The system seems to work roughly as follows.* Part of our brain constructs a crude map of the characteristics of the information that we need in terms of low-level features. Suppose I enter a supermarket produce section looking for oranges. My brain will tune my low-level feature receptors so that orange things send a stronger signal than patches of other colors. From this, a rough map of potential areas where there may be oranges will be constructed. Another part of my brain will construct a series of eye movements to all the potential areas on this spatial map. The eye movement sequence will be executed with a pattern processor checking off those areas where the target happened to be mangoes, or something else, so that they are not visited again. This process goes on until either oranges are found, or we decide they are probably hidden from view. This process, although efficient, is not always successful. For one thing, we have little color vision at the edges of the visual field, so it is necessary to land an eye movement near to oranges for the orange color-tuning process to work. When we are looking for bananas a shape-tuning process may also come into play so that regions with the distinctive curves of banana bunches can be used to aid the visual search.*

IMPLICATIONS FOR DESIGN

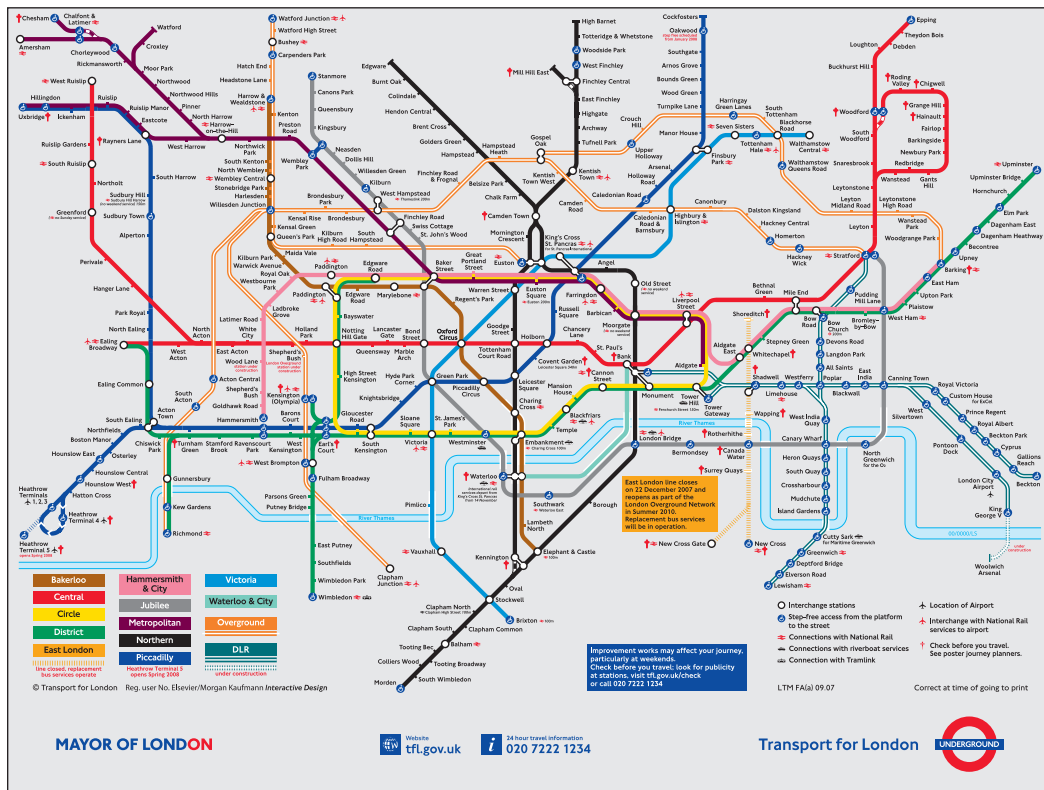
If we understand the world through just-in-time visual queries, the goal of information design must be to design displays so *that visual queries are processed both rapidly and correctly for every important cognitive task the display is intended to support*. This has a number of important ramifications for graphic design. The first is that in order to do successful design we must understand the cognitive tasks and visual queries a graphic is intended to support. This is normally done somewhat intuitively, but it can also be made explicit.

*J.R. Duhamel, C.L. Colby, and M.E. Goldberg, 1992. The updating of the representation of visual space in parietal cortex by intended eye movements. *Science*. Jan. 3; 255(5040):90–92.

*The book *Active Vision* by John Findlay and Ian Gilchrist (Oxford University Press, 2003) is an excellent introduction to the way eye movements are sequenced to achieve perception for action.

A map is perhaps the best example illustrating how graphic design can support a specific set of visual queries. Suppose we are lodged in a hotel near Ealing in West London, and we wish to go to a pub near Clapham Common where we will meet a friend. We would do well to consider the underground train system and this will result in our formulating a number of cognitive tasks. We might like to know the following:

- Which combination of lines will get us to the pub?
- If there is more than one potential route, which is the shortest?
- What are the names of stations where train changes are needed?
- How long will the trip take?
- What is the distance between the hotel and its nearest underground station?
- What is the distance between the pub and its nearest underground station?
- How much will it cost?
- Many of these tasks can be carried out through visual thinking with a map.



The famous London Underground map designed by Harry Beck is an excellent visual tool for carrying out some, but not all, of these tasks. Its clear schematic layout makes the reading of station names easy. The color coding of lines and the use of circles showing connection points make it relatively easy to visually determine routes that minimize the number of stops. The version shown here also provides rapidly accessible information about the fare structure through the grey and white zone bars.

Since we have previously used maps for route planning, we already have a cognitive plan for solving this kind of problem. It will likely consist of a set of steps something like the following.

STEP ONE is to construct a visual query to locate the station nearest our hotel. Assuming we have the name of a station, this may take quite a protracted visual search since there are more than two hundred stations on the map; if, as is likely, we already have a rough idea where in London our hotel is located, this will narrow the search space.

STEP TWO is to visually locate a station near the pub, and this particular task is also not well supported. The famous map does support the station-finding task by spacing the labels for clear reading, but unlike other maps it does not give an index or a spatial reference grid.

STEP THREE is to find the route connecting our start station and our destination station. This visual query is very well supported by the map. The lines are carefully laid out in a way that radically distorts geographical space, but this is done to maintain clarity so that the visual tracing of lines is easy. Color coding also supports visual tracing, as well as providing labels that can be matched to the table of lines at the side.

Suppose that the start station is Ealing on the District Line (green) and the destination station is Clapham Common on the Northern Line (black). Our brains will break this down into a set of steps executed roughly as follows. Having identified the Ealing station and registered that it is on the green-colored District Line, we make a series of eye movements to trace the path of this green contour. As we do so, our top-down attentional mechanisms will increase the amplification on neurons tuned to green so that they “shout louder” than those tuned to other colors, making it easier for our mid-level pattern finder to find and connect parts of the green contour. It may take several fixations to build the contour, and the process will take about two seconds. At this point, we repeat the tracing operation starting with Clapham Common, our destination station, which is on the black Northern Line. As we carry out this tracing, a second process may be operating in parallel to look for the crossing point with the green line. These operations might take another second or two.

Because of the very limited capacity of visual working memory, most information about the green contour (District Line) will be lost as we trace out the black contour (Northern Line). Not all information about the green line is lost, however; although its path will not have been retained explicitly, considerable savings occur when we repeat an operation such

as this. Repeating a tracing operation will take less cognitive effort, and require fewer fixations, than finding it in the first place. A hallmark of visual thinking is that it is often easier to redo some cognitive operation than to remember it.

Of course, the process may not be quite as straightforward as described. Much can go wrong. We may be visually sidetracked by the wrong branch of the Northern Line. The process is flexible and adaptive, and good visual thinkers have error-checking procedures. A final scan of the entire route will confirm that it is, indeed, a valid solution.

STEP FOUR is a visual query to get a rough estimate of how many stations there are on the route. This is unlikely to involve actual counting; instead, it will be a judgment that will be used together with prior experience to produce an estimate of the journey duration. This will naturally lead to misjudgments because the lengths of lines on the map do not correspond to travel times.

Our analysis shows that the London Underground map supports the visual route-finding step well but almost completely fails to support other planning steps such as finding a tube station near to the pub and estimating journey time. This is not to say that it is a bad design; the map is justifiably famous. The designers sacrificed spatial accuracy in favor of clear labels and routes, and therefore the map is very poor for providing information about the distance covered on the ground.

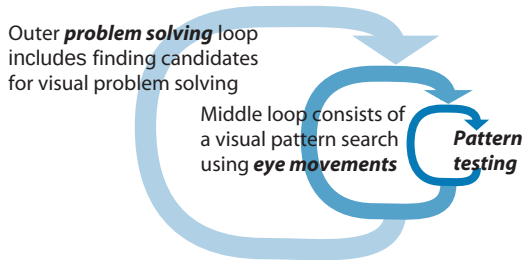
Having a computer behind an *interactive* graphic display takes the capability of visual cognitive tools to another level. A program called *tubeguru*[™] is an Internet-based computer program that adds many of the cognitive supports that the underground map lacks; a local street map in the vicinity of a selected station helps with the start and end of journeys, and a journey planner provides information about how to get from a specific address to another specific address, using London's busses and trains.

Tubeguru and other computer programs with visual interfaces add an additional dimension to the cognitive process. They allow for some parts of the computation process to occur in the brain, and other parts to occur in the computer program. Human and computer together form a *problem-solving system* with the screen display and the keyboard providing the interface. In later chapters, we further explore this rather mind-bending idea. For now we return to the simpler case of visual thinking with a static graphic display.

NESTED LOOPS

A useful way of describing the way the brain operates to solve problems is as a set of nested loops. Outer loops deal with generalities. Inner loops process the details. In the outer loop, the brain constructs a set of steps to solve

the problem and then executes them: find a map, find end point stations, trace lines, find intersections. This sequencing of the problem components is not in and of itself visual in nature; it is more likely to occur through the operations of verbal language subsystems of the brain. However, some problem components may be identified as having visual-thinking solutions, and these are used to construct visual queries on the display.



The middle loop on the diagram is a visual search executed to find patterns that address the visual query. This involves executing a sequence of eye movements. In the previous example, this involved a visual tracing of the colored contours that represent train lines.

The inner loop is activated when the eye arrives at a point of fixation. A process of visual testing begins, and patterns within the central region of the visual field are evaluated at a rate of about twenty per second; although since the eye only stays in one place for less than two-tenths of a second, roughly one to four simple patterns may be evaluated on each fixation. In the example of finding the green line, this evaluation might have the following form. Is this contour green? Is it a continuation of the section of contour that was just registered?

Of course we are being metaphoric here. Nothing in the brain is making up questions and answering them. Rather, a neural process produces a signal that results in a change in the status of objects held in visual working memory. One such change in status would be that a neural representation of a green contour object becomes “bound” to another section of green contour in the temporary nexus and held together through the process called visual working memory.

Nested processes are characteristic of computer programs, but the ones executed by the brain are far more flexible and adaptive than those executed in a computer. They rely on patterns of action that have been built up over experience. We already have encoded in our memories visual search patterns for a huge number of situations, such as what we need to

look for when entering a restaurant, looking at a web page, and driving a car toward an intersection. These are not, however, rigid rules, but rather like flexible plans that can be adapted to particular circumstances.

DISTRIBUTED COGNITION

This chapter has alluded to the existence of various brain structures and processes. In later chapters, specific areas of the brain where processes occur will be discussed in more detail. We will encounter visual areas, V1 and V2 specialized for parallel feature processing, V4 specialized for pattern processing, the Fusiform Gyrus specialized for object processing, and the frontal lobes as well as midbrain areas specialized for controlling high-level attention. The control of eye movements is mediated by mechanisms in the lateral interparietal area and superior colliculus. There are several dozen areas of the brain that have been mapped in animals using recordings from single neurons. Recent advances in brain scanning have allowed researchers to see which areas of the brain are most active when people perform specific visual tasks. The latest techniques even reveal the sequence in which different areas become active.

This is not, however, a book about neurophysiology, but a book about visual thinking. Its purpose is to provide a theoretical understanding of how we perceive in order to inform the design process. The names are useful but ultimately it does not matter exactly where in the brain something happens; instead it is important to know what kinds of visual information the brain can process efficiently. The timing and sequencing of visual operations is also important. Therefore, the approach taken in this book is functional and modular. Specific brain structures are mentioned because each has a specialized function in the visual thinking process. We could just talk about these brain structures as abstract functioning modules, but giving them their proper names preserves the link with the underlying science. Overall the goal is to steer a path between oversimplification and overwhelming neurophysiological detail.

There is a general point to be considered regarding the modular structure of the brain. The brain is not a holistic undifferentiated processor of information, as was once thought. Instead, it is made up of a number of specialized regions, each devoted to carrying out some specific processing task, such as scheduling eye movements or processing features. Visual thinking comes about through the coordinated activities of these processing regions. There is no central processing unit in the brain; rather the whole functions as a kind of *distributed* computer. Also, there is no central conductor coordinating all of the distributed parts. Each component

coordinates interactions with the other components to which it is connected. This is the reason why we can sometimes carry out more than one task simultaneously—talking and driving a car, for example. These tasks use regions of the brain that are mostly independent.

The idea of distributed cognition is no longer particularly controversial, but the idea of distributed cognition begs to be extended outside the head, and this is truly radical. Distributed cognition holds that cognition is the result of a set of interconnected processing modules, each doing something relatively simple and sending signals on to other modules. But why must these modules be inside the head? What is the difference between information stored in human memory and information stored in a book or picture? After all, we can store images in a photo album far better than we can store them in our head. We can store them even better on a computer if it has a effective search capability. Thinking involves a constant interplay between new patterns and old patterns, and patterns can come from both inside and outside the skull. As Don Norman famously noted,* *“The power of the unaided mind is highly overrated. Without external aids, memory, thought and reasoning are all constrained. But human intelligence is highly flexible and adaptive, superb at inventing procedures and objects that overcome its own limits. The real powers come from devising external aids that enhance cognitive activities. How have we increased memory, thought and reasoning? By the invention of external aids: it is things that make us smart.”*

Graphs, diagrams, and illustrations have only become widely available as visual thinking tools over the past two hundred years. More recently there has been an explosive development of diagramming techniques driven first by color printing technology and currently by the Internet. Graphic PowerPoint slides and the like have become a ubiquitous tool for information presentation. Increasingly, the tools that support cognition are computer-based, and increasingly they incorporate images and visualizations as well as words. The term *visualization* as it is used in the previous sentence is actually quite new. Visualizations used to be mental images that people formed while they thought. Now the term more often means a graphical representation of some data or concepts. Visualizations are becoming important in most areas of science and commerce. Advertisements have long been visualizations. All of these artifacts are tools for visual thinking.

CONCLUSION

Perceiving is a skilled active process. We seek out what we need through frequent eye movements, so that critical information falls on the

*Donald A. Norman, 1993. Cognition in the head and in the world. *Cognitive Science*. 17: 1–6.

high-resolution fovea. Eye movements are executed to satisfy our need for information and can be thought of as a sequence of visual queries on the visual world. Each time the eye briefly comes to rest, the pattern-processing machinery goes to work sorting out what is most likely to be relevant to our current cognitive task. Almost everything else is either not seen at all or retained for only a fraction of a second. A few fragments are held for a second or two, and a tiny percentage forms part of our long-term memory. To be sure we do have visual memories built up over our lifetimes and these are what make the scraps of information we capture meaningful and useful. These memories are not detailed, but they do provide frameworks for fleshing out the fragments. They also provide frameworks for planning eye movements and other kinds of actions.

The idea of the visual query is shorthand for what we do when we obtain information either from the world at large or some kind of information display. We make visual queries every time we search for some visual information that we need to carry out our cognitive task of the moment. Understanding what visual queries are easily executed is a critical skill for the designer. The special skill of designers is not so much skill with drawing or graphic design software, although these are undoubtedly useful, but the talent to analyze a design in terms of its ability to support the visual queries of others. This talent comes from hard-won pattern analysis skills that become incorporated into the neural fabric of perception, as well as the skill to execute a cognitive process that takes into account a variety of competing considerations.

One reason design is difficult is that the designer already has the knowledge expressed in the design, has seen it develop from inception, and therefore cannot see it with fresh eyes. The solution is to be analytic and this is where this book is intended to have value. Effective design should start with a visual task analysis, determine the set of visual queries to be supported by a design, and then use color, form, and space to efficiently serve those queries. Skilled graphic designers already do this intuitively. It is my earnest hope that this book will help in the transition from unskilled to skilled designer by providing visual analytic tools derived from an up-to-date understanding of human perception.

At this stage, it should be clear that this book is not about the kind of visual thinking that goes into fine art where the goals are frequently the opposite of clarity, but rather beauty, visual impact, or an investigation of a new vocabulary of expression. Because of its exploratory, pioneering nature, leading-edge art often speaks only to small cliques of insiders, collectors, critics, and gallery owners. This book is about graphic design

that provides a channel for clear communication that supports visual thinking and acts as an interface to the vast information resources of the modern world. This should not, however, imply that its message is the enemy of creative expression. There is never a single optimal solution to a design problem, but rather a huge variety of alternative clear and effective designs.

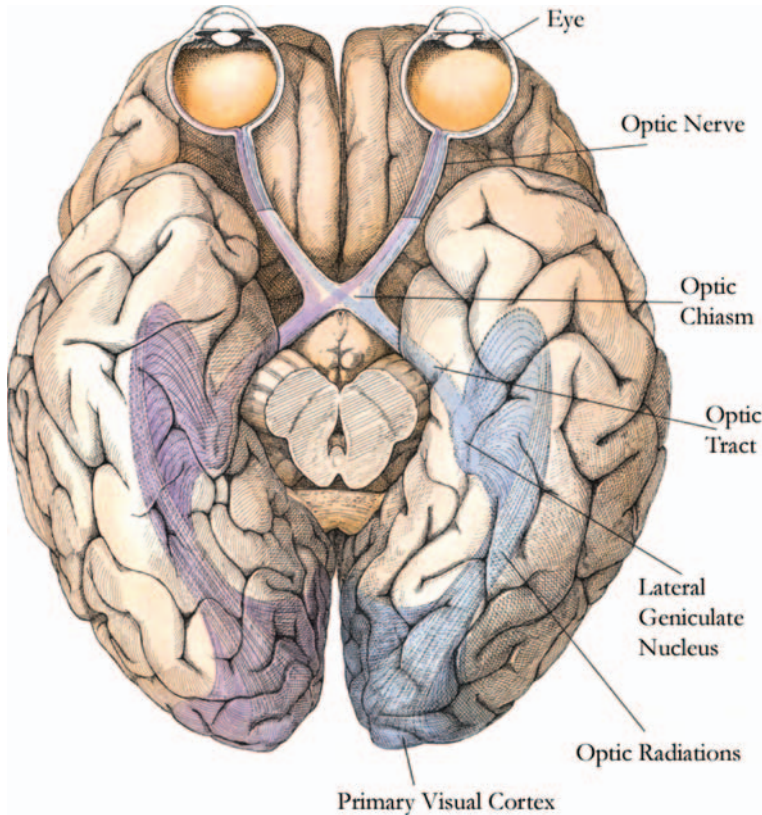
Chapter 2

What We Can Easily See

Imagine a thief searching a moonlit room with a flashlight. He can see the vague outlines of furniture, but no details at all. He uses the narrow beam of the flashlight to pick out the tops of dressers and side tables where he expects valuables may be. Our visual attention actually works something like this; we move the spotlight of our attention by moving our eyes from point to point, picking out details. Generally we only have vague information to plan each eye movement, so we often fail to find the information we seek on a particular eye fixation. But some things are very easy to find even at the edge of the visual field, like a blinking light over the water, or a bright patch of red sweater in a crowd of people wearing black. We can sense them in the periphery and then make an eye movement so that they become the center of both vision and attention.

This chapter is about the theory of vision that describes what makes something small easy to see. It is also about the nitty-gritty details of design. What does it take to make a graphic symbol that can be found rapidly? How can something be highlighted? The problem for the designer is to ensure all visual queries can be effectively and rapidly served. In practice, this means

ensuring that semantically meaningful graphic objects that make up a design each has the right amount of salience. The most important and frequent visual queries should be supported with the most visually distinct objects. The perceptual laws of visual distinctness are based on the low-level early-stage processing in the visual system. The elementary pattern-processing systems we find there provide the substrate on which all graphical interpretation is built.



A view of the brain from beneath. Light is transformed into neural signals by the retina. Information is then passed along the optic nerve, via the waystation of the lateral geniculate nucleus to the primary visual cortex located at the back of the brain.

In the primary visual cortex billions of neurons process the entire image, providing the elements of form, color, texture, stereoscopic depth, and motion all at once.

In the following set of random letters, two p's have been highlighted with a blue background. Finding those two p's is easy. They seem to pop out from the page. In contrast, it is more difficult to find the two q's. To find the q's every letter must be scanned and it will take at least ten times longer than finding the p's. Further, finding the q's will impose a much greater cognitive burden, certainly disrupting one's train of thought.

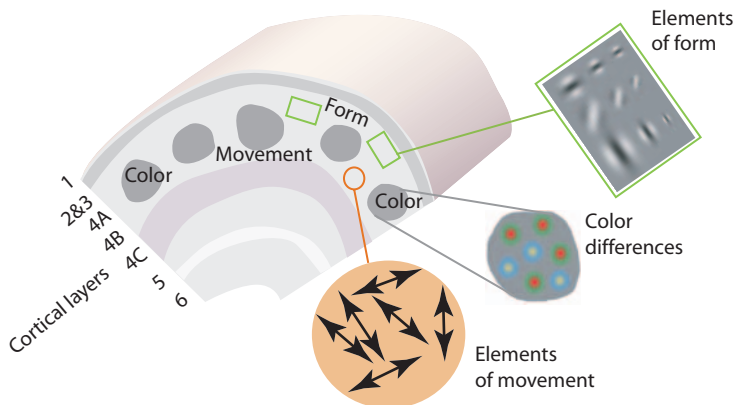
ehklhfdiyaoryweklblkhockxlyhirhupwerlkhlkuyxoiasusifdh
lksajdhflkihqdaklljerlajeslselusdslfjsalsuslcljlsdsjaf;ljdulafjluj
oufojrtpjhklghqlkshlkhldshflymcvciwopzlsifhrmckreieui

The fact that the p's are easy to see seems straightforward and unsurprising. But why do the q's take longer to find? After all, we all spend thousands of hours reading and writing with this alphabet. So the q shape is not unfamiliar.

THE MACHINERY OF LOW-LEVEL FEATURE ANALYSIS

In order to understand why the p's are easy and the q's are hard on the previous page we need to dig more deeply into what is going on during the earlier stages of visual processing. The neural architecture of the primary visual cortex has been mapped in detail through experiments in which neurophysiologists inserted very small electrical probes into single neurons in the brains of various animals. David Hubel and Torsten Wiesel pioneered this method in the 1960s. They discovered that at the back of the brain in a region called the primary visual cortex (also called visual area 1, or V1) cells would “fire,” thereby emitting a series of spikes of electrical current when certain kinds of patterns were put in front of an animal's eyes.* Through many hundreds of such experiments, neuroscientists have discovered that V1 is a kind of tapestry of interlocking regions where different kinds of information are processed. The elements of color, shape, texture, motion, and stereoscopic depth are processed by different interwoven areas. Visual area 2 (V2) receives input from V1. V2's neurons respond to slightly more complex patterns, based on the processing already done in V1.

*David Hubel and Torsten Wiesel obtained a Nobel prize for this groundbreaking research in 1981.



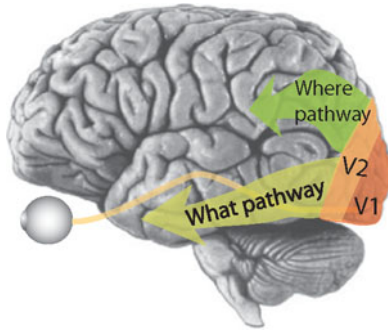
A fold of the primary visual cortex (V1) contains many layers of neurons. Signals from the retina, via the lateral geniculate nucleus, come up through several layers, dense with cross connections to layers 3 and 4 where individual neurons each respond to a specific simple feature.

There are a million nerve fibers providing input from each eye to the V1 area where several billion neurons process information. Information is then passed on to area V2 where several billion more neurons process it at a more complex level. Areas V1 and V2 can each be thought of as a parallel computer, far more complex and powerful than anything humans

have built to date. Parallel computing is the name given to computer systems that are made up of many small computers, each doing essentially the same thing on a different piece of data. These kinds of computers are used to solve problems such as weather forecasting, where each processor deals with a little bit of the atmosphere. These cortical areas are *parallel* computers because they process every part of the visual image simultaneously, computing local orientation information, local color difference information, local size information, and local motion information.

WHAT AND WHERE PATHWAYS

In this chapter, we are mostly concerned with what goes on in the primary visual cortex, but it is useful to look ahead a bit to see where the information is going. V1 and V2 provide the inputs to two distinct processing systems called the *what* and the *where* systems, respectively.*



*Area V3 is also part of both what and where pathways but its processing role is still uncertain.

The what pathway sweeps forward from V1 and V2 along the lower edge of the brain on each side, processing information about the identity of an object.

The where pathway sweeps forward higher up on the brain and processes information about where objects in the world are located.

The what pathway is concerned with the identification of objects in the environment. It helps identify if a particular pattern of light and color represents a chicken, a Volkswagen, or Aunt Mabel. The where pathway is concerned with the location of information and with guiding actions in the world, such as reaching out and grabbing things, moving from place to place, or making eye movements. What makes an object easy to find is how easily we can direct a rapid eye movement to focus our attention on it.

EYE MOVEMENT PLANNING

Visual search is not random; however, there is a chicken and egg problem. If we are looking for something smallish, we can only see it when we are looking at it. But how do the eyes get directed to the right locations if the information has not been processed? The answer is that there is very limited pre-processing that is used to direct attention. To understand this is to understand what is easy to see.*

Part of the machinery is a mechanism called *biased competition*. If we are looking for tomatoes, then it is as if an instruction has been issued.

*The theory of visual search presented here is from J.M. Wolf, 1998. Visual search. In *Attention*. H. Paschler, Ed. Psychology Press.

“All you red-sensitive cells in V1, you all have permission to shout louder. All you blue- and green-sensitive cells, try to be quiet.” Similar instructions can be issued for particular orientations, or particular sizes—these are all features processed by V1. The responses from the cells that are thereby sensitized are passed both up the *what* pathway, biasing the things that are seen, and up the *where* pathway, to regions that send signals to make eye movements occur. Other areas buried deeper in the brain, such as the hippocampus, are also involved in setting up actions.

In a search for tomatoes all red patches in the visual field of the searcher become candidates for eye movements, and the one that causes red-sensitive neurons to shout the loudest will be visited first. The same biased shouting mechanism also applies to any of the feature types processed by the primary visual cortex, including orientation, size, and motion. The important point here is that knowing what the primary visual cortex does tells us what is easy to find in a visual search.

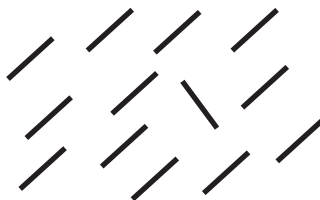
This is by no means the whole story of eye-movement planning. Prior knowledge about where things are will also determine where we will look. However, in the absence of prior knowledge, understanding what will stand out on a poster, or computer screen, is largely about the kinds of features that are processed early on, before the where and what pathways diverge.

WHAT STANDS OUT = WHAT WE CAN BIAS FOR

Some things seem to *pop out* from the page at the viewer. It seems you could not miss them if you tried. For example, **Anne** name almost certainly popped out at you the moment you turned to this page. Ann Triesman is the psychologist who was the first to systematically study the properties of simple patterns that made them easy to find. Triesman carried out dozens of experiments on our ability to see simple colored shapes as distinct from other shapes surrounding them.*

Triesman's experiments consisted of visual search tasks. Subjects were first told what the target shape was going to be and given an instant to look at it; for example, they were told to look for a tilted line. \

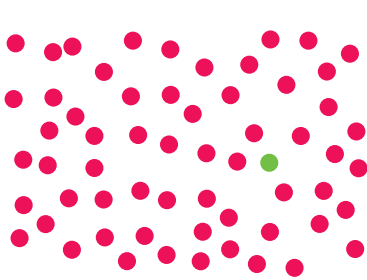
Next they were briefly exposed to that same shape embedded in a set of other shapes called distracters.



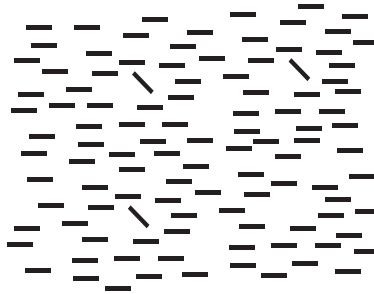
*A large number of popout experiments are summarized in A. Triesman, and S. Gormican, 1988. Feature analysis in early vision: Evidence from search asymmetries, *Psychological Review*. 95(1): 15–98.

They had to respond by pressing a “yes” button if they saw the shape, and a “no” button if they did not see it. In some trials, the shape was present, and in others it was not.

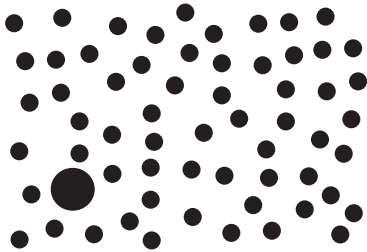
The critical finding was that for certain combinations of targets and distracters the time to respond *did not* depend on the number of distracters. The shape was just as distinct and the response was just as fast if there were a hundred distracters as when there was only one. This suggests a parallel automatic process. Somehow all those hundred things were being eliminated from the search as quickly as one. Triesman claimed that the effects being measured by this method were *pre-attentive*. That is, they occurred because of automatic mechanisms operating prior to the action of attention and taking advantage of the parallel computing of features that occurs in V1 and V2.



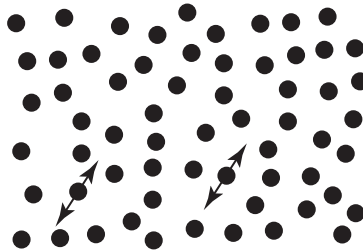
The green dot pops out



The oblique lines pop out



The large circle pops out



If two dots were to oscillate as shown they would pop out

Pop-out effects depend on the relationship of a visual search target to the other objects that surround it. If that target is distinct in some feature channel of the primary visual cortex we can program an eye movement so that it becomes the center of fixation.

Although these studies have contributed enormously to our understanding of early-stage perceptual processes, pre-attentive has turned out to be an unfortunate choice of term. Intense concentrated attention is required for the kinds of experiments Triesman carried out, and her subjects were all required to *focus their attention on the presence or absence of*

a *particular target*. They were paid to attend as hard as they could. More recent experiments where subjects were not told of the target ahead of time show that all except the most blatant targets are missed.* To be sure, some things shout so loudly that they pop out whether we want them to or not. A bright flashing light is an example. But most of the visual search targets that Triesman used would not have been seen if subjects had not been told what to look for, and this is why pre-attentive is a misnomer.

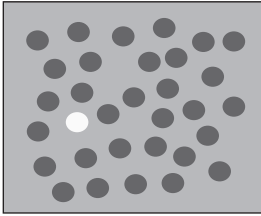
A better term would be *tunable*, to indicate those visual properties that can be used in the planning of the next eye movement. Triesman's experiments tell us about those kinds of shapes that have properties to which our eye-movement programming system is sensitive. These are the properties that guide the visual search process and determine what we can easily see.

The strongest pop-out effects occur when a single target object differs in some feature from all other objects and where all the other objects are identical, or at least very similar to one another. Visual distinctness has as much to do with the visual characteristics of the *environment* of an object as the characteristics of the object itself. It is the degree of feature-level *contrast* between an object and its surroundings that make it distinct. From the purposes of understanding pop-out, contrast should be defined in terms of the basic features that are processed in the primary visual cortex. The simple features that lead to pop out are color, orientation, size, motion, and stereoscopic depth. There are some exceptions, such as convexity and concavity of contours, that are somewhat mysterious, because primary visual cortex neurons have not yet been found that respond to these properties. But generally there is a striking correspondence between pop-out effects and the early processing mechanisms.

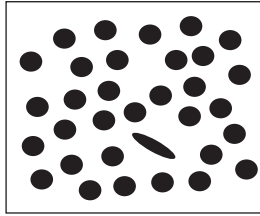
Something that pops out can be seen in a single eye fixation and experiments show that processing to separate a pop-out object from its surroundings actually takes less than a tenth of a second. Things that do not pop out require several eye movements to find, with eye movements taking place at a rate of roughly three per second. Between one and a few seconds may be needed for a search. These may seem like small differences, but they represent the difference between visually efficient at-a-glance processing and cognitively effortful search.

So far we have only looked at patterns that show the pop-out effect. But what patterns *do not* show pop-out? It is equally instructive to examine these. At the bottom of the following page, there is a box containing a number of red and green squares and circles. When you turn the page, look for the green squares.

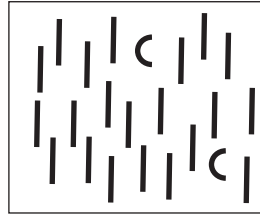
*The phenomenon of not seeing things that should be obvious is called inattentional blindness. A. Mack and I. Rock, 1998. *Inattentional Blindness*. MIT Press, Cambridge, MA.



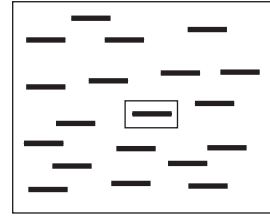
Grey value



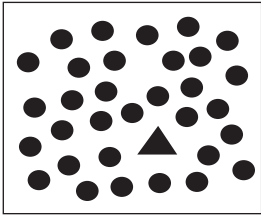
Elongation



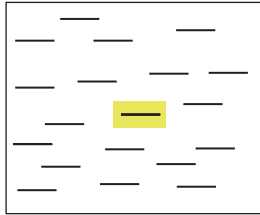
Curvature



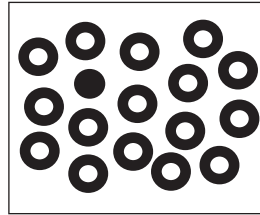
Added surround box



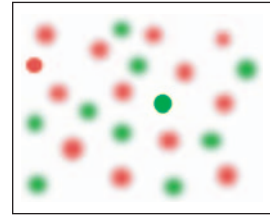
Shape



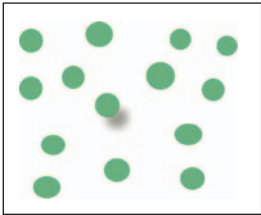
Added surround color



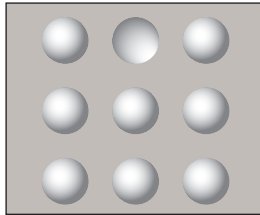
Filled



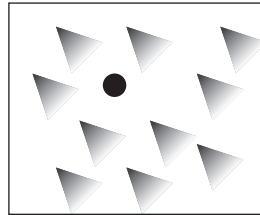
Sharpness



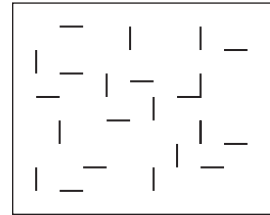
Cast shadow



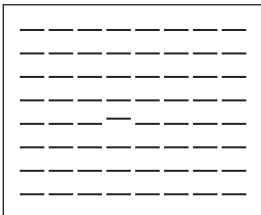
Convex and concave



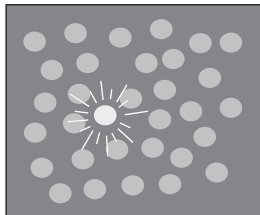
Sharp vertex



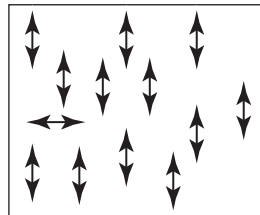
Joined lines



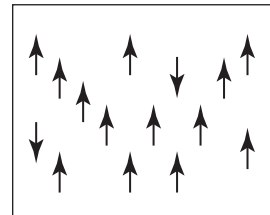
Misalignment



Blinking

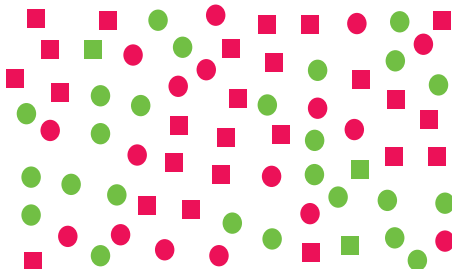


Direction of motion



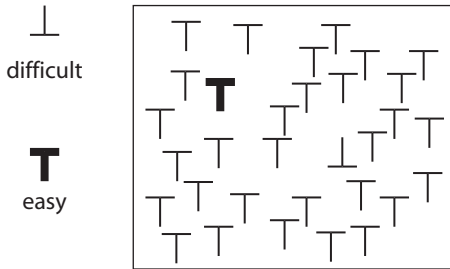
Phase of motion

What makes a feature distinct is that it differs from the surrounding feature in terms of the signal it provides to the low-level feature-processing mechanisms of the primary visual cortex.

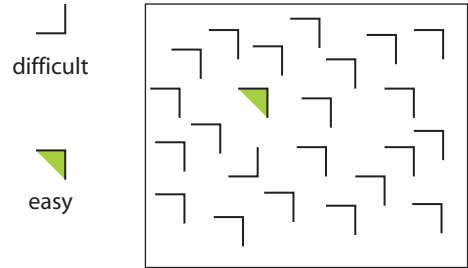


There are three green squares in this pattern. The green squares do not show a pop-out effect, even though you know what to look for. The problem is that your primary visual cortex can either be tuned for the square shapes, or the green things, but not both.

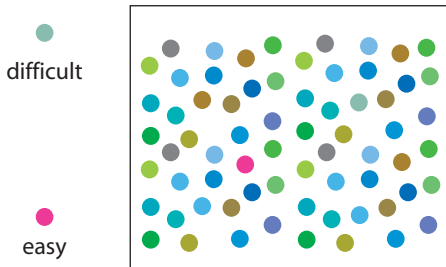
Trying to find a target based on two features is called a visual conjunction search, and most visual conjunctions are hard to see. Neurons sensitive to more complex conjunction patterns are only found farther up the *what* processing pathway, and these cannot be used to plan eye movements. In each of the following examples, there is something that is easy to find and something that is not easy to find. The easy-to-find things can be differentiated by V1 neurons. The hard-to-find things can only be differentiated by neurons farther up the *what* pathway.



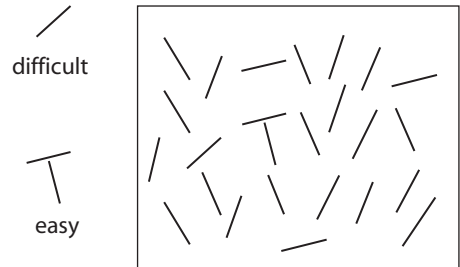
The inverted T has the same feature set as the right-side-up T and is difficult to see. But the bold T does support pop-out and is easy to find.



Similarly the backwards L has the same feature set as the other items, making it difficult to find. But the green triangle addition does pop out.



A color that is close to many other similarly colored dots cannot be tuned for and is difficult to find.



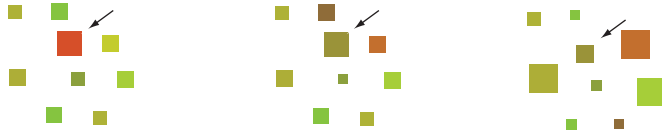
Similarly, if a line is surrounded by other lines of various similar orientations it will not stand out.

For the pop-out effect to occur, it is not enough that low-level feature differences simply exist, they must also be sufficiently large. For example, as a rule of thumb a thirty-degree orientation difference is needed for a feature to stand out. The extent of variation in the background is also important. If the background is extremely homogenous—for example, a page of twelve-point text—then a small difference is needed to make a particular feature distinct. The more the background varies *in a particular feature channel*—such as color, texture, or orientation—then the larger the difference required to make a feature distinct.

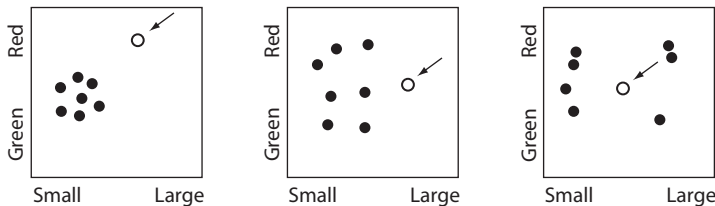
We can think of the problem of seeing an object clearly in terms of *feature channels*. Channels are defined by the different ways the visual image

is processed in the primary visual cortex. Feature channels provide a useful way of thinking about what makes something distinct. The diagrams below each represent two feature channels. You can think of one axis representing the color channel and the other axis representing the size channel. The circle represents a target symbol having a particular color and size. This is what is being searched for. The filled circles represent all of the other features on the display.

Objects to be searched



Corresponding feature space diagrams



TOP ROW. Three sets of objects to be searched. In each case an arrow shows the search target. LEFT BOTTOM ROW. The corresponding feature space diagrams. If a target symbol differs on two feature channels it will be more distinct than if it differs only on one. On the left panel the target differs in both color and size from non-targets. In the middle panel the target differs only in size from non-targets. A target will be least distinct if it is completely surrounded in feature space as is shown in the right panel.

One might think that finding things quickly is simply a matter of practice and we could learn to find complex patterns rapidly if we practiced enough. The fact is that learning does not help much. Visual learning is

6
difficult

●
easy

2359807754321
5478904820095
3554687542558
558932450●452
9807754321884
3554387542568
2359807754321



difficult



easy



The number 6 cannot be picked out from all the other numbers despite a lifetime's experience looking at numbers; searching for it will take about two seconds. By contrast the dot can be seen in a single glance, about ten times faster. The features that pop out are hardwired in the brain, not learned.

Individual faces also do not pop out from the crowd even though it may be our brother or sister we are looking for. However, the pinkish round shapes of faces in general may be tunable, and so our visual systems can program a series of eye movements to faces. This does not help much in this picture because there are so many faces. The yellow jacket in the image on the right can be found with a single fixation because there is only one yellow object and no other similar colors.

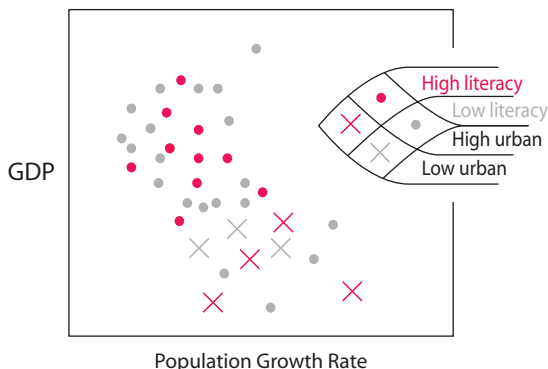
a valuable skill and with practice experts can interpret patterns that non-experts fail to see. But this expertise applies more to identifying patterns once they have been fixated with the eyes, and not to finding those patterns out of the corner of the eye.

LESSONS FOR DESIGN

The lessons from these examples are straightforward. If you want to make something easy to find, make it different from its surroundings according to some primary visual channel. Give it a color that is substantially different from all other colors on the page. Give it a size that is substantially different from all other sizes. Make it a curved shape when all other shapes are straight; make it the only thing blinking or moving, and so on.

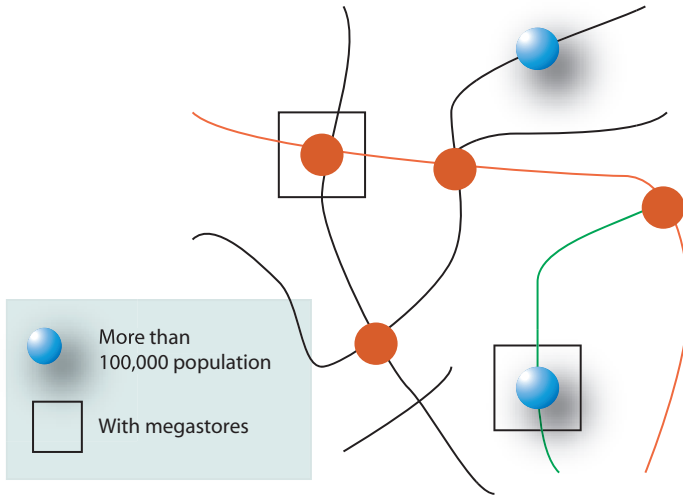
Many design problems, however, are more complex. What if you wish to make several things easily searchable at the same time? The solution is to use different *channels*. As we have seen, layers in the primary visual cortex are divided up into small areas that *separately* process the elements of form (most importantly, orientation and size), color, and motion. These can be thought of as semi-independent processing channels for visual information.

A design to support a rapid visual query for two different kinds of symbols from among many others will be most effective if each kind of query uses a different channel. Suppose we wish to understand how gross domestic product (GDP) relates to education, population growth rates, and city dwelling for a sample of countries. Only two of these variables can be shown on a conventional scatter plot, but the other can be shown using shape and color. We have plot the (GDP) against population growth rate, and we use shape coding for literacy and color coding for urbanization.



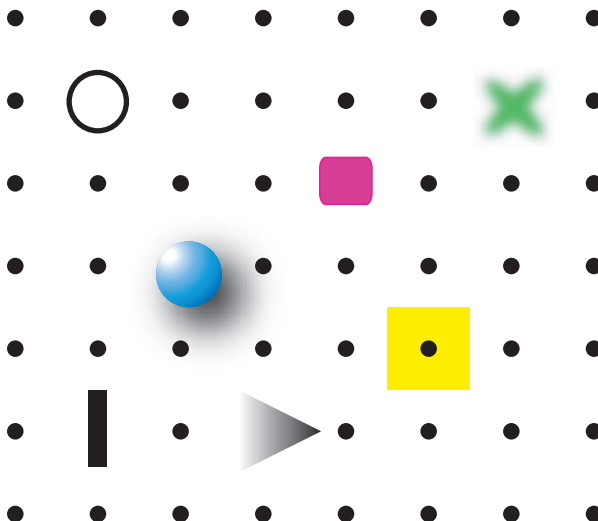
In this scatter plot, two different kinds of points are easy to find. It is easy to visually query those data points representing countries with a high level of literacy. These use color coding. It is also easy to visually query the set of points representing countries with a low urban population. These are distinct on the orientation channel because these symbols are made with **X**s containing strong oblique lines.

It is not necessary to restrict ourselves to a single channel for each kind of symbol. If there are differences between symbols on multiple channels they will be even easier to find. Also, tunability is not an all-or-nothing property of graphic symbols. A symbol can be made to stand out in only a single feature channel. For example, the size channel can be used to make a symbol distinct, but if the symbol can be made to differ from other symbols in both size and color, it will be even more distinct.



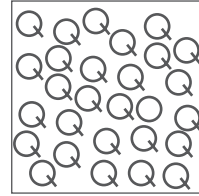
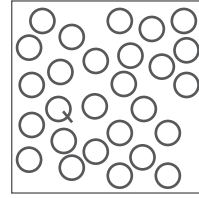
Suppose that we wish to display both large towns and towns with megastores on the same map. Multiple tunable differences can be used. The blue symbols are shaded in 3D and have cast shadows. The outline squares are constructed from the only bold straight vertical and horizontal lines on the map. This means that visual queries for either large towns or towns with megastores will be efficiently queried.

Any complex design will contain a number of background colors, line densities and textures, and the symbols will similarly differ in size, shape, color, and texture. It is a challenge to create a design having more than



A set of symbols designed so that each would be independently searchable. Each symbol differs from the others on several channels. For example, there is only one green symbol; it is the only one with oblique lines and it is the only one with no sharp edges.

two or three symbols so that each one will support rapid pop-out searching. Creating a display containing more than eight to ten *independently* searchable symbols is probably impossible simply because there are not enough channels available. When we are aiming for pop-out, we only have about three difference steps available on each channel: three sizes, three orientations, three frequencies of motion, etc. The following set is an attempt to produce seven symbols that are as distinct as possible given that the display is static.



There is an inherent tradeoff between stylistic consistency and overall clarity that every designer must come to terms with. The seven different symbols are stylistically very different from each other. It is easy to construct a stylistically consistent symbol set using either seven different colors or seven different shapes, but there is a cognitive cost in doing so. Visual searches will take longer if only color coding is used, than when shape, texture, and color differentiate the symbols, and in a complex display the difference can be extreme. What takes a fraction of a second with the multi-feature design might easily take several seconds with the consistent color-only design.

Many kinds of visibility enhancements are not symmetric. Increasing the size of a symbol will result in something that is more distinctive than a corresponding decrease in size. Similarly, an increase in contrast will be more distinctive than a decrease in contrast. Adding an extra part to a symbol is more distinctive than taking a part away.



There is a kind of visual competition in the street as signs compete for our attention.



Blinking signs are the most effective in capturing our attention. When regulation allows it and budgets are unlimited, Times Square is the result.

MOTION

Making objects move is a method of visibility enhancement that is in a class by itself. If you are resting on the African Savannah, it pays to be sensitive to motion at the edge of your visual field. That shaking of a bush seen out of the corner of your eye might be the only warning you get that something is stalking you with lunch in mind. The need to detect predators has been a constant factor through hundreds of millions of years of evolution, which presumably explains why we and other animals are extremely sensitive to motion in the periphery of our visual field. Our sensitivity to *static* detail falls off very rapidly away from the central fovea. Our sensitivity to motion falls off much less, so we can still see that something is moving out of the corner of our eye, even though the shape is invisible.

Motion is extremely powerful in generating an *orienting response*. It is hard to resist looking at an icon jiggling on a web page, which is exactly why moving icons can be irritating. A study by A.P. Hillstrom and S. Yantis suggests that the things that most powerfully elicit the orienting response are not simply things that move, but things that *emerge* into the visual field.* We rapidly become habituated to simple motion; otherwise, every blade of grass moving would startle us.



In the design of computer interfaces, one good use of motion is as a kind of human interrupt. Sometimes people wish to be alerted to incoming email or instant messaging requests. Perhaps in the future virtual agents will scour the Internet for information and occasionally report back their findings. Moving icons can signal their arrival. If the motion is rapid, the effect may be irritating and hard to ignore, and this would be useful for urgent messages. If the motion is slower and smoother, the effect can be a gentler reminder that there is something needing attention. Of course, if we want to use the emergence of an object to provoke an orienting response on the part of the computer user it is not enough to have something emerge only once. We do not see things that change when we are in the midst of making an eye movement or when we are concentrating. Signaling icons should emerge then disappear every few seconds or minutes to reduce habituation.*

The web designer now has the ability to create web pages that crawl, jiggle, and flash. Unsurprisingly, because it is difficult for people to suppress the orienting response to motion, this has provoked a strong aversion

*A.P. Hillstrom and S. Yantis, 1994. Visual attention and motion capture. *Perception and Psychophysics*. 55(4): 109–154.

*I introduced the idea of using motion as a human interrupt in a paper I wrote with collaborators in 1992. C. Ware, J. Bonner, W. Knight and R. Cater. Moving icons as a human interrupt. *International Journal of Human-Computer Interaction*. 4(4): 173–178.

among users to the web sites where these effects flourish. The gratuitous use of motion is one of the worst forms of visual pollution, but carefully applied motion can be a useful technique. This is not to say that motion per se is bad. We can be outdoors where trees sway, clouds move, and people pass to and fro without feeling irritated. It is especially high-frequency rapid motion or blinking that induces the unavoidable orienting response. Another caveat with respect to motion is that many people like the energy and stimulation that motion imparts—one person's amusement arcade is another person's hell. We should make the distinction between situations where the goal is entertainment and situations where the goal is providing information in the most effective possible way.

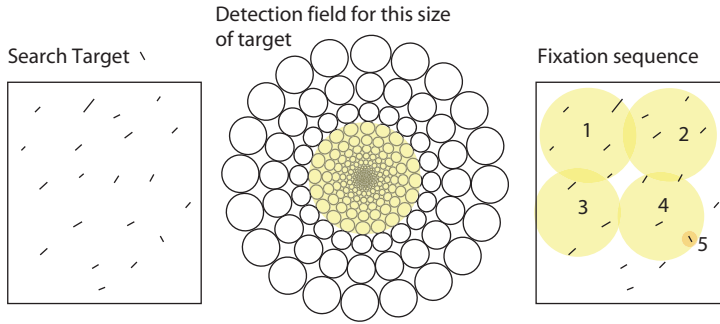
VISUAL SEARCH STRATEGIES AND SKILLS

So far we have focused on the specific feature properties that are used in the very short-term planning of the next eye movement, but this is by no means the whole story. Finding a small object is a skilled strategic process. At every instant in time part of the brain is planning the next eye movement based on the information available from the current fixation of the eye as well as some small amount that is retained from the previous few fixations. In addition, our previous experiences with similar visual environments also influence the visual search pattern and this is where skill comes in. Our prior experience tells us where to look for something, and this information is stored as a pattern of eye movement tendencies. When we read a newspaper, for example, we are likely to scan the main headlines and images first. But the patterns of eye movements that occur in response to a particular scene are never inevitable; it is a fluid just-in-time process, and the potential search targets are continually being reassessed.

A graphic design that has visual structure at several scales can aid the search process. Large-scale structure is needed as a means for finding important mid-scale and small-scale information.

THE DETECTION FIELD

Remember that the size of processing units (brain pixels) increases with distance from the center of gaze (see Chapter 1). Small features cannot be detected by large brain pixels at the edge of visual space. If the targets are small and near the current center of fixation, the brain can apply the selective feature-based biasing mechanism we have been discussing. But if small targets are off to the side, some other process must kick in. Several eye movements will be required to scan the visual field until one lands close enough to the target.



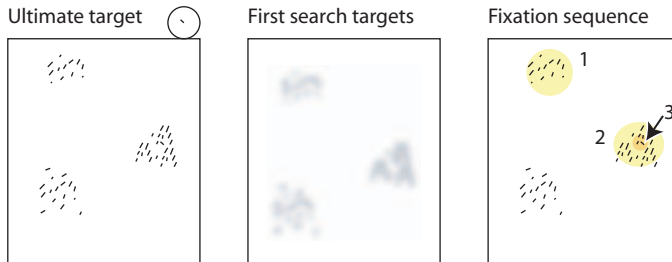
Brain pixels get larger the farther away they are from the point of fixation, thus several eye movements are needed before the target can be detected. On the fourth fixation, the target is detected at the edge of the tunable region. A final eye movement brings it to the center of the fovea.

The area around the center of the fovea where the presence of a particular target may be detected can be thought of as a *detection field*. If a search target is within the detection field, this does not necessarily mean that it can be positively identified; it only means that this can become a target for the next fixation, and after that it may be identified. In the absence of a candidate target in the detection field, the purpose of a *scanning strategy* is to get the eye *in the vicinity* of the target so that the feature-based pop-out mechanism can function as a final step.

A typical scanning strategy is the one we use when reading; we start in the upper-left-hand corner and move our eyes across from left to right starting at the top and progressing down. In most real-world search tasks, the visual field is not uniformly filled with potential candidates. Sometimes the likely candidates are grouped in clusters, and our search strategy will take advantage of this. Big patterns have a much bigger detection field, and this allows for a hierarchical search strategy. If small target patterns are embedded in specific types of large patterns, then this information can be used. First we make an eye movement to the likely neighborhood of a target, based on the limited information in our peripheral vision; next, the local pattern information provides a few candidates for individual detailed fixations.

Our everyday environment has structure at all scales. Rooms contain large furniture objects such as desks, chairs, bookshelves, and cupboards, as well as small hand-sized objects such as books, telephones, and coffee cups. These are arranged in predictable relationships; the books are on the shelves and the desktops, and the coffee cup is on the desk. This allows for a much more efficient visual search. For the thief in the night, the big patterns

may be dressers and side tables. The small patterns may be pieces of jewelry. Sometimes it can be enough that the small targets are arranged in clumps, enabling a strategy based on looking first for the clumps, then searching within each clump.



In this example, the targets are clustered and a two-stage search strategy is used. The first stage involves tuning for and making an eye movement to a cluster of targets (seen as fuzzy blobs in the periphery). After the fixation is made, the blob is resolved into individual targets. The second stage involves tuning for and making an eye movement to a particular candidate target within a blob. The final tuning is based on orientation.

The process we have described so far is missing one essential property. How does the search control system avoid revisiting places that have already been examined? Unless there is a memory for where eye fixations have recently been placed, the point just visited will be the best candidate for the next fixation. Without a blocking mechanism the eye would be trapped, flicking back and forth between the two most likely target areas. The brain must somehow mark the locations of recent fixations and inhibit the tendency to revisit them. It is thought that a structure on the *where* pathway called the *lateral intraparietal* area performs this function.* Experimental evidence suggests that between four and six locations recently visited with eye movements are retained. In some cases, the identity of the object at a particular location may also be retained in visual working memory, but in other cases the brain simply flags the fact that a particular location has been visited.

THE VISUAL SEARCH PROCESS

We can now elaborate on the inner loops of visual problem solving that were introduced in Chapter 1, concentrating on the eye movement control process. We will start with the outermost loop and work in.

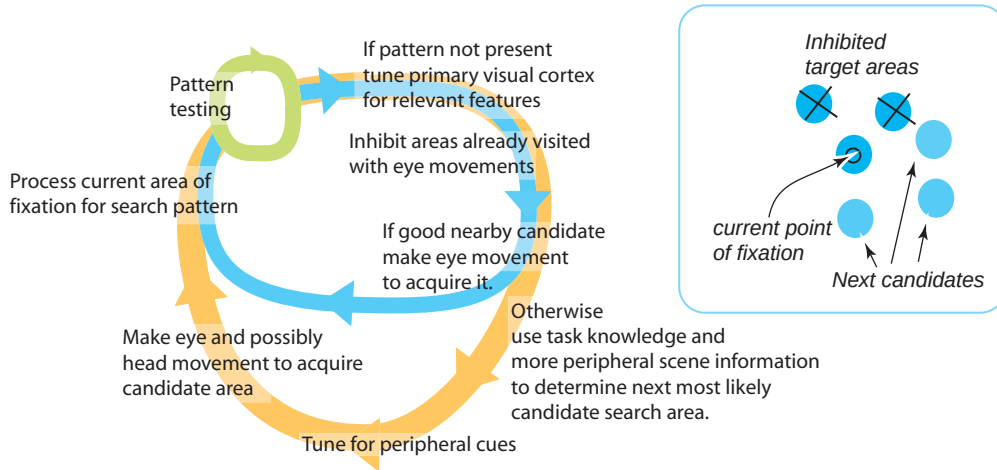
Move and scan loop. Assuming that we know what we are looking for, when we enter an environment, our initial search strategy will involve orienting the head and perhaps walking to get the best viewpoint. From this vantage, we will initiate a sequence of fixations. If the target is not found we move to a new vantage point to continue the search.

Eye movement control loop. Planning and executing eye movements occurs between one and three times per second. This involves both the biasing mechanism, so that new candidate targets can be determined

*See James W. Bisley, and Michael E. Goldberg, 2003. Neuronal activity in the lateral intraparietal area and spatial attention. *Science*. 299: 81–86.

based on their elementary properties of form (orientation, size, color, motion), and a simple map of what regions have been recently visited by means of eye movements.

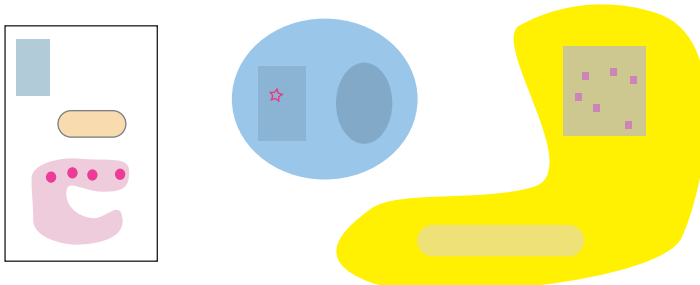
Pattern-testing loop. When the eye alights on a promising target area, the inner loop function is executed. This involves testing the pattern to see if it is the search target or not. The brain takes about one-twentieth of a second to make each test; typically between one and four tests are made on each fixation.



USING MULTISCALE STRUCTURE TO DESIGN FOR SEARCH

To support efficient visual search, a design should be given large-scale as well as small scale structure. All too often, artificial information displays lack visual structure at different scales. Menus and windows tend to all look the same. One rectangular box is much like another. Adding multiscale visual structure will make search much more efficient, as long as the smaller objects of search can predictably be associated with larger visual objects.

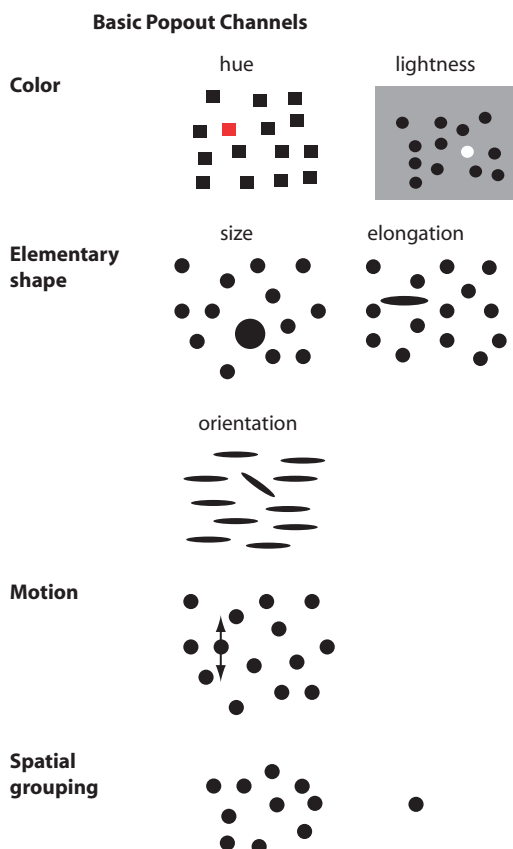
Multi-scale structure



Having structure at multiple scales is most important for designs that will be used over and over again. It permits visual search skills to develop in the form of eye movement sequences that occur in response to the general properties of a particular scene. But even for designs that are used only for a few minutes, high-level structure supports location memory and makes it easier to revisit places that have been looked at only seconds ago.

CONCLUSION

Visual search is not an occasional activity only occurring when we have lost something. Because we have so little information in our heads, search is something that is fundamental to almost all seeing. Even though we are mostly unaware that we are doing it, the visual world is being reassessed in terms of where we should look next on every fixation. Two seconds is a long time in a visual thinking. There is a world of difference between something that can be located with a single eye movement and one that takes five or ten. In the former case, visual thinking will be fluid, in the latter, it will be inefficient and frustrating.



One key to making efficient visual search is through the use of pop-out properties. If a visual object is distinct on one or more of the visual channels, then it can be processed to direct an eye movement. The strongest pop-out differentiators are the basic feature channels found in V1. These include the elements of form, size, elongation, and orientation; the elements of color, including hue and lightness (these are discussed further in Chapter 4); and motion, and spatial layout. The figure on the previous page gives a summary.

Large-scale graphic structure can also help with visual search, but only if the searcher *already knows* where in a large structure an important detail exists. Such knowledge is built up as we scan an image and graphic structure is always helpful in enabling us to return to something we have recently seen. Knowledge also comes, however, from our previous experience with similar graphics and this is a two-edged sword. If the design being searched conforms to stereotype then search will be easy because habitual visual scan patterns, guided by spatial structure, will support search. If a design violates a learned pattern then searches using habitual eye movement strategies will result in frustration.

Chapter 3

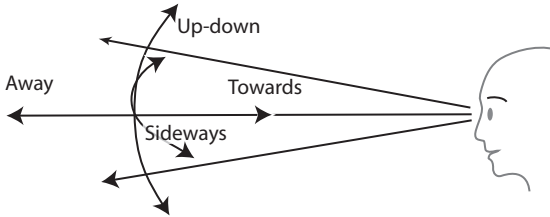
Structuring Two-Dimensional Space

Patterns are formed where ideas meet the evidence of the world. Top-down attentional processes cause patterns to be constructed from the retinal image that has been decomposed, like an impressionist painting, into a myriad of features in V1. Patterns are formed in a middle ground of fluid dynamic visual processing that both pulls meaningful patterns from features and imposes order. Patterns are also the building blocks of objects, which themselves can be thought of as more complex patterns.

Understanding how patterns are constructed and reconstructed can tell us a lot about the design problems of organizing space, either in ways that are unambiguous and clear, or in ways that are subject to multiple interpretations. Both have their uses. We shall start with a discussion of the dimensions of space to show why image plane patterns are so special. Then we move on to discuss how patterns are formed from the low-level features discussed in the previous chapter.

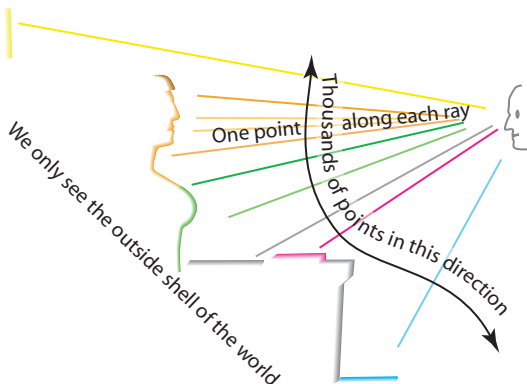
2.5D SPACE

Although we live in a three-dimensional world, these dimensions are not equal in terms of human perception of space. The two dimensions laid out “in the picture plane” have little in common with the third dimension towards and away from the viewpoint.



If we direct an imaginary ray away from the pupil of one of our eyes into the world, it will eventually encounter a point on the surface of an object; except for those cases when the ray reaches the empty sky. That surface point has a color and a distance from the eye. Of course, in the real world, light travels in the reverse direction and information about the surface color is recorded by cone receptors at the back of the eye, but the point is that there is only one color in the away direction corresponding to each retina location. About a million fibers in each optic nerve transmit to the brain a million pieces of information about the world in the combined *up-down* and *sideways* dimensions (defined by the diagram above). For each of those points of information there is, at best, one additional piece of distance information relating specifically to the *away* dimension, and this extra information must be indirectly inferred. It is for this reason that people sometimes refer to visual space as having 2.5 dimensions, where the 0.5 refers to the *away* dimension. However,

It is useful to consider visual space using radial coordinates, centered on the eye-point of an observer. We can usefully call the three dimensions the up-down dimension, the sideways dimension, and the towards-away dimension. For convenience we shall shorten these to *up-down*, *sideways*, and *away*.



Since most things are not transparent, only facing surfaces are visible and only one point of color is available on each ray from a particular brain pixel. But there are millions of rays that can be distinguished in the up-down and sideways dimensions.

this substantially overstates the case. It would be more accurate to say that visual space has 2.05 dimensions. Sometimes the term *image plane* is used to refer to *up-down* and *sideways* dimensions combined because this is the information that can be captured on the image plane of a camera. The term *depth* is used to refer to away information.

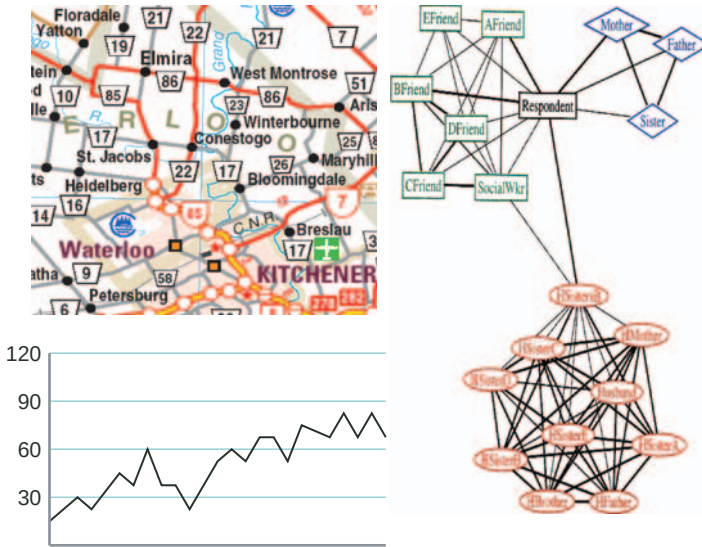
Our ability to get a whole new sample of visual space is also radically different for the different dimensions. We can sample the *up-down* and *sideways* dimensions of space with rapid eye movements and get a mostly new sample of a million points of data on each retina in less than one-tenth of a second. By contrast, in order to get new information in the *away* direction we have to walk to a new location, so that information that was blocked becomes revealed. This can take from several seconds to a few minutes or more. Head-turning movements are intermediate in time costs and allow us to access parts of the up-down and sideways dimensions using eye movements. The point here is that image-plane sampling through eye movements is ten to a hundred times more efficient than depth sampling by walking and provides much greater cognitive efficiency.

Pattern-processing resources in the brain are mostly devoted to information in the image plane, as opposed to depth. These two-dimensional patterns are fundamentally important for two reasons. First, they are the precursors of objects. Second, a pattern can also be a relationship between objects. In some ways, pattern finding is the very essence of visual thinking, and often to perceive a pattern is to solve a problem.



Finding the boundaries of objects is an important function of the pattern-processing systems. In order for the brain to find an object, it must somehow be distinguished from other objects in the environment, and often the most important piece of information is that it has a continuous contour running all around it. The figure on the left is distinguished from its background by many different kinds of contours.

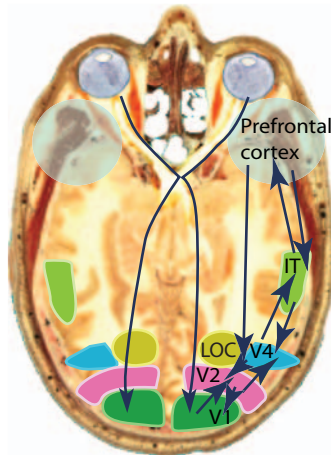
Later in this chapter, we will examine the design implications of the brain's pattern-finding mechanisms. First, we consider the nature of the pattern-processing machinery.



When we look at a map the task of finding a route from one city to another is a problem of pattern finding. When an investor examines a graph showing how a company's revenue has changed over the past year, she is looking for patterns that may predict future performance. When an anthropologist draws a diagram of the relationships of all of the people in a remote village, the patterns help him understand the social structure.

THE PATTERN-PROCESSING MACHINERY

There are two kinds of processes involved in pattern perception. One has to do with dividing up-down space through the binding of continuous contours and areas. An area called the *lateral occipital cortex* is implicated in this. The other has to do with a sequence of ever more complex patterns processed through the *what* pathway, culminating in an object.



Incoming information first arrives in visual area V1 at the back of the brain. The information then passes successively through V2 and V4 and then to the infero-temporal cortex (IT). The lateral occipital cortex (LOC) is involved in region finding. Signals from the prefrontal cortex are sent back to consolidate task-relevant patterns.

This is sometimes called the "what" pathway, as opposed to the "where" pathway. The what pathway has the function of identifying objects. The where pathway has more to do with visually guided actions.

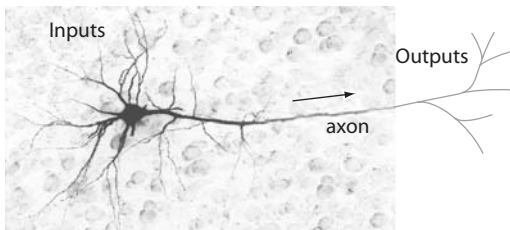
THE BINDING PROBLEM: FEATURES TO CONTOURS

We shall start with the way the brain constructs contours and areas from the millions of fragmented pieces of information in V1. The process

of combining different features that will come to be identified as parts of the same contour or region is called *binding*. There is no such thing as an object embedded in an image; there are just patterns of light, shade, color, and motion. Objects and patterns must be discovered, and binding is essential because it is what makes disconnected pieces of information into connected pieces of information.*

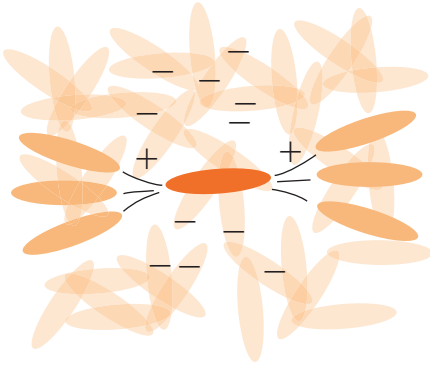
To explain how the responses of individual feature-detecting neurons may become *bound* into a pattern representing the continuous edge of an object, let us indulge in a little anthropomorphism and imagine ourselves to be a single neuron in the primary visual cortex. As a neuron, we are far more complex than a transistor in a computer—our state is influenced by incoming signals from thousands of other neurons; some excite us, some inhibit us. When we get overexcited we *fire*, shooting a convulsive burst of electrical energy along our axon. Our outgoing axon branches out and ultimately influences the firing of thousands of other neurons. Supposing we are the kind of neuron in the primary visual cortex that responds to oriented edge information, we get most excited when part of an edge or a contour falls over a particular part of the retina to which we are sensitive. This exciting information may come from almost anything, part of the silhouette of a nose or just a bit of oriented texture.

*It is never acceptable to simply say that a higher level mechanism *looks at* the low-level information. That is called the homunculus fallacy. It is like saying that there is an inner person (a homunculus) that sees the information on the retina or in the primary visual cortex. A proper explanation is that a set of mechanisms and processes produce intelligent actions as an outcome of interactions between processes operating in sub-systems of the brain.

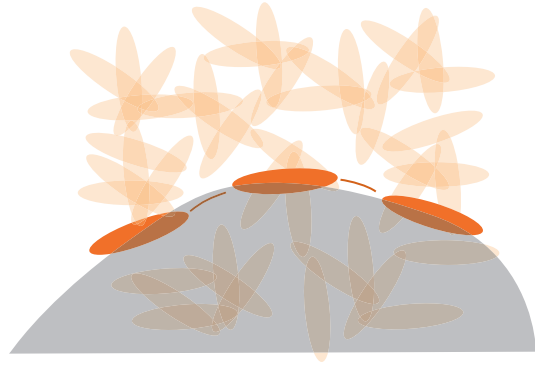


Thousands of inputs are received on the dendrites. These signals are combined in the cell body. The neuron fires an electrical spike of energy along the axon, which branches out at the end to influence other neurons, some positively, some negatively. Neurons fire all the time, and information is carried by both increases and decreases in the rate.

Now comes the interesting bit. In addition to the information from the retina, we also receive input from neighboring neurons and we set up a kind of mutual admiration society with some of them, egging them on as they egg us on. To form part of our edge-detecting club, they must respond to part of the visual world that is *in-line* with our preferred axis, and they must be attuned to roughly the same orientation. When these conditions are met, we start to pulse in unison sending out pulses of electrical energy together.



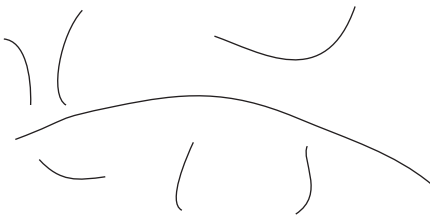
Edge detector neurons have a positive excitatory relationship with other edge detectors that are nearby and aligned. There is an inhibitory relationship with cells responding to non-aligned features.



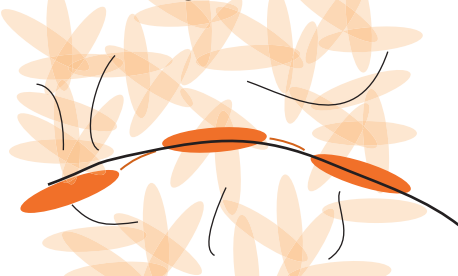
When light from an edge falls across the receptive fields of positively connected neurons a whole chain of neurons will start pulsing together. They are bound together by this common activity.

Our neuron is also surrounded by dozens of other neurons that respond to other orientations and, like some human groups, it actively discourages them. It sends inhibitory signals to nearby neurons that have different orientation preferences. This has the effect of damping their enthusiasm. The result is that neurons that are stimulated by long unbroken edges in the visual image mutually reinforce one another and all begin to fire in unison. Neurons responding to little fragments of edges have their signals suppressed.

A noisy edge



The response to the main contour is enhanced through mutual excitation.



The response to the short contours with other orientations is suppressed.

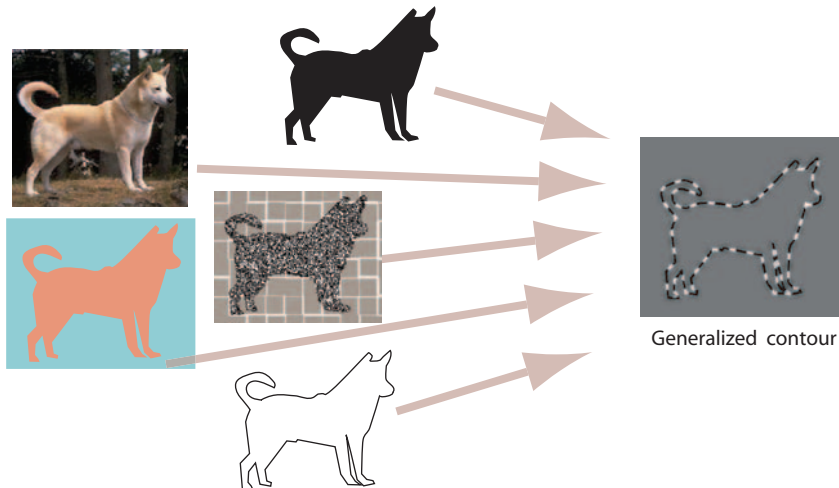
In addition to the signals received by the neuron from the retinal image and from its neighbors, we should not forget the top-down actions of attention. If a particular edge is part of an object that is related to some task being performed, for example, the edge belonging to the handle of a

dog that someone is reaching for, then additional reinforcement and binding will occur between the pattern information and an action pattern involving the guidance of a hand. The mental act of looking for a pattern makes that pattern stand out more distinctly. On the other hand, if we are looking for a particular pattern and another pattern appears that is irrelevant to our immediate task, we are unlikely to see it.

Binding is not only something required for the edge-finding machinery. A binding mechanism is needed to account for how information stored in various parts of the brain is momentarily associated through the process of visual thinking. Visual objects that we see in our immediate environment must be temporarily bound to the information we already have stored in our brains about those objects. Also, information we gain visually must be bound to action sequences, such as patterns of eye movements or hand movements. This higher-level binding is the subject of Chapter 6.

THE GENERALIZED CONTOUR

An object may be separated from its background in many different ways including luminance changes at its silhouette, color differences, texture boundaries, and even motion boundaries. Therefore the brain requires a generalized *contour extraction mechanism* in the pattern-processing stage of perception. Emerging evidence points to this occurring in a region called the lateral occipital cortex (LOC) that receives input from V2 and V3 allowing us to discern an object or pattern from many kinds of visual discontinuity.*



One of the mysteries that is solved by the generalized contour mechanism is why line drawings are so effective in conveying different kinds of information. A pencil line is nothing like the edge of a person's face, yet a simple

*Zoe Kourtzi of the University of Tuebingen in Germany and Nancy Kanwisher of MIT in the U.S. recently found that the lateral occipital cortex has neurons that respond to shapes irrespectively of how the bounding contours are defined.

Kourtzi, Z. and Kanwisher, N. 2001. Representation of perceived object shapes by the human lateral occipital cortex. *Science*. 293: 1506–1509.

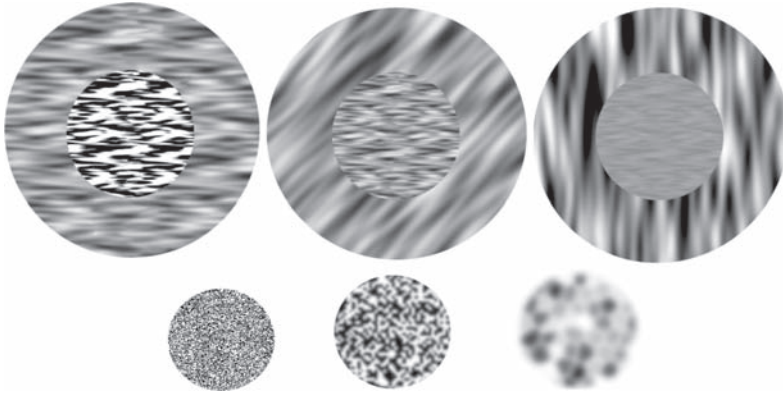
Many different kinds of boundaries can activate a generalized contour.

A generalized contour does not actually look like anything. It is a pattern of neural activation in part of the brain.

sketch can be instantly identifiable. The explanation of the power of lines is that they are effective in stimulating the generalized contour mechanism.

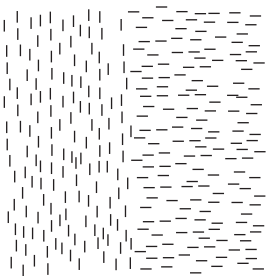
TEXTURE REGIONS

The edges of objects in the environment are not always defined by clear contours. Sometimes it is only the texture of tree bark, the pattern of leaves on the forest floor, or the way the fur lies on an animal's back that makes these things distinct. Presumably because of this, the brain also contains mechanisms that rapidly define regions having common texture and color. We shall discuss texture discrimination here. The next chapter is devoted to color.

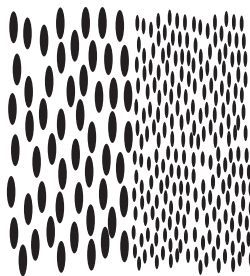


Excluding color and overall lightness, the primary factors that make one texture distinct from another are grain size, orientation, and contrast.

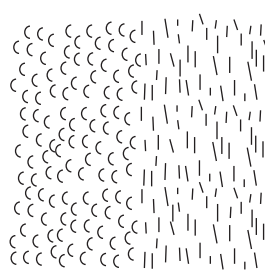
In general, the same properties that make individual symbols distinct (discussed in Chapter 2) will also make textures distinct. Primarily these are texture element orientation, size contrast, and color—the properties of V1 neurons in the primary visual cortex. In a way, this is hardly surprising; if large numbers of symbols or simple shapes are densely packed we stop thinking of them as individual objects and start thinking of them as a texture. The left- and right-hand sides of these texture patches are *easy to discriminate* because they contain differences in how they affect neurons in the primary visual cortex.



Orientation



Grain size

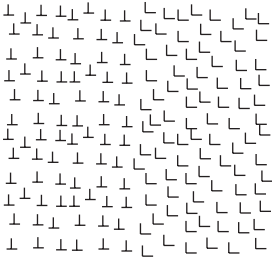


Curve versus Straight

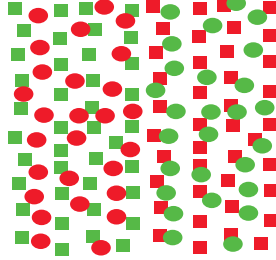


Blur

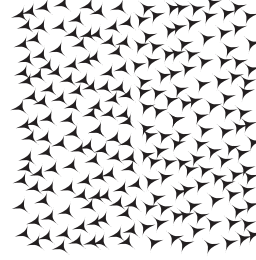
When low-level feature differences are not present in adjacent regions of texture, they are more difficult to distinguish. The following examples show *hard-to-discriminate* texture pairs.



Ts and Ls have the same line components.



Red circles and green rectangles versus green circles and red rectangles.

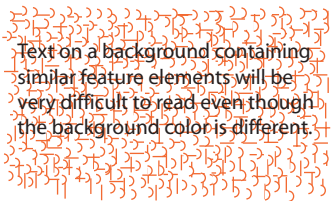


The spikes are oriented differently; the field of orientations is the same.

Although the mechanisms that bind texture elements into regions have been studied less than the mechanisms of contour formation, we may presume that they are similar. Mutual excitation between similar features in V1 and V2 explains why we can easily see regions with similar low-level feature components through spreading activation. More complex texture patterns, such as the Ts and Ls, do not allow for the rapid segmentation of space into regions. Only basic texture differences should be used to divide space.

INTERFERENCE AND SELECTIVE TUNING

The other side of the coin of visual distinctness is visual interference. As a general rule, like interferes with like. This is easy to illustrate with text.



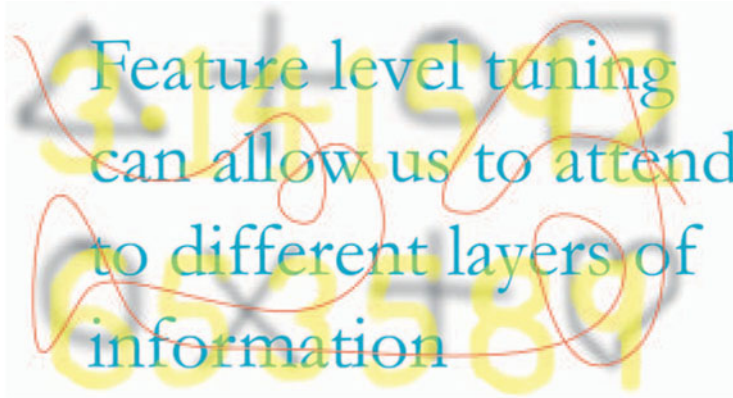
The more the background differs in element granularity, in feature similarity, and in the overall contrast, the easier the text will be to read.

Subtle, low-contrast background texture with little feature similarity will interfere less.

To minimize this kind of visual interference (it cannot be entirely eliminated), one must maximize feature-level differences between patterns of information. Having one set of features move, and another set static is probably the most effective way of separating overlaying patterns, although there are obvious reasons why doing this may not be desirable.

PATTERNS, CHANNELS, AND ATTENTION

As we discussed in Chapter 2, attentional tuning operates at the feature level, rather than the level of patterns. Nevertheless, because patterns are made up of features we can also choose to attend to particular patterns if the basic features in the patterns are different. The factors that make it easy to tune are the usual set and include color, orientation, texture, and the common motion of the elements. However, some pattern-level factors, such as the smooth continuity of contours, are also important. Once we choose to attend to a particular feature type, such as the thin red line in the illustration below, the mechanisms that bind the contour automatically kick in.



With this figure you can choose to attend to the text, the numbers, the thin red line, or the fuzzy black symbols. As you attend to one kind of representation, the others will recede.

Representing overlapping regions is an interesting design problem that can be approached using the channel theory introduced in the previous chapter. An example problem is to design a map that shows both the mean temperature and the areas of different vegetation types. The goal is to support a variety of visual queries on the temperature zones, or the vegetation zones, or both. Applying what we know about low-level feature processing is the best way of accomplishing this. If different zones can be

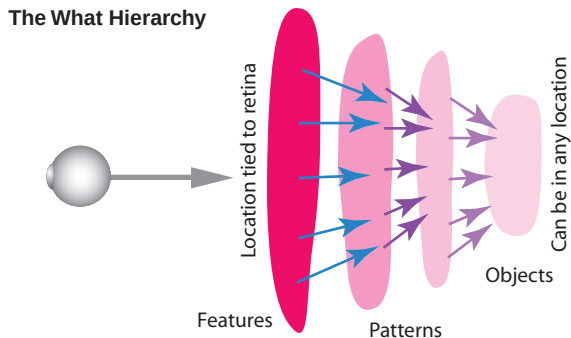


If many overlapping regions are to be shown then a heterogeneous channel-based approach can help. On the left is a diagram in which every combination of five overlapping regions is shown. The central version using color is clearer, but still confusing. The version on the right uses color, texture, and outline. Take any point on this last diagram and it is easy to see the set of regions to which it belongs.

represented in ways that are as distinct as possible in terms of simple features, the result will be easy to interpret, although it may look stylistically muddled.

INTERMEDIATE PATTERNS

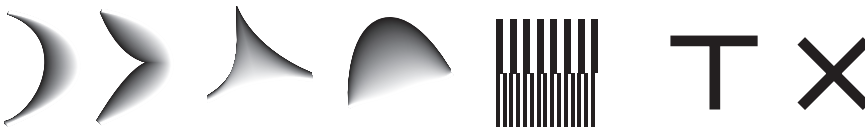
It is thought that the *what* pathway identifies objects in a series of brain regions responding to increasingly complex patterns. In area V2, neurons exist that respond to patterns only slightly more complex than the patterns found in V1 neurons. At the V4 stage, the patterns are still more complex chunks of information. V4 neurons then feed into regions in the inferotemporal cortex where specific neurons respond to images of faces, hands, automobiles, or letters of the alphabet. Each of the regions provides a kind of map of visual space that is distorted to favor central vision as discussed in Chapter 1. However, as we move up the hierarchy the mapping of visual imagery on the retina becomes less precise.



As information flows up the what hierarchy, it becomes increasingly complex and tied to specific tasks, but also decreasingly localized in space. In the inferotemporal cortex, a neuron might respond to a cat anywhere in the central part of the visual field.

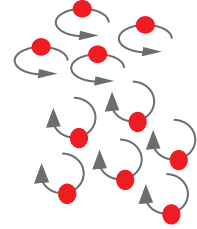
Currently, there is no method for directly finding out which of an infinite number of possible patterns a neuron responds to best. All researchers can do is to find, by a process of trial and error, what they respond to well. The problem increases as we move up the what hierarchy. As a result much less is known about the mid-complexity patterns of V4 than the simple ones of V1 and V2. What is known is they include patterns such as spikes, convexities, concavities, as well as boundaries between texture regions, and T and X junctions.* A selection is illustrated here.

*Anitha Pasupathy and Charles Connor, 2002. Population coding of shape in area V4. *Nature Neuroscience*. 5(12): 1332–1338.



In addition to static patterns, V4 neurons also respond well to different kinds of coherent motion patterns. Motion patterns are as important as static patterns in our interactions with the world. As we shall see in Chapter 5, motion can tell us about the layout of objects in space.

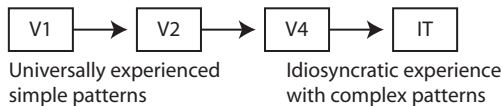
Motion patterns can also be especially important in telling us about the intentions of other humans and animals. Indeed there is emerging evidence that *biomotion* perception may have its own specialized pattern detection mechanisms in the brain. We will return to this in Chapter 7, which is about language and communication.



Objects moving together are perceived as a group. With this motion pattern, two groups would be seen.

PATTERN LEARNING

As we move up the processing chain from V1 to V4 and on to the IT cortex, the effects of individual experience become more apparent. All human environments have objects with well-defined edges, and this common experience means that early in life everyone develops a set of simple oriented feature detectors in their V1. An exception to this is people with uncorrected large optical problems with their eyes when they are babies. They never learn to see detailed features, even if their eyesight is corrected optically later in life. There is a critical period in the first few years of life when the neural pattern-finding systems develop. The older we get, the worse we are at learning new patterns, and this seems to be especially true for the early stages of neural processing. There are also some differences in how much experience we have with simple patterns. A person who lives in New York City will have more cells tuned to vertical and horizontal edges than a person who exists hunting and gathering in the tangle of a rainforest. But these are relatively small differences. Leaving aside people with special visual problems, and concentrating on urban dwellers, at the level of V1 everyone is pretty much the same, and this means that design rules based on V1 properties will be universal.



The more elaborate patterns we encounter at higher levels in the visual processing chain are to a much greater extent the product of an individual's experience. There are many patterns, such as those common to faces, hands, automobiles, the characters of the alphabet, chairs, etc., that almost everyone

experiences frequently, but there are also many patterns that are related to the specialized cognitive work of an individual. All visual thinking is skilled and depends on pattern learning. The map reader, the art critic, the geologist, the theatre lighting director, the restaurant chef, the truck driver, and the meat inspector have all developed particular pattern perception capabilities encoded especially in the V4 and IT cortical areas of their brains.

As an individual becomes skilled, increasingly complex patterns become encoded for rapid processing at the higher levels of the what pathway. For the beginning reader, a single fixation and a tenth of a second may be needed to process each letter shape. The expert reader will be able to capture whole words as single perceptual chunks if they are common. This gain in skill comes from developing connections in V4 neurons that transforms the neurons into efficient processors of commonly seen patterns.

SERIAL PROCESSING

The pop-out effects discussed in the previous chapter are based on simple low-level processing. More complex patterns such as those processed by levels V4 and above do not exhibit pop out. They are processed one chunk at a time. We can take in only two or three chunks on a single fixation.

VISUAL PATTERN QUERIES AND THE APPREHENDABLE CHUNK

Patterns range from novel to those that are so well learned that they are encoded in the automatic recognition machinery of the visual system. They also range from the simple to the complex. What is of particular interest for design is the complexity of a pattern that can be apprehended in a single fixation of the eye and read into the brain in a tenth of a second or less.

Apprehendability is not a matter of size so long as a pattern can be clearly seen. For example, the green line snaking across this page can easily be apprehended in a single fixation. The visual task of finding out what lies at the end of the green line requires only a single additional eye movement once we have seen the line. But apprehendability does have to do with both complexity and interference from other patterns. Finding out if the line beginning at the red circle ends in the star shape will require a sequence of fixations. That pattern is made up of several apprehendable chunks.

For *unlearned* patterns the size of the apprehendable chunk appears to be about three feature components. Even with this simple constraint, the variety is enormous. A few samples are given on the right. Each illustrates roughly the level of unlearned pattern complexity that research suggests we can take in and hold from a single glance.

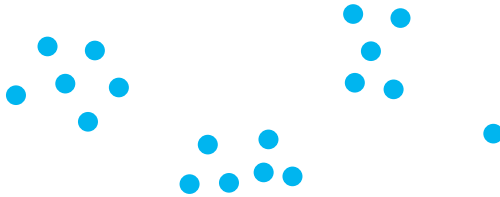


MULTI-CHUNK QUERIES

Performing visual queries on patterns that are more complex than a single apprehendable chunk requires substantially greater attentional resources and a series of fixations. When the pattern is more complex than a single chunk the visual query must be broken up into a series of subqueries, each of which is satisfied, or not, by a separate fixation. Keeping track of apprehendable chunks is the task of visual working memory, and this is a major topic of Chapter 6.

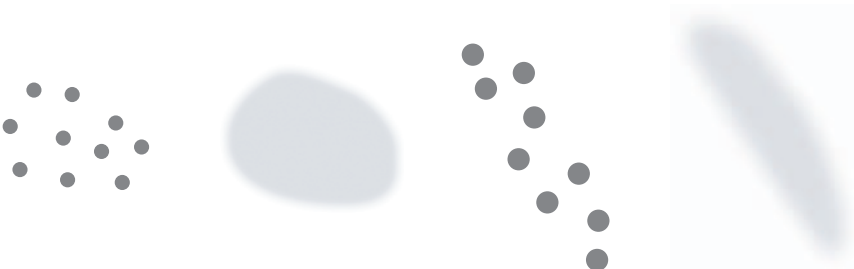
SPATIAL LAYOUT

Pattern perception is more than contours and regions. Groups of objects can form patterns based on the proximity of the elements.



Both the perception of groups and outliers are determined by proximity.

We do not need to propose any special mechanism to explain these kinds of grouping phenomena. Pattern detection mechanisms work on many scales; some neurons respond best to large shapes, others respond to smaller versions of those same shapes. A group of smaller points or objects will provoke a response from a large-shape detecting neuron, while at the same time other neurons, tuned to detect small features, respond to the individual components.



Both of the patterns shown above will excite the same large-scale feature detectors giving the overall shape. The pattern on the left will additionally stimulate small-scale detectors corresponding to the grey dots.

Large-scale oriented feature detectors will be stimulated by the large-scale structure of these patterns.

Objects in the real world have structure at many scales. In a garden, for example, individual flowers provide small-scale patterns, and these are organized into patches of color depending on the design of a flower bed. The entire structure of lawns, flower beds, trees, and paths form a large-scale pattern.



These patterns stimulate mechanisms responding to both small-scale feature and medium-scale features. They are patterns of pattern.

V4 neurons still respond strongly to patterns despite distortions that stretch or rotate the pattern, so long as the distortion is not too extreme. This is a critical property that allows us to recognize shapes and patterns even when they are seen from somewhat different angles and from somewhat different distances. A single V4 neuron might respond well to all of these different T patterns.



The ability to respond to a distorted pattern is the kind of thing that neurons do well but has been very difficult to reproduce with computers. It is not an exaggeration to say that our ability to perceive patterns under distortions and when partially hidden is fundamental to visual intelligence.

HORIZONTAL AND VERTICAL

The up-down direction and the sideways direction are special. They are perceived differently from each other and from other orientations. We are very sensitive to whether something is exactly vertical or horizontal and can make this judgment far more accurately than judging if a line is oriented at forty-five degrees. The reason for this has to do with two things: one is maintaining our posture with respect to gravity; the other is that our modern world contains many vertically oriented rectangles and so more of our pattern-sensitive neurons become tuned to vertical and horizontal contours. This makes the vertical and horizontal organization of information an effective way of associating a set of visual objects.



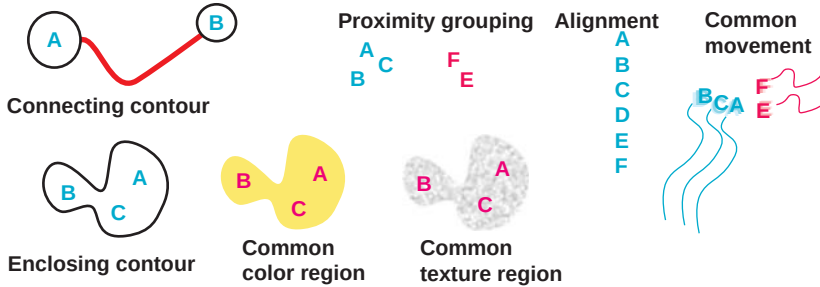
The line on the right is only rotated 3 degrees from the vertical, yet this departure from parallel is clearly evident.

People are extremely sensitive when judging small end-to-end misalignments of straight lines.

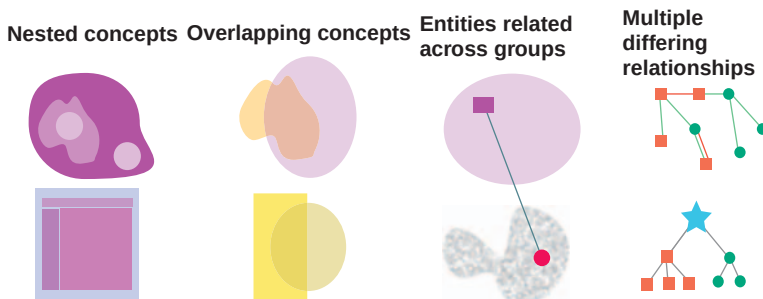
PATTERN FOR DESIGN

Most designs can be considered hybrids of learned symbolic meanings, images that we understand from our experience, and patterns that visually establish relationships between the components, tying the design elements together. A design can be made visually efficient by expressing relationships by means of easily apprehended patterns.

Relationships between meaningful graphical entities can be established by any of the basic pattern-defining mechanisms we have been discussing: connecting contours, proximity, alignment, enclosing contour, color, texture, and common movement.



Graphic patterns defined by contour or region can be used in very abstract ways to express the structure of ideas. The relationship between concepts can be defined by graphically using techniques that are fundamental to the way diagrams work.



Relationship-defining patterns can also be used in combination. For example, a texture region can be used to define one kind of relationship, and this can be used in combination with connecting contours that define another kind of relationship. The design challenge is to use each kind of design device to its greatest advantage in providing efficient access to visual queries.



Road Map. Supports route planning.

Example: Find an efficient route between Glen Allan in the upper left corner and Cambridge in the lower right. The cognitive process involves the following complex set of visual pattern queries.

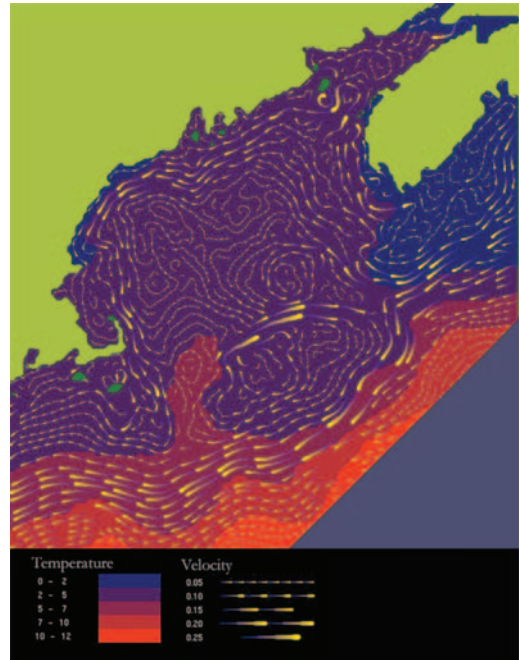
Locate end points corresponding to Glen Allan and Cambridge.
[Query, visual search for text labels].

Find path from Glen Allan to main road. [Query, search for where grey line intersects with red line].

Follow highway in direction of Cambridge. [Query, visually trace red and orange lines towards the lower left].

Detect highway exit point [query white oval] near to road connecting to Cambridge. At this point the point of fixation will be close enough to Cambridge that the remainder of the route can be grasped in a single apprehendable chunk.

Find path from exit to Cambridge. [Query, discover red path from exit symbol to Cambridge symbol].



Map of water currents and temperature.

Supports reasoning about currents in the Western North Atlantic. The ocean current patterns are represented using streaklets of different length and width. The background color provides information about water temperature.

Example: Find where the strongest currents are located.
[Query, find locations of the fat long streaklets].

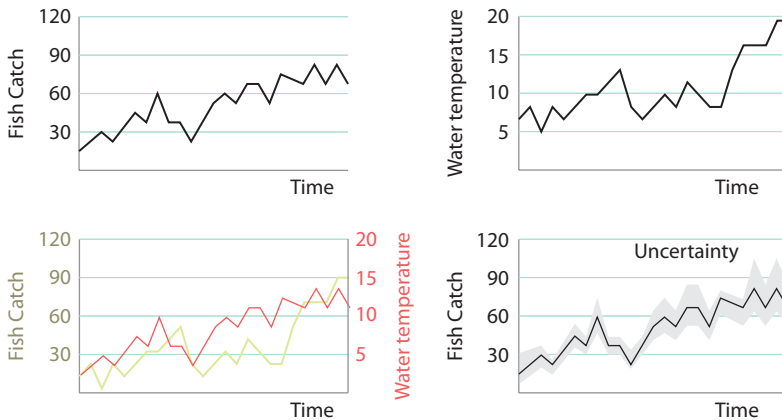
Example: Find out where something that was dropped in the water (for example, fish larvae) might end up after a period of time. [Query, find where a particular train of streaklets leads].

Example: Find the regions with the coldest water.
[Query, find locations of darkest blue color].

Design comment: In the regions where the current velocity is slow (thin lines), the direction is unclear.

EXAMPLES OF PATTERN QUERIES WITH COMMON GRAPHICAL ARTIFACTS

To elaborate these ideas, the remainder of this chapter is devoted to a set of examples of common graphical information displays. Each of them supports a form of visual reasoning; in each case the quality of the design depends on its supporting a specific set of visual pattern queries. We begin with two maps shown on the previous page and continue with a selection of charts, a news page, and some other diagrams.



Line Graph: Supports reasoning about how one measured variable changes with another.

Example: Determine fish catch as it varies over time. Common tasks are to determine when the maximum and minimum catches occurred, and whether the catch is trending up or down. *[Queries, find the highest point; find the lowest point; find whether the line slopes up or down overall.]*

Example: Often people reason by comparing one graph against another. In this case, the task might be to see if the fish catch is correlated with water temperature. *[Queries, discover if the high points and low points match for the two graphs; find if there is a correspondence in the overall trend for the two graphs.]*

Design note: Visual queries to find corresponding patterns are much easier to make if the two graphs can be combined.

Example: Often there is a degree of uncertainty in measurements, and this can be represented by a broad swath showing the uncertainty range at each point. *[Query, determine the vertical extent of the grey band at a particular point on the black line.]*

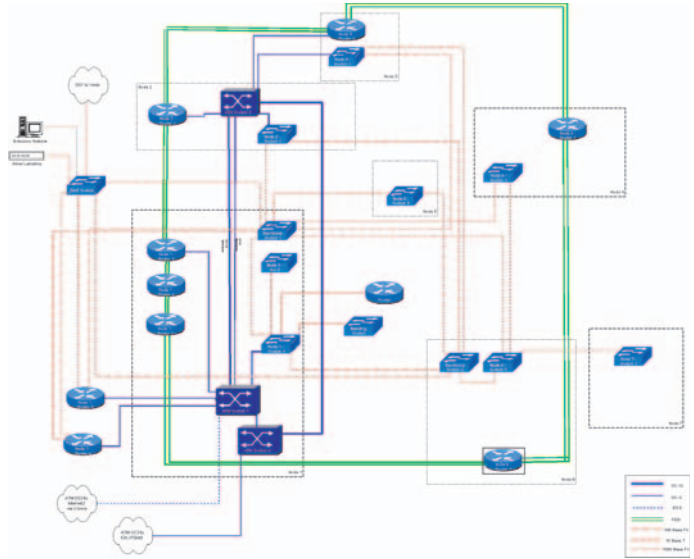
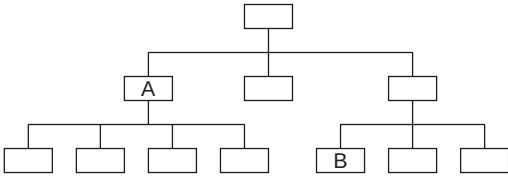
The point of the preceding examples has been to show that a wide variety of graphical display can be analyzed in terms of visual queries, each of which is essentially a visual search for a particular set of patterns. It is easy to construct such examples for any kind of graphical display. Indeed, it is a central thesis of this book that this kind of visual analysis is a critical design skill.



Web page for news site: Supports the search for various classes of current news including business, sports, politics, etc., as well as headline stories. The page has been deliberately blurred to bring out the overall spacial structure of the page.

Example: Discover major categories of news supported by this site. [*Query to locate top bar and left bar to find menus of alternatives*]. This design benefits from the fact that most web pages have adopted similar layout patterns.

Example: Discover current headline stories. [*Query to locate large text headlines. Read headlines. Query to locate pictures. Apprehend gist*]. The rapid apprehension of the gist of images is discussed in Chapter 5.



Node link diagrams: Support reasoning about relationships between entities. Typically the boxes represent entities, and the linking lines represent relationships. One variation on the node-link diagram is the organization chart.

Example: If A and B should work together on a project, who in the organization should be involved in setting this up. [Query, trace upward paths from both A and B noting all the boxes and the individuals these boxes represent].

Node link diagrams (second example): Shows network links between computers. Different styles of lines represent different rates of data communication.

Example: Can the network function when a node fails? [Query, is node X the sole link connecting any pair of components].

Example: Is data bandwidth adequate between a given pair of nodes? [Query, locate end points and trace paths between them, using line style representing the capacity of alternative links.] This becomes broken down into subqueries for colored connected contours, and is similar to the problem of route planning using maps.

SEMANTIC PATTERN MAPPINGS

For the most part, when we see patterns in graphic designs we are relying on the same neural machinery that is used to interpret the everyday environment. There is, however, a layer of meaning—a kind of natural semantics—that is built on top of this. For example, we use a *big* graphical shape to represent a *large* quantity in a bar chart. We use something graphically attached to another object to show that it is *part of* it.

These natural semantics permeate our spoken language, as well as the language of design. Indeed, the philosopher of language, George Lakoff, argues persuasively that spatial metaphors are not just ways of making language more vivid, they are fundamental to the way language works in communication and reasoning.[♦] Even when we talk about abstract, inherently

[♦]George Lakoff, 1980. *Metaphors we live by*. University of Chicago Press.

non-spatial ideas, phrases such as *connected to*, *built on*, *contained within* are so common that we do not even think of them as metaphoric. According to Lakoff, these kinds of spatial analogies are fundamental to human reasoning, which is ultimately grounded in human experience gained by interacting with the environment.

Lakoff is interested in the use of spatial metaphors in natural language. Here we are concerned with spatial metaphors and their use in graphic design, where they are just as ubiquitous. Below is a list of the most basic ones.

Graphical Code	Semantics
Small shapes defined by closed contour, texture, color, shaded solid.	Object, idea, entity, node.
Spatially ordered graphical objects.	Related information or a sequence. In a sequence the left-to-right ordering convention borrows from the western convention for written language.
Graphical objects in proximity.	Similar concepts, related information.
Graphical objects having the same shape, color, or texture.	Similar concepts, related information.
Size of graphical object Height of graphical object.	Magnitude, quantity, importance.

In this chapter, we have used the term *pattern* in its everyday sense to refer to an abstract arrangement of features or shapes. There is a more general sense of the word used in cognitive neuroscience, according to which every piece of stored information can be thought of as a pattern. The sequence of processing from V1 through the inferotemporal cortex can be thought of as a series of ever more complex pattern processors, with the patterns at each level being made up of the simpler patterns processed by the preceding levels. Thus complex patterns are *patterns of patterns*.

The generalized idea of a pattern applies to more than just shapes. Temporal patterns can also be found in changing sequences of visual shapes. Behavioral responses to visual patterns can be regarded as action patterns. Finding patterns is the essence of how we make sense of the world. In a fundamental sense, the brain can be thought of as a set of interconnected pattern-processing modules. Input patterns from the visual

Graphical Code		Semantics
Shapes connected by contour.		Related entities, path between entities.
Thickness of connecting contour.		Strength of relationship.
Color and texture of connecting contour.		Type of relationship.
Shapes enclosed by a contour, or a common texture, or a common color.		Contained entities. Related entities.
Nested regions, partitioned regions.		Hierarchical concepts.
Attached shapes.		Parts of a conceptual structure.

world activate patterns of responding, such as sequences of eye movements. The processing of visual thinking also involves the interplay between patterns of activity in the non-visual verbal processing centers of the brain.

Although pattern finding is something that everyone can do because it is a fundamental part of the process of seeing, designers must have an additional skill. They must see critically. This is very different from simply seeing that a pattern is present and using that pattern for some cognitive task like finding a route using a map. The map designer must critically analyze which combinations of patterns will provide the best support for the set of cognitive tasks a map supports. This critical seeing may seem intuitive to the skilled designer, but this intuition is hard-won through years of experience honing critical perception. Because pattern perception is so central, we will return to this and many of the other issues raised in this chapter.

Chapter 4

Color

Most large animals have worse color vision than humans. Color vision is of little benefit to grass eaters like zebras and cows—these animals have only two dimensions of color vision. The motion of a tiger’s prey is more critical than its color, and although cats, like grazing animals, have the physiological basis for two dimensions of color, it is extremely difficult to train them to respond to color. For the most part, they behave as if they



Color vision makes it much easier to see the fruit of the West African Akee tree.

live in a grey-scale world. On the other hand, color is invaluable to fruit eaters; color lets them see the oranges in the tree or the red berries in the bush. Birds have even better color vision than humans, being able to distinguish four or more dimensions. Humans and the other great apes, omnivorous hunter-gatherers that we are, have three dimensions of color.

The purpose of this chapter is to develop a theory-based approach to how color should be used in design. This will be somewhat incomplete because color is a complex technical subject, although a surprisingly small amount of theory is needed to derive the most important aesthetic principles. The parts of color theory not covered include colorimetry and color reproduction. Colorimetry is the discipline dealing with the precise measurement and specification of color.* Color reproduction is the discipline dealing with how colors are printed on paper, as well as methods for transforming colors from some display medium, such as a computer monitor, to another, such as printed paper through color gamut mapping.†

THE COLOR-PROCESSING MACHINERY

There are two basic types of light receptors in the retina at the back of the eyeball: rods and cones. Rods, the most numerous type, are specialized for very low light levels. Rods are wasted on modern humans because they are overloaded at the light levels of our artificially lit world. These days most people rarely try to get around under starlight, and this is no longer an important survival skill. Unfortunately, we have no way of trading our rod receptors for cone receptors.

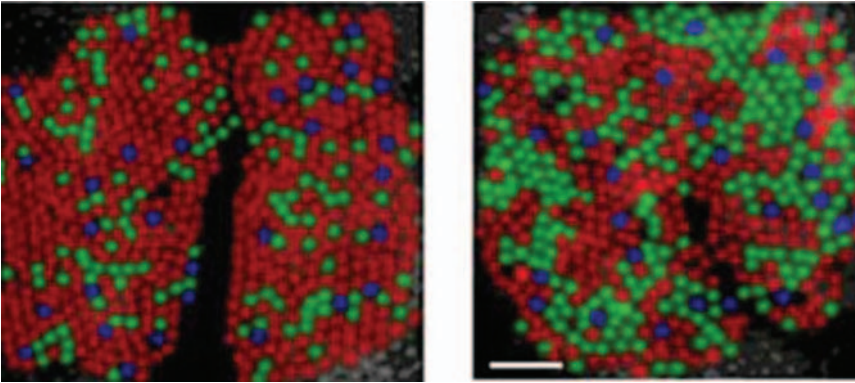
Cone receptors are the basis for normal daytime vision, and they come in three subtypes—short-wavelength sensitive, middle-wavelength sensitive, and long-wavelength sensitive. These three different types of cones mean color vision is fundamentally three-dimensional. This is the reason that televisions and monitors have three types of liquid crystal filters, or three different-colored light-emitting phosphors in older cathode-ray tube monitors.

David Williams, at the University of Rochester, has recently succeeded in obtaining images of the human retina and classifying the cones. Here are examples from two different people.

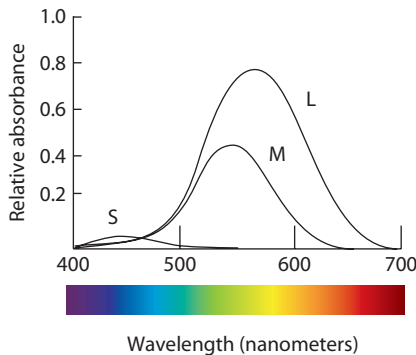
What is immediately apparent in these images of retinas is that there are far fewer short-wavelength-sensitive cones (blue) than middle- or long-wavelength-sensitive cones. Compounding this, the few short-wavelength-sensitive cones are less sensitive to light than either the middle- or long-wavelength-sensitive cones.

*The classic text dealing with color measurement as well as basic results is G. Wyszecki and W. S. Styles, 1982. *Color Science: Concepts and Methods, Quantitative Data and Formulae* (2nd ed.), John Wiley & Sons, Inc., New York. As of 2007 it is still in press. Charles Poynton's site <http://www.poynton.com/ColorFAQ.html> is a very useful source of technical color information.

†See Maureen Stone, 2003. *A Field Guide to Digital Color*. A.K. Peters. The effects of mixing paints and printing inks are complex because of the ways light interacts with pigments—mixing colored lights is much simpler. Still, a reasonably complete range of colors can be produced with the colors cyan, magenta, and yellow. The reason these are different from the red, green, and blue of monitors is that these printing dyes subtract light reflected by underlying white paper, as opposed to emitting it.



Cones in the fovea where there are no rods. Those sensitive to short, medium, and long wavelengths are colored blue, green, and red, respectively (Images from David R. Williams, University of Rochester.)



Sensitivity functions for the three classes of cone receptors. Cone receptors are sensitive to light having wavelengths between 400 and 700 nanometers ($1 \text{ nm} = 10^{-9} \text{ meters}$).

Note that the sensitivity distributions of long (L) and medium (M) wavelength-sensitive cones overlap considerably. The brain takes the difference between the two signals to get useful hue information.

It is impossible to reproduce pure spectral colors with printing inks so the color bands below should only be taken as rough approximations to the actual hues.

A consequence of our having few and weak blue-sensitive cones is that it is a mistake to show text or anything else with detail using blue on a dark background. The result is usually illegible.

Showing small blue text on a black background is a bad idea.
There is insufficient luminance contrast.

Showing small blue text on a black background is a bad idea.
There is insufficient luminance contrast.

A related problem occurs when using yellow for text. A pure yellow hue excites both the numerous middle- and long-wavelength cones, making yellow the lightest of all pure hues. Yellow can be almost as light as

white, and small yellow text on a white background is extremely hard to read, but pure yellow is very distinct on a black background.

Showing small yellow text on a white background is a bad idea.
There is insufficient luminance contrast.

Showing small yellow text on a white background is a bad idea.
There is insufficient luminance contrast.

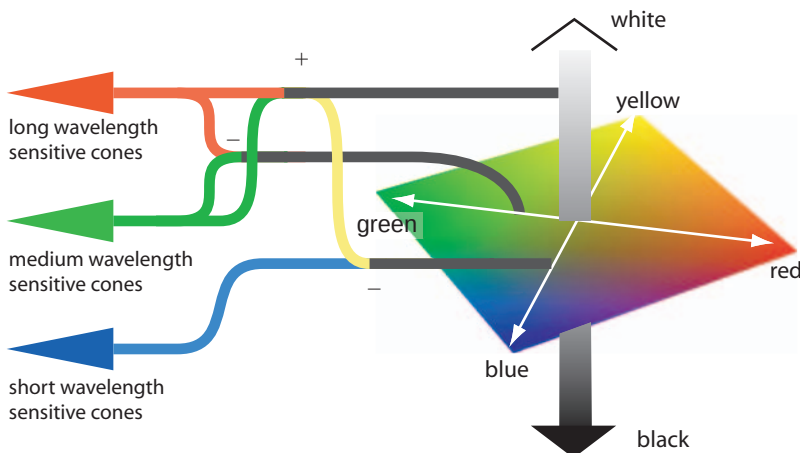
Another interesting feature of the images of retinas on the previous page is the way the red- and green-colored cones are clumped together. We can see considerable detail in black and white images because for black-white detail it is only necessary for black and white parts of the image to fall on two or more different cone receptors, no matter which type. But the patchy nature of the different cone types means that we are far less able to see detail where the differences are purely chromatic.

OPPONENT PROCESS THEORY

A major transformation in the color signal from the receptors occurs in area V1, where information traveling along the optic nerve first arrives at the cortex. Neural networks add and subtract the cone signals in different ways, transforming them into what are called the *color-opponent channels*.[♦] There are three channels, designated *red-green*, *yellow-blue*, and *black-white* (or luminance), respectively. The red-green channel represents the *difference* between the signal from the middle- and long-wavelength-sensitive cones.

This allows us to be highly sensitive to subtle red-green contrasts despite the overlap in the cone-sensitivity functions. The luminance channel *combines* the outputs of long- and middle-wavelength-sensitive cones. The yellow-blue channel represents the *difference* between the luminance

[♦]Color-opponent theory can be traced back to Ewald Hering who published his ideas in Vienna in 1878.



In V1, the raw signals from the cones in the retina are transformed. Some neurons compute differences between red- and green-sensitive cone signals. Some neurons compute the sum of red- and green-sensitive cones. Still others compute yellow-blue differences.

The result is three kinds of color signals that are called color-opponent channels.

channel and the blue cone signals. Note that although the black-white channel combines two type of cone information, it also differentiates in a spatial sense. Black-white differences are calculated simultaneously between all adjacent regions in the retina.

CHANNEL PROPERTIES

Most of the important principles for effectively using color in design can be derived from an understanding of the red-green, yellow-blue, and black-white color channels. What follows is a kind of annotated list of channel properties.

Contrast. A phenomenon known as simultaneous contrast occurs in each of the channels. The effect of simultaneous contrast is distortion of the appearance of a patch of color in a way that increases the difference between a color and its surroundings. This is called lightness or brightness contrast when it occurs in the black-white channel and chromatic contrast when it occurs in either the red-green or yellow-blue channel.



The two gray bars are exactly the same shade, but because of simultaneous lightness contrast the bar on the left seems darker than the one on the right.

Although contrast is often thought of as an *illusion*, a kind of visual error, the mechanisms of contrast help us see surface colors in the real world. Contrast comes about because the visual system is better at determining the *differences* between patches of light in the world than how much light is reflected from them as measured by a light meter. Difference information helps us discount the amount of light in our environment, enabling us to perceive how light or dark the surfaces of objects are whether they are seen on a dull winter day or a bright sunny day at the beach. It is because the brain is sensitive to differences and not absolute values that we can reproduce a reasonable facsimile of a beach scene at the movie theatre, despite the fact that there may be one-hundredth the amount of light reflecting from the screen compared to the real-world scene.

Chromatic contrast occurs in the red-green and yellow-blue channels. Its effects are much more complex and harder to predict than grey-scale contrast.



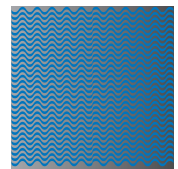
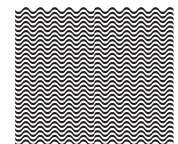
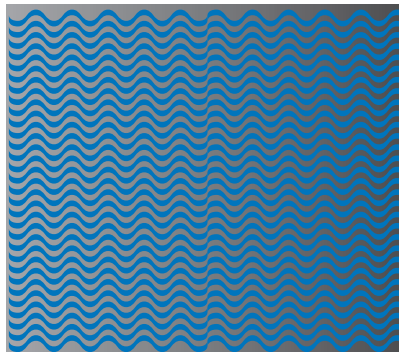
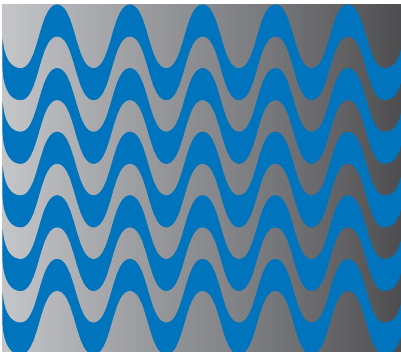
The two yellowish bars have the same hue, but because of simultaneous chromatic contrast the one on the left seems more yellow and the one on the right greener.

Unique hues. The opponent-process theory suggests that the colors black, white, red, green, yellow, and blue are special. Each occurs when there is a strong signal (either positive or negative) on one of the channels and a neutral signal on the other two channels. Confirming their importance, a remarkable study of more than one hundred human languages by Brent Berlin and Paul Kay carried out in the 1960s showed these basic color terms are the most commonly used no matter what the language.*



*Brent Berlin and Paul Kay, 1969.
Basic Color Terms: Their Universality and Evolution. University of California Press, Berkeley.

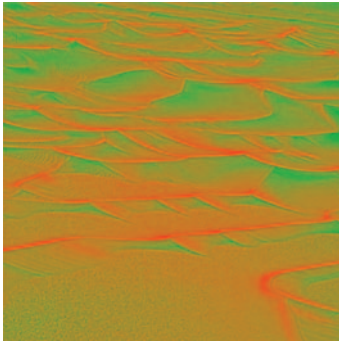
Sensitivity to spatial detail. The luminance channel has much greater capacity to convey detailed information than the chromatic channels. The patterns show that we can still see large shapes represented chromatically, but the fine pattern is much harder to see expressed through chromatic differences than when expressed through black-white differences.



Motion. The luminance channel can convey motion information much more effectively than the chromatic channels. When moving shapes are shown in red and green only, their motion appears to slow down.

Stereoscopic depth. This is the kind of depth information we get from having two eyes. The brain's processing of stereoscopic depth information is done via the luminance channel.

Shape-from-shading. We understand the shape of three-dimensional surfaces from the pattern of shading that comes from the way different facets of a surface are oriented to the light. The luminance channel can process shape-from-shading, whereas the chromatic channels cannot. In the illustration below, a black and white photograph of a shaded surface has been doctored so that the black-white sequence has been transformed into a red-green sequence. In this way, the original image that stimulated only the luminance channel has been modified so that it only stimulates the red-green channel. Although sharp boundaries are still clear, the topography of the shape has become virtually meaningless.



This image has had all of its grey-scale values transformed into red-green values. Because shape-from-shading processing is based on luminance channel information it is difficult to interpret.

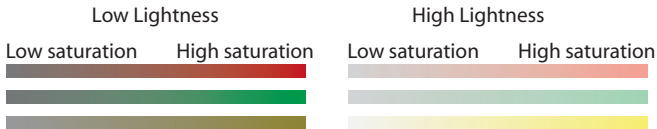


This is a NASA image of sand dunes on Mars taken by the Viking Lander. Note how readily we see the different grey values as making up a 3D undulating surface.

Saturation. The more vivid a color, the more saturated it is said to be. More saturated colors are those that have strong signals on one or both of the chromatic channels.

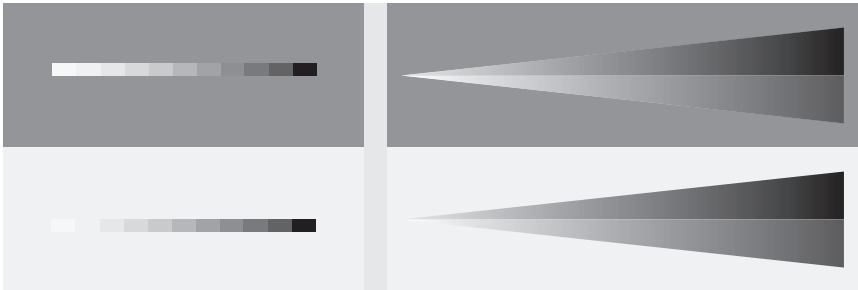


Maximum saturation varies with luminance. When colors are dark there is a smaller difference signal on the chromatic channels, and saturations are lower for dark colors. When colors are light there is also usually a reduction in saturation for reasons having to do with color reproduction technology rather than perception. Colors on a computer monitor are limited by the maximum amount of light coming from the red, green, and blue pixels. The lightest (or brightest) color is necessarily white because this occurs when all the pixels are fully on. The next lightest colors are low-saturation pastel hues. A similar effect occurs for printing inks.



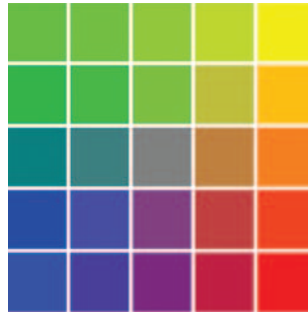
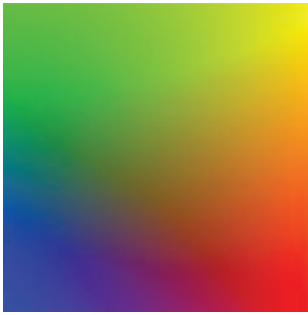
Luminance, lightness, brightness. *Luminance* refers to the physical amount of light as it might be measured by an instrument such as a photometer. The term *lightness* is used when discussing surface colors, and *brightness* is generally used when discussing emitted light. Both terms, however, apply to the amount of light coming from a particular part of the visual field and therefore to signals that affect the luminance channel.

Luminance nonlinearity. The luminance channel is nonlinear. We are more sensitive to dark gray differences than to light differences, although this effect also depends on the background. On top of this, the background is important. With a dark gray background, our sensitivity is increased to gray differences in the midrange. With a light background, our sensitivity is increased to differences near white. This effect is sometimes called *sharpening*.



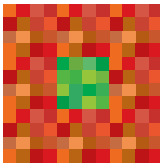
The top and bottom inner patterns are the same. Only the background is different.

Color segmentation. The visual system has a strong tendency to segment smoothly changing colors into regions consisting of the unique hues.

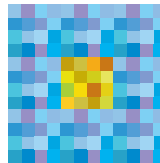


Notice how the eye tends to divide the smoothly varying colors in the square on the left into regions of red, green, yellow and blue. To a lesser extent, purple and orange can be seen. It is almost impossible to see the neutral gray in the center. In the segmented version (on the right), more colors are clearly evident even though there are actually fewer present.

Color blindness. People who are color blind (about 8 percent of males) are missing the red-green channel, and therefore have a two-dimensional rather than a three-dimensional color space. They can still distinguish colors that differ in the yellow-blue direction, and their ability to see gray-scale is unaffected. Yellow-blue color blindness is much less common.



Color-blind individuals will have difficulty distinguishing the patch of green colors from the surrounding red colors.



They will have no difficulty distinguishing a patch of yellow colors from the surrounding blue colors.

Chromatic color as a property of surfaces. A major function of visual processing is to let us perceive the properties of surfaces in the environment, rather than the amount of light coming from those surfaces. Many of the mechanisms of color vision exist to support this function. Adaptation in the receptors means that we are relatively insensitive to overall changes in light levels. Contrast mechanisms make us more sensitive to local changes in surface colors, rather than differences in light and shade across a scene.

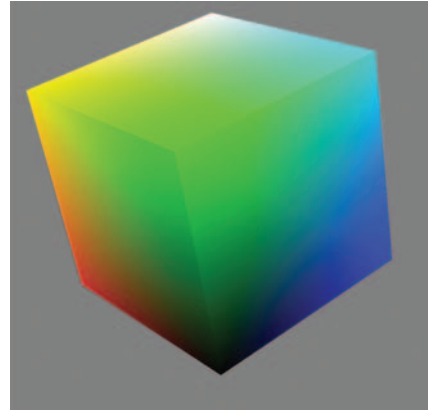
Color and attention. As discussed in Chapter 2, the pattern-finding mechanisms of the brain can be tuned for color. This allows for rapid visual searches for things that are of particular colors. But color search also

depends how many other colors there are in the environment. If all the world is gray, a patch of vivid color pops out. If the world is a riot of color then a visual search for a particular hue becomes difficult.

Color solids. The three-dimensional nature of color space is a straightforward consequence of having three cone types in the retina. But if we wish to view this three-dimensional space, how exactly should the colors be laid out? Here are just two of the many possible solutions.

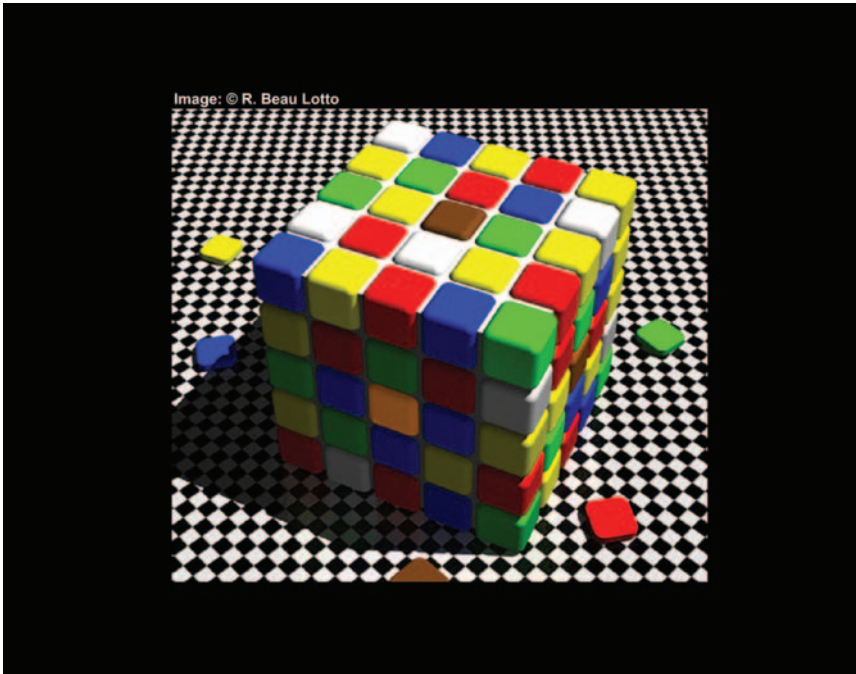


This is a set of calibrated color samples. Each is designed to be equally different from the adjacent sample, making it an example of a *uniform color space*. (From Science Museum/ Science and Society Picture Library, London.)



This is the outside of the cube of colors that can be represented on a computer monitor by combinations of the outputs of the red, green, and blue guns. The space is not perceptually uniform.

Color appearance. It only takes a combination of three lights (red, green, and blue) to create a full range of colors (for example, a photographic image reproduced on a computer monitor), and this is why color vision is said to be three-dimensional. Nevertheless, color *appearance* has more than three dimensions. The reason is spatial. A patch of color is never seen in isolation, and its appearance is affected by the colors that surround it, its orientation with respect to the light source, whether it is perceived as lying in or out of shadow, the texture of the surface on which it lies, and so on. Colors like brown and olive green only occur through contrast with surrounding colors. A color patch that seems brown when viewed on a white background will be perceived as orange when viewed in isolation in a dark room.



This remarkable demonstration by R. Beau Lotto of University College, London, shows how dramatically our perception of light and shade can affect our perception of color. The (apparently) brown square at the top center has exactly the same color as the (apparently) yellow square in the center of the front face. If you don't believe it, cut a small hole in a piece of white paper and use it to examine each patch of color with the surroundings blocked out. (Image created by R. Beau Lotto, www.lottolab.org.)

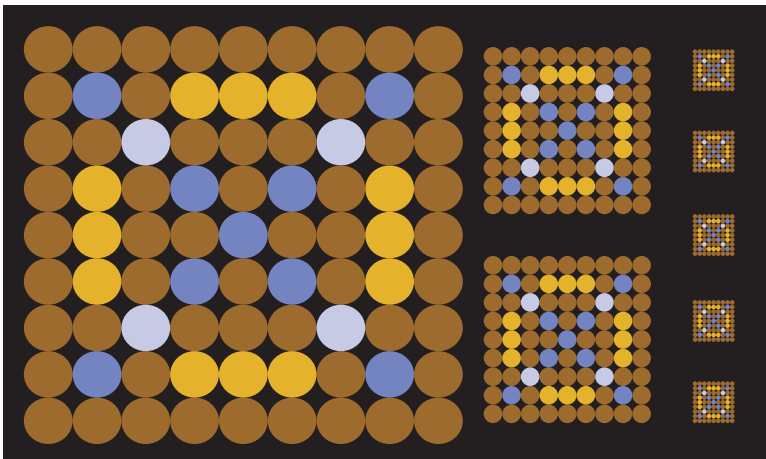
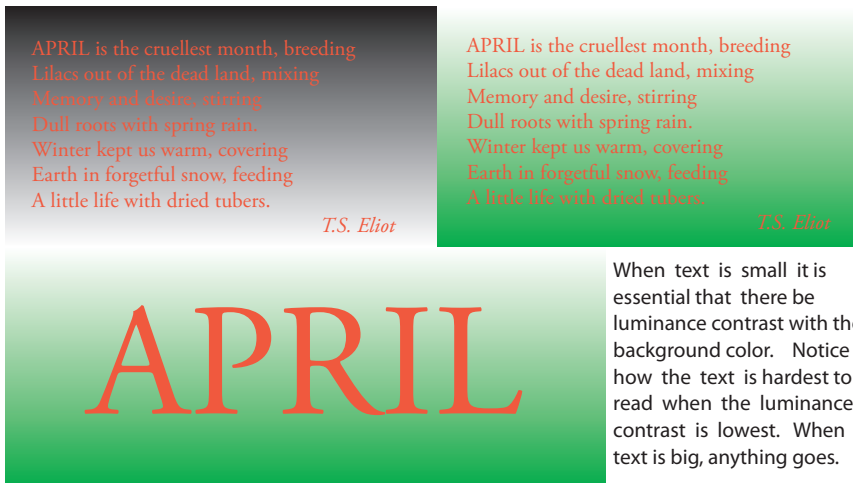
PRINCIPLES FOR DESIGN

The remainder of this chapter is devoted to applying color theory in elaborating a set of principles for the use of color in design. The approach is utilitarian, emphasizing clarity and support for visual tasks, but should not prevent designers from creating graphics that are visually delightful. There is a huge scope for creativity within the functional framework developed here.

SHOWING DETAIL

The most important single principle in the use of color is that whenever detailed information is to be shown, luminance contrast is necessary. Of course, black and white give the most extreme contrast possible. However, it is also possible to get excellent luminance contrast with yellow on black or dark blue on white. As graphical features get larger, so the need for extreme luminance contrast declines.

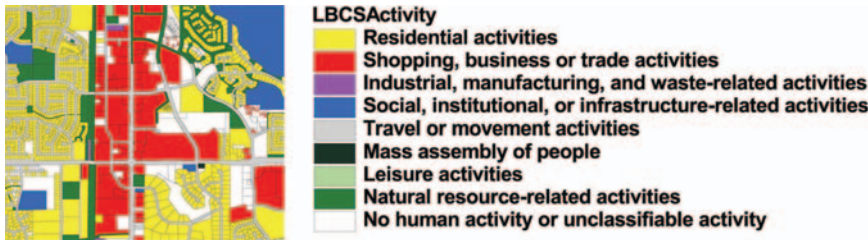
Luminance contrast is especially critical for small text. The International Standards Organization (ISO) recommends a ratio of at least 3:1 between the luminance of text and the luminance of the background. This severely restricts the range of colors that can be used for text. We can have black (or very dark-colored text) on a background that is white, or consists of light pastel hues; or we can have the reverse. When design elements are very large, for example in a poster title, luminance contrast is less critical and design can be given free rein. The red-green and yellow-blue channels do not convey fine detail, but they are just as effective as black and white for making large patterns distinct.



In the larger version of this pattern, the colors are the most salient features. In the much smaller versions, to the right, the luminance values dominate so that a circular O pattern becomes the most distinct feature.

COLOR-CODING INFORMATION

Perhaps the most important use of color is to indicate categories of information. In this land-use map, color codes are used to show different land-use zones.



In the design of color codes, the two primary considerations must be *visual distinctness*, to support visual search operations, and *learnability*, so that particular colors come to “stand for” particular entities. When learnability is the issue, it is important to use the unique hues first—red, green, yellow, and blue—followed by other colors that have relatively consistent names: pink, brown, orange, grey, and purple.*



There are strict limits as to how many colors can be used effectively as codes. If a design is complex and the symbols are quite small, then no more than a dozen codes can be used with complete reliability.* The main reason why we cannot use more colors is that the backgrounds can distort the appearance of a small patch of symbol color leading to the confusion of one symbol with another.

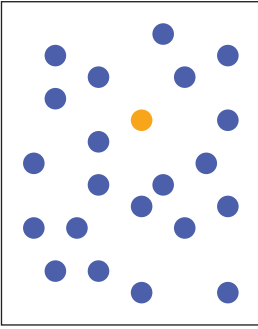
To effectively support *visual search* the colors of the background and the other symbols in the set are as important as the symbol being searched. When we conduct a visual search for an object having a particular color our brains tune our visual systems so that neurons sensitive to that hue get to “shout louder” than neurons tuned to other colors. There are, however, limits to this capability. A strong pop-out effect depends both on the other colored objects in the scene and on the background color.

LARGE AND SMALL AREAS

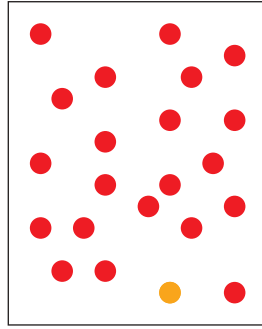
More complex designs often include color-coded background regions as well as color-coded symbols. Because of different channel properties,

*Although there may be quite large individual differences in how people perceive colors, when it comes to labeling them with words we appear to have a shared system. B. Salim, K.A. Jameson, N. Alvarado, and M.K. Szeszel, 2005. Semantic and perceptual representations of color: Evidence for a shared color naming function. *The Journal of Cognition and Culture*. 5 (3–4): 427–486.

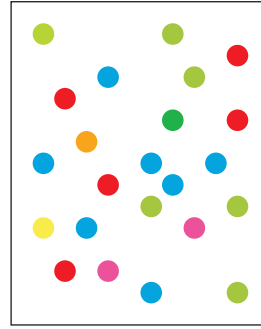
*There have been many studies designed to answer the question “How many colors can be used reliably in a symbol set?” yielding a range of answers, depending on exactly how they were conducted. But the number of colors recommended is always between 6 and 12.



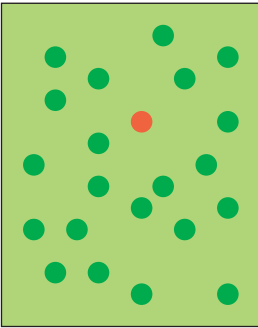
● The larger the chromatic difference between the target symbol and the other symbols, the easier the search.



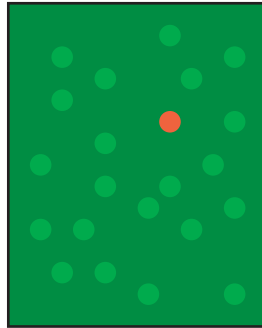
● When there is only a small color difference from non-target symbols, the search is difficult.



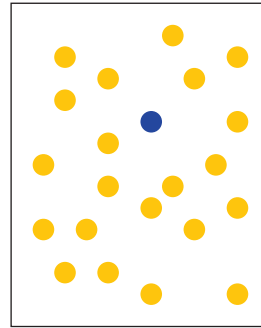
● When there are many non-target symbol colors, the search is the most difficult.



● If non-target symbols are similar to the background, they are easy to exclude from the visual search.



● A luminance difference plus a chromatic difference from other symbols and the background leads to the easiest search.



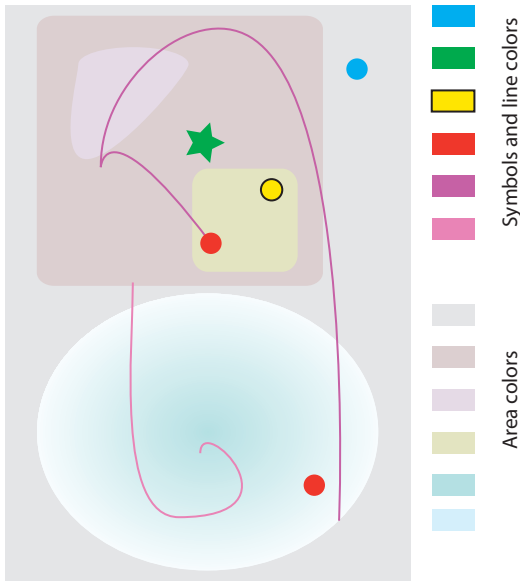
● A dark target on a light background with light non-target symbols can be as effective as the reverse.

small areas should be strongly colored and have black-white channel contrast with larger background areas if they are to be distinct. Large areas can have more subdued colors.

When small symbols must be displayed against a colored background it can be difficult to come up with symbol colors that are both distinct from each other and from the background. Using low-saturation colors for large areas generally makes the task easier. Because we are far more sensitive to color differences between larger areas, strongly saturated colors are unnecessary for large objects or backgrounds. Also, color contrast will distort the perceived color of small symbols less if the background colors are low in saturation.

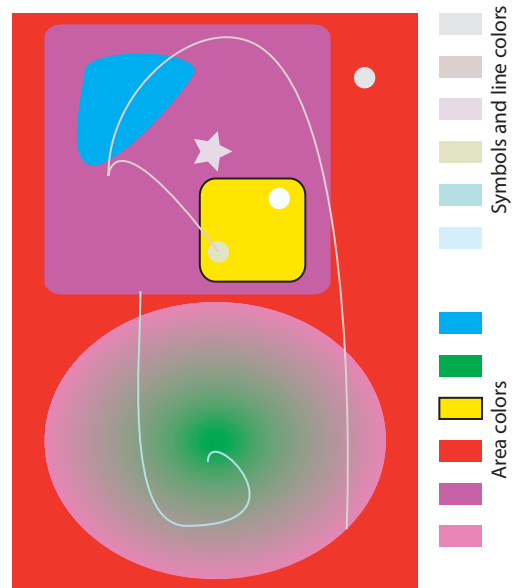
EMPHASIS AND HIGHLIGHTING

A strong signal on any of the opponent channels will attract attention better than a weak signal. Also, the strength of the signal depends on the



GOOD. This example uses high saturation (strong colors) to code small areas such as symbols and lines. Larger background areas are all light and low saturation.

A black contrasting border is used to separate the yellow circle from its background because both have similar luminance.



BAD. This example shows what happens when the colors used for small areas and large areas are switched. The color codes used for the symbols and lines are difficult to discriminate because of contrast effects.

Besides being completely ineffective, the second example would generally be regarded as unattractive.

difference between a visual object and its background. Objects that have reduced contrast with their backgrounds are naturally less salient.

Liberty consists in the freedom to do everything which *injures no one else*;
hence the exercise of the natural rights of each man has no limits except those
which assure to the other members of the society the enjoyment of the same rights.
These limits can only be determined by law.

Law can only prohibit *such actions as are hurtful to society*.
Nothing may be prevented which is not forbidden by law,
and no one may be forced to do anything not provided for by law.

*Declaration of the Rights of Man
National Assembly of France 1789*

A common mistake, often seen in PowerPoint slides, is to highlight something using color in such a way that luminance contrast is reduced. The highlighted text shown above is *less* distinct than the other text even though it uses a saturated hue.

Liberty consists in the freedom to do everything which **injures no one else**;
hence the exercise of the natural rights of each man has no limits except those
which assure to the other members of the society the enjoyment of the same rights.
These limits can only be determined by law.

Law can only prohibit **such actions as are hurtful to society**.
Nothing may be prevented which is not forbidden by law,
and no one may be forced to do anything not provided for by law.

*Declaration of the Rights of Man
National Assembly of France 1989*

It is often more effective to emphasize text by reducing the contrast of the other text. Care must be taken to maintain sufficient luminance contrast for clarity.



This illustration of a lumbar spinal fusion technique, by medical illustrator Christy Krames, has highlighted the critical area of interest by de-emphasizing the background through the use of neutral tones with some low-saturation blues. The critical area of interest is rendered in higher contrast and with the color red appearing only in this area. (Image courtesy of Christy Krames.)

COLOR SEQUENCES

A particular specialized use of color is the color-coding of maps. A map need not be geographical; we might have a map of the stock market, a map of radiation levels in the cosmos, or generally speaking, of any quantity that varies over a plane. Some maps have well-established color sequences already defined by convention. Usually in these cases it is unnecessary and unwise to promote an alternative. An example is the convention used by geographers to show the height of land with a color sequence that starts with green for the land transitions through light and dark brown in the mountains and often ends with white at the peaks.



Color is most effective as a highlighter when there is only a single color used, other than black and white. (Image courtesy of National Park Service.)

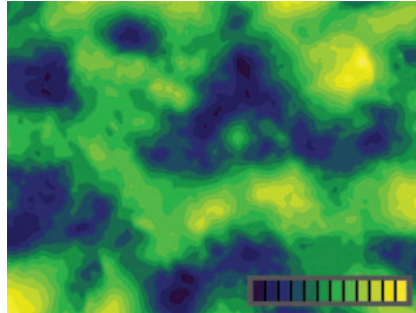
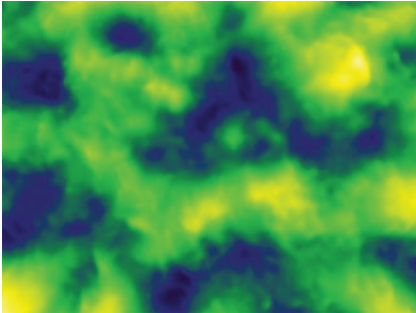
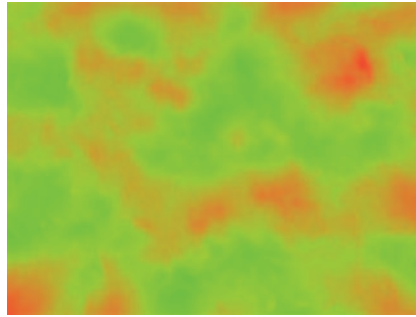
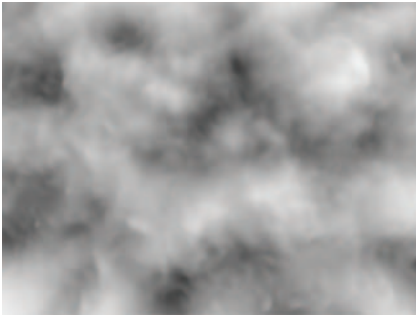
There are two primary tasks supported by color sequences used with maps. First, it allows people to perceive the spatial layout of *patterns in the data*, such as the shape and location of the hills, ridges, and valleys. Second, we can read *quantitative values* from the map; for example, judging the height or energy level at a particular location.

Because shape perception is based mainly on luminance channel information, color sequences that vary mainly in luminance will be the most effective in revealing patterns in the data. For example, contrast effects in gray-scale sequence can lead to very large errors when it comes to reading off data values using a key.* Different colors in the sequence along with a color key makes for more accurate readings. The best solution may be to provide a sequence that spirals up through color space, starting off with a dark color and selecting a series of colors, each of which has higher luminance than the previous one.

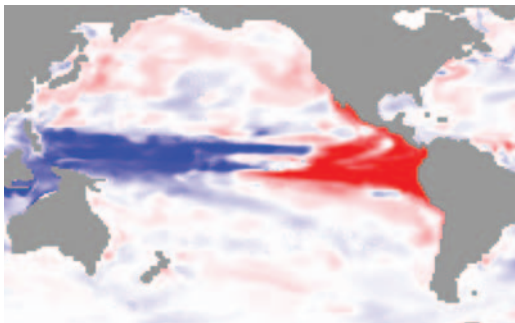
Zero is often an important number in data analysis, and sometimes a color sequence is needed that represents zero in an intuitive way. A good way of showing zero is to use a neutral value on the red-green and yellow-blue channels. Increasingly saturated values can be used to represent values above and below zero.

*In a study published in 1988 I measured errors as large as 20% of the data range for grey-sequence codings. I proposed the use of color sequences that spiral up through color space, always increasing in lightness.

Ware, C. 1988. Color sequences for univariate maps: Theory, experiments and principles. *IEEE Computer Graphics and Applications*. Sept. 41–49.



A black-white (top left) sequence reveals the features of the map much better than a red-green sequence (top right). Even better is a sequence of colors that consistently increases in lightness while cycling through a range of hues (bottom left). A stepped sequence allows for more precision, especially when reading from a key, although less detail is revealed (bottom right).



This map shows temperature differences (from normal) for the Pacific Ocean at 100 meters depth during an El Niño period. Warmer than usual is shown as red. Cooler than usual is shown as blue. Zero difference is shown as white.

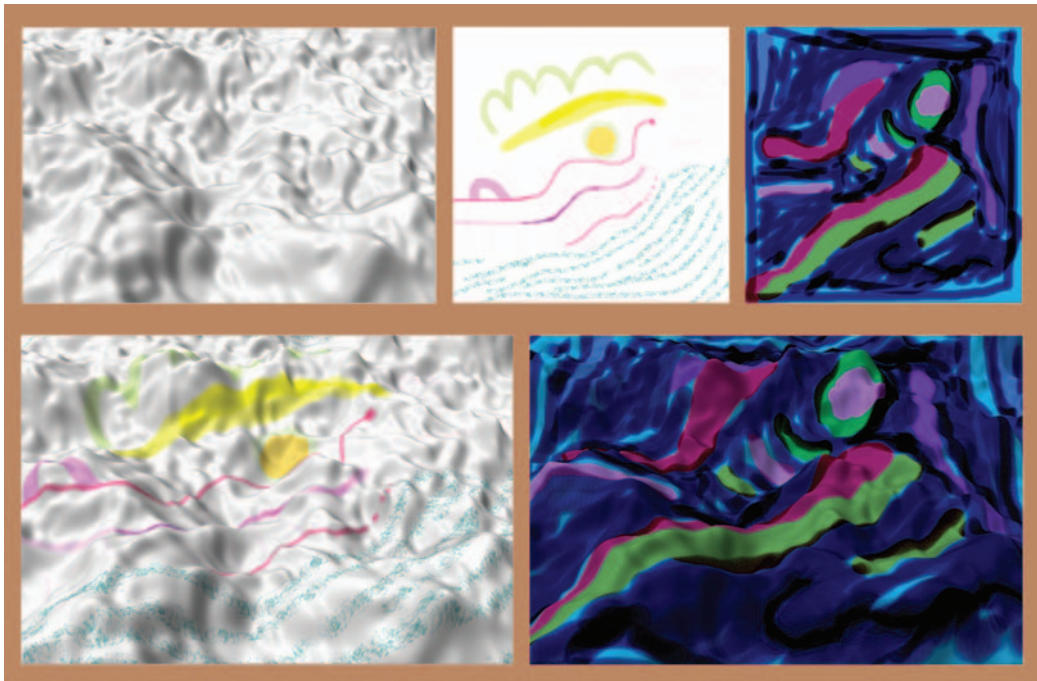
One kind of color sequence that is often used by scientists is an approximation to the spectrum of light (purple-blue-cyan-green-yellow-red). Because this represents a sequence of wavelengths of light, it is often erroneously presumed to be a perceptual sequence. A simple experiment will show this not to be the case. Give a number of people a set of color

chips consisting of red, green, yellow, and blue. Ask them to put them in a sequence. You will get many orderings, and the spectrum sequence is no more likely than any other. However, if you give them more chips, they may come up with a circle arrangement because each small section of the spectrum has a perceptual ordering. For comparison, ask those same people to order a set of gray chips ranging from black to white. The result will always be either a black-to-white ordering or a white-to-black ordering.

Parts of the spectrum are perceptually ordered, but the whole spectrum is not. Although the spectrum may be useful for reading values from data using a color key, it will not be the best for revealing patterns.

COLOR ON SHADED SURFACES

What does it take to see both the shape of a shaded surface and the colored patterns that are painted on that surface? Shape from shading information is conveyed by the luminance channel, and any colored pattern that interferes with luminance channel information destroys our ability to see the shape of the surface.

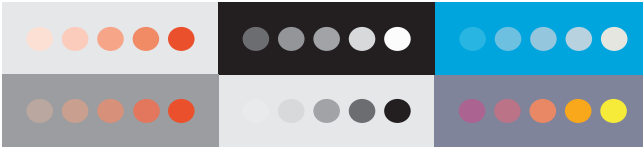


Adding the light-colored pattern (top center) to a shaded surface (top left) interferes little with our perception of surface shape as the combined image (lower left) shows. Adding a dark-colored pattern with strongly saturated colors (top right) to the same shaded surface almost completely destroys our ability to see the surface shape (lower right).

SEMANTICS OF COLOR

Colors are often used symbolically. In western culture, red represents danger, heat, and stop; green represents go, safety, and renewal; blue represents cold; white represents purity, and so on. Some of these meanings are well entrenched. Changing stop signs to the color green would have disastrous consequences. Such color meanings are culturally determined, and so a particular choice of colors may have different a meaning in another country. For example, in China red is used to symbolize good fortune and renewal. In most of Asia, white is the color for mourning, whereas in the west it is black. Designers for international markets must take such color symbolisms into account.

Other semantic mappings relating to quantity may be more universal. Quantities that are larger should generally be mapped to colors that are more distinct. This visually means that more saturated colors should be used to represent greater quantity. Similarly, darker colors on a light background or lighter colors on a dark background can be used to indicate greater quantity.



Six different ways of showing greater quantity. All involve increasing the relative difference from the background color.

For all but the simplest designs, the choice of a set of colors is a complex problem that typically involves tradeoffs. Every piece of information cannot be maximally distinct. Considerations such as the consistency of design across multiple graphics used in a document or a web site may also restrict design options. As a rule of thumb, the most common and most important visual queries should be given the most weight. This means that the most distinctive hues should be reserved for the most commonly accessed symbols and patterns in a design.

CONCLUSION

Based on color-opponent theory, the functional aesthetics of color can be reduced to a remarkably small number of guidelines. The image on the retina is transformed at every point into red-green, yellow-blue, and black-white difference signals. Every region of an image is evaluated

with respect to its adjoining regions in terms of how much it differs in each of the channels. One consequence of this is that colors are changed in appearance by adjacent colors. This is called simultaneous lightness or chromatic contrast. As a result, we can reliably use only a handful of very different colors symbolically to label information. If we try to use more, contrast effects mean that colors will be confused. Also, contrast effects mean that the appearance of a specific hue can only be judged in the context of a particular set of surrounding colors. Not only that, but color appearance depends on the amount of area covered by each patch of color; large areas change the color appearance of small areas, much more than small areas change the appearance of large areas. Because of this it is generally better if large areas have subdued colors. Also, to make small areas distinct they should be strongly colored to counter the lack of spatial resolution in the red-green and yellow-blue channels.

Another consequence of opponent-channel theory is that red, green, yellow, blue, black, and white are special. They each produce the most extreme signal on one channel while being neutral with respect to the other channels. These are the most distinct and easily identified colors and form the ideal labels, all other considerations being equal.

Only the black-white channel is capable of conveying much detail and so dark-light contrast is *essential* if detail is to be shown. This is true for text, fine lines, and fine textures. The black-white channel is also essential to show the shape of curved surfaces through shading, and this means that smooth graduations of darkness and lightness will be perceived largely as shading differences.

Unlike gray-scale gradients, smooth graduations of color tend to be perceptually categorized into regions of red, yellow, green, and blue. If it is necessary to present a hue so that it can be properly judged it is useful to isolate it with a neutral black, white, or gray border. Even better is to see it in context. Computer graphics allows this; the color of a graphical object can be adjusted *in situ* in a developing design.

We have just listed the main properties needed to make design decisions leading to visual clarity. Not surprisingly, there is a lot more to the use of color than this. Color design is subtle and can be a source of beauty and pleasure or disgust and irritation. Mostly this is a matter of socially constructed taste—there are no absolute standards. Our goal in this book is to find the bedrock of functional aesthetics in the wiring of the brain, and sadly, we have little to say about beauty.

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Chapter 5

Getting the Information: Visual Space and Time

“It is one thing to say with the psalmist ‘I lift mine eyes up to the hills’ —it is another thing to drag one’s carcass up there.”

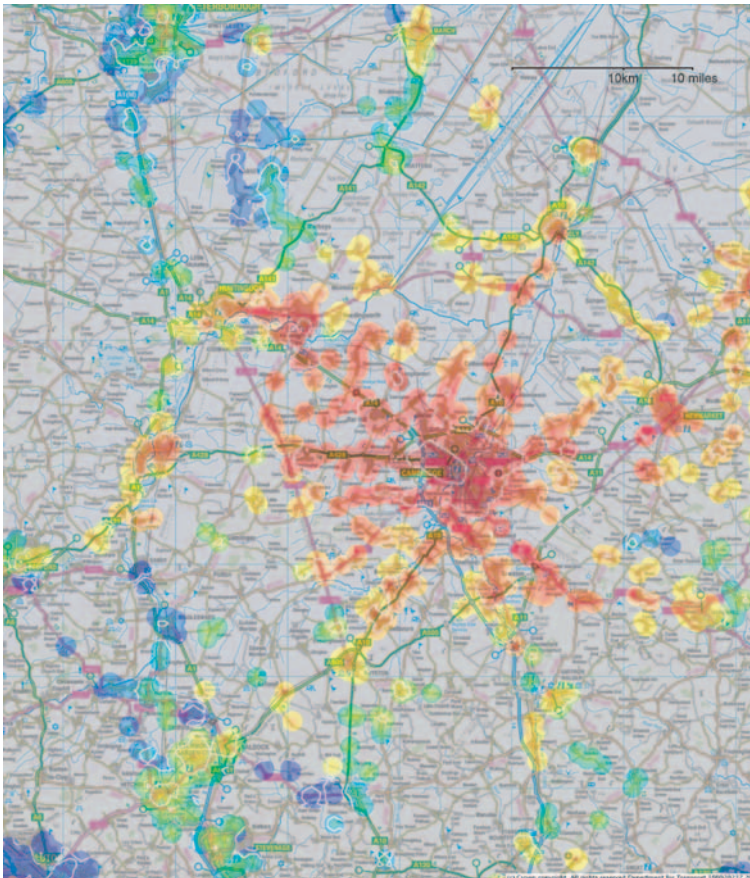
GK Chesterton paraphrased by John Prange.

Moving our eyes is the lowest-cost method we have for getting additional information about our environment. It has such a low cognitive cost that we are not even aware that we are making eye movements several times a second. In visual data searches “drag[ing] one’s carcass up there” is what we do when we search through a jumbled filing cabinet to find information, often entirely losing our train of thought in the process. It is also what we do when we travel a thousand miles to an information-sharing meeting.

Often when we engage in business travel, our goal is to meet people and discuss a project or gain access to concentrated information, for

example, at a conference. These are essentially cognitive activities, and so the cost of travel can be thought of as a cognitive cost. From a business perspective, even the social alliances that are formed are communication channels for the flow of valuable information and ideas. Although sound can be transmitted cheaply, sincerity, or the lack of it, is conveyed by subtle facial expressions, which is why face-to-face meetings are often worth the cost of travel.

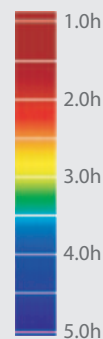
Eye movements, rummaging through filing cabinets, and business travel are ultimately all methods for traversing space to get information into our working memory systems, though they differ enormously in efficiency. It is in this light that we consider the structure of three-dimensional space.



This map of the region around Cambridge, England, shows the times to reach various regions by means of public transportation.

Some outlying villages can be reached as quickly as places on the outskirts of the city. Others are inaccessible. If the reason for travel were to obtain information, these times should be regarded as cognitive costs.

The colors give the travel times in hours.



chris@mysociety.org

With technology we can radically change the costs of access, storage, and transmission for both visual and auditory information. For example,

high-quality video conferencing systems can provide all of the visual and sound cues for the social dynamics of meetings to be preserved. The result can be substantial cost savings from reduced air travel, and meeting expenses, as well as greater overall productivity since more time can be spent in environments conducive to cognitive work.



High-quality video conferencing systems can transmit many of the social cues that make face-to-face meetings important.

Conversely, low-quality video conferencing systems fail to convey the subtle cues of expression and gesture important for judging intentions. (Image from Polycom.)

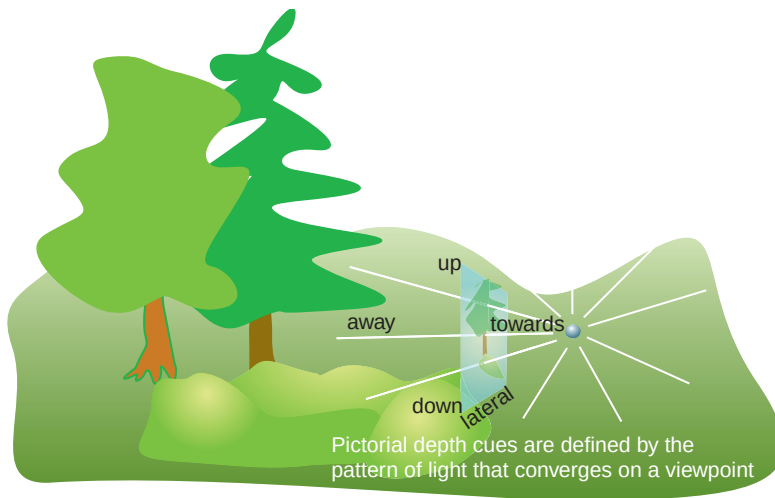
This chapter is about the perceptual and cognitive structure of space taking into account both how depth is perceived and the costs of navigation to access information. Next we will look at the ways in which depth is perceived.

DEPTH PERCEPTION AND CUE THEORY

It seems incontestable to say that we live in a three-dimensional world. But “live in” is a weasel phrase. The *physical* world is indeed three-dimensional, for most practical purposes. Although some theoretical physicists believe that all matter is constructed of strings in twelve-dimensional space, this only matters at a subatomic scale. The cognitive organism, however, acts on visual information that is imaged on the retina. This leads to perceived space having dimensions that are very different from the spaces mathematicians and physicists consider because each dimension has different affordances. The perceptual *egocentric* space is commonly thought of in terms of the *up* dimension, the *sideways* dimension, and the *towards-away* dimension.

In Chapter 3, we considered visual pattern processing that occurs on information captured by the *up* and *sideways* dimensions. Now we turn our attention to the information available to the eye and brain from the *towards-away* dimension. This information is contained in what psychologists

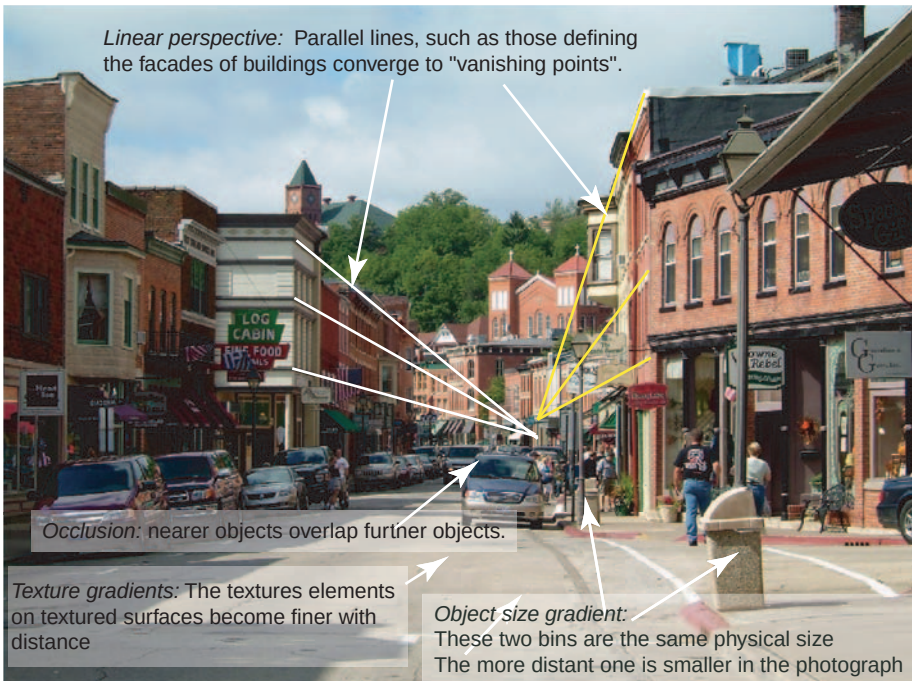
call *depth cues*. Most depth cues consist of environmental information that we use to judge distances from our particular viewpoint. There is much less information available in the *towards-away* directions than *up* and *sideways* directions, since the latter is directly available from the image on the retina. This is why perception is sometimes called a 2.5-dimensional phenomenon with the 0.5 referring to the *towards-away* information. The 0.5 should not be taken literally; *towards-away* information is much less than a half of what is provided by the other dimensions because we cannot see through most objects and so for every one of the million brain pixels recording up and sideways information there is only, at best, one point of depth information.



Some of the depth cues are called pictorial, because they can be used in static pictures.

One way of thinking about pictorial cues is in terms of a pane of glass which we look through with one eye holding steady at a particular viewpoint. If we were to put dabs of paint on the glass pane, corresponding to the colors in the world behind it, we would capture all of the pictorial depth cues because pictorial cues are defined by the projection of points in space onto a plane.

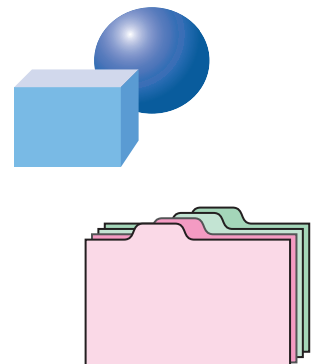
Depth cues can be divided into those that are *pictorial* and those that are *non-pictorial*. Pictorial depth cues can be reproduced in a photograph or a realistic painting. First and most powerful among these is occlusion; objects that visually block other objects appear closer. There is a whole set of pictorial depth cues that derives from the geometry of perspective including size gradients, texture gradients, and linear perspective. Other pictorial depth cues are cast shadows, height on the picture plane, shape from shading, depth of focus, size relative to known objects, and atmospheric contrast reduction. The non-pictorial depth cues are stereopsis (arising from our having two eyes), structure from motion (which we get as we move through the world), as well as lesser effects like focusing of the lens in the eye, and the convergence of the two eyes that occurs when we fixate near objects.



Pictorial depth cues do not have to be applied as a complete package in an all-or-nothing fashion; they can be applied independently according to the goals of the design. This means that the choice of the designer is not three-dimensional versus two-dimensional, but which of the depth cues to apply. Each of them has unique properties that can support different kinds of visual queries.

Occlusion: Objects near to us block or visually *occlude* objects farther away. An object that occludes another appears closer. Occlusion is the strongest depth cue, and if it is placed in competition with another, such as size constancy, occlusion wins.

Occlusion provides both a method and a metaphor for rank-ordering information with the most important partially occluding the less important. When designing with occlusion it is critical that the partially occluded objects can still be identified. Interactive computer applications allow for occluded information to be brought to the front with a mouse click.



Perspective: *Size gradients:* More distant objects are smaller on the picture plane than similarly sized nearby objects.

Size changes can be used to provide a visual metaphor for the relative importance of different information objects. Using this cue has the advantage that less important information takes up less space than more important information.



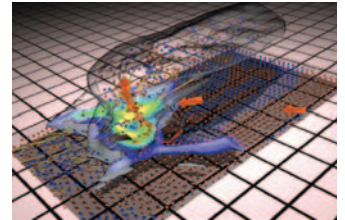
Perspective: *Texture gradients:* There is a reduction in size and increase in density of texture elements with distance. The texture elements on a ground plane provide a size reference for objects, although most textures provide less support than a grid in making size judgments. Unfortunately, fine textures are usually not reproduced on computer displays, which can be a major shortcoming.



Linear perspective: Projections of parallel lines converge on the picture plane.

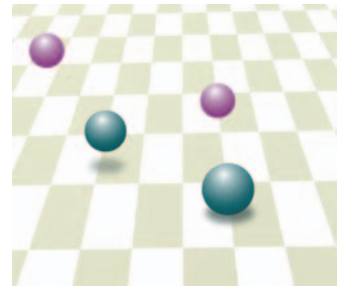
A grid of lines is a common device used in scientific illustrations to provide a reference plane for judging the three-dimensional layout of objects. A grid also provides a ubiquitous ruler for visual queries aimed at judging the sizes of objects.

Robert B. Wilhelmsen et al. A study of the evolution of a numerically modeled severe storm. *International Journal of Supercomputing Applications*. 4, 20–36, 1990.



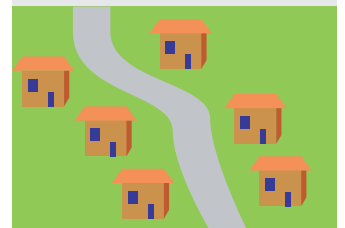
Cast shadows: The shadow cast by one object on another provides information about the distance between them. The height of the two cyan balls above the checkerboard can be seen from the shadows. The actual depth information in this image is provided by the checkered ground plane. The depth of the balls is perceptually inferred from its relationship to the perspective grid.

The magenta balls, having no shadows, have nothing to tie them to the ground plane; they seem to float at an indeterminate distance.



Height on picture plane: Because our visual world is dominated by the ground plane, objects higher up in the visual field are generally farther away. Placing an object higher up on a picture plane can be used to represent greater distance.

This simple method is often used in nonperspective maps to indicate relative distance from a certain point, typically at the bottom of the map. (Note that sometimes this is called a *parallel* or *axonomic projection*.)



Shading: The surfaces of objects reflect more or less light depending on how they are oriented towards the light source.



Depth of focus: The human eye, like a camera lens, focuses objects at a specific distance. Objects that are farther away or nearer are blurred. This effect provides inexact depth information. It is also called depth of field by photographers.

Sharp images more readily attract the eye than blurred ones, and so degree of blur can be used as a design technique to direct the attention of a viewer.



Reference to nearby known objects: Objects of known size provide a reference against which other objects are judged. Perspective cues only give cues to relative size, but with a reference object absolute size can be judged. Note that although this effect is knowledge-based and not pure geometry like the rest, it is one of the most important cues to distance.

The pink cow has been placed next to a boulder in this image of Martian rocks to provide a size reference. It would work even better if the cow had been given a cast shadow to anchor it to the ground.

Image courtesy of Kurt Schwehr and NASA.



Degree of contrast: Because neither air nor water is completely transparent, the contrast between an object and its background is *reduced* as distance increases.

Air is relatively clear, and most atmospheric contrast effects only occur when considerable distances are involved. Water, being less transparent, generates larger effects.

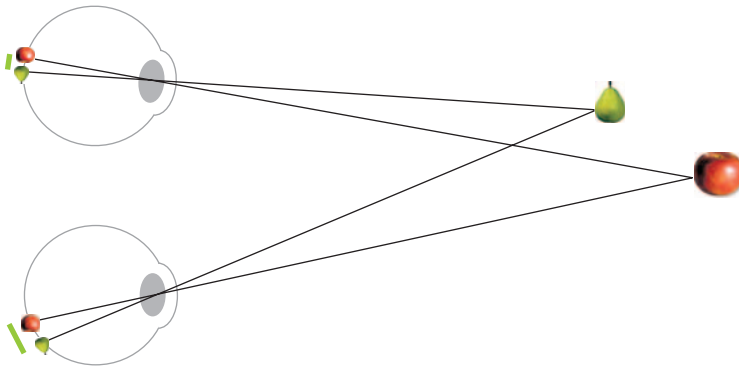
Designers can reduce contrast artificially to exaggerate atmospheric contrast and create an enhanced sense of depth. In addition, reducing contrast, like depth of focus or size scaling, can be used to direct attention away from less important objects.



STEREOSCOPIC DEPTH

Some depth cues are not captured in a static image, and are therefore not pictorial. One kind of depth cue comes from the fact that humans, like many other animals, have two front-facing eyes. Visual area 1 contains specialized mechanisms for using the small differences in the images in the two eyes to extract distance information; this ability is called *stereoscopic depth perception*.

Different depth cues have different uses depending on the tasks we are trying to perform. Stereoscopic vision is optimal for *visually guiding our hands* as we reach for nearby objects. It works best in making judgments of the relative distances of *nearby* objects, within a meter or two of our heads. Stereoscopic depth judgments are also most precise for objects that are at *nearly the same depth*. The brain is not good at using stereo information to judge large relative distances. Because of these properties, people who have little or no stereo depth perception (20 percent of the population) still have no difficulty driving cars or walking around, although they will be clumsy when trying to thread needles.



The brain uses differences between the images in the two eyes to determine the relative distance to objects.

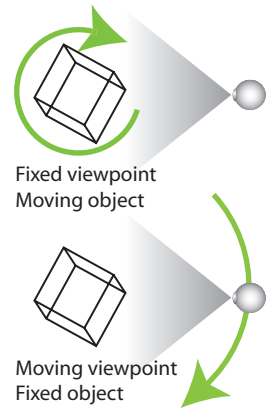
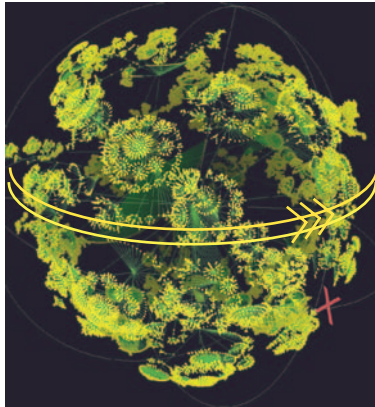
Technologies that support stereoscopic viewing present slightly different perspective views of a three-dimensional scene, each based on a slightly different viewpoint.

Stereo technology has, for the most part, been a story of fads that faded, and this is largely because of a failure to understand that the main virtue of stereo vision is the precise guidance of hand movements. The Victorians were fascinated with stereo photos and sold thousands of stereo images. Now they clutter the stands of flea markets. Stereo cameras had a heyday in the 1930s and 1940s but now are almost unobtainable. Many inventors and entrepreneurs have lost their shirts on stereographic movies and television systems. The problem is that none of these technologies allowed for manual interaction with the visual three-dimensional objects that were represented. If we had virtual three-dimensional environments that allowed us to reach in and move things, then we would appreciate stereo technology more.

STRUCTURE FROM MOTION

Another non-pictorial depth cue that is more useful and powerful than stereoscopic depth is called *structure from motion*. The brain can interpret the moving pattern that arises when objects rotate with respect to a viewpoint to derive information about depth. No special technology is needed beyond simple animation; structure from motion information is provided in conventional movies whenever the camera moves or whenever an object rotates in the frame of the camera.

It is hardly surprising that structure from motion provides better depth information than stereopsis. Stereoscopic depth comes from just two views, one for each eye. With motion the brain can take advantage of a continuous series of views to interpret depth information.



If a rigid three-dimensional computer graphics model, such as this network diagram, is rotated about a vertical axis, its three-dimensional structure will become dramatically more apparent. (Copyright 2002. The Regents of the University of California.)

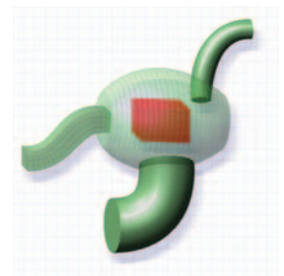
2.5D DESIGN

2.5D design is a phrase intended to encapsulate a set of design concepts that arise once we start thinking of space in human egocentric coordinates.* The 2.5-dimensional design concept is not new. Low-relief friezes on temples, the Bayeux tapestry, and many medieval manuscript illustrations are 2.5-dimensional designs.

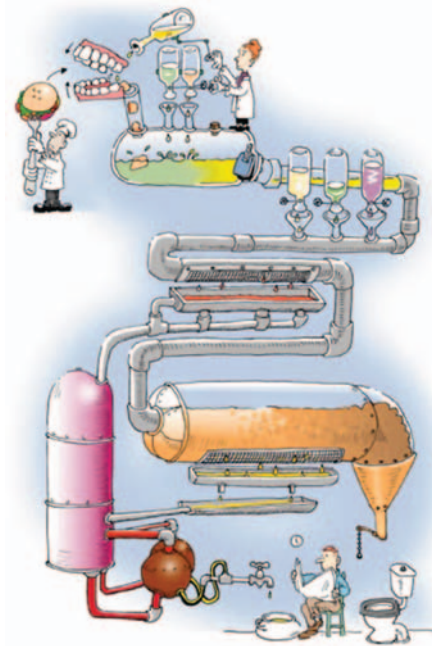
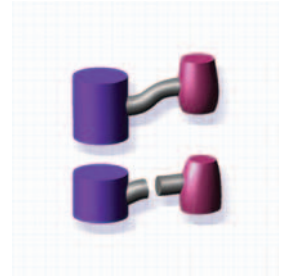
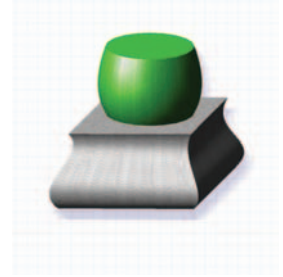
The key idea in 2.5-dimensional design is to treat depth in the picture plane very differently from the other two dimensions. Its chief tenets are as follows:

- Depth cues should be used selectively to support design goals. It does not matter if they are combined in ways that are inconsistent with realism.
- Objects should be laid out with minimum occlusion. Usually this means that overall depth in the scene should be limited.

*The term *2.5D design* originates with the work of David Marr who introduced a 2.5D "sketchpad" as one of the stages of visual processing [Marr, D, 1982. *Vision*. Freeman, NY,]. However, the specific design principles presented here are not based in any direct way on Marr's theory.



- When occlusion is necessary, ensure critical information is not occluded. Sometimes transparency can help reveal what is behind an occluded object.
- Joints between compound multipart objects should be made clearly visible.
- Make clear spatial relationships, such as *on top of*, *attached to*, and *inside of*.
- Depth should be minimized, but when there are different depths the layout should be carefully designed to make relative distances clear. Simple cast shadows, depth-of-focus effects, and transparency can be used for this.
- Display text in the image plane. Text that is slanted in perspective views is harder to read.
- Give common views of objects for rapid identification.



This illustration of the human digestive system is an example of 2.5-dimensional design. Although many of the individual components are rendered in three dimensions, the layout is almost entirely in the picture plane. The connections between components are made very clear. There is occasional occlusion but it never destroys object identity. (Image from David Dickson, [illustrator.www.daviddicksonillustration.com](http://www.daviddicksonillustration.com).)

Note that some of the principles listed above particularly the ones relating to joint and object views, are based on concepts that are presented later in Chapter 6. These have been included here to make a more coherent and complete list of 2.5-dimensional design concepts.

HOW MUCH OF THE THIRD DIMENSION?

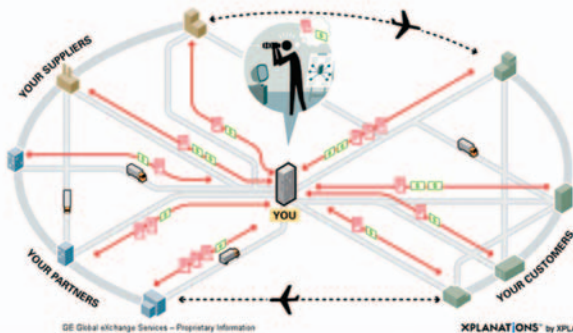
An important issue in determining whether a three-dimensional or two-dimensional design is appropriate is the nature of what is to be displayed. Some data already has three-dimensional spatial properties; for example, architectural designs, and data from the physical and biological sciences.

On the other hand, many kinds of data and concepts are not inherently spatial. Typical examples are business statistics, social networks, and the myriad of abstract concepts people have developed to cope with the world (education, economic value, causality, to name a few). Some people have tried to use three-dimensional representations for these kinds of things, believing that because we live in a three-dimensional world, three dimensions must be better than two. This argument fails to take into account the fact that we do not perceive three dimensions. Even saying that we perceive 2.5 dimensions is a considerable exaggeration because the information we pick up from the *towards-away* direction is much less than half of what we get from the other two. Attempts to show abstract non-spatial data using three-dimensional views have not been successful mostly because the information is not displayed clearly and because getting around in three dimensions is much harder than navigation in two dimensions. The cost of getting a good viewpoint in three dimensions is almost always higher than clicking to follow a hypertext link or zooming in two dimensions. This makes visual queries expensive and slows down the cognitive process. Three-dimensional information mazes are especially to be avoided since it takes people a long time to generate a mental map of such a space. On the other hand, a primarily two-dimensional design with a judicious use of depth can be useful because it allows for effective ways of layering information.

Even when designing a truly three-dimensional virtual information space is justified because the data to be displayed has important three-dimensional properties, 2.5-dimensional design principles should still be applied. Any time we take the viewpoint into account we are making a 2.5-dimensional design decision. This is because the viewpoint affects the visibility of things in the image plane, which is a two-dimensional representation. Often it is useful to mix two-dimensional design elements with three-dimensional design elements. Navigation controls should always be visible on the screen and have a size such that they can be easily selected. Therefore, many three-dimensional interfaces actually have two-dimensional controls placed in the margin. Text labels should be laid out in the image plane, even though they are attached to

Transactions are the lifeblood of your business

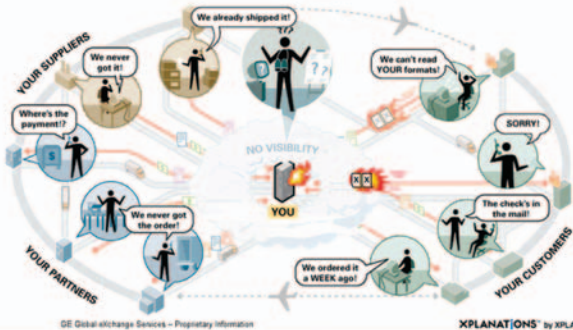
For the right things to happen, the flow of information must be reliable. To ensure accountability, you must know who has what information, and when they got it.



PPT slide: The old way (without GX)

How many different document formats are in the system?

Is the flow of information reliable and consistent?
Can you handle other people's errors and exceptions?
Do you know who has what information?
Do you know when they got it?

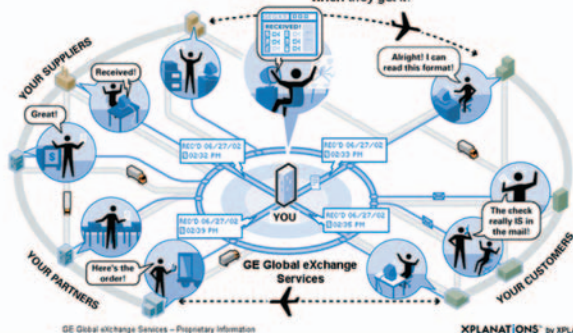


PPT Slide: The new way (with GX)

Outsource your transactions to GE Global eXchange Services

You don't have to buy software or hardware or hire people, and you won't have to upgrade. Your information flow is reliable.

Other people's errors and exceptions are efficiently managed. Trading partners are accountable because you know who got what, and when they got it.



This design of a business process was developed by the company XPLANE™ by xplane.com©.

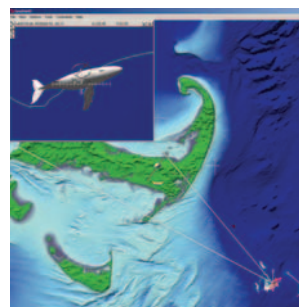
It is a good example of the application of 2.5-dimensional design principles to the visualization of a complex business system. It makes careful use of occlusion, height on the picture plane, and layering. No important objects are obscured through occlusion. All the important junctions in the information network are clearly visible.

three-dimensional objects. When designing an interface to a large data space, a two-dimensional overview map should be provided in combination with one or more perspective views showing details. This mitigates the common problems of being lost in a data space. The two-dimensional overview map will be easier to use if it has the same general orientation as the magnified image since people find it difficult to cognitively integrate data seen from radically different viewpoints.

AFFORDANCES

The 2.5-dimensional design principles discussed in the previous pages mostly apply to static, non-interactive designs. They are useful ideas, but they have taken us away from the main thread of this chapter—the connections between perception and action—to which we now return. A useful starting place is the concept of *affordances* developed by the psychologist James Gibson in the 1960s.* Gibson started a revolution in the study of perception by claiming that we perceive physical affordances for actions, and not images on the retina, which had been the basis of prior theories. A flattish firm ground surface *affords* passage, through walking or running or maneuvering a vehicle. A horizontal surface at waist height *affords* support for objects and tools we are working with. Objects of a certain size, shape, and weight *afford* use as tools. For example, a rock can be substituted for a hammer.

Perception of space is fundamentally about perception of action potential within the local environment. Negative affordances are as important as positive ones since they rule out whole classes of activity. A brick wall is a negative affordance completely ruling out easy access to large areas of space.



The small inset perspective view of the whale combined with an overview map allows for details to be understood in a spatial context.

*James Gibson's book, *The Senses Considered as Perceptual Systems*, stresses the structure of information in the environment as the foundation of perception.

Boston, Houghton Mifflin. 1966.12

A pedestrian perceives the places that afford safe walking. A driver perceives the places that afford vehicle navigation. Buildings provide the negative affordances in that they restrict travel. On the other hand, if a pedestrian's goal is shopping for shoes, then affordances relating to the likely presence of shoe stores will be perceived.



Workbenches and kitchens are structured to afford good support for manual construction tasks.

Stereoscopic depth perception, cast shadows and texture gradients are important depth cues for interacting in such an environment. (Image courtesy of Joshua Eckels. josh@jeckels.com.)

Gibson conceived affordances as *physical properties of the environment*, but in reality many of the most useful affordances have to do with access to information. To make the concept of affordances more broadly applicable, computer interface designers have changed its meaning. As it is used now, it has a cognitive dimension that departs from Gibson's definition. *Cognitive affordances* are readily perceived possibilities for action. For example, a computer interface may have a number of on-screen buttons that are available for metaphorically pressing with the mouse cursor.

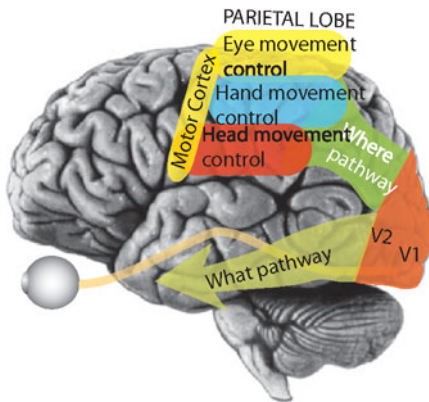


This office space, although appearing messy, is well structured for cognitive work. The computer is the nearest object to the worker and at the focus of attention. Other information carrying objects are papers arranged so that the most important are close to hand and near the tops of piles. Less frequently accessed sources of information, such as books, are more distant. The work space has affordances structured for efficient access to information.

THE WHERE PATHWAY

Affordances are ultimately about linking *perception and action*, and there are pathways in the brain that are specialized for this task. The *where* pathway runs forward from V1 and V2 to the parietal lobe in the middle of the upper part of the brain. The parietal lobe contains a number of

coarse *action maps* of visual space. There are maps with retinal coordinates, body coordinates, and external world coordinates. Each map links visual information to some form of guided movement. The lateral intraparietal (LIP) region has a representation of visual space in retinal coordinates, with areas weighed based on attentional priority. LIP is critically involved with planning the next eye movement. The medial intraparietal (MIP) region contains another distinct representation of nearby space that is used to guide arm movements. Other regions provide a head-centered representation of space and a body-centered representation of space, used for locomotion. Each of these regions has strong onward connections to the motor cortex—a part of the brain involved in setting up the sequence of muscle contractions needed to set part of the body in motion.



The parietal lobe is the upper-central part of the cortex. It contains pathways that link vision to action.

Various maps of visual space are contained in the parietal lobe, each supporting a different kind of action pathway. These regions are actually much smaller than shown on this diagram.

The use of the term “map” in the above description implies something rather static, detailed, and permanent. This is misleading because these maps are, in fact, temporary and have very little detail. For example, the eye-space map is updated every time we move our eyes. It contains a rough representation of the locations of a few objects in space that are task-relevant. These representations function to support the activities we are currently engaged in, and they contain just enough information for those actions.

Much of what goes on in the *where* pathway is unconscious. Most of the time, we are completely unaware that we are making eye movements. We are also unaware of the hand movements we make in grasping moving things. When we walk on rough ground, we constantly adjust our foot placements without conscious attention. Our motor-planning system registers the presence of small obstacles, adjusts the spacing of our footsteps,

and sets off a train of signals to muscles that causes our foot placement to change by just the right amount.

A basic ability to move our eyes to a simple target is probably innate, although it becomes much more skilled with time. But other visually guided movements must be learned from scratch, often with considerable effort.* Visual motor skills, such as walking, reaching for an object, driving a car, or playing golf, go through well-documented stages, from being extremely difficult and demanding complete attention to effortless and being done automatically. Increasing levels of skill means that the pathways linking perception and action are stronger and require less and less high-level attention. Baseball players have highly developed pathways specialized for catching or hitting fast-moving balls. Teenagers who play video games for hundreds or thousands of hours have highly developed visual-motor pathways for navigating through artificial three-dimensional spaces. These pathways are optimized, mapping the specific configuration of buttons found in game controllers into movements through virtual computer graphics-generated spaces.

ARTIFICIAL INTERACTIVE SPACES

In interactive video games, the critical affordances are artificial, and in a sense, metaphoric. Because our mental models of space and gravity come first from the physical world, we assume that objects in the virtual world of computer games will stay on tabletops. The real is a metaphor that guides the design of the artificial.

It is only necessary for things to exhibit *roughly* the right physical behavior for objects to appear normal when we interact with them. For example, the way balls ricochet from each other can be altered in ways that substantially distort physical laws and people will not notice. Vision researchers call this *the naïve physics of perception*. It means that the models that are embedded in our nervous systems are only crude approximate representations of physical reaction patterns. One consequence of this is that we can make virtual worlds of video games in which the basics of physics are distorted without people noticing. Video games often distort the physics of action and reaction in extreme ways and even when we do notice we can very rapidly adapt to such distortion so long as the pattern linking perception and action is preserved. For example, game designers usually reduce the force of gravity by a factor of two or more and still players are able to control the characters with little or no difficulty even though they jump and fall more slowly than they should according to strict realism.

*Elizabeth Spelke and her co-workers at Cornell University measured babies' looking behaviors, and used the results to argue that we are not born with an innate appreciation for gravity, although we are born with an appreciation for the permanence and solidity of objects.

E.S. Spelke, K. Breinlinger, J. Macomber, and K. Jacobsen, 1992. Origins of Knowledge, *Psychological Review*. 99(4): 605–632.

Efficient designs for information access generally build on visual motor skills we already possess. For example, although a computer mouse is completely disconnected from the cursor on the screen, the cursor moves to the right when we move our hand to the right. This builds on the very early skills a baby develops when initially random hand movements cause objects in the visual field to move. Try rotating your mouse 90 degrees and selecting some object on your screen to see how difficult things can be when we violate the basic correspondence between hand movement and cursor movement.

The idea that our mental models of space perception are grounded in real-world interaction requires some qualification. The neural architecture and the most basic human capabilities are innate. The world is changing in profound ways. As younger and younger children work with computers, purely invented consequences of actions, such as cursors that move with the mouse, become basic. Pull-down menus are common and have come to be as real in an important way as physical balls that roll when kicked. With extended use, the motor pathways that connect perception to action can become just as strong for the computer interaction as for the physical action.

Many individuals spend significant parts of their lives interacting with computers containing simulated environments. If these virtual worlds contain virtual laws that differ substantially from the real world, then these patterns will become the basis for an alternative naïve physics, not based on the real world, but upon artificial worlds. Human perception is, over time, becoming less grounded in real-world interactions. Skill with a computer mouse is already assumed for computer-interface designers. Skill with common video game controllers is becoming increasingly ubiquitous.



As natural as walking?

SPACE TRAVERSAL AND COGNITIVE COSTS

We now return to the point that moving through space can be regarded as a cognitive cost. We weigh the time to go to the library against the time to access information online, taking into account the relative quality of information we are likely to get. There are cognitive tradeoffs other than time to consider. We choose to travel by train, as opposed to driving a car, because we can read and write while on the train but not in the car, although even on the train the work we can do is far less than we can accomplish in an office that is highly optimized for cognitive work.

In this light, it is useful to review the basic costs of some common modes of information access. A list of rough values for various costs is set out in the following table. Some designers of computer interfaces literally count such costs for the execution of a particular task—how many mouse

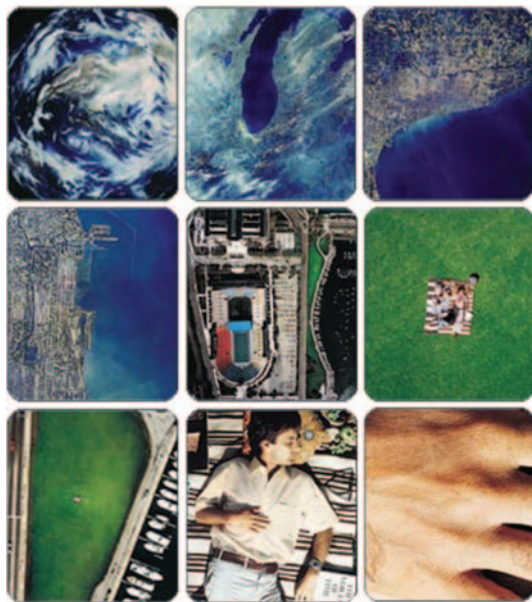
movements, how many eye movements, how many keystrokes, and so on. In this way, the efficiency of one design can be compared against another. The table is far too rough an approximation to be used for this purpose. Rather it is intended as food for thought. A computer interface design can often be dramatically improved by changes that involve, for example, making it possible to access a frequently used piece of information via an eye movement instead of a mouse movement. Although pure cognitive efficiency is never the only design issue, it is always something that should be taken into account.

Internal pattern comparison	0.04 seconds	Occurs whenever our eyes alight.
Eye movements	0.1 seconds	Tightly coupled with visual cognitive system.
Mouse hover: A hover query is what occurs when the cursor is placed over an information object and information appears. No clicking is required.	1.0 seconds	If they are highly optimized, hover queries can be invaluable techniques for rapid exploration. As the mouse cursor moves over a particular object, related data objects become highlighted.
Mouse select	1.5 seconds	A mouse selection of a web link can cause an entirely new page of information to appear.
Zooming: Discussed shortly	$\text{Dist} = 4t$ (seconds)	The most efficient way of traversing large distances with visual continuity.
Walk	4 kph/dist (minutes)	Good affordances for contemplative thought. Poor affordances for interactive cognitive work.
Drive	80 kph/dist (minutes-hours)	Moderate affordances for contemplative thought. Poor affordances for interactive cognitive work.
Fly	400 kph/dist (hours)	Moderate affordances for contemplative thought. Depending on class of travel moderate affordances for interactive cognitive work.
Train	100 kph/dist (hours)	Moderate affordances for contemplative thought. Depending on class of travel moderate affordances for interactive cognitive work.

The advent of high-performance three-dimensional computer graphics has led to the invention of navigation methods that are non-metaphoric in that they are not based on real-world interactions. One of these is the zooming interface. As it is usually implemented (for example, in Google Earth), zooming causes a constant change in scale per unit time. If we are zooming in, in one second we might see objects four times as big. In two seconds, we will be able to see four times as big again, or sixteen times in total. In three seconds we can see $4 \times 4 \times 4$, or sixty-four times the detail.

At this rate, we can zoom from an overview of the entire earth showing down to a view of a person's hand, in about fourteen seconds. Fourteen more seconds will get us down to the size of a single molecule. This is the power of exponential change.

Smooth zooming at a rate of four times per second will still give perfect visual continuity.



Part of the extraordinary image sequence created by Charles and Ray Eames, showing a spiraling in from space to a point on earth.

The hyperlink is another example of a new non-metaphoric method of navigating information spaces. The click of a mouse button selecting an object on one screen causes a new screen of information, sometimes from across the world, to appear in less than a second. Interface techniques such as zooming and hyperlinks are causing such radical changes in the cost of navigating information spaces that human society is being altered. Because information access is increasingly no longer a matter of traversing physical space students rarely go to libraries, people meet one another online, and gamblers can avoid casinos.

CONCLUSION

The main point of this chapter has been that the functional aesthetics of space are related to the cost of accessing information. Although we live in three dimensions, we always see the world from a particular viewpoint, and it is always faster to redirect our gaze to get new information than to move our heads or our whole bodies. Perceptual space is also flattened in

terms of its information content, with much less information being rapidly available in the *towards-away* (depth) direction than the *sideways* and *up-down* directions because we cannot see through most objects. The pattern-processing mechanisms of the brain operate on the two-dimensional retinal image which means that two-dimensional image plane information is analyzed much more efficiently than depth information.

This is not to deny depth perception, but depth is perceived in ways that are radically different from the way patterns in the image plane are perceived. The brain determines relative distance by means of depth cues such as occlusion, shape from shading, and linear perspective. Information from the depth cues is used in a piecemeal fashion to help perform the task at hand. Sometimes motion parallax will be most important, sometimes relative size and sometimes shading information, it all depends on what we are trying to accomplish.

The set of design principles that takes the space-time structure of perception into account is called 2.5-dimension design. The fact that the brain is capable of interpreting depth cues in a piecemeal fashion frees the designer to break the rules of perspective and make use of depth cues in different combinations as they seem most suitable to serve the purposes of a design. There is no need to use them all together in a consistent way unless visual realism is, in and of itself, a design goal. The piecemeal use of depth cues is part of 2.5-dimension design; however, the layout of objects and patterns as they appear in the image plane should be the primary design consideration.

The designer of interactive computer-based applications has options relating to the cost of information access. The cost of eye movements can be compared together with the cost of a mouse click. The cost of virtually walking can be compared to the cost of flying or zooming. Having an understanding of the various tradeoffs involved in making these choices is critical in designing effective cognitive tools.

Chapter 6

Visual Objects, Words, and Meaning

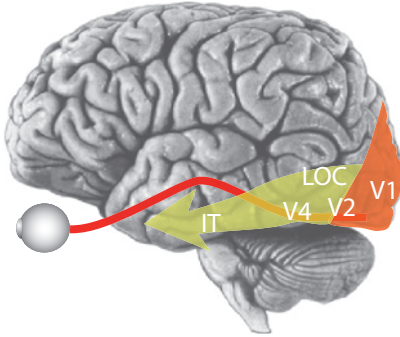
In the movie *A Clockwork Orange*, the antihero, Alex, a sadistic punk of the future, is a guinea pig in an experiment to “reprogram” him and thereby make him a useful member of society. Ultimately, it is hoped, the method will remove the need for prisons. The reprogramming involves pairing *images* of sex and violence with nausea-inducing drugs. It is a plausible idea that this should work because images can evoke strong emotional responses and are often associated with obsessive behavior; however, this kind of treatment has not been successful in reality.

The activation of meaning from an image generally occurs in a small fraction of a second, much less time than it takes to read a paragraph of text. This activation through a single glance makes images far more efficient than words at conveying certain kinds of information.

In this chapter, we examine the brain processes involved in object and scene perception and consider the links between visual and non-visual (especially verbal) information. We shall begin with an overview of the neurophysiology, and end with a discussion of the role of images in obsessive thinking.

THE INFEROTEMPORAL CORTEX AND THE WHAT CHANNEL

We have already discussed the processing of increasingly complex patterns through brain areas V1, V2, V4, and LOC. Now we move up to the stage of the inferotemporal cortex, where we will encounter patterns that are meaningful in an everyday sense.



The “what” channel consists of a series of strongly interconnected brain areas that contain neurons responding to progressively more complex patterns. At the end of the chain is the inferotemporal (IT) cortex.

The inferotemporal cortex is an area at each side of the brain (behind the temples) containing neurons specialized for complex visual patterns corresponding to recognizable objects and scenes. It contains a number of subareas. There is, for example, a subregion called the fusiform gyrus that contains cells responding specifically to faces. Other regions are specialized for cars and houses. There are also links to areas in the forebrain and the midbrain that provide *multimodal* connections between visual information and information processed by other sensory channels.

In order to recognize objects and scenes, the brain must solve a number of very difficult problems. Consider face recognition; studies of human pattern perception show that humans are much better at identifying objects when the viewpoint is a familiar one. Recognizing an upside down face is difficult and so is recognizing alterations to a familiar face as the following image demonstrates.



This modified picture of Marilyn Monroe looks much worse when you turn it right side up! This effect is called the Thatcher Illusion because it was first illustrated with the face of the British Prime Minister Margaret Thatcher.

Spatial memory for scenes is also viewpoint specific. Suppose someone has always entered a room from a particular door and has always stayed in that room near to that entrance so they have never experienced it from a substantially different viewpoint. Generally they will fail to recognize the room when entering from a door on the opposite side. We can still recognize objects and scenes when there are quite large variations in the viewpoint from which we see them; the difficulties occur only with extreme changes of view. What applies to scenes also applies to individual objects. As a rule of thumb, we can rotate an object by about 20 degrees and scale it by a factor of two or three and still identify it rapidly. How does the brain do it?

GENERALIZED VIEWS FROM PATTERNS

The IT object identification areas receive bottom-up information from the V4 patterns processing region. V4 neurons are capable of responding to patterns that are rotated and distorted in variations of a prototype.



A neuron in area V4 might respond to all of these pattern variations.

But a pattern cannot be rotated or distorted too much or the neuron will stop responding. This tolerance of distortion makes the task of the IT cortex much easier. Complex objects, such as human faces, cats, teapots, or automobiles (all things that neurons in the IT cortex are likely to be sensitive to) can be thought of as patterns of patterns. The identifiable object patterns are constructed from simpler V4 patterns, which in turn are made up of elementary pattern components coming from V1 and V2.* In order to identify a face, for example, several sets of neurons must respond to a few *generalized views* such as a full face view, a three-quarters view, and a profile view. Each of these views provides a quite distinct visual pattern of patterns. Most other views of a person's face will provide only minor variations of one of these generalized views. Because the component pattern recognizers are tolerant of distortion, the overall mechanism is too.

*Neuroscience has yet to decide whether it is single neurons or assemblages of neurons that fulfill this function.



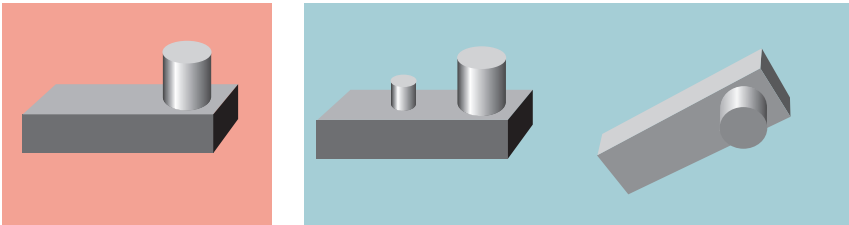
A face is a pattern of patterns.

Some theorists, beginning with the philosopher Plato, have held that we recognize objects with reference to other objects that are somehow *canonical*, abstract ideal forms. Superficially, the generalized views mentioned above might seem like a similar idea. This is not at all what is intended. We recognize faces and other things because clusters of neurons that receive inputs from other clusters of neurons respond strongly when a certain face becomes imaged on the retina. This is not a matching process, and there is no idealized form, simply a response to a range of patterns.

STRUCTURED OBJECTS

There has been an intense theoretical debate in the community of vision researchers as to whether or not we have three-dimensional “models” in our heads.* Most of the evidence suggests that most of us do not and that the “generalized views” account of recognition is sufficient to explain how people normally perceive objects in most cases. There are exceptions; for instance, people like carpenters and engineers develop specialized skills to mentally manipulate 3D structures. Such people do have some limited ability to build models in their head, albeit quite simple ones. The fact that there are very large individual differences in this skill suggests that it is not basic to all perception.

Nevertheless, there is clear evidence that we all have some limited ability to perceive the three-dimensional structure of objects. For some objects composed of component parts, the way those parts are connected is critical.



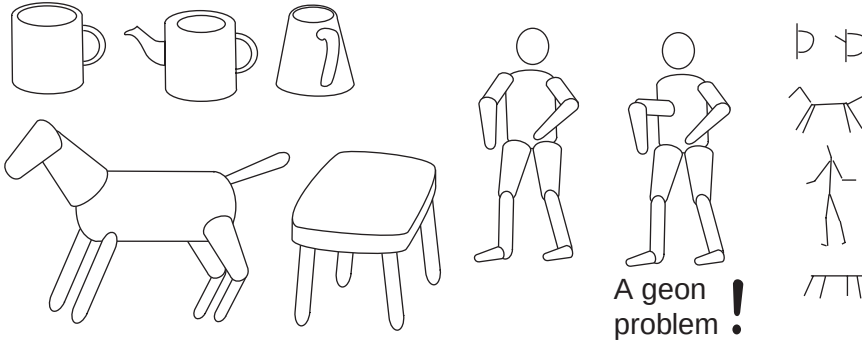
If asked which object in the blue box matches the one in the pink box most people will choose the one on the right. But according to the theory of object perception, based on two-dimensional generalized views, the one on the left of the blue box should be chosen because the image is mostly identical.

In order to account for our ability to identify structured objects, Irving Biederman proposed *geon* theory.* This is the idea that the brain contains mechanisms to identify three-dimensional structural components of objects that he calls *geons*. Geons, in his theory, are three-dimensional shapes such as cones and cylinders that can be curved or straight. The brain stores information about the geon components of objects, together

*A model in the head would be a neural representation of the three-dimensional spatial layout of an object's parts.

*I. Biederman, 1987. Recognition by components: A theory of human image understanding. *Psychological Review*. 92(2): 115–147.

with a description of the way they are connected—a kind of structural skeleton. When the object we see is a horse, the geons are its four legs, each composed of sub-geons, its head, its neck, and its tail. The way these parts are connected forms the geon skeleton.



These kinds of objects are defined by the way their distinct component parts are attached together. If one of the attachments changes, our perception of the object will change into a different object, possibly something unrecognizable, or an object that is in some way bizarre, like the figure with an arm coming out of his stomach.

The geon theory is probably the most controversial of the theories that are discussed here. Few researchers believe that neurons respond to exactly the set of three-dimensional primitives that Biederman proposed. However, there is strong evidence that we perceive the way parts of objects are interconnected—the structural skeleton in Biederman's theory. Objects can be identified far more rapidly if they are presented in views that clearly reveal the connections between component parts. For this reason, in design, it is always a good idea to make the connection points between parts of objects as clear as possible.



In the left-hand pair of images, the critical connections between the parts of both the teapot and the hand are evident in the silhouette. In the right-hand pair these connections are not as clearly shown and in the case of the teapot the handle is invisible. The right-hand pair are less likely to be identified correctly from a brief presentation.

There is no reason for the brain to have only a single process for object identification and it is likely that both the generalized two-dimensional

views theory, and the structured object theory are correct. The brain appears to do most identification through the mechanism of two-dimensional patterns, but for certain kinds of objects a kind of higher-level structural analysis of the kind Biederman proposed may provide the most critical information. Both theories suggest ways of making clear graphic representations.

GIST AND SCENE PERCEPTION

Identifying scenes provides another kind of problem for the visual system. People can easily identify scenes, such as “a busy road,” “a rural landscape,” or “a fast-food restaurant,” and they can make this kind of categorical judgment in less than a tenth of a second even if they have never seen that particular road, landscape, or restaurant before. This is about the same time it takes for a person to identify that an object is a person, a dog, a car or belonging to some generic class. This rapid identification is a problem for traditional theories of scene perception because they held that a scene is identified by means of the objects it contains, which implied that objects would have to be recognized first and faster.



We can get the gist of a scene in less than a tenth of a second. This is as fast as we can identify individual objects.

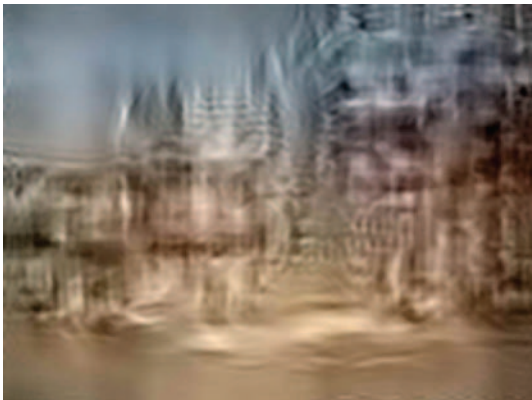
The rapid characterization of a scene is called getting its *gist*. Neither the generalized two-dimensional views theory nor the structured object recognition theory can account for why it is so rapid and efficient. Gist perception can, nevertheless, be thought of as another example of a process whereby patterns of patterns are the key. Two researchers at the Massachusetts Institute of Technology, Antonio Torralba and Aude Oliva, have sketched out how this might occur.* They showed how common scenes have typical spatial feature components, distributed in characteristic ways. These can include large-scale pattern arrangements of patches of textures and colors. There is no need for objects to be identified for a scene to be recognized.

*A. Oliva and A. Torralba. 2006. Building the gist of a scene: The role of global image features in recognition. *Progress in Brain Research*. 155: 527–532.

For example, beach scenes are generally characterized by a large blue area at the top of the image (the sky) with striated white and blue-grey areas either to the left or the right of the image (the sea with waves), and a large beige area (sand) containing smaller vertical and horizontal shapes (people) on the remaining side. Recognizing such patterns of patterns is exactly the kind of thing that neurons in the IT cortex are good at.



This is the average of hundreds of pictures containing a person, all taken in Mallorca, Spain. Notice that even though there are no individual small-scale features present, it is clear that there is a standing person in an exterior environment with a quite flat terrain. It is not necessary to recognize objects to perceive the scene. A. Oliva and A. Torralba, 2006. Building the gist of a scene: The role of global image features in recognition. *Progress in Brain Research*, 155: 527–532.

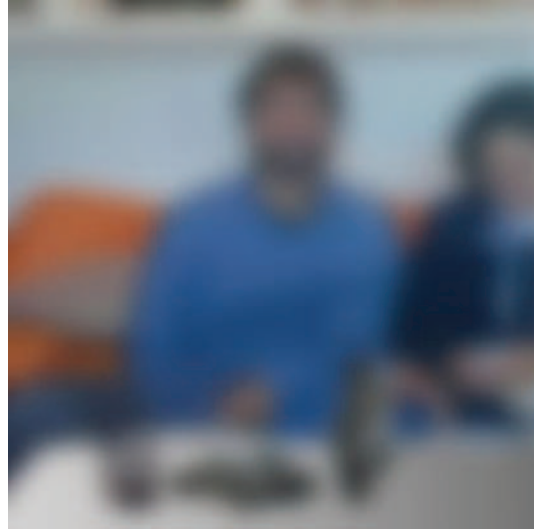
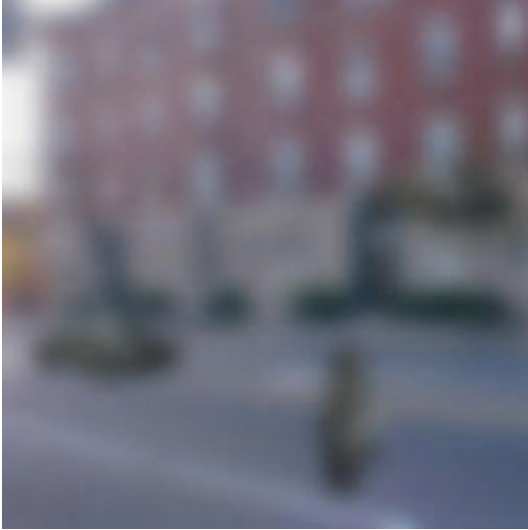


The statistical properties of a medieval cityscape are represented here.



The statistical properties of a beach scene are represented here.

Scene gist can powerfully influence how we see an object embedded in a scene. In the blurred street scene below left there is an automobile visible on the street. In the picture below right a bottle is standing on the table. Closer evaluation reveals these two blobby objects to be the same.*



*Antonio Torralba, 2003. Context priming for object detection. *International Journal of Computer Vision*. 53(2): 169–191.

Although they seem very different, objects and scenes can be thought of as high-level patterns; both consist of patterns of patterns. The IT cortex has numerous specialized object identification subregions, each of which has developed to process a different class of objects or scenes. These pattern processing clusters develop through our lifetimes, reflecting our visual skill development. The more we need to be able to appreciate certain kinds of visual signs, the more neurons become recruited to that task and the more tuned they become to specific patterns that are meaningful to us. The process starts when we are babies, and it never stops. Some specialists, including race car drivers, visual designers, chefs, dance critics, and those who fly fish, continue to develop and refine very specialized pattern identification skills.

VISUAL AND VERBAL WORKING MEMORY

Up to this point, we have been discussing objects and scenes as pure visual entities. But scenes and objects have meaning largely through links to other kinds of information stored in a variety of specialized regions of the brain. This information includes action patterns, eye-movement scanning strategies, and semantic content stored in the language systems of the brain. These links become activated through the operation of high-level attention-guiding mechanisms that are called *working memories*.

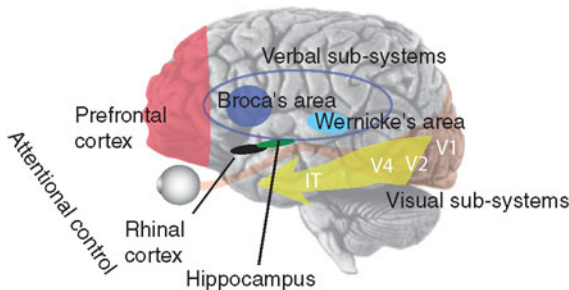
•The account given here has a number of sources. The term *nexus* describing a temporary binding together of information comes from Ronald A. Rensink. 2000. The dynamic representation of scenes. *Visual Cognition*. 7: 17–42. But the idea also has roots in the concept of a cognitive *object file* developed earlier by Daniel Kahneman and Anne Triesman, 1992. The reviewing of object files: Object specific integration of information. *Cognitive Psychology*. 24: 175–219.

The operative term in *working memory* is *work*, not *memory*. Information is only held in working memory from between one-tenth of a second and, at most, a few seconds, and it is only held to support some ongoing cognitive process. An entity in visual working memory can be thought of as a temporary grouping or *nexus*, whereby links are formed between active visual patterns derived from the visual image on the retina scene information and information relating to non-visual stored meanings. The most important of these non-visual links is information held in verbal working memory.

VERBAL WORKING MEMORY

The brain has a specialized set of neural subsystems that process language. These are situated in parts of the brain that are separate and distinct from the visual processing subsystems. They include Wernicke's area, where language is interpreted, and Broca's area, where speech is produced. Verbal working memory, like visual working memory, is a temporary transitory store.

Verbal working memory can hold about *two seconds'* worth of speech information in what is sometimes called an *echoic loop*. It is also useful to think of this as about three chunks of information. Much of what is conventionally taken to be "thinking," namely the internal monologue that we sometimes become aware of, is a form of internalized speech.



CONTROL OF THE ATTENTION AND THE COGNITIVE PROCESS

Information in visual and verbal working memory is often temporarily bound together. A visual chunk may be momentarily bound to a verbal chunk. The image of a stray cat and the phrase "get it out of here" become combined. Other temporary bindings are also formed. Visual chunks may trigger eye-movement plans and cognitive action plans needed to execute the next few mental operations. A plan is formed to chase the cat. These temporary bindings can be thought of as acts of attention, the shifting focus of the mind.

The prefrontal cortex has long been considered critical to the temporary bindings that occur as part of more complex plans. In the words of neuroscientists Earl Miller and Jonathan Cohen, “*Cognitive control stems from the active maintenance of patterns of activity in the prefrontal cortex that represent goals and the means to achieve them. They provide bias signals to other brain structures whose net effect is to guide the flow of activity along neural pathways that establish the proper mappings between inputs, internal states, and outputs needed to perform a given task.*”♦

♦E.F. Miller and J.D. Cohen. 2001. An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*. 24: 167–202.

LONG-TERM MEMORY

When we look at something, perhaps 95 percent of what we consciously perceive is not what is “out there” but what is already in our heads in long-term memory. The reason we have the impression of perceiving a rich and complex environment is that we have rich and complex networks of meaning stored in our brains. This meaning becomes activated by the quite small amount of information that actually makes it through the visual processing system to visual working memory. This is why many husbands do not notice when their wives have new hairstyles and wives do not notice when their husbands shave off beards or moustaches. What is perceived of wives and husbands is mostly what we have previously learned about them.

Long-term memory should not be thought of as a separate storage area in the brain, like a computer disc drive, where information is dumped after it has been processed by the sensory systems. It is not a distinct unit; rather, the brain is a system where every neuron both *processes* and *stores* information. V4 processes and stores visual pattern information and as we learn to identify new patterns neural circuits are formed and strengthened to process those patterns. Similarly, the IT cortex both processes and stores higher-level cognitive sequencing operations that relate, for example, to the sequence of steps we carry out when performing a visual task, such as reading a map.

The processes of visual thinking are very diverse, because what we think about is very diverse. Most visual thinking involves *situated interactions* with the world. Consider for a moment the following three examples: reading a statistical bar chart, merging onto an expressway in an automobile, and discussing a project in a face-to-face meeting. These seem very different, but each requires the interpretation of visual information and a structured sequence of actions depending on the information. Each is a skilled activity involving patterns of cognitive action that are refined and consolidated every time we perform the act successfully. They are also refined and consolidated when mistakes are made, recognized, and corrected. Visual

thinking procedures re-apply memory patterns that have been stored as action sequences. This is what constitutes the skill of visual thinking.

There are two different classes of long-term memory information and these are called *explicit* and *implicit*. Explicit memory is what we can explicitly recall and describe when we are given a long-term memory test and only a little of what we see and do makes it into this form of memory.* Implicit memory is whatever is retained every time we see, hear, or do something. Very little of this can be explicitly recalled but it does leave a trace. There is a lot more implicit memory than explicit memory, but explicit memory is special because of the role it plays in communication. We cannot communicate something if we cannot explicitly recall it.

For explicit learning of visual patterns and associations to occur, these patterns must be the focus of attention, at least momentarily, and make it into visual working memory. It is also necessary for an association to be made between visual information and verbal information. In addition, in order for long-term memory traces to be formed it is necessary for a twenty-four-hour consolidation to occur.

The midbrain structure, the hippocampus, is key to the formation of long-term memories. The evidence for this is striking; some unfortunate people have had their hippocampi damaged on both sides of the brain, and as a result they are completely unable to consolidate memories from events that happened after the damage occurred. These people will greet their relatives as if there had been a long absence, even if they have been out of the room for only a few minutes.

For long-term memory traces to occur there must be either an internal or external consequence associated with a cognitive act. An internal consequence might be that a particular sequence of neural processes leads to a solution to a problem. For example, making it onto the freeway smoothly will enhance and consolidate the neural patterns that lead to the particular sequence of eye movements, head movements, visual processing of the pattern of moving vehicles, and steering movements that yielded the successful outcome. Interpreting a bar chart and coming to a useful conclusion, perhaps about a business strategy, will yield an internal cognitive reward and similarly enhance the neural patterns that occurred during the interpretation. A successful dialogue with a co-worker (involving visually interpreting facial expressions) may yield an external social reward in the form of a smile. In each case, the reward strengthens the pathways supporting a neural activation sequence that underlies the processing of information. The long-term memory that is gained supports particular cognitive action sequences and so skill is acquired. This is very unlike the static “data” that

*D. Schacter, 1992. Implicit knowledge: New perspectives on unconscious processes. *Proceedings of the National Academy of Sciences*. 89: 11113–11117.

is stored in computer disk drives. In visual thinking, perception is a kind of cognitive action sequence, not some static registering of the external world. Long-term memory results from a strengthening of neural connections, increasing the likelihood that a particular neural chain reaction will occur.

PRIMING

To illustrate a point in a presentation, I might look through a few hundred images in a database. But little is retained for explicit memory access; the next day I will be unlikely to be able to explicitly recall having seen more than two or three. Nevertheless, some trace of the others remains in my brain. If I need to revisit the same database to find a new image, I will be able to skim through the images quite a bit faster. Somehow I can rule out all of the irrelevant images faster because I have seen them before. This effect is called *long-term priming*, or *perceptual facilitation*, and it is a form of implicit memory. Any images that we see and process to some extent prime the visual pathways involved in their processing. This means that those images, and similar images, will be processed faster the second time around. Repeat processing is easier, at least for a day or two.

Priming is the reason artists and designers often prepare for a particular creative bout by reviewing relevant images and other materials for a day or two. This gets the relevant circuits into a primed state.

GETTING INTO VISUAL WORKING MEMORY

We are now in a position to sketch out the entire process whereby a visual object comes to be represented in the mind. Of course, the cognitive process is ongoing; what we have just seen influences where we will look next and the information we pull out of the retinal image. In turn, that information influences where we will look next, and so on.

But we must start somewhere, so we will start, arbitrarily, at the bottom. The information (biased by attentional tuning as described in Chapter 2) sweeps up through the various stages of the *what* hierarchy, retina $\rightarrow$ V1 $\rightarrow$ V2 $\rightarrow$ V4 $\rightarrow$ IT, with more and more complex patterns being formed at each level. The simple features of V1 and V2 become continuous contours and specific shapes in V4. More complex patterns are activated in specialized regions of the IT cortex that correspond to meaningful objects we have experienced in our lifetimes.

When the wave of activation reaches the IT cortex *all* similar patterns are activated, although by differing amounts depending on how good a fit there is between the image information and the particular pattern detection unit. For example, if we are seeing the front of a person's face all

stored patterns corresponding to frontally viewed faces will be activated, only some more than others depending on the degree of match. Ongoing cognitive activities will also have an influence. For example, looking for a particular person will prime those patterns that have links to other information already in verbal working memory about that person. Those patterns that have been primed will be more readily activated than those that have not.

Many patterns become activated to some extent in the IT cortex, but we actually perceive only very few. A second wave of activation, this time from the top-down, determines which of the many patterns actually makes it into working memory. At this point, a kind of neural choice occurs and so we perceive not all faces but the face of a particular person. The selection process is based on the pattern that is responded to most strongly. This is called a *biased competition model*.^{*} The biasing has to do with priming and the task relevance of the visual information. Many patterns compete, but only between one and three win. Winning means that all competing pattern matches to other faces are suppressed. The result of winning is that a top-down wave of activation both enhances those lower-level patterns and suppresses all other pattern components.

The overall result is a nexus binding together the particular V4 patterns that make up the winning objects. Only one to three visual objects make the cut and are held in visual working memory.

Those visual objects that win will generally be linked to other information that is nonvisual. Relevant concepts may become active in verbal working memory; action sequences controlling the eyes or the hands may be activated or brought to a state of readiness.

How fast can we extract objects from a visual image? In an experiment carried out in 1969 at MIT psychologist, Mary Potter, and her research assistant, Ellen Levy, presented images to people at various rates.^{*} They were asked to press a button if they saw some specific object in a picture, for example, a dog. They found that they could flash a sequences of pictures, sometimes with one containing a dog, at rates of up to ten per second and still people would guess its presence correctly most of the time. But this does not mean that they could remember much from the pictures. On the contrary, they could remember almost nothing, just that there was a dog present. It does, however, establish an upper rate for the identification of individual objects. As a general rule of thumb, between one and three objects are rapidly identified each time the eye alights and rests at each fixation point usually for about one-fifth of a second. Even when we study individual objects closely, this typically involves a series of

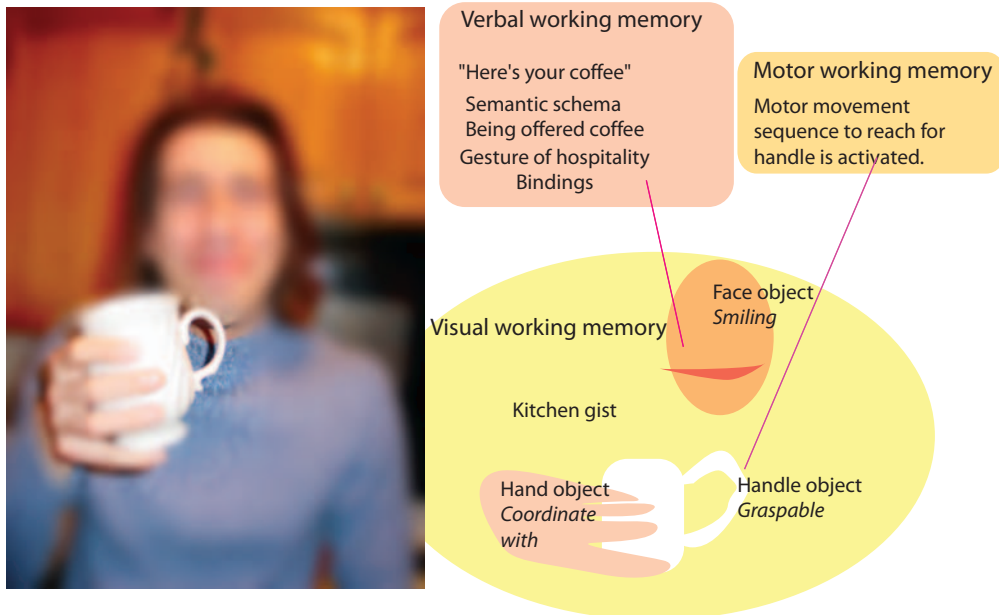
^{*}Many theorists have developed variations on the biased competition idea. See, for example, Earl Miller, 2000. The prefrontal cortex and cognitive control. *Nature Reviews: Neuroscience*. 1: 59–65.

^{*}Mary C. Potter and Ellen Levy E.I. 1969. Recognition memory for a rapid sequence of pictures. *Journal of Experimental Psychology*. 81: 10–15. The technique of rapidly presenting image sequences is called RSVP for *Rapid Serial Visual*.

fixations on different parts of the object to pick out specific features, not a prolonged fixation.

THINKING IN ACTION: RECEIVING A CUP OF COFFEE

To put some flesh on these rather abstract bones, consider this example of visual thinking in action. Suppose we are being given a cup of coffee by a new acquaintance at a social gathering. There are a number of visual tasks being executed. The most pressing is the hand coordination needed to reach for and grasp the cup by the handle. Fixation is directed at the handle, and the handle is one of the objects that makes it into visual working memory. In this context the meaning of the handle has to do with its graspability and a link is formed to a motor sequence required to reach for and grasp the handle. The location of the other person's hand is also critical because we must coordinate our actions with hers and thus some information about the hand makes it into visual working memory. This makes two visual working memory objects.



Our fixation immediately prior to the present one was directed to the face of the acquaintance and this information is retained as a third visual

working memory object. Face information has links to verbal working memory, information important for the dialogue we are engaged in. The visual working memory objects also have links to several action plans. One plan involves the motor sequence required to grasp the handle. Another plan schedules the next eye movement. Once the handle has been grasped, the eye will be directed back to the face. Yet another plan is concerned with the maintenance of the conversation. None of these plans is very elaborate, very little is held in each of the working memories, but what is held is exactly what is important for us to carry out this social dialogue that includes the receipt of a mug of coffee. The whole thing can be thought of as a loosely coupled network of temporarily active processes, a kind of dance of meaning.

ELABORATIONS AND IMPLICATIONS FOR DESIGN

We now have the outline of what it means to have a visual object in mind. The remainder of this chapter is devoted to elaborating parts of this theory and discussing some of its implications for design.

MAKE OBJECTS EASY TO IDENTIFY

The theory of objects as patterns of patterns means that some objects will be easier to identify than others. The most typical exemplars of an object class are identified faster because the corresponding visual patterns are more strongly encoded. We will be able to see that a member of the Labrador breed is a dog faster than we can identify a Daschund or a Doberman.

There are also views that are more canonical. These include both typical views and views that show critical relationships between structural parts. Showing the joints clearly in a structured object will make it easier to identify that object. This can be done by posing the object in such a way that the joints are clear in the silhouette, as is common in some Egyptian low-relief carvings. It can also be done by selecting a viewpoint that makes the critical joints and component parts clear.

To summarize, if we want objects to be rapidly and reliably identified, they should be typical members of their class and shown from a typical viewpoint. In addition, for any compound object, connections between component parts and all critical component parts should be clearly identifiable as discussed earlier in this chapter.



This view of a low-relief carving in the Kom Ombo Temple in Egypt has figures arranged so that all their important parts are visible. Hands, feet, heads, and shoulders are all given in stereotyped views. Likewise, the connections between these components are all clear. (Image from Suresh Krishna.)

NOVELTY

Humans have come to dominate the world partly because of our curiosity. Seeking visual novelty is one of the fundamental abilities of newborns.* In later life we continue to actively seek novelty, either to exploit it to our advantage or to avoid it if it presents a danger. Novelty seeking manifests itself when we are not intensely focused on some cognitive agenda. At such times people use their free cognitive cycles to scan their environments, seeking mental stimulation. We are usually not aware that we are doing this.

This provides an opportunity for the advertiser. Because image gist is processed rapidly, providing an image in an advertisement ensures that at least some information will be processed on the first glance. Holding the viewer's attention can be achieved through novelty. One method for triggering further exploratory cognitive activity is by creating a strong gist-object conflict. If a scene with clearly expressed gist is combined with an object that is incompatible with that gist, the result will be a cognitive effort to resolve the conflict in some way. The advertiser may thereby capture a few more cognitive cycles.

Gist-object mismatches are easy to create and can be as straightforward as putting someone in his underwear in a room full of suits, although

*Novelty seeking in babies is so strong that it has become one of the basic tools used to understand the perceptual capacities of the very young. For example, this method has been used to show that an understanding of the *persistence of objects* seems to be innate. Babies are most curious about objects that disappear, or reappear without cause. They look longer at novel objects. Elizabeth Spelkey, and co-workers 1992. *Origins of knowledge Psychological Review*. 99, 605-632.

the novelty of this particular device is fast wearing off. Computer tools such as Photoshop make it almost trivial to add an object to a scene. The trick is to add some witty twist and at the same time get people to think of some product in a more positive light. Gist-object mismatch is actually an old device that has been used by artists throughout the centuries. The technique became the dominant method of the twentieth-century surrealist painters such as Rene Magritte (men in suits falling from the sky, a train emerging from a fireplace) and Salvador Dali (soft clocks melting from trees, a woman with desk drawers emerging from her body).

Another way of holding attention is to create a visual puzzle. The image below, created by London photographer, Tim Flach, uses an extreme form of non-canonical view to accomplish this. Because of the view, gist perception will not be immediate, although there is probably enough information to discern that the image is of animate forms. The strong, unfamiliar outlines of the forms will be enough to capture a second glance and further exploration will likely ensue.



IMAGES AS SYMBOLS

Some graphic symbols function in much the same way as words. They have commonly agreed-on meanings. Examples are “stop” and “yield” traffic signs. Many religious symbols have become embedded in our visual

culture. What is important about these symbols is that the visual shapes have become cognitively bound to a particular non-visual cluster of concepts. Viewing these symbols causes an automatic and rapid activation of these concepts, although the degree to which this occurs will depend on the cognitive activity of the moment.

Symbols with the ability to trigger particular associations can have huge value. Companies often spend vast amounts in getting their symbols to the state of automatic recognition by a large proportion of the population. In extreme examples, such as soft drinks manufacturing, the value of the company is closely linked to the value of the company's symbol. Coca Cola has been estimated to be worth thirty-three billion dollars. Most of that is branding. Producing cola for a few cents a bottle is easy. What is difficult is getting people to pay \$2.00 for it. Coca Cola's famous trademark itself is estimated to be worth more than ten billion dollars.

There is a conflict between the need of advertisers to provide novelty in order to capture the attention of potential buyers and the need to be consistent to establish a well-defined brand image. Companies can only change a logo at huge cost and with enormous care and planning. Generally it is much safer to improvise variations on an old symbol than to establish a new one. This allows for attention grabbing novelty, while maintaining that all-important instant brand recognition.

The poster for the movie *Walk the Line*, advertising a biography of country and western singer, Johnny Cash, is full of images that function as symbols. The guitar symbolizes that its wearer is a musician. The particular shape of the finger plate denotes the country and western style. The flames in the background symbolize the personal torment of the singer and also, a famous Cash song "Ring of Fire." The vertical bar symbolizes the title of another song, also given in words, "Walk the Line." For anyone already somewhat familiar with the life and work of Johnny Cash, the poster will automatically trigger these rich layers of meaning.



MEANING AND EMOTION

Certain kinds of scenes and objects have strong emotive associations. Some are universal. Most people have positive associations for cute children and animals. But many other images have far more varied individual associations. For one person an image of a lake might be associated with good times at the family cottage. For another, the lake might be a foreign place evoking discomfort, lack of sanitation, and insects.



Mothers and children have positive associations for almost everyone. Such images can be used to support messages about family values, health care products, or the need for education. (From andipantz/stockphoto.)

Beyond the most superficial level, understanding a communication requires cognitive effort and unless people have strong self-motivation, information will not be attended to and therefore not processed. Without some emotional valence many presentations will not be effective simply because the audience will not care. Adding emotive images can turn a disinterested audience into an attentive audience.



A message about the danger to humpback whales from ship strikes is far more likely to result in actions and donations if accompanied by images such as this. It shows a whale killed by a ship's propeller.

IMAGERY AND DESIRE

Mental imagery often accompanies feelings of desire. Daydreams about food, sex, material goods, or experiences such as lying on the beach consist, in large part, of a stream of mental imagery. In a paper entitled “Imagery, Relish and Exquisite Torture,” a group of researchers from the University

of Queensland in Australia and the University of Sheffield in England set out a theory of how the process works.* They argue that imagery provides a short-term form of relief for a psychological deficit. Unfortunately, the effect is short-lived and a heightened sense of deficit results. This leads to more imagery seeking behavior. The effect is something like scratching an itch. At first it helps, but a short time later the itch is worse and more scratching results. The imagery involved can be either internal, purely in the mind, or external, from magazines or web pages. An important point is that the process can be started with external imagery. Their studies also suggested a way of getting out of this cycle of imagery-mediated obsession. Having people carry out cognitive tasks using alternative imagery, unrelated to the object of desire, was effective in reducing cravings.

CONCLUSION

A visual object is a momentary nexus of meaning binding a set of visual features from the outside world together with stuff we already know. Perhaps 95% of what we “see” from the outside world is already in our heads. Recognizing an object can cause both physical and cognitive action patterns to be primed, facilitating future neural activation sequences. This means that seeing an object biases our brain towards particular thought and action patterns, making them more likely.

The name psychologists give to the temporary activation of visual objects is *visual working memory*, and visual working memory has a capacity of between one and three objects depending on their complexity. This means that we can hold one to three nexii of meaning simultaneously, and this is one of the main bottlenecks in the visual thinking process. A similar number can be held in verbal working memory and often the two kinds of objects are bound together. Some objects are constructed and held only for the duration of a single fixation. A few objects are held from fixation to fixation but retaining objects reduces what can be picked up in the next fixation.

Visual working memory capacity is something that critically influences how well a design works. When we are thinking with the aid of a graphic image we are constantly picking up a chunk of information, holding it in working memory, formulating queries, and then relating what is held to a new information chunk coming in from the display. This means that the navigation costs discussed in the previous chapter are critical. For example, suppose a visual comparison is required between two graphic objects; they might be faces, pictures of houses, or diagrams. We will be much

*D.J. Kavanagh, J. Andrade and J. May, 2005. Imagery, relish and exquisite torture: The elaborated intrusion theory of desire. *Psychological Review*. 112: 446–467.

better off having them simultaneously on the same page or screen than having them on different pages. In each case visual working memory capacity is the same, but we can pick up a chunk or two, then navigate to a new point of comparison at least ten times faster with eye movements than we can switch pages (either web pages or book pages), so side-by-side comparison can be hugely more efficient.

In this chapter, we have begun to discuss the separation of visual and verbal processing in the brain. This is a good point to comment on a commonly held misconception. There is a widely held belief that there are right-brain people who are more visually creative and left-brain people who are more verbal and analytic. The actual evidence is far more complex. There is indeed some lateral specialization in the brain. Language functions tend to be concentrated on the left-hand side of the brain, especially in men. Rapid processing of the locations of objects in space tends to be done more on the right. But the available evidence suggests that creativity comes as a result of the interaction between the many subsystems of the brain.* If there is a location for creativity it is in the frontal lobes of the brain, which provide the highest level of control over these subsystems.

*For a thorough review of the evidence see Joseph B. Hellige's 1993 book, *Hemispheric Asymmetry: What's Right and What's Left?* Harvard University Press.

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Chapter 7

Visual and Verbal Narrative

Forget about that old cliché, “A picture is worth a thousand words.” Try to express the following in pictures.

If halibut is more than ten euros a kilo at Good Food, go to the fish market on 5th Street.

It is unlikely that it could be done, even if you are capable of drawing a halibut. Good design is not about pictures versus words. The real issues are as follows. When are images most effective? When are words and other formal symbols most effective? If both images and words are used, how should they be combined?

We begin this discussion by considering sign languages used by deaf people. It may seem strange that sign languages have much to tell us about the fundamental nature of language, but thinking about them enables us to make clear distinctions about what is best expressed through words and images and diagrams. Sign language also gives us an incontestable

example of something that actually *is* a visual language, so we know what one looks like.

It is easy to assume that sign languages were invented by hearing people as ways of translating spoken languages, like English and French, into something the deaf could easily understand. In fact, nothing could be further from the truth. Sign languages were invented by deaf people. Mostly they arose spontaneously in the nineteenth century when previously isolated deaf children came together in schools that were created by philanthropists. Sign languages were developed by the students in those schools, not the teachers. Indeed, up until the 1980s many schools for the deaf actually prohibited signing because they believed that lip reading would better help integrate the pupils into society at large. The attempt to suppress sign language was the cause of considerable suffering since signing provides the deaf with a means of expression comparable in richness and expressive power to spoken language.*

Sign languages are not translations of spoken languages. They are distinct languages with their own grammar and vocabularies. American Sign Language (ASL) is quite distinct from British Sign Language (BSL) and neither has the grammar of spoken English. New sign languages are still being born. In Nicaragua, linguists have watched, fascinated, as a new sign language has emerged over the past thirty years. This began to develop with the establishment of a center for special education in 1977. It has rapidly developed its own grammatical rules, demonstrating, among other things, that language is an innate capability of humans.*

Although sign languages are gestural and not verbal, there are great similarities between the two forms. Both sign language and spoken language have a critical period of development. If children are not provided with the social environment for their language skills to develop in the first few years of life they will never develop fluency. Children who have grown up with animals, *enfants sauvage*, never develop fluency because the critical period is missed.

The parts of the brain used in the understanding and production of sign languages are the same areas used in verbal language listening and talking—namely Wernicke’s area and Broca’s area respectively. Sign language uses the neural sub-systems normally associated with verbal processing, even though signs must pass through the visual pattern-finding machinery of the brain to get to these areas.

VISUAL THINKING VERSUS LANGUAGE-BASED THINKING

We need a clear way of distinguishing between the *visual thinking* that goes with graphic expression, and other forms of thinking, especially

*In his book on the history of sign language in America, Douglas Baynton describes how signing was banned in schools for the deaf in favor of speaking and lip reading, using techniques pioneered by Alexander Graham Bell. 1996. See *Forbidden Signs: American Culture and the Campaign Against Sign Language*. University of Chicago Press.

*For more on sign languages read Karen Emmorey’s *Language, Cognition, and the Brain: Insights from Sign Language Research*, Mahwah, N.J.: Erlbaum, 2002.

language-based thinking. The existence of sign languages clearly demolishes the idea that one mode is fundamentally visual and the other mode is fundamentally auditory. A more basic distinction between the modes of expression has to do with symbolization and logic. We shall consider these in turn.

LEARNED SYMBOLS

First and foremost, language is a socially developed system of shared symbols, together with a grammar. It can be expressed through speech, or written down or it can be signed. Natural language contains a set of nouns that denote concrete or abstract objects. These were invented. At some time in the past, a group of individuals came to a common understanding that the vocalization “dog” denoted a class of animals. Once the word was established new individuals outside the initial group learned the symbol. For the most part these symbols accord to the principle of arbitrariness. There is nothing about the word dog that is dog-like; we could just as easily use the word “zab,” or “cat,” the important point is that we be consistent.

Sign languages also adhere to the principle of arbitrariness for the most part. Some gestures bear similarities to the actions they represent, especially those relating to actions, but most signs do not. Also, as a sign language becomes more sophisticated the proportion of arbitrary signs is increased.

The basis of visual thinking is pattern perception, not learned symbols. Our understanding of meaning through patterns does not come from social convention. Our perceptual machinery comes partly from evolution and partly from our visual experiences as we interact with the world; in other words, our pattern perception is partly innate and partly learned. When we see designed graphical patterns, objects and connections are perceived using this combination of perceptual processes. Patterns convey meaning in ways that are not arbitrary and not socially determined.

Visual designs are almost always hybrids; they have aspects that support visual thinking through pattern finding, and they have aspects that are conventional and processed through the language system. Some visual symbols, for example, ♥ ♪ € , are the same as words in the sense that they are agreed-on arbitrary symbols and they bear little or no resemblance to the things they represent. Actual hearts are not shaped like the symbols on valentine cards. Some realist paintings are almost entirely visual and pattern based; they do not rely on arbitrary visual symbols. Diagrams are firmly in the middle, combining meaning from agreed-on arbitrary symbols and meaning from the patterns used to show relationships.



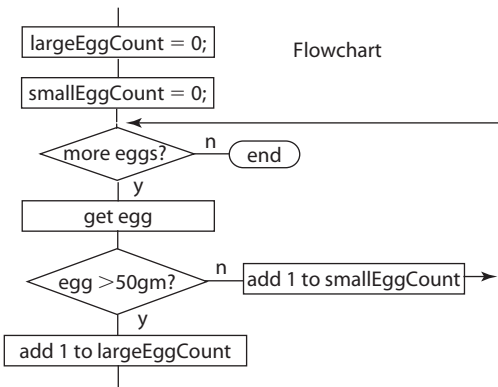
The series of positions from the phrase “Good Morning” in American Sign Language. The actual sign is not a series of poses, however, but a gesture that is fluid and dynamic. This sign accords with the principle of arbitrariness; that is, there is no actual resemblance between the sign and a good morning.

www.lifeprint.com

GRAMMAR AND LOGIC

Natural language, whether it is sign language or spoken language, incorporates a form of logic that is quite distinct from the logic of visual representation. Natural language is full of qualifiers such as “if,” “and,” “but,” “otherwise,” “nevertheless,” and “while.” This is not formal mathematical logic, but it does allow for a kind of abstract reasoning. Such terms enable us to give conditional instructions to people, such as the directions to buy halibut with which we started this chapter. The most widely used computer programming languages are based on natural language. Statements beginning with “while,” “if,” and “else” provide their basic structure and this is why entry-level programming is easy to learn.

There have been many attempts to make graphical computer programming language alternatives to written languages. These have mostly failed, sometimes spectacularly. In the 1980s, a diagramming technique called the flowchart was mandated for large commercial software projects. It used symbols to express programming logic for “if” and “while” statements. An alternative technique called pseudo-code has proven to be a much better way of expressing computer code in a more readable form. Pseudo-code provides an explicit bridge between natural language and computer code; in pseudo-code “if” and “while” statements are used much in the same way as they are in natural language. The following program counts how many large eggs and small eggs there are in a batch. It is shown using both a flowchart and pseudo-code to illustrate the differences.



Pseudocode

```

largeEggCount = 0;
smallEggCount = 0;
while(more eggs)
{
    get egg;
    if (egg greater than 50 gm)
        add 1 to largeEggCount
    else
        add 1 to smallEggCount
}
  
```

Notice that although the flowchart is called a visual language it contains almost as many words as the pseudo-code. The critical difference is that the diamond-shaped symbols are used to represent program logic

in the flowchart, whereas pseudo-code uses a form similar to the natural language sentence, “If an egg is greater than 50 gm then add 1 to the large egg count, otherwise add 1 to the small egg count.”

Flowcharts were a very expensive mistake. Hundreds of thousands of dollars were spent documenting computer programs in this way, only for the results to languish on shelves never to be consulted. It turned out that it was easier to read a pseudo-code description, or even the program logic itself, than the flowchart. Flowcharts now stand as a reminder of the limitations of visual representation. There are some things that words do much better.

Visual representations can incorporate a kind of logic, but it is very different from the more abstract logic of natural language. Visual logic is the logic of pattern, object, and space that was discussed in Chapters 3, 5, and 6.

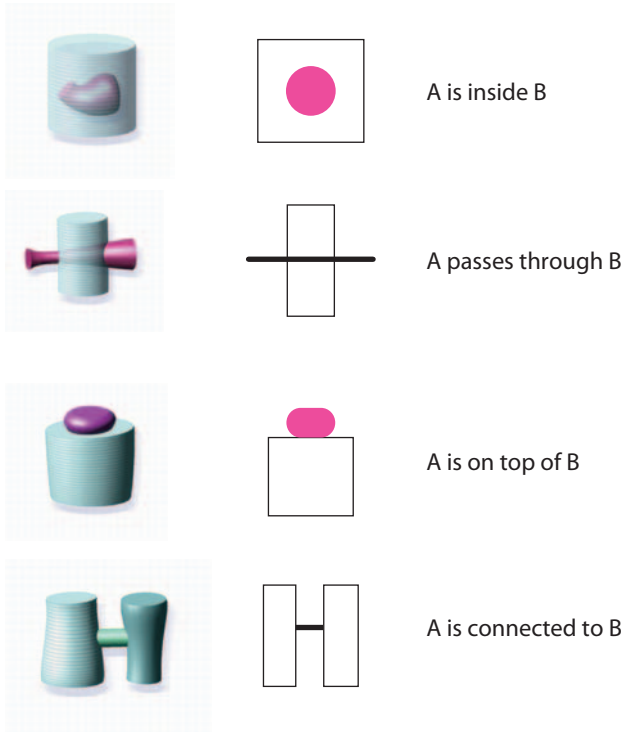
COMPARING AND CONTRASTING THE VERBAL AND WRITTEN MODES

Let’s summarize and compare the relative strengths of verbal and visual media. Verbal, sign, and written language provides a socially designed tool for communication. The symbols and grammar are shared by hundreds of millions of people in the case of the most common languages. Verbal, sign, and written languages use the same specialized areas of the brain for processing this information. What many people refer to as “thought,” the kind of inner dialogue whereby people plan for the future and rehearse conversations with others, is mostly a form of internalized speech, sometimes called *mentalese*. This internalized language form shares the kind of rough and ready logic that comes from specialized conditional operators such as “if,” “and,” “but,” “maybe,” and “perhaps.” These enable people to reason about consequences of action in ways that are very flexible.

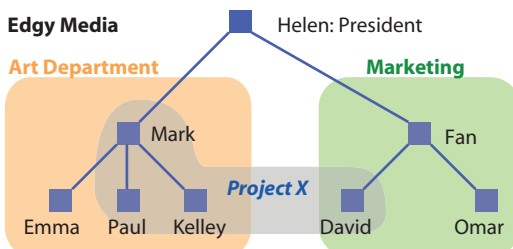
Visual representations of information are processed by the visual system. This is an extremely powerful pattern finding system, which is very good at finding structure in diagrams. Its logic is very different from verbal logic and consists mostly of structural relationships, such as those shown below.

As the diagram on the following page shows, the graphic expressions of the various kinds of structural relationships work well when radically simplified and each has a corresponding verbal expression, so it cannot be claimed that this form of expression is uniquely graphic. Nevertheless,

when multiple relations are combined in a diagram the graphical form can be far more succinct and easier to reason with.



Almost all graphic designs are hybrids of the visual and language modes of expression and the amount of each should depend on what is to be conveyed. Words, images, and spatial patterns should be used for what they express best. In most cases images do not make good labels, and describing complex patterns of relationships with words is confusing. In the following example the structure of a small company is described in two ways, by diagram and by words. Both contain roughly the same information.



Helen is the president of the Edgy Media company. Mark, Emma, Paul, and Kelley work in the art department. Mark is manager of the art department. Fan, David, and Omar work in Marketing. Fan is manager of the Marketing department. Mark, Paul, Kelly, and David all work on Project X.

It is difficult to get an overall understanding of the Edgy Media company structure from the verbal description, but with the diagram a number of structural relationships become immediately evident; for example, that the company personnel structure is a three-level hierarchy. It is also much easier to carry out simple reasoning tasks with the aid of the diagram. Questions such as “Who is the most senior person working on Project X?”, “Which department has the most people working on Project X?”, and “Who does Fan report to?” can all be addressed with simple visual queries.

LINKING WORDS AND IMAGES THROUGH DIXIS

Even before she can speak an infant can communicate by pointing. A reaching gesture can be interpreted as “give me that.” Add some facial expression, indicating interest or dislike, and the gesture becomes modified, “that scares me” or “what is that?”

Later in life, pointing is still an important part of communication, especially when it is necessary to link visual and verbal information. Phrases like “look at that one,” “put it over there,” and “it’s here” gain their meaning through the linking of objects to speech by gestures such as pointing. Actions that provide the subject or object of a spoken sentence by directing attention to objects in the world are called *deictic gestures*.

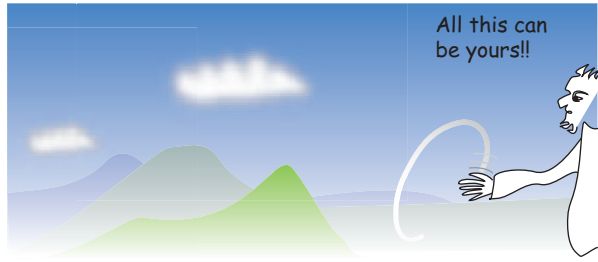
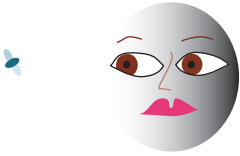
Richard Bolt of the MIT Media Lab captured this concept succinctly with the phrase, “put that there.”* This phrase is uttered with two pointing gestures. The first denotes the object to be moved and the second denotes the intended location. What is special about this sentence is that the words “that” and “there” gain their referents only by being combined with pointing.

*R.A. Bolt. 1980. Put-that-there: voice and gesture at the graphics interface. *Proceedings of the 7th Annual Conference on Computer Graphics and Interactive Techniques. SIGGRAPH*. 262–270.



Deictic gestures can be accomplished in many more ways than pointing with a finger. The direction of someone’s gaze, or how he orients his body, can also serve to denote the subject of an utterance.

Get it away from me!



Deictic gestures can have more elaborate semantics than simple designation of an object. A large circular gesture can indicate a group of objects. The vagueness or precision of a gesture can express the degree of uncertainty in what is being denoted.

POWERPOINT PRESENTATIONS AND POINTING

In formal meetings people often stand up in front of screens, run through a set of slides, and present their ideas. A good PowerPoint lecture is a combination of imagery presented on a screen and words provided by the presenter. The perceptual and cognitive theory given in this and the previous chapter suggests the following guidelines. PowerPoint slides should be primarily devoted to images and diagrams. Only essential text should be placed on the slide; for example, labels to make a diagram comprehensible and perhaps a title to set up a question that the narrator can answer. Narrative should be provided by the speaker through spoken words. Deictic gestures should link the two media. A clear deictic gesture allows the speaker to direct the visual attention of audience members to different parts of the visual imagery.

Although these points may seem quite obvious, PowerPoint slides are often poorly designed. In many cases people put far too many words on the slides, causing the viewer to read the slides rather than listen to the presenter. Words that match what the speaker is saying can be the worst of all. People will often read ahead and then mentally wander off as the speaker catches up. It is also a common mistake for the slide to contain a set of section “bullets” listing the main points. Generally these belong in the speaker’s notes, not on the screen. They also have the effect of weakening narrative tension because the viewer will look ahead and reason about the conclusion bullet long before the speaker gets there.

A major shortcoming in current PowerPoint software is the lack of an effective pointer to link the words of the speaker with the graphic content



In the movie *An Inconvenient Truth*, former Vice President Al Gore uses visual imagery to powerfully convey the potential dangers of global warming. There are very few words on his slides. He is clearly illuminated so that his speech and gestures can be effective.

of the screen. The mouse cursor is one option, but it is too small to be clearly visible. Another option is the laser pointer, but laser spots are also too small and the devices amplify hand tremor, which is distracting.

To realize the full value of PowerPoint presentations good support for diexis is essential. What is needed is a bold on-screen non-jittery pointer to link the screen images with the words of the speaker.

Gestures are an important part of communication and there should be light on the speaker so that his or her gestures are clear. Unfortunately this, too, is usually lacking. Most lecture rooms are dimmed to make the slides clear and there is no provision to illuminate the speaker.

MIRROR NEURONS: COPYCAT CELLS

In the early 1990s, an Italian team of researchers, led by Giacomo Rizzolatti, made a discovery that relates to one of the most fundamental cognitive capacities we have.* He was carrying out a study of the motor cortex of a monkey and had inserted a tiny micro electrode into a neuron in its motor cortex. This neuron, like others in the same region of the brain, sent out electrical signals whenever the monkey reached out to move an object.

The experiment was proceeding much as usual when something remarkable happened. A graduate student had entered the lab eating an ice cream cone. This got the monkey's attention and when the student raised the cone to his lips, the monkey's neuron responded just as it had when it moved its own hands.

This event marked the discovery of *mirror neurons* that have since been discovered in many areas of the brain. Mirror neurons exhibit activation in the *same neural pathways* that would normally be used to execute a particular action. Thus they provide a basis for learning by imitation. They have been found in many primates and evidence suggests that they are especially prevalent in humans. They have been called *empathy neurons* because they may allow us to read the emotions of others through the same pathways that generate our own facial expressions when we are expressing emotions.†

Mirror neurons support a pre-language form of communication that maps action onto action. They probably provide the basis of a "gut feeling" level of empathic communication. Indeed, some scientists speculate that the mirror neuron system may be critical to the complex social structure maintained by humans. Mirror neurons support a relatively unsophisticated non-symbolic moment-to-moment form of communication. We now turn our attention to the much more complex forms of communication that are involved in storytelling.

*G. Rizzolatti and L. Craighero. 2004. *Annual Review of Neuroscience*. 27: 169–192.

†According to the research of psychologist Paul Ekman there are seven basic and universal facial expressions expressing emotion: anger, sadness, fear, surprise, disgust, contempt, and happiness. P. Ekman. 2003. *Emotions Revealed. Recognizing Faces and Feelings to Improve Communication and Emotional Life*. Times Books, New York.





We may feel the emotions of others via our own facial muscles through the mediation of mirror neurons. These illustrations of anger, fear, joy, and sadness are from Gary Faigin, *The Artist's Complete Guide to Facial Expression*. Watson Guptill Publications. New York, 1990.

VISUAL NARRATIVE: CAPTURING THE COGNITIVE THREAD

The purpose of narrative, whether it is visual or verbal, is to capture the cognitive thread of the audience. The *cognitive thread* is the sequence of concepts that are held actively in visual and verbal working memories, together with the links between them.* If a movie or presentation is successful, the cognitive thread of audience members will roughly follow the narrative thread that the author has designed. This means that designing for narrative is very different from designing for information seeking.

Information seekers have highly individual cognitive threads that are shaped moment-to-moment by the specific demands of the cognitive process of solving a problem. This process is internally driven. Conversely, the audience of a narrative presentation will, if they are attending, have cognitive threads that are much more similar to each other, although still far from identical because of the variety of their prior experiences. A professional film critic and a regular movie fan will see many things differently in a movie, but in each frame they both will look mostly at the same visual objects in the same order and this will result in many of the same concepts being activated in the same order.

The broad framework around which narratives are built has three components: establishing a problem, elaborating a problem, and resolving a problem. The kind of problem and the nature of its resolution depend on the genre. In a mystery thriller, finding a murderer is often the problem. A lecture about science will generally have some unsolved puzzle about the nature of the world as the problem. In a business presentation the problem may be opening up a new market for a particular type of product.

This pattern of setting up, elaborating, and resolving is repeated at many levels in a narrative. The same pattern can be found in a single short

*We are using the term *concept* here to refer to a temporary nexus of meaning, as described in the previous chapter.

dialogue as well as longer episodes. It makes the audience seek for cognitive resolution and this provides the motivation for sustained attention. As we shall see in this chapter narrative can be carried either visually or through verbal language.

Q&A PATTERNS

An example of short time scale visual narrative is the question and answer (Q&A) pattern used in cinematography. In his book on cinematography Steven Katz discusses the three-shot sequence as a method for accomplishing narrative tension on a small scale. He calls these Q&A patterns. The first two shots provide seemingly unrelated information and this leads the viewer to question the connection between these ideas. The third shot then provides the link and the resolution.*

A simple example has become a cinematic cliché. First we see a woman walking along a street towards the right side of the screen; she is carrying a pile of parcels. Because the camera follows her our eyes follow her. Next we follow a man also walking along a street. He is moving quickly towards the left side of the screen. In the third shot, the two collide and the parcels spill on the sidewalk. This provides the resolution of the three-shot sequence, but opens up a new question: what will be the future relationship of these two people?

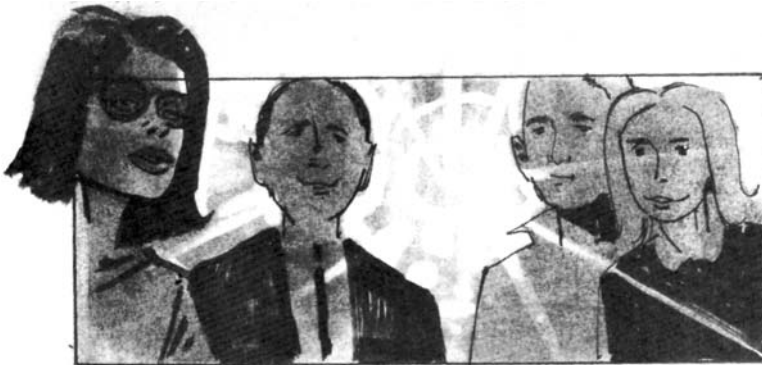
FRAMING

At the most basic level the main task for the author of visual narrative is to capture and control what the audience is looking at, and hence attending to, from moment to moment. Part of this is done simply by carefully framing each shot and by designing the transitions between shots so that intended object attention becomes inescapable. Cinematographers have worked out how to do this with great precision.

The camera frame itself is a powerful device for controlling attention. Anything outside of it cannot be visually attended and so framing a scene divides the world into the class of objects that can be attended and those that cannot. Within the frame objects can be arranged to bias attention. Having the critical objects or people at the center of the screen, having the camera follow a moving object, or having only the critical objects in focus are all methods for increasing the likelihood that we will attend according to the cinematographer's plans.

Cinematographers pay great attention to the background of a shot. Anything that is visually distinctive, but irrelevant, is like an invitation to viewers to switch their attention to that item and away from the narrative thread. One mechanism for making the background less likely to attract

*Most of the points relating to cinematography come from a book by S.D. Katz. 1991. *Film Directing Shot by Shot: Visualizing from Concept to the Screen*. Michael Weiss Productions and Focal Press.



83- SAY HELLO TO MRS. ROBINSON, BENJAMIN



84- HELLO, MRS. ROBINSON



84- HELLO BENJAMIN.

Storyboards are sketches used to plan a sequence of shots in movie making.

This is a sequence of storyboards for the movie *The Graduate* by Harold Michelson. It shows the first time that Benjamin, a recently graduated student, meets Mrs. Robinson, with whom he will later have an affair.

Because of the way the camera frames the shots, a viewer has little choice but to attend first to Benjamin's father, next to Benjamin lazing on a float in the pool and finally to Mrs. Robinson.

(From S.D. Katz. 1990. *Film Directing Shot by Shot: Visualizing from Concept to the Screen*. Michael Wiese Productions, Studio City, California.)

attention is through the careful application of depth of focus. Increasing the lens aperture of a camera has the effect that only objects at a particular depth are in sharp focus. Everything that is closer or farther away then becomes blurred and less likely to capture the viewer's attention.

FINSTS AND DIVIDED ATTENTION

The human eye has only one fovea and so we can only fixate one point at a time, or follow a single moving object. The brain, however, has mechanisms that allow us to keep track of several objects in the visual field,

although we do not perceive much of these secondary objects. Zenon Pylyshyn, a psychologist at the University of Western Ontario in Canada, studied this capability and found that the maximum number of objects people could track was four. He called the mental markers that are placed on the moving objects FINSTs for *fingers of instantiation*.*

In Pylyshyn's experiments keeping track of moving objects was all the participants had to do, but the task required intense concentration; under more normal circumstances our cognitive activity is much more varied. It is unlikely that our brains usually track more than one or two objects in addition to the one we are focally attending.

Suppose we are watching children in a crowded playground. The FINST capability means that we can maintain fixation on one child and simultaneously, in the periphery of our vision, keep track of one or two others. We do not get much information about the actions of the other children, but if we need to know about them the FINST marker means that a visual query, by means of an eye movement taking less than a tenth of a second, is all that is required to check on them.

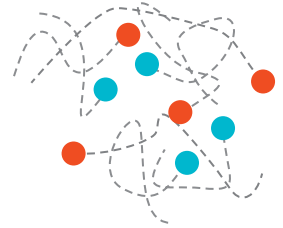
The practical implication of FINSTs for cinematographers or video game designers is that it is possible to create a visually chaotic scene, and for the audience or player to keep track of up to four actors to some extent, even though only one can be the immediate focus of attention at any instant. More than this and they will lose track.

SHOT TRANSITIONS

Because of saccadic eye movements, perception is punctuated; the brain processes a series of distinct images with information concentrated at the fovea. The eye fixates on a point of interest, the brain grabs the image and processes it, the eye moves rapidly and fixates on another point of interest, the brain processes that image, and so on. The low capacity of visual working memory means that only a few points of correspondence are retained from one fixation to the next. These points are only what is relevant to the task at hand, typically consisting of the locations of one or two of the objects that have been fixated recently together with information about their rough layout in space in the context of scene gist.

The fact that perception itself is a discrete frame-by-frame process accounts for why we are so good at dealing with shot transitions in movies. Cinematographers often shift abruptly from one camera's view to another camera's view, causing an abrupt jump in the image. This is a narrative device to redirect our attention within a scene or to another scene. Transitions from shot to shot must be carefully designed to avoid disorientation.

*Z.W. Pylyshyn. 1989. The role of location indexes in spatial perception: A sketch of the FINST spatial-index model. *Cognition*. 32: 65–97.



The brain can keep track of three or four moving objects at once.

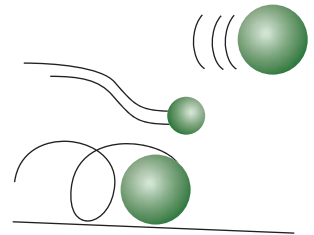
The visual continuity of ideas can be as useful and effective as the verbal continuity of ideas. One common technique is the *establishing shot*, followed by the focus shot. For example, from a distance we see a character on the street, talking to another person. Then we switch to a closer image of the two people and hear their dialogue. The first establishing shot provides gist, and the layout of objects that will later be the target of several focal shots. Manuals for cinematographers such as the Katz book provide detailed catalogues of various shot transitions. These can all be regarded as devices for manipulating the cognitive thread of the audience. If shot transitions are done well, most members of the audience will look at the same objects in the same sequence.

CARTOONS AND NARRATIVE DIAGRAMS

Strip cartoons are a form of visual narrative that have much in common with movies. They substitute printed words for spoken words, and they substitute a series of still images for moving pictures. Like movie shot transitions, the frames of a cartoon strip rely on the brain's ability to make sense of a series of discrete images. Although there are enormous differences in the visual style of a cartoon strip compared to a Hollywood movie, it is remarkable how similar they are cognitively.

The lack of real motion in a cartoon strip frame is compensated for by the use of action lines. These are graphical strokes that show movement pathways. In one respect action lines are arbitrary symbols; real objects do not have lines trailing them when they move. In another respect, they are not arbitrary because they explicitly show the recent history of an object, and by extrapolation, the immediate future.

Short-term prediction is one of the most important functions of the brain. In a sense, the brain is constantly making predictions about the state of the world that will follow from actions such as eye or hand movements. When the state of the world is not as expected, neural alarm bells go off or, in the words of computer science, exceptions are triggered.* This short-term prediction can be thought of in terms of pattern processing. The patterns that are picked up at various levels of the visual system have a temporal dimension as well as a spatial dimension. Once a temporal pattern is detected by a set of neurons the abrupt cessation of that pattern will trigger an exception. For example, one of the most basic temporal patterns is the assumption of object persistence discussed in the previous chapter. When an object marked by a FINST disappears, an exception is triggered and a shift of attention will likely ensue. It is remarkable that the simple forms of cartoon action lines can capture and express this basic function;



*In his book, *On Intelligence* (Time Books, 2004), Jeff Hawkins argues that prediction is a fundamental operation found universally in brain structures. When temporal patterns fail to play out, special *exception* signals are triggered to be handled by higher level processing units.

the ball seen in the static illustrations shown opposite is projected forwards in time, like a ball that we see in the real world. In a subsequent frame we would expect these moving objects to continue their motion paths.

Another form of narrative can be found in assembly diagrams. Certain purchases like shelving and playground toys often come disassembled. Assembly instructions have the classic form of a narrative in that a problem is posed (assembling something), elaborated (the detailed steps), and hopefully resolved (the fully assembled object).

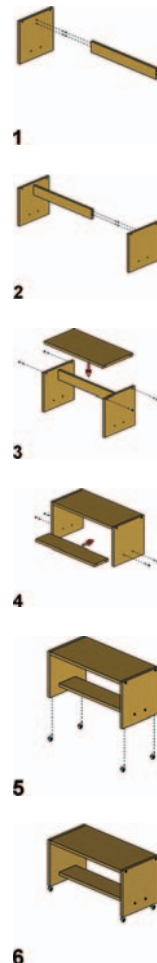
A group of researchers at Stanford University led by Barbara Tversky studied the cognitive process involved in following assembly instructions.* They measured how successful people were at assembling furniture using a variety of diagrammatic instruction styles. As the result of a series of such studies, they developed a set of cognitive principles for assembly instructions together with exemplary sequences of diagrams. The main principles were as follows.

- **A clear sequence of operations should be evident to maintain the narrative sequence.** This might be accomplished with a cartoon-like series of frames, each showing a major step in the assembly.
- **Components should be clearly visible and identifiable in the diagrams.** This is largely a matter of clear illustration. It is important to choose a viewpoint from which all components are visible. With more complex objects a transparent representation of some components may help accomplish this goal.
- **The spatial layout of components should be consistent from one frame to the next.** It is easier cognitively to maintain the correspondence between parts in one diagram frame and the next if a consistent view is used.
- **Actions should be illustrated along with connections between components.** Diagrams that showed the actions involved in assembly, rather than simply the three-dimensional layout of the parts, were much more effective. Arrows were used to show the placement of parts and dotted lines to show the addition of fasteners.

On the right is an example of a set of graphic instructions they produced according to their principles. People did better with these than with the manufacturer's instructions.

One might think that the narrative assembly diagram is a somewhat impoverished version of instructions that could be better presented using a videotape showing someone assembling the furniture correctly. This is not the case. The advantage of the narrative diagram is that it supports two

*J. Heiser, D. Phan, D. Agrawala, B. Tversky and P. Hanrahan. 2004. *Identification and Validation of Cognitive Design Principles for Automated Generation of Assembly Instructions*. *Advance Visual Interfaces*.



different cognitive strategies. It supports the strategy of narrative, where the audience is taken through a series of cognitive steps in a particular order. It also supports a freer information-gathering exploratory strategy, where the person using the diagram constructs and tests hypotheses about which part is which and how they fit together. With this strategy the diagram user may look at parts of the diagram at any time and in any order. A video presentation is less effective in supporting information gathering because navigating through time in a video sequence is much more difficult than simply looking at different parts of a diagram sequence. The multiframe pictorial narrative is an important form because it supports both cognitive styles.

SINGLE-FRAME NARRATIVES

Some single-frame diagrams are designed to perform a narrative function; they lead the interpreter through a series of cognitive steps in a particular order. One example is given below in the form a poster designed by R.G. Franklin and R.J.W. Turner (Reproduced with permission of the Minister of Public Works and Government Services Canada, 2007 and courtesy of



Natural Resources Canada, Geological Survey of Canada. <http://geoscape.nrcan.gc.ca/>). Its goals are to educate the public on the way water is used in a modern house and to encourage conservation.

The narrative thread in this diagram is less demanding than in the multi-frame cartoon strip; nevertheless, the intention that the ideas be explored in a particular order is expressed by the sequence of arrows connecting the different major components.

CONCLUSION

We began this chapter by discussing the fundamental differences between visual and linguistic forms of expression. Linguistic forms of expression are characterized by the use of a rich set of socially invented arbitrary symbols as well as a form of logic exemplified by the “ifs,” “ands,” and “buts” of natural language. The grammar of visual representation is quite different having to do with pattern relationships such as “connected to,” “inside,” “outside,” or “part of.” This means that certain kinds of logic are best left to verbal or written language.

Narrative, however, can be carried either through language or purely visual techniques or both. This is because narrative is about leading the attention of an audience and this can be achieved through either modality, although what can be expressed within a narrative is very different depending on which one is used. Language can convey complex logical relationships between abstract ideas and support conditional actions. Visual media can support the perception of almost instantaneous scene gist, rapid explorations of spatial structure and relationships between objects, as well as emotions and motivations. Both can maintain and hold the thread of audience attention, which is the essence of narrative.

One of the first choices to be made when designing a visual communication is the desired strength of the narrative form. Movies and animated cartoons are the strongest visual narrative forms because they exert the most control over serial attention to visual objects and actions. Graphic cartoons can also provide narrative but exert an intermediate-level control over the cognitive thread; cartoon frames are intended to be perceived in a particular order but allow for random access. Diagrams of various kinds as well as pictures and photographs are capable of conveying some sense of narrative, although this is less than movies and cartoon strips. Usually the viewer must actively discover the narrative in a diagram of still images, unlike the inescapable narrative train of a movie.

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Chapter 8

Creative Meta-seeing

When Michelangelo planned the frescoes for the Sistine Chapel ceiling he did not do it in his head or on the plaster. Instead the initial phases of creative work were done through the medium of thousands of sketches. These ranged from the highly speculative initial sketches to the finished paper designs that his assistants pricked through with black charcoal dust to transfer them to the still wet fresco plaster. Studies of how artists and designers work suggest that although the germ of an idea may often comes in a reverie as a purely cognitive act, the major work of creative design is done through a kind of dialogue with some rapid production medium. Sketches have this function for the visual artist; rough clay or wax maquettes have the same function for the sculptor. Sketching is not confined to artists. Hastily drawn diagrams allow the engineer and scientist to ‘rough out’ ideas; computer programmers sketch out the structure of computer programs. Many people, when they wish to organize their ideas, use pencil and paper to literally *draw* the links between labeled scribbles standing for abstract concepts.

This chapter is about the psychology of constructive visual thinking in the service of design. In addressing this subject we will encounter a number of fundamental questions. What is the difference between seeing and imagining? What is the difference between the visual thinking that occurs when we are making a sandwich, the visual thinking involved when we are reading a map, and the visual thinking we do when we are *designing* an advertising poster, a painting, or a web site?

Although creative visual thinking can be almost infinitely varied, studies of designers, artists, and scientists have identified some common elements no matter what the tasks. We begin with a generalized view of the steps involved in the stages of the creative process.

Step 1—The visual concept is formed: Depending on the application, a graphic idea may be very free and imaginative or very stereotyped. An art director doing design for an advertising firm may be open to almost any new visual concept, so long as it can be somehow linked to the product and yield a positive association. Conversely, an architect designing an apartment building on a fixed budget is likely to have his visual imagination constrained to a small set of conventional alternatives. In either case the initial concept may be quite abstract and not particularly graphical. A “three-story, L-shaped building with 12 units” might be sufficient in the case of the architects.

Step 2—Externalization: A loose scribble is drawn on paper to externalize the concept and provide a starting point for design refinement.

Step 3—The constructive critique: The scribble is visually critiqued; some elements are visually tested. The designer performs a kind of informal cognitive task analysis, executing a series of visual queries to determine if the design meets requirements. As part of the process, new meanings may be bound with the external imagery and potential additions imagined.

Step 4—Consolidation and extension: The original scribble is modified. Faint existing lines may be modified or strengthened, consolidating the aspect of the design they represent. New lines may be added. Other lines may be erased or may simply recede as other visual elements become stronger.

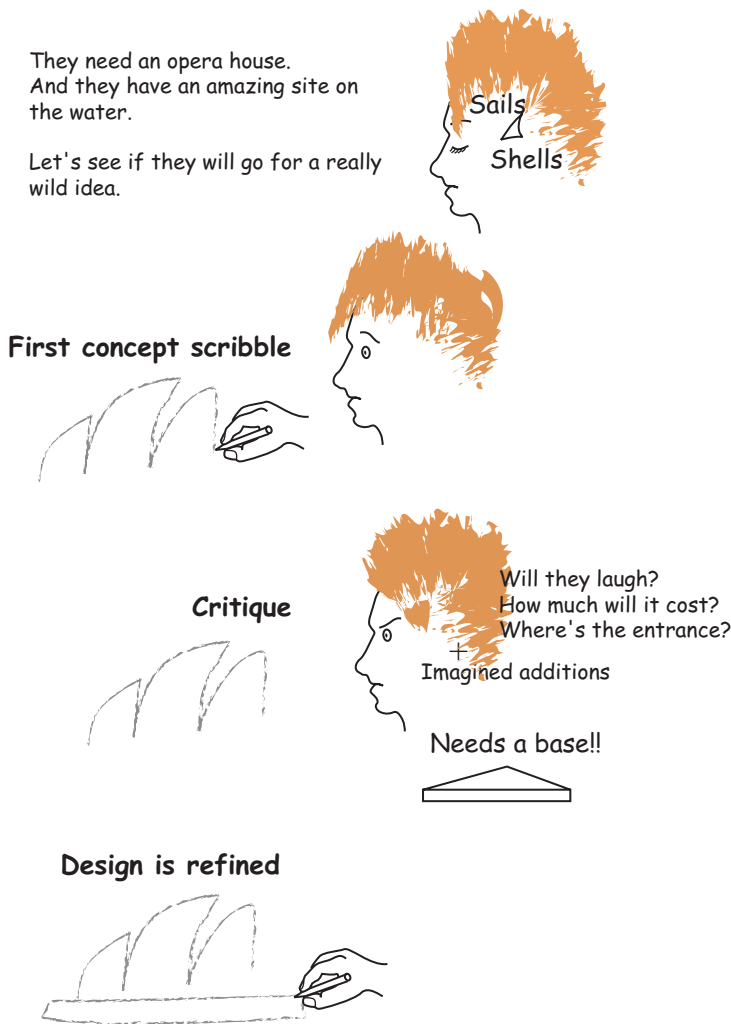
MENTAL IMAGERY

From the 1980s, two main theories of mental imagery have dominated. One, theory, championed by Zenon Pylyshyn, holds that mental imagery is purely symbolic and non spatial.[•] Spatial ideas are held as *logical propositions* such as “the cup is on the table” or “the picture is to the left of the door.” According to this view there is *no spatial representation* in the brain in the sense of patterns of neural activation having some spatial correspondence to the arrangement of the objects in space.

The second view, championed by Stephen Kosslyn, holds that mental imagery is constructed using the same neurological apparatus responsible for normal seeing.[•] These phantom images are constructed in the spatial neural maps that represent the visual field in various areas of the visual

[•]Z. Pylyshyn. 1973. What the mind's eye tells the mind's brain: A critique of mental imagery. *Psychological Bulletin*. 80: 1–25.

[•]S. Kosslyn and J.R. Pomeranz. 1977. Imagery, propositions, and the form of internal representations. *Cognitive Psychology*. 9: 52–76.



cortex, forming an internal sketch that is processed by higher levels, much as external imagery is processed.

More recently a third view has emerged based on the idea that mental images are mental activities. This view has been set out clearly by a philosopher, N.J.T. Thomas of California State University. This is part of the revolution in thinking about perception that is sweeping cognitive science, and which inspired the creation of this book. According to this account, visual imagery is based on the same cognitive activities as normal seeing, hence it is sometimes called activity theory.*

Normal seeing is a constructive, task-oriented process whereby the brain searches the environment and extracts information as needed for the task

*N.J.T. Thomas. 1999. Are theories of imagery theories of imagination? An active perception approach to conscious mental content. *Cognitive Science*. 23: 207–245.

at hand. Of course, mental images are not based on any immediate external information — that is their very definition – but they are based on the same constructive mental processes. They consist of the activities without the external information.

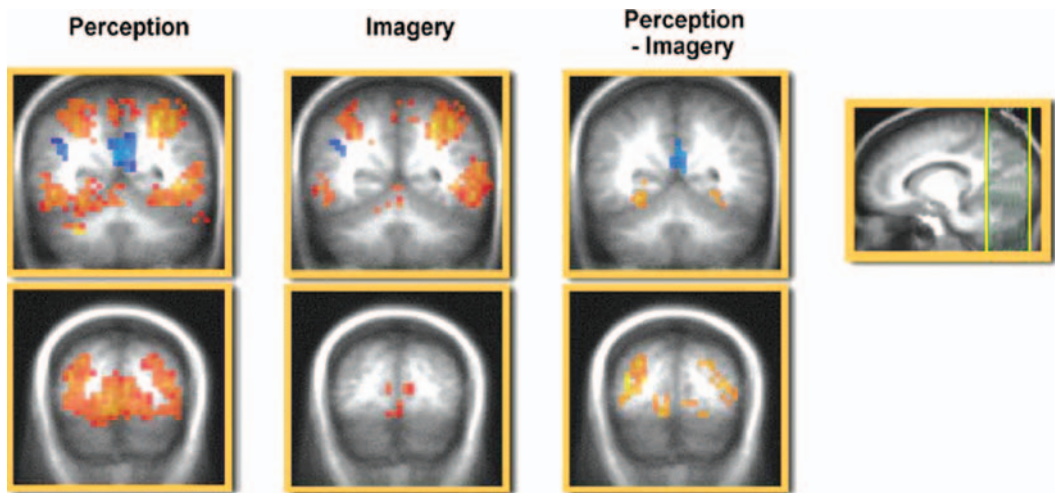
Understanding mental images is critical because design is a creative process wherein some parts are done as mental images, and some parts are done by a sort of hybrid between mental imagery and normal seeing where design elements are cognitively added to incomplete sketches.

Supporting the activity theory of mental imagery is the finding that eye movements occur when people “look at” mental images that are entirely in their heads. Moreover, the pattern of locations (although not the sequence) that the eyes fixate when viewing a scene is closely matched when a scene that has been previously viewed is mentally imaged.*

Other support for activity theory comes from using functional Magnetic Resonance Imaging (fMRI) to reveal which areas of the brain are active during mental imaging. fMRI studies show that much of the same brain machinery is active when people imagine something visual as is active during normal seeing with the eyes open, at least in the higher visual processing areas.* On the other hand fMRI also shows that the lower, primary visual cortex is *not* active during mental imagery. Since these lower areas are the obvious place for Kosslyn’s phantom image to be created, this has been taken as evidence against the spatial sketchpad theory.

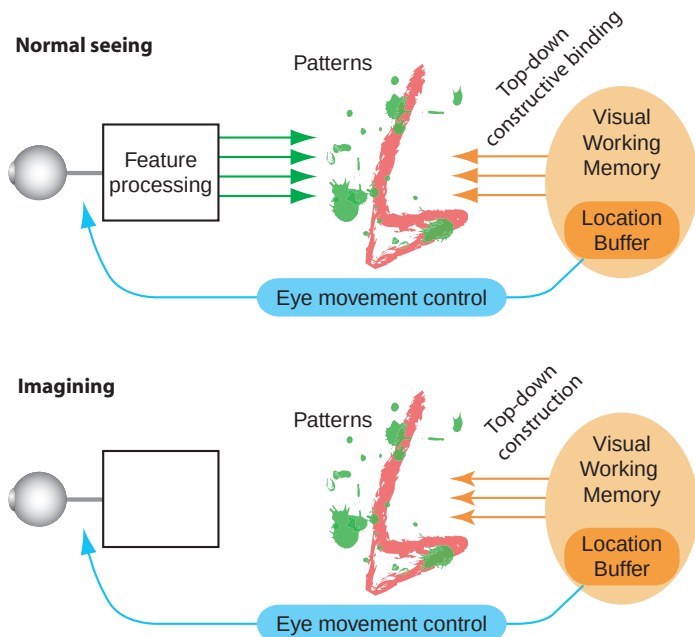
*B. Laeng and D.S. Teodorescu. 2002. Eye scanpaths during visual imagery reenact those of perception of the same visual scene. 26: 207–231. AUTHOR: Publication name????

*G. Ganis, W.L. Thompson and S.M. Kosslyn. 2004. Brain areas underlying visual mental imagery and visual perception: an fMRI study. *Cognitive Brain Research*. 20: 226–241.



This fMRI image shows that in the higher levels of visual processing, for example, in the parietal lobe area associated with active looking, very similar patterns of brain activation occur whether the person is actually looking at an image or imagining that same image. In the earlier stages of visual processing, at the back of the brain (shown below) there is much less activation in visual cortical areas.

As discussed in Chapter 6, certain kinds of visual objects have their specialized brain areas. A brain area called the fusiform gyrus is essential for human face perception. fMRI studies show this area is active both when people see faces and when they mentally image them. Likewise, a region of the midbrain called the para-hippocampal area is associated with perception of spatial locations. This area is also active in the creation of mental imagery.



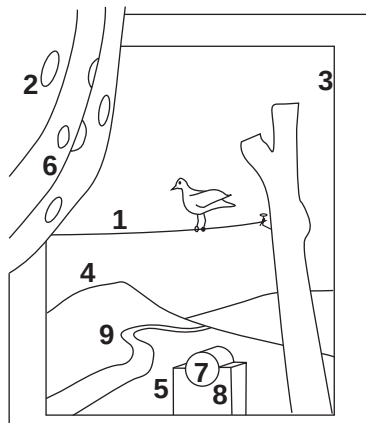
According to activity theory the process of visual imagination uses the same high-level mechanisms as normal vision. Eye movements, although seemingly unnecessary, occur as part of the activity of imagining.

Taken together the fMRI studies show that the same higher-level neural machinery is active during mental imaging as is active during seeing. But the lower-level machinery of the primary visual cortex (discussed in Chapter 2) is not active in mental imagery and this is why Kosslyn's account would seem to be at least partially wrong. According to Kosslyn's view, visual imagery is held at lower levels of the visual system and examined by higher levels and the primary visual cortex is exactly the area where one would expect such a mental sketch to be held. Visual imagery seems to occur entirely in the higher levels of visual processing, using, among other things, the active spatial processes normally involved in the planning and execution of eye movements.

Mental imagery can therefore be thought of as an internalized active process; much as our inner dialogue is internalized speech, visual imagery is based on the internalized activities of seeing. Just as internalized speech can be used to plan future actions and interpret past ones in an active process, so internalized seeing can be used to plan and interpret as part of the design process.

However, there is a key difference: everyone uses internalized speech as a thinking tool but the constructive internalization of mental imagery is a skill that is more specialized. Experienced designers will internalize the dialogue with paper, others who do not use sketching as a design tool will not.

Mostly, though, the process of design combines some of the cognitive activities of normal seeing with the activities of visual imagination. The particular unique process of design is a kind of *constructive* perception. This leads us to a critical cognitive tool of the designer, the line sketch.



Lines representing:

- 1) A thin wire
- 2) A surface color boundary
- 3) An aperture
- 4) A silhouette contour for a rounded surface
- 5) A silhouette of a sharp edge
- 6) An internal contour
- 7) A crack between two abutting objects
- 8) An edge where two planes meet
- 9) A material boundary

THE MAGIC OF THE SCRIBBLE

Lines are capable of representing many different things as the figure illustrates.* How can this chameleon quality be explained? The answer lies in the critical importance of edges in visual processing. As discussed in Chapter 3, a major function of early and middle stage visual processing is the extraction of continuous contours. An object may be separated from its background in many different ways including luminance changes at its silhouette, color differences, texture boundaries, and even motion boundaries. A generalized contour extraction mechanism in the pattern processing stage of perception allows us to discern an object from many kinds of visual discontinuities. Lines on paper provide a strong signal to this

*An extensive discussion of the expressive power of lines can be found in John M. Kennedy's book *The Psychology of Picture Perception*, Jossey Bass, 1974. Kennedy was my PhD thesis advisor.

generalized contour mechanism with the consequence that something as abstract as a circle on paper can have five different meanings.



Five different objects with the same visual representation.

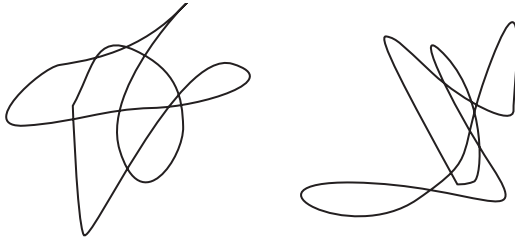


In this cartoon, Saul Steinberg exploits the chameleon quality of lines to represent many kinds and properties of objects. He also uses line quality as a way to express the whimsical chain of a child's monologue and the harsh abstract logic of the adult. (S. Steinberg, *Untitled*, 1959. Ink on paper. First published in *The New Yorker*, March 14, 1959. © The Saul Steinberg Foundation/Artists Rights Society (ARS), New York.)

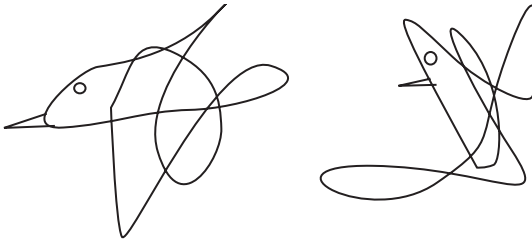
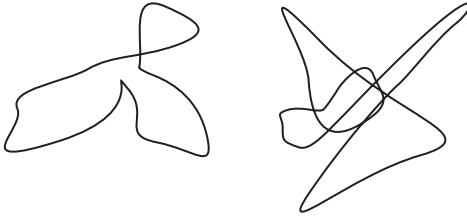
The Italian psychologist Manfredo Masseroni devised the following exercise for the reader to understand directly how the magic of lines can be used as part of a creative design process.* Make a simple scribble by moving a pen around in a large irregular looping pattern. Don't think about representing anything while you do it. You should end up with a few scribbles like the ones on the next page. To transform your meaningless scribbles into birds just add a small circle ○ and a < shape to some loop of the scribble. Magically, a bird will appear and different parts of the scribble will suddenly become head, wings, tail, and body. What is remarkable about this exercise is that almost any scribble can become a bird.

The point of this exercise is twofold. First, it emphasizes the constructive power of seeing. Simple looping lines become wings and bodies. Second, and more important, it illustrates an important creative design technique that many artists and designers use and that, as we shall see, points at the essence of the way sketches function as creative tools.

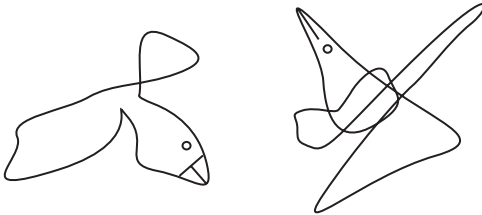
*M. Masseroni. 2002. *The Psychology of Graphics Images*. Lawrence Erlbaum. The above is a side box for Page 9.



Random irregular loop scribbles.

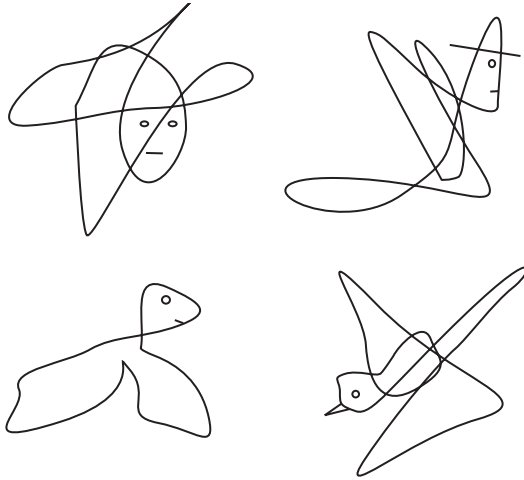


The same scribbles with only minimal additions become fanciful birds!



Alternatively, by adding slightly different features you should be able to create people or other animals.

This mode of drawing can be used as a kind of visual brainstorming. You will undoubtedly come up with designs that are different from anything you would have deliberately produced had you proceeded more methodologically. And you may well find out that the results are livelier and visually more interesting. The deliberately “random” scribble is often used by artists and designers as a way of liberating their creative process from stereotyped visual thinking. The perceptual mechanisms are exquisitely tuned to find meaning often from slender evidence, and this is why meaning is so readily found in meaningless scribbles.



Different small additions can radically change the interpretation

DIAGRAMS ARE IDEAS MADE CONCRETE

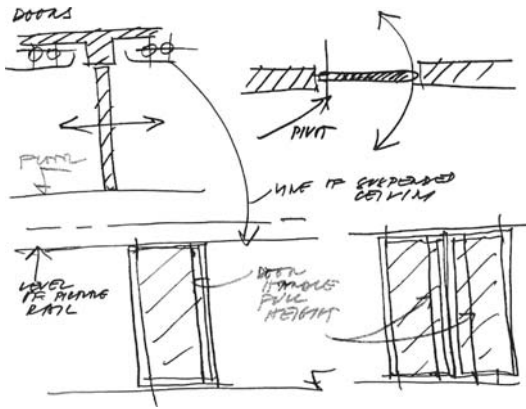
A diagram differs from a sketch in that its purpose is to express concepts that need not be spatial in nature. Usually some of the concepts are denoted by words as well as graphical forms. Lines and other spatial elements are used as abstractions to connect more basic concepts into concept structures.

Diagrams are used to plan, design things, and structure ideas. Architects, computer software designers, engineers, and scientists use them as essential cognitive tools. Artist's notebooks are often full of diagrams that differ fundamentally from their sketches in that they are not visual *prototypes* for a finished piece of art; rather they are *expressions of concept structures*. The entities of a diagrammatic design will, in most cases, have very little resemblance to the things they represent. Diagrams often contain formalized components that have precise meanings. Examples are transistors and resistors in circuit diagrams, doors and windows in architectural drawings. Any specialized field typically has its own set of conventions. It is impossible to discuss the universe of such diagramming symbols because there are thousands of them in existence; often a single technical field will have several hundred specialized symbols. There is one kind of symbol, however, that merits special comment, because it is an abstraction used in many ways in many different kinds of diagrams. This is the arrow.

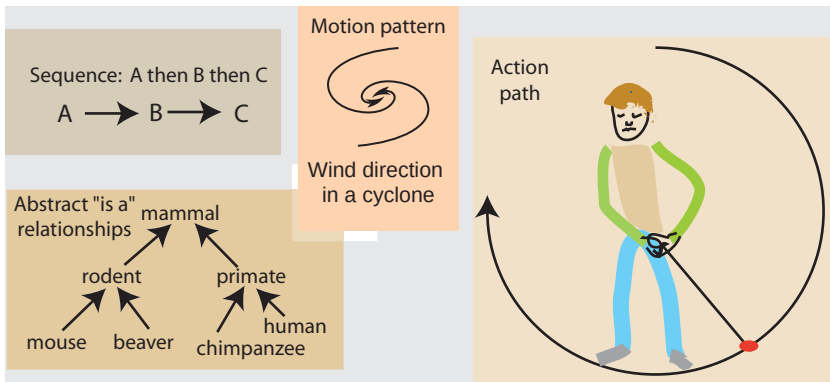
Arrows are interesting because of the way they allow the application of the powerful pattern finding mechanisms of the visual system to be combined with a measure of abstract meaning. They can express motion paths, both historical and planned, as well abstract conceptual relationships of various types, including causal relationships, sequences, and taxonomic



Darwin Notebook. Original is in Cambridge University Library.



In this design diagram for doors, architect Eva Jiricna uses arrows in multiple ways including the motion path of a door, labeling of parts connecting verbally expressed ideas to parts of the graphic, and connecting different parts of the compound diagram. (sketches by Eva Jiricna.)



A few of the abstractions are expressible through arrows.

hierarchies. They can also simply direct attention to a particular feature of a diagram or photograph. The arrowhead has the purely symbolic function of showing directionality either in an abstract sense (the phrase “a mouse *is a* rodent” has a directionality in the sense that the converse is not true) or in the concrete sense of a direction of motion. In showing motion, in which the shaft of the arrow can be either curved or shows a motion path, the arrowhead is purely symbolic and shows which way along the path the motion is headed. In either case this hybrid of a line pattern with a directional arrowhead can be rapidly scribbled, supporting the tight cognitive loop that is the essence of visual thinking through diagramming.

REQUIREMENTS AND EARLY DESIGN

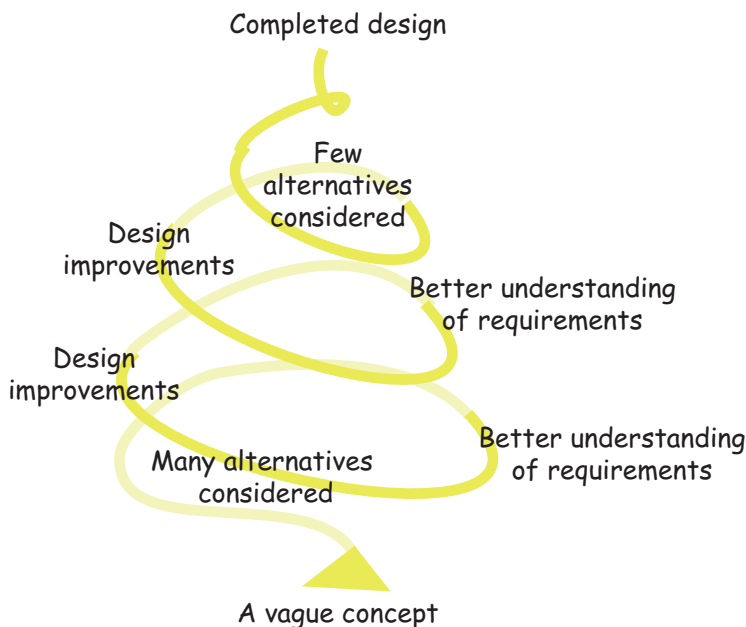
Having considered the details of how lines function in visual thinking it is necessary to return to the larger design process wherein the workload is distributed between the brain and visual tools.

The first part of any design process is an understanding of the requirements of the design. In some disciplines, such as engineering, this stage is explicit and well defined. In less constrained disciplines, a designer must still establish

requirements and goals before examining potential solutions although the process may be quite informal. The aim is to understand and define a problem before attempting to solve it. This can avoid the pitfall of becoming married to visually seductive but inappropriate or ineffective solutions.

Sometimes, however, the articulation of requirements goes hand in hand with the presentation of early design solutions. The process is called a *spiral design* methodology. Many clients can only appreciate design constraints when they are presented with partially realized alternatives, so having examples prepared is necessary to effective dialogue.

When spiral design is used, the requirements are only vaguely specified at first. Both the requirements and the design become more refined as the project proceeds. This has the advantage of enabling insights during later stages of the project to be incorporated. The major disadvantages are that it can be difficult to incorporate changes into a contract and deadlines may need to be altered.



Often clients do not appreciate what is possible until they see an example.

Prototype designs can serve as a discussion point, bringing in the opinions of many stakeholders in the project. At the next meeting the design is refined and presented again for further refinements.

Much specification of requirements is not inherently visual, but for those that are, a *visual task analysis* is required. This is an understanding of the visual cognitive operations the design is intended to support.

VISUAL TASK ANALYSIS

A visual task analysis consists of the identification of cognitive tasks that must be supported by a design. Once a set of tasks has been identified

they can be broken down further into a set of visual queries that will be executed by an individual in performing a cognitive task with the product. For example, a firm of landscape architects might produce design sketches to reason about how the garden will function for the people who will ultimately use it. Tasks include reasoning about affordances for walking along paths and over lawns, the need for patio space and flower beds. Visual queries relating to walking or strolling will be best supported by a design allowing for paths, lawns, and patios to be perceivable as visual gestalt through the careful use of color, texture, or line.

THE CREATIVE DESIGN LOOP

In a landmark study of the creative design process, Masaki Suwa from the Hitachi Research Laboratory in Japan and Barbara Tversky of Stanford University asked a set of advanced architecture students to design an art museum. The students were videotaped while they sketched designs and later asked to watch their own tapes and comment on what they had been thinking. This study revealed that the designers would often start with a very loose sketch, usually of a ground level layout, then *interpret what they had set down*. They did not just formulate ideas in their heads and then set these ideas on paper. The process of sketching was itself a constructive act. The sketch had the function of dividing the paper into regions of *potential* architectural meaning. A line might be considered at one moment as the boundary of a room, or it might come cognitively to represent the edge of a sculpture garden at one side of a building. Suwa and Tversky later called this *constructive perception* and suggested that it is fundamental to the design process.*

Experienced architects are also prolific sketchers. Bryan Lawson conducted a series of interviews with six leading architects to gain an understanding of how they worked.* Of the Spanish architect, Santiago Calatrava, he wrote: “Calatrava is an accomplished artist with a fine sense of line and texture, but one senses that his graphical output is never the result of a wish to produce a drawing *but rather to understand a problem*.” [emphasis mine]

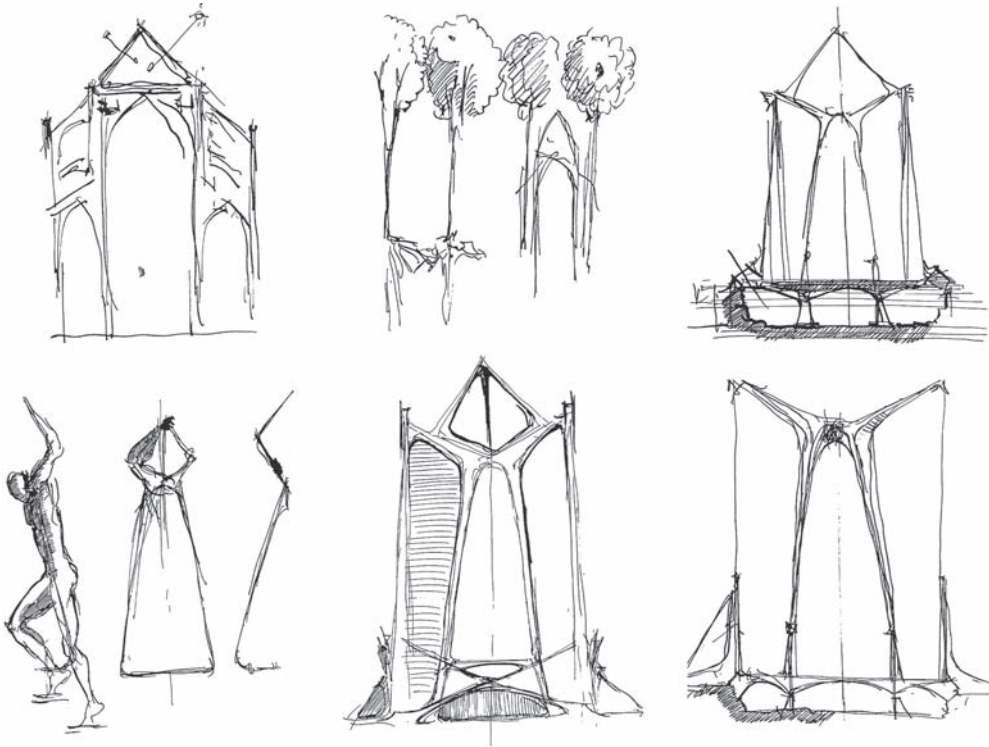
One of the most critical characteristics of a sketch is the speed with which it can be produced and this means that it can also be discarded with little cost. A designer may easily produce twenty or thirty sketches to find the solutions to various problems and retain only one.

COGNITIVE ECONOMICS OF DESIGN SKETCHING

Which cognitive operations are carried out purely in the head through mental imagery, and which cognitive operations result in marks on paper, can be understood partially in terms of cognitive economics. We can shuffle visual concepts in and out of visual working memory several times

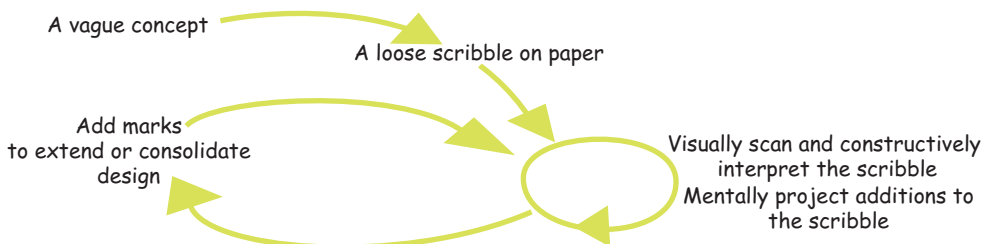
*M. Suwa and B. Tversky. 2003. Constructive Perception: A metacognitive skill for coordinating perception and conception. *Cognitive Science Society Proceedings*. 1140–1144.

*B. Lawson. 1994. Design in mind. *Architectural Press*.



A few sample sketches from Calatrà's notebooks show trees, the human form, and traditional architectural forms, all being used as sources of inspiration for the design of the roof truss system of a modern cathedral. Each of the examples explores a different idea. In some a metaphorical comparison of forms (the male figure and the tree) is the primary focus. Others are explorations of abstract forms that can meet the structural requirements of a building. (sketches by Santiago Calatrà.)

a second. Adding a mark to a sketch takes longer, a second or two to preserve an idea that would otherwise be lost. Visual working memory is the most flexible high-speed sketchpad, but has very limited capacity and is very impermanent. Rough sketches on paper have less flexibility, but they also support greater complexity and can be kept or rapidly discarded. Permanent detailed designs have the least flexibility, cost a lot to produce, but can be very elaborate. As the design process progresses the



kinds of design artifacts that are produced may change from many low-cost sketches to a few high-cost detailed and finished drawings.

It is known from human memory research that recognition is vastly easier than recall. In other words, we can recognize that we have seen something before far more easily than we can reconstruct a memory. It is similarly true that identifying an effective design is vastly easier than creation of that same design. In a sense this is not surprising; it takes but a few seconds to appreciate, at least superficially, that one is in the presence of an interesting and visually exciting design. But this suggests that it is useful to have a design production method that can produce lots of designs at least semi-automatically. Then our job is to select from them, equally rapidly.

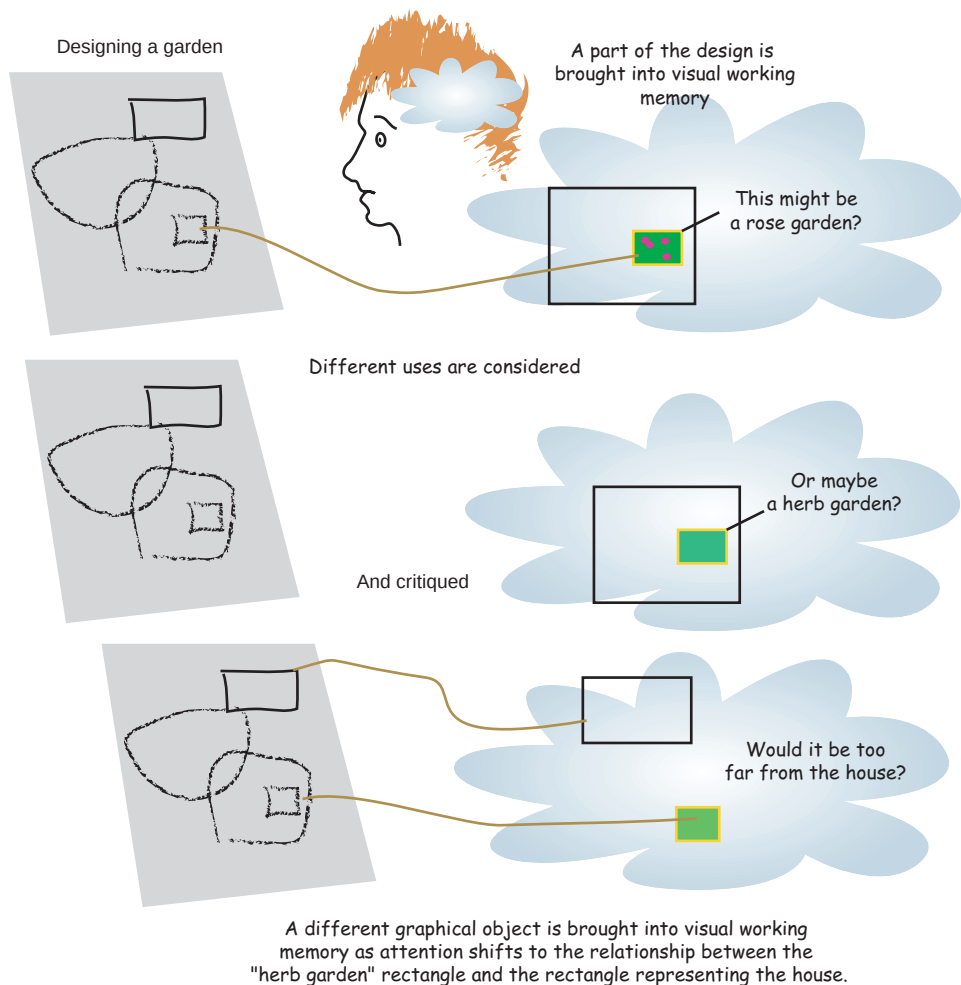
THE PERCEPTUAL CRITIQUE

As the designer quickly creates a conceptual design sketch an ongoing perceptual critique is occurring. This can be thought of as a form of meta-seeing in that it is critical and analytic in a way that goes beyond the normal process of visual thinking associated with everyday tasks. It involves the interpretation and visual analysis of the marks on the paper that have just been put down.

There are two kinds of preliminary design artifacts that we have been considering and each involves different forms of meta-seeing. The first, the *concept diagram*, is about design concepts and usually has little in common with the appearance of any physical object. An example of this would be Darwin's sketch of the evolutionary tree. The second is the *prototype design* which is a rough version of a final product. An example of this would be a rough sketch of a poster or a web page. Many of the most interesting diagrams are combinations of the two. For example, a landscape architect will create sketches that have a rough spatial correspondence to where flower beds and lawns will appear in a finished garden. Many parts of the sketch and the cognitive operations it supports will be highly abstract and conceptual (for example, arrows showing strolling routes).

Designs are tested by means of functional visual queries. For example, a visual query can establish the distance between design elements and thereby discover if an element representing an herb garden is an appropriate distance from an element representing the house. Similar visual queries might be made regarding the area of the lawn, the location of pathways, and the shapes and distributions of flowerbeds. In each case, the corresponding elements are brought into visual working memory, and tested in some way, by means of judgment of distance, area, or shape.

The graphical marks that represent a rose garden in one instant may be reconsidered in the next as representing a space for an herb garden. The cognitive operation is a change in the bindings between visual working memory objects and verbal working memory objects.



Purely cognitive visual imagery can be added to the visual working memory representation of parts of the design. A paved path might be imagined as part of the design. In this case part of what is held in visual working memory corresponds to marks on the paper, whereas the additions are pure mental images. The very limited capacity of visual working memory requires that imagined additions are simple. In order to extend the design in an elaborate way, the imagined additions must be added to the scribble. However, because mental additions have a far lower cognitive

cost than adding to the drawing, many additions can be imagined for every graphical addition to the sketch. Mental deletions can similarly be proposed, and occasionally consolidated by erasing or scribbling over some part of the sketch.

META-SEEING WITH DESIGN PROTOTYPES

The design prototype is a rapid sketch that shows in some respects what a finished produce will actually look like. In the case of a proposed layout for a web page it might suggest a possible set of colors and the locations of various information boxes. The perceptual critique of a design prototype can be used for direct tests of how well the design will support cognitive tasks. These have to do with the perception of symbols, regions, and contours, and the details of this have been extensively covered in the previous chapters. Now we are concerned with how the low-level pattern finding is integrated into the higher level process of meta-seeing.

The designer who is doing a quick conceptual sketch has a problem. Critiquing something that you yourself have produced is much more difficult than critiquing something someone else has produced. You, the designer, already know where each part of the design is located—you put those elements in place and you know what they are intended to signify. Perceiving it from the point of view of a new viewer is a specialized skill. This, in essence, is a kind of cognitive walkthrough in which you must simulate in your own mind what other people will see when they view the finished product. This involves simulating the visual queries that another person might wish to execute and asking of yourself questions such as, “Is this information sufficiently distinct?” and “Does the organization support the proper grouping of concepts?”

To some extent self-critiquing can improve simply through the passage of time. After two or three days the priming of the circuits of the brain that occurred during the original design effort will have mostly dissipated and fresh alternatives can be more readily imagined.

Sometimes a more formal analysis of a design may be warranted. This requires a detailed breakdown of cognitive tasks, to be performed with a design. Each task is then used as the basis for a *cognitive walkthrough* of the prototype design using people who are not designers but potential consumers of the design. The process involves considering each task individually, determining the visual queries that result, then thinking through the execution of those queries. For example, if the design is a road map, then the queries might involve assessing whether the major roads connecting various cities can easily be traced.

Doing cognitive walkthroughs with other participants has a very high cost of time and effort. The combined amount of cognitive work is high and typically only justified for more finished designs. Many early design sketches may not be good solutions and must be rejected. The principle of cognitive economy dictates that designers must develop visual skills that help them to assess designs at an early stage without involving other participants.*

VISUAL SKILL DEVELOPMENT

Many designers are not aware of their critical analytic processes, anymore than we are aware of the skills we have when buttering a piece of bread. It is so long since we first learned to do it that we have completely forgotten that stage at which the task required intense concentration. This is why the expert is often intolerant of the novice. He simply does not understand how an apparently obvious design flaw can be overlooked. But the ability to do a visual critique is an acquired skill.

All skilled behavior proceeds from the effortful to the automatic. For example, at the early stages of learning to type it is necessary to conduct a visual search for every key, and this consumes so much visual, procedural, and motor capacity that nothing is left over for composing a paragraph. At the expert level the finger movement sequences for whole words have become automatic and composing on the keyboard seems completely natural. Very little high-level cognitive involvement is needed and so visual and verbal working memory are free to deal with the task of composing the content of what is to be written.

The progression from high-level effortful cognition to low-level automatic processing has been directly observed in the brain by Russel Poldrack and a team at the University of California who used fMRI to study the development of a visual skill.* For a task, they had subjects read mirror-reversed text, which is an early-stage skill for most people. They found that parts of the brain involved in the task were from the temporal lobe areas of the brain associated with conscious effortful visual attention. In contrast, reading normal text, which is a highly automated learned skill, only caused brain activity in lower-level pattern recognition areas of the cortex.

It takes time to convert the hesitant effortful execution of tasks into something that is done automatically in the brain but it is not simply practice time that that counts. A number of studies show that sleep is critical to the process. Skills only become more automatic if there is a period of sleep between the episodes where the skill is practiced. This has shown to be the case for many different tasks: for texture discriminations that might

*For a full of account of the cognitive walkthrough methodology see C. Wharton, J. Reiman, C. Lewis and P. Polson. 1994. *The Cognitive Walkthrough Method: A Practitioner's Guide*. New York: Wiley.

*R.A. Poldrack, J.E. Desmond, G.H. Glover and J.D.D. Gabrieli, 1998. The neural basis of visual skill learning: and fMRI study of mirror reading. *Cerebral Cortex*. 8(1): 1–10.

be important in judging the quality of cloth, in typing letter sequences, and in tasks, like riding a bicycle, that require physical dexterity. Thus, we may practice a task for hours and gain little benefit; but if we practice for only half an hour, then wait a day, some measure of *automaticity* will be gained.

It is not enough that we simply go through the motions in learning visual analytic skills. Whatever is to be learned must be the focus of attention. A couch potato may watch television soap operas for hours every day over years but learn nothing about the skills of film directing. However, a director will see, and *interpret the value of*, lighting, camera angles, and shot sequences. The important skills for the designer are interpreting and constructive criticism. It is not enough to look at many visual designs; each must be subjected to a visual critique for these skills to develop. This book is intended to provide perceptually based design rules that form one basis for critical assessment.

CONCLUSION

The skill of thinking through sketching has nothing to do with ability to draw in the conventional sense of drawing a portrait or a landscape. Almost any scribble will do. The power of sketching in the service of design comes from the visual interpretive skills of the scribbler. Prototype design sketches allow for preliminary design ideas to be critiqued, both by the creator and by others. A concept sketch differs from a prototype sketch in that it is a method for constructing, organizing, and critiquing ideas. It provides abstract representations of ideas and idea structures. Most sketches are hybrids of prototype and concept sketches, partly appearance and partly ideas.

The power of sketching as a thinking tool comes from a combination of four things. The first is the fact that a line can represent many things because of the flexible interpretive pattern-finding capability of the visual system. The second has to do with the way sketches can be done quickly and just as easily discarded. Starting over is always an option. The third is the critical cognitive skill of interpreting lines in different ways. Part of this skill is the ability to project new ideas onto a partially completed scribble. Part of it is the ability to critique the various interpretations by subjecting them to functional visual queries. The fourth is the ability to mentally image new additions to a design. Despite the fact that no significant drawing skill is needed, thinking with sketches is not easy. The skill to visually analyze either prototype designs or idea structures is hard won and it is what differentiates the expert from the novice designer. A whole lifetime's experience enriches a scribble and transforms it from a few meaningless marks to a thinking tool.

Chapter 9

The Dance of Meaning

Meaning is what the brain performs in a dance with the external environment. In this dance tokens of meaning are spun off into electronic and social media and tokens of meaning are likewise picked up. New meaning is constructed when patterns already stored within the brain are combined with patterns constructed from external information. Increasingly, new meaning is also constructed by inanimate computers that do at least partial analysis, and synthesis of patterns and tokens of meaning, then present the results using a visual display.

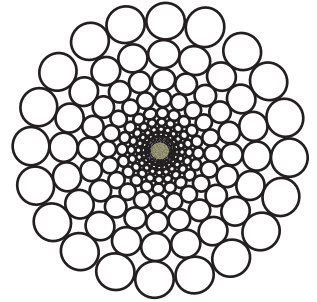
In this book, the dance partners are considered to be individual people interacting with visual displays. In reality the dance is far more intricate and elaborate; there is a constant stream of new meaning being developed by people interacting with one another. Visual thinking is but a small part of the dance. Nevertheless, because of the special power of the visual system as a pattern-finding engine, visual thinking has an increasingly important role. This book itself is part of the dance, as is everything that is designed to be accessed visually.

One of the main themes we have explored in this book is that at every level of description visual thinking can be thought of as active processes operating through the neural machinery of the brain as well as through the external world. The purpose of this chapter is to review and summarize what we have covered so far and then discuss some of the broader implications of how the theory of perception applies to design.

REVIEW

The following few pages give a twelve-point summary of the basic machinery and the major processes involved in visual thinking. With the twelfth point we shall segue into observations that go beyond what has been said before.

1. The eye has a small high-resolution area of photoreceptors called the fovea. We see far more detail in the fovea than off to the side and we sample the world by making rapid eye movements from point to point. Eye movements rotate the eyeball so that imagery from different parts of the visual world falls on the fovea to be analyzed by the brain.
2. Our brains construct visual queries to pick up what is important to support what we are doing cognitively at a given instant. Queries trigger rapid eye movements to enable us to pick up information that answers the query.



How affluent?



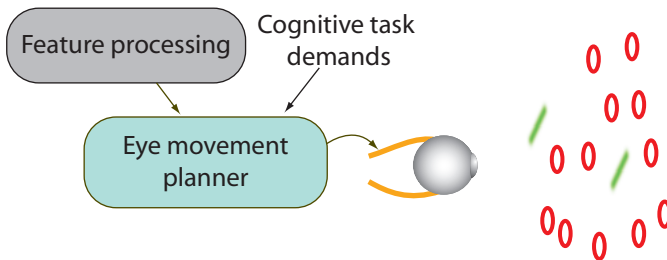
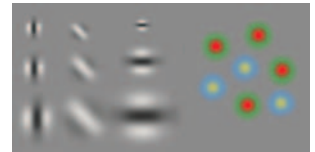
What are their ages?

"Unexpected Returns" by Ilya Repin, taken from A. Yarbus, 1967. Eye movements during perception of complex objects. L.A. Riggs, ed., *Eye Movements and Vision*, Plenum Press. NY. Chapter VII, 171–196.

On page 166 is a picture by the Russian realist painter Ilya Repin, titled, “They did not expect him.” To the right, shown in red, are eye movement traces from one person asked to perform different analytic tasks. When asked about the material well-being of the family, the eye movements fixate on clothes, pictures on the walls, and other furnishings (top). When asked about the ages of the family members, the eye movements fixate on faces almost exclusively (bottom).

These records were made by the Russian psychologist Alfred Yarbus, who in 1967 used a mirror attached by means of a suction cup to his subjects’ eyeballs to reflect light onto photographic paper and thereby record eye movements.

3. The first stage of cortical visual processing is a local feature analysis done simultaneously for every part of the visual field. The orientation, size, color, and motion of each part of the image falling on the retina is determined all at once by feature-selective neurons. Smaller scale features are only analyzed in the fovea at the center of vision. All other visual processing is based on the initial division of the visual world into features.



Understanding feature-level processing tells us what will stand out and be easy to find in a visual image. In the image above, the green bars are distinct in terms of several feature dimensions: color, orientation, curvature, and sharpness. We can easily execute eye movements to find the green bars because orientation, color, and curvature are all feature properties that are processed at an early stage, and early stage properties are the ones that can be used by the brain in directing eye movements.

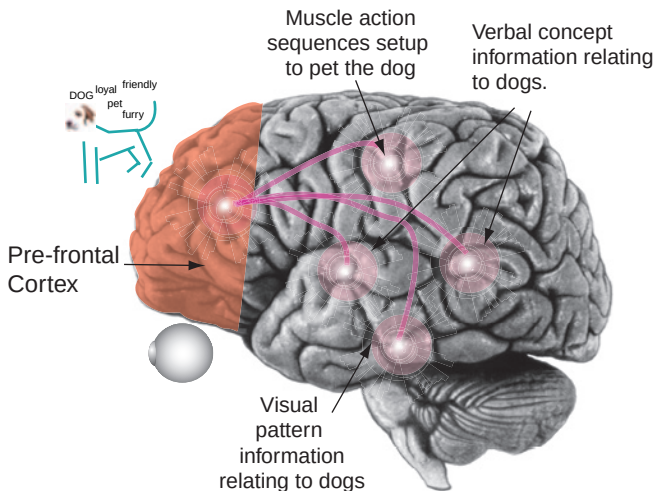
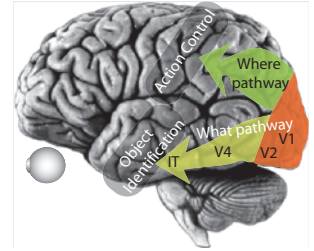
A simple motion pattern can also be thought of as a feature. Something that moves in a field of mostly static things is especially distinctive and easy to find.

4. There are two major processing pathways called the *where* and *what* pathways.

The *where* pathway has connections to various regions in the parietal lobe responsible for visually guided actions, such as eye-movements, walking, and reaching to grasp objects.

The *what* pathway is responsible for identifying objects through a series of stages in which increasingly complex patterns are processed, each stage building on the previous one.

Between the *low-level* feature analysis and *high-level* object recognition is an intermediate pattern-finding stage. This divides visual space into regions bounded by a contour and containing similar textures, colors, or moving features. In V4 more complex compound shapes are identified from patterns of features. In the IT cortex neurons respond to specific meaningful patterns such as faces, hands, letters of the alphabet, and automobiles.



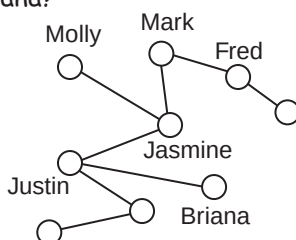
5. Various kinds of information are combined in a temporary nexus of meaning. This information can include visual pattern information, language-based concepts, and action patterns. This nexus is short lived and is what makes up the contents of the working memories.

From one to three meaningful nexuses can be formed every time we fixate on part of a scene. Some of these may be held from one eye fixation to the next, depending on their relevance to the thought process. This three-object visual working memory limit is a basic bottleneck in visual thinking.

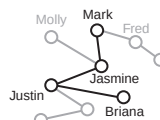
6. Language processing is done through specialized centers in the left temporal lobe. Language understanding and production systems are specialized for a kind of informal logic exemplified by the “ifs,” “ands,” and “buts” of everyday speech. This is very different from the visual logic of pattern and spatial arrangements.
7. The natural way of linking spoken words and images is through diexis (pointing). People point at objects just prior to, or during, related verbal statements, enabling the audience to connect the visual and verbal information into a visual working memory nexus.
8. One way that visual displays support cognition is by providing aids to memory. Small images, symbols, and patterns can provide *proxies* for concepts. When these proxies are fixated, the corresponding concepts become activated in the brain. This kind of visually triggered activation can often be much faster than retrieval of the same concept from internal long-term memory without such aids. When an external concept proxy is available, access to it is made by means of eye movements which typically take approximately one-tenth of a second. Once the proxy is fixated, a corresponding concept is activated within less than two-tenths of a second. It is possible to place upwards of thirty concept proxies in the form of images, symbols, or patterns on a screen providing a very quickly accessible concept buffer. Compare this to the fact that we can hold only approximately three concept chunks in visual or verbal working memory at a time. There is a major limitation to this use of external proxies—it only works when there are learned associations between the visual symbols, images, or patterns and particular concepts.



How did the information
get from Mark to Briana?



We just need
to see the pattern
of links



9. Another way in which visual displays support cognition is through pattern finding. Visual queries lead to pattern searches and seeing a pattern, such as a connection between two objects, often provides the solution to a cognitive problem.
10. One basic skill of designers can be thought of as a form of constructive seeing. Designers can mentally add simple patterns to a sketch to test possible design changes before making any changes to the sketch.
11. Long-term memories are cognitive skills, rather than fixed repositories like books or CD ROMs. The pathways that are activated when a cognitive task is carried out become stronger if that task is successfully completed. These pathways exist in all parts of the brain and on many levels; they are responsible for feature detection, pattern detection, eye movement control, and the sequencing of the cognitive thread. Activated long-term memories are essentially reconstructions of prior sequences of neural activity in particular pathways. Certain external or internal information can trigger these sequences. In the case of pattern recognition the sequences are triggered by visual information sweeping up the ‘what’ pathway.

Some visual skills, such as seeing closed shapes bound by contours as “objects” or understanding the emotional expressions of fellow humans, are basic in the sense that they are to some extent innate and common to all humans, although such skills are refined with practice and experience.



Understanding this scene requires the ability to perceive social interactions. (Image courtesy of Joshua Eckels.)



Other perceptual tasks, such as reading a contour map, understanding a cubist painting and interpreting an X-ray image, require specialized pattern recognition skills. These higher-level skills are much easier to acquire if they build on more basic skills, which means that an artist cannot be too radical and still expect to be widely understood.

12. How does the mind control itself? In a sense, it doesn't — there is no control center of the mind. Intelligence emerges from the activities of different parts of the brain and the information that passes back and forth internally, as well as activities involving the external world via vision and touch and other senses. Although the prefrontal cortex appears to be the part of the brain exerting the highest level of control over cognitive processes, it would be a mistake to think of it as “smart.” The brain is made up of fifty or so specialized regions, none of which on its own can be thought of as intelligent. Examples are the regions V1, V2, V4, and the IT cortex. Patterns of neural activation beget other patterns of activation as one patterned cascade of firing naturally triggers other different cascades of activation in a kind of chain reaction of neural activity. The chain reaction of cognition, one pattern of activation leading to the next, is what we have been calling the cognitive thread.

Visual thinking is based on a hierarchy of skills. Sophisticated cognitive skills build on simpler ones. We cannot begin to play chess until we can identify the pieces. We will not become expert until we have learned patterns involving whole configurations relating to strategic advantage or danger. As we get skilled at a particular task, like chopping onions, the operation eventually becomes semi-automatic. This frees up our higher-level control processes to deal with higher level problems, such as how to deal with an extra person coming to dinner. The process whereby cognitive activities become automated is absolutely critical in the development of expertise because of the fundamental limitations of visual and verbal working memory capacities. If a set of muscle movements involved in drawing a circle on paper becomes automated, then the designer has free capacity to deal with arrangements of circles.

Patterns of neural activation are not static configurations, but sequences of firing. At the highest levels, involving the pre-frontal cortex at the front of the brain and the hippocampus in the middle, these sequences can represent action plans. This, too, is hierarchical. Complex tasks, like cooking a meal, involve high-level plans that have the end goal of getting food on the table, together with mid level plans, like peeling and mashing potatoes, as well as with low level plans that are semi-automatic, such as reaching for and grasping a potato.

A cognitive tool can be a map or a movie poster, but increasingly cognitive tools are interactive and computer based. This means that every visual object shown on the screen can be informative in its own right, as well as be a link, through a mouse click, to more information. We may also be able to manipulate that information object with our computer mouse, to literally organize our ideas.

The sum of the cognitive processing that occurs in problem solving is moving inexorably from being mostly in the head, as it was millennia ago before writing and paper, to being a collaborative process that occurs partly in the heads of individuals and partly in computer-based cognitive tools. Computer-based cognitive tools are developing with great speed in human society, far faster than the human brain can evolve. Any routine cognitive task that can be precisely described can be programmed and executed on a computer, or on millions of computers. This is like the automation of a skill that occurs in the brain of an individual, except that the computer is much faster and less flexible.

IMPLICATIONS

The active vision model has four broad implications for design.

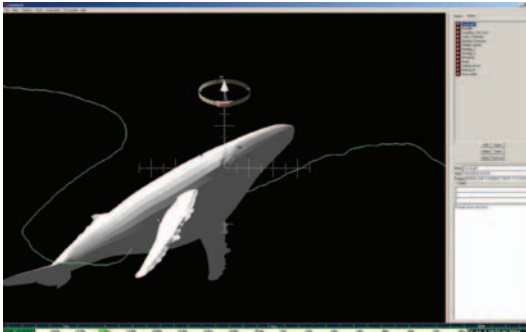
1. To support the pattern-finding capability of the brain; that is, to turn information structures into patterns.
2. To optimize the cognitive process as a nested set of activities.
3. To take the economics of cognition into account, considering the cost of learning new tools and ways of seeing.
4. To think about attention at many levels and design for the cognitive thread.

DESIGN TO SUPPORT PATTERN FINDING

Properly exploiting the brain's ability to rapidly and flexibly discover visual patterns can provide a huge payoff in design. The following example, from

my own work, illustrates this. Over the past few years I have been fortunate to be involved in a project to discover the underwater behavior of humpback whales. The data was captured by a tag attached to a whale with suction cups. When the tag came off it floated to the surface and was retrieved. Each tag provided several hours of data on the position and orientation of the whale as it foraged for food at various depths in the ocean. This gave us an unprecedented opportunity to see how humpback whales behave when they are out of sight underwater.

Our first attempt to provide an analysis tool was to create a program that allowed ethologists to *replay* a moving three-dimensional model of the whale at any desired rate.* They were initially thrilled because for the first time they could see the whale's underwater behavior. Some fascinating and previously unknown behavioral patterns were identified by looking at these replays. Nevertheless, although the visualization tool did its job, analysis was extremely time consuming. It took many hours of watching to interpret an hour's worth of data.



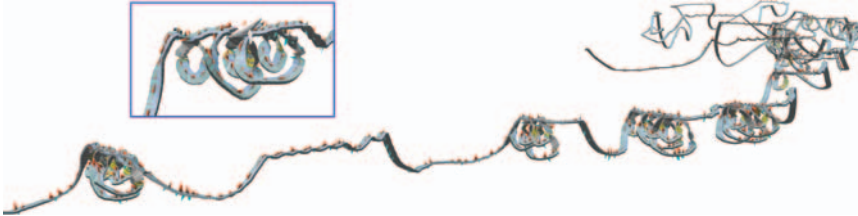
*This work is described in C. Ware, R. Arsenault, M. Plumlee and D. Wiley. 2006. Visualizing the underwater behavior of humpback whales. *IEEE Computer Graphics & Applications*, July/August issue. 14–18.

GeoZu4D software allows for the underwater behavior of a humpback whale to be played back along with any sounds recorded from a tag attached to the whale by means of suction cups.

Using this replay tool, the visual thinking process can be roughly summarized as follows. The whale movements were replayed by an ethologist looking for *stereotyped temporal sequences of movements*. When something promising was observed the analyst had to remember it in order to identify similar behaviors occurring elsewhere in a record. The ethologist continued to review the replay looking for novel characteristic movement patterns.

Although this method worked, identifying a pattern might take three or four times as long as it took to gather the data in the first place. An eight-hour record from the tag could take days of observation to interpret. If we consider the problem in cognitive terms the reason becomes clear. We can remember at most only a half dozen temporal patterns in an hour of video, and these may not be the important or stereotyped ones. Further,

we are not nearly as good at identifying and remembering motion patterns as we are at remembering spatial patterns. Also, every time an ethologist formed the hypothesis that some behavior might be stereotypical, it was necessary to review the tracks of all the other whales, again looking for instances of the behavior that might have been missed.



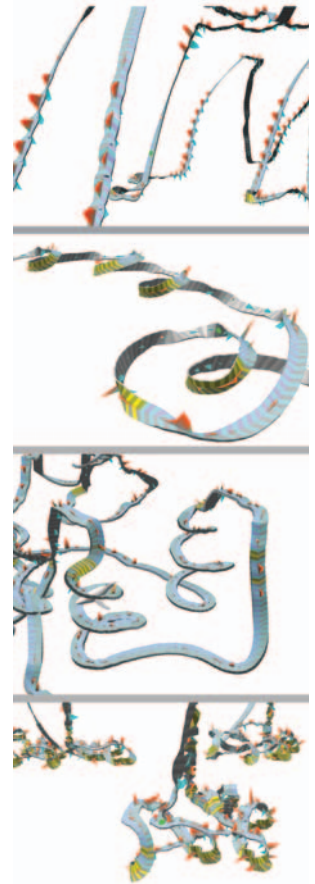
Our second attempt at an analysis tool was far more successful because it took much greater advantage of the brain's pattern-finding ability. We transformed the track of the whale into a 3D ribbon. We added a saw-tooth pattern above and below the ribbon that was derived from calculated accelerations indicating the fluke strokes of a whale.

Transforming the whale track into a ribbon allowed for much more rapid identification of behavior patterns. This new tool enabled a much different visual thinking process. The ethologist could quickly zoom in on a region where feeding behaviors were seen and stop to view a static image. Behavior patterns could be identified by visually scanning the image looking for repeated graphic patterns. Because eye movements are so fast and static patterns can be picked out efficiently, this method enabled analysts to compare several patterns *per second*. This process was hundreds of times faster than the replay method. Naturally, a complete analysis still took a great deal of work, but now we could scan the data for familiar or new patterns within minutes of extracting the data files from the tag. The new visual thinking process was hundreds of times faster than the old one based on simple replay.

The whale behavior study clearly demonstrates the general principle that patterns can be identified and compared very rapidly if we can turn information into the right kind of spatial display. The design challenge is to transform data into a form where the important patterns are easy to interpret.

OPTIMIZING THE COGNITIVE PROCESS

Good design optimizes the visual thinking process. The choice of patterns and symbols is important so that visual queries can be efficiently processed by the intended viewer. This means choosing words and patterns

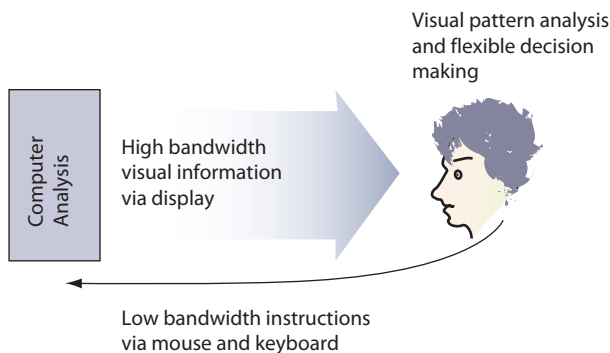


Ribbons make the underwater behavior of a humpback whale accessible through pattern perception.

each to their best advantage. When designing the visual interface to an interactive computer program we must also decide how the visual information should change in response to every mouse click.

Extraordinarily powerful thinking tools can be made when a visual interface is added to a computer program. These are often highly specialized—for example, for stock market trading or engineering design. Their power comes from the fact that computer programs are cognitive processes that have been standardized and translated into machine executable form. They offload cognitive tasks to machines, just as mechanical devices, like road construction equipment, offload muscle work to machines. Once offloaded, standardized cognitive tasks can be done blindingly fast and with little or no attention. Computer programs do not usually directly substitute for visual thinking, although they are getting better at that too; instead they take over tasks carried out by humans using the language processing parts of the brain, such as sophisticated numerical calculations. A visual interface is sometimes the most effective way for a user to get the high volume of computer-digested information that results. For example, systems used by business executives condense large amounts of information about sales, manufacturing, and transportation into a graphical form that can be quickly interpreted for planning and day-to-day decision making.

It is useful to think of the human and the computer together as a single cognitive entity, with the computer functioning as a kind of *cognitive co-processor* to the human brain.* Low-bandwidth information is transmitted from the human to the computer via the mouse and keyboard, while high-bandwidth information is transmitted back from the computer to the human for flexible pattern discovery via the graphic interface. Each part of the system is doing what it does best. The computer can pre-process vast amounts of information. The human can do rapid pattern analysis and flexible decision making.



*The term 'cognitive co-processor' comes from a paper from Stuart Card's famous user interface research lab at Xerox Palo Alto Research Center (PARC) although I am using it in a somewhat different sense here. Robertson, S. Card and J. Mackinlay. 1989. The cognitive co-processor for interactive user interfaces. *Proceedings of the ACM UIST Conference*. 10–18.

Peter Pirolli and Stuart Card developed a theory of information access to help with the design of interfaces to cognitive tools. They began with the *foraging theory* developed by ethologists to account for animal behavior in the wild. Most wild animals spend the bulk of their time in a highly optimized search for food. To survive they must balance the energy expended in finding and consuming food—just eating and digesting has a high cost for grazing animals—with the energy and nutrients obtained. They found that people forage for information on the Internet much as animals forage for food; they are constantly making decisions about what *information scent* (another term from Stuart Card's Influential laboratory) to follow, and they try to minimize how much work they must do, offloading tasks onto the computer wherever possible.*

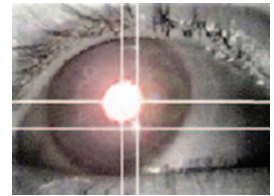
The ideal cognitive loop involving a computer is to have it give you exactly the information you need when you need it. This means having only the most relevant information on screen at a given instant. It also means minimizing the cost of getting more information that is related to something already discovered. This is sometimes called *drilling down*.

It might be thought that an eye tracker would provide the ideal method for drilling down, since eye movements are the natural way of getting objects into visual working memory. Eye tracking technology can determine the point of gaze within about one centimeter for an object at arm's length. If the computer could have information about what we are looking at on a display, it might summon up related information without being explicitly asked.

There are two problems with this idea. First, tracking eye movements cheaply and reliably has proven to be technically very difficult. Eye trackers required careful and repeated calibration for each user. Second, when we make eye movements we do not fixate exactly; we usually pick up information in an area of about one centimeter around the fovea at normal computer screen viewing distances and this area can contain several informative objects. This means the computer can only 'know' that we might be interested in any of several things. Showing information related to all of them would be more of a hindrance than a help.

The quickest and most practical method for drilling down is the mouse-over *hover query*. Imagine that by moving the mouse over a part of a diagram all the other on-screen information *relating to the thing the mouse is moving over* becomes highlighted and the relevant text enhanced so that it can be easily read. This is exactly what was done by Tamara Munzner and a team at Stanford University in their experimental *Constellation* system seen on the next page.* Their application was a kind of network diagram

*P. Pirolli and S. Card. 1999. Information foraging. *Psychological Review*. 106: 643–675.



*T. Munzner, F. Guimbretiere and G. Robertson. 1999. Constellation: A visualization tool for linguistic queries from MindNet. *Proceedings of IEEE InfoVis. Conference*, San Francisco. 132–135.

showing the relationships between words. The diagram was far too complex to be shown on a screen in any normal way, but by making it interactive hundreds of data objects could be made usable. Compare this to a typical network diagram, in which only between ten and thirty nodes are represented, and the advantage becomes clear. This kind of mouse-over clickless hover query is the next best thing to moving the eyes around an information space. Clickless hover queries can be made only about once a second, slow compared to three per second mouse movements, but this still leads to a very rapid interaction where the computer display seems like part of the thinking process, rather than something to be consulted.

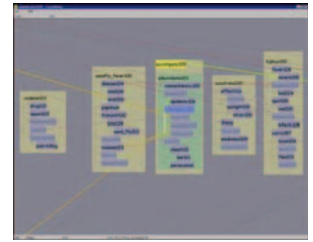
Constellation was a single example of an experimental user interface, but it provides a more general lesson. The ideal computer system that supports visual thinking should be extremely responsive, presenting relevant information just at the moment it is needed. This is not easy to achieve, but something for which to strive.

It should be noted that the most common application of hover queries is in so-called “tool-tips.” These are the presentation of additional information about a menu item or icon when the mouse is placed over it. Generally these are implemented so that a delay of a second or two occurs between the query and the information display. This is probably appropriate for tool-tips because we do not want such information to be constantly popping up, but it would not be appropriate for more tightly coupled human-computer systems. Such a long delay would do serious damage to the efficiency of the cognitive loop that is supported by Constellation.

LEARNING AND THE ECONOMICS OF COGNITION

Many of the visual problems we solve in life seem completely mundane: walking across a room, preparing a salad, looking for a road sign. They are mundane only because we have done them so often that we no longer consider them as requiring thinking at all. When we are born we have little visual skill except the basic minimum needed to identify that something is an “object” and a special propensity to fixate on human faces. Newborns do, however, have the neural architecture that allows the capabilities to develop.* Everything else is a learned skill and these skills make up who we are. Our visual skills vary from the universal, like the ability to reach for and grasp an object, to the very specialized, like determining the sex of chicks, a very difficult visual pattern matching task that can only be done by highly trained experts.

All cognitive activity starts off being difficult and demanding attention. As we develop skills the neural pathways involved in performing the



*One of the most important fixed architectural cognitive capacities is the three-item limit in visual working memory. Even this is somewhat mutable. A recent study showed that video game players can enhance their visual working memory capacities from three to four items. But we do not find people with a working capacity of ten items.

See C.S. Green and D. Bavelier. 2003. Action video game modifies visual selective attention. *Nature*. 423: 534–537.

task are strengthened. These neural pathways carry particular patterns of activation, and strengthening them increases the efficiency of sequences of neural firing (the cognitive activities). The cognitive process becomes more and more automatic and demands less and less high-level attention. The beginning typist must use all his cognitive resources just to find the letters on the keyboard; the expert pays no attention to the keys. The use of drawing tools by an engineer or graphic designer is the same.

Sometimes we have a choice between doing something in an old, familiar way, or in a new way that may be better in the long run. An example is using a new computer-based graphic design package, which is very complex, having hundreds of hard-to-find options. It may take months to become proficient. In such cases we do a kind of cognitive cost-benefit analysis weighing the considerable cost in time and effort and lost productivity against the benefit of future gains in productivity or quality of work. The professional will always go for tools that give the best results even though they may be the hardest to learn because there is a long-term payoff. For the casual user sophisticated tools are often not worth the effort. This kind of decision making can be thought of as cognitive economics. Its goal is the optimization of cognitive output.

A designer is often faced with a dilemma that can be considered in terms of cognitive economics. How radical should one make the design? Making radically new designs is more interesting for the designer and leads to kudos from other designers. But radical designs, being novel, take more effort on the part of the consumer. The user must learn the new design conventions and how they can be used. It is usually not worth trying to redesign something that is deeply entrenched, such as the set of international road signs because the cognitive costs, distributed over millions of people, are high. In other areas, innovation can have a huge payoff.

The idea of an economics of cognition can be fruitfully applied at many cognitive scales. In addition to helping us to understand how people make decisions about tool use, it can be used to explain the moment-to-moment prioritization of cognitive operations, and it can even be applied at the level of individual neurons. Sophie Deneve of the *Institute des Sciences Cognitives* in Bron, France, has developed the theory that individual neurons can be considered as *Bayesian* operators, “accumulating evidence about the external world or the body, and communicating to other neurons their certainties about these events.” It her persuasive view each neuron is a little machine for turning prior experience into future action.*

There are limits, however, to how far we can take the analogy between economic productivity and cognitive productivity. Economics has money

*In 1763 the Reverend Thomas Bayes came up with a statistical method for optimally combining prior evidence with new evidence in predicting events.

For an application of Bayes' theorem to describe neural activity, see S. Deneve, *Bayesian Inference in Spiking Neurons*. Published in *Advances in Neural Information Processing Systems*. Vol. 17. MIT Press, 1609–1616.

as a unifying measure of value. Cognitive processes can be valuable in many different ways and there is potentially no limit on the value of an idea.

ATTENTION AND THE COGNITIVE THREAD

A revolution is taking place in the science of human perception and attention is its core concept. In some ways attention is a grab bag of an idea, operating differently in different parts of the brain. At the early stage of visual processing, attention biases the patterns which will be constructed from raw imagery. Different aspects of the visual images are enhanced or suppressed according to the top-down demands of the cognitive thread. An art critic's visual cortex at one moment may be tuned to the fine texture of brush strokes, at another to the large-scale composition of a painting, and at yet another to the color contrasts. Attention is also the very essence of eye movement control because looking is a prerequisite for attending. When we wish to attend to something, we point our eyeballs at it so that the next little bit of the world we are most interested in falls on the fovea. The planning of eye movements is therefore the planning of the focus of attention, and the sequence of eye fixations is intimately tied to the thread of visual thinking.

Visual working memory is a process that is pure attention. It is a momentary binding together of visual features and patterns that seem most relevant to the cognitive thread. These constructions are short lived, most lasting only a tenth of a second, or less than the duration of a single fixation. A few persist through a series of successive fixations, being reconnected to the appropriate parts of the visual image after each one. Some visual working memory constructions are purely imaginary and not based on external imagery. When an artist is contemplating a stroke of a pen, or an engineer is contemplating the alteration of a design, mentally imaged potential marks on the paper can be combined with information from the existing sketch. The combination of imaginary with the real is what makes visual thinking such a marvel and is a key part of the internal-external dance of cognition.

There is a basic cycle of attention. Between one and three times a second our brains query the visual environment, activate an eye movement to pick up more information, process it, and re-query. The information picked up becomes the content of visual working memory. This cycle provides the basic low-level temporal structure of the cognitive thread. At a higher level the cognitive thread can have a wide variety of forms. Sometimes the cognitive thread is largely governed by external information coming through the senses. In the case of a movie the cinematographer, directors,

and actors can, to a large extent, control the cognitive thread of the audience. This is not to deny each person in an audience their own thoughts and opinions about what they are seeing, but they will look at the same images, and even the same parts of those images in the same order. The objects constructed in a hundred visual and verbal working memories will be similar and occur in the same order.

Most of the time our visual cognitive thread is occupied with mundane things, like finding the path in front of us, picking up an object or watching someone's face as they talk to integrate information from their facial expression with what they are saying. These tasks are highly learned and so most of the time we are not aware of doing them; nevertheless, they do take up our attentional capacity.

Sometimes the cognitive thread is up for grabs. Someone walking, bored, along a flat, unobstructed sidewalk only requires a small amount of visual attention to keep to the path and the rest of his attentional capacity is a kind of free resource. Such a person is likely to give a second glance to anything that is even slightly novel, especially if it is moving.

There is also the visual thinking process that occurs when someone reasons with a graphic design as an external aid. A map reader is carrying out a very goal directed query loop that involves both the rough logic of the language processing system together with the pattern-finding capability of the visual system.

To add to the diversity of attention, the cognitive thread shifts back and forth between visual processing and language-processing modalities. Sometimes it is primarily visual, or visual and motor, as when we are doing something like tracing out a line with a pen. Sometimes it is driven by the language modules of the brain. We cannot do more than one visual task at a time, and we cannot do more than one verbal task at a time. But we can carry out a verbal task, such as talking on a cell phone, and another visual motor task, like folding laundry, at the same time, if one of them is a highly learned skill. The separation is not perfect and people driving while talking on phones are undoubtedly more dangerous, but these are examples of real multi-tasking using visual and verbal channels to semi-independently carry out two cognitive tasks at once. Nevertheless, it is when visual and language modalities are combined that the brain is most effective. A well-designed presentation, for example, will use words and graphics, each to convey different kinds of information and the two kinds of information will be linked using pointing or simple proximity in space and time.

WHAT'S NEXT?

The science of human perception is continuing to advance and the account of visual thinking given in this book will, to some extent, be obsolete before it hits the bookstores. As research moves forward, our increasing knowledge of the human brain and perception will lead us in new and exciting directions. I believe, however, that the broad picture of how visual thinking works is essentially correct and will stand the test of time. There is a feeling among vision scientists that the secrets of the brain are at last becoming unlocked and that the outlines of lasting theories are beginning to emerge.

One way forward for the science and art of design is for computers to get in on the act. Already programs such as John Anderson's ACT-R can simulate the activities of the human brain as it carries out cognitive tasks.* These simulations incorporate ideas of attention and visual working memory, although they lack more subtle concepts such as novelty and emotional content. Nevertheless, these programs are being used to test computer interface designs for information systems. A virtual human with an ACT-R brain can be used to test a particular computer interface design that includes, to a limited extent, its graphic design. The design can be changed depending on its results and the program run again to see if the new design leads to more efficient cognition. Because computers are fast, many more interface designs can be tested faster than using real humans for the tests, and computers can be used to automatically make design alterations ultimately eliminating the human designer.

Of course, we do not have to go the ACT-R route. The science of perception can and should enrich the craft of design, not take it over. As someone who loves both the science of perception and the art of design, I believe that they should not be antagonistic; instead the two disciplines should learn from and mutually support one another.

Ultimately the science of perception must take design into account because the designed world is changing people's thinking patterns. Real world cognition increasingly involves computer-based cognitive tools that are designed to support one mode of thinking or another. This cognitive support environment is developing and evolving from year to year in a process that is happening much faster than evolution. Designed tools can change how people think.

The human brain evolved in a world made up of natural objects: plants, rocks, earth, sky, other humans, and animals. None of these were explicitly designed by early humans, although like other animals humans have

*J.R. Anderson. 1996. ACT: A simple theory of complex cognition. *American Psychologist*. 51: 355–365.

always shaped their environments. Most people now live in cities where almost everything has been designed. Our visual thinking skills are shaped by how we interact with this world of designed objects from the moment we first open our eyes. Objects in our modern environment incorporate computer programs, and some of them are explicitly designed as cognitive tools. At work or at play, many of us are in front of computer screens for a significant percentage of our lives. For many people the manipulation of a mouse or video game controller is as skilled and entrenched as any basic life skill. Many modern teenagers are more skilled at navigating three-dimensional virtual worlds with a game controller than they are at running over rough real-world terrain. Visual design must take into account both the relatively fixed capacities of the human brain as well as the evolving skill sets of people who use sophisticated and powerful cognitive tools. Interactive design is becoming ever more important as the loop coupling humans and computers tightens. The visual system will always be the highest bandwidth sense by far, and making full use of its flexible pattern-finding capabilities can provide great benefits.

Index

A

- Abstractions, and diagrams, 155–156
- Action maps, in parietal lobe, 101
- Active vision model, and design, 172
- Actors, change detection experiment, 1–2
- ACT-R software, function, 181
- Advertisements
 - images as symbols, 124
 - as visualizations, 20
 - and visual novelty, 122
- American Sign Language (ASL), 130
- Anderson, John R., 181
- Apprehendability
 - definition, 55
 - multi-chunk pattern queries, 56
- Architects
 - and diagrams, 155
 - as sketchers, 158
- Arrows, in diagrams, 155–156
- Artificial interactive spaces
 - information access design, 103
 - naive physics of perception, 102
- Artists, and diagrams, 155
- ASL, *see* American Sign Language (ASL)

Attention

- basic cycle, 179–180
- and color-opponent channels, 73–74
- and early processing, 179
- and eye movements, 9
- and graphic design, 3
- and mundane tasks, 180
- in pop out experiments, 28–29
- and seeing, 3
- street signs, 35
- and top-down processes, 12
- in two-dimensional space, 52–53
- visual *vs.* language processing, 180
- in visual narrative, 140–141
- and working memory, 115–116, 179

Away dimension, in 2.5D space, 44–45

Axonometric projection, definition, 92

B

- Ballard, Dana, 13
- Bayes, Thomas, 178
- Bayesian operators, neurons as, 178
- Baynton, Douglas, 130
- Beach scenes, perception, 112–113

Beck, Harry, 15–16

Bell, Alexander Graham, 130

Biased competition model

- definition, 119
- in eye movement planning, 26–27

Biederman, Irving, 110

Binding problem, two-dimensional space, 46–49

Biomotion perception, patterns, 54

Black-white channel

- and color-opponent theory, 84–85
- and maps, 82
- in opponent process theory, 68

Blakeslee, Sandra, 11

Blind spot, and visual field, 8

Bolt, Richard, 135

Bottom-up processes

- definition, 8, 9
- feature processing stage, 10
- objects, 10–11
- pattern construction, 10
- patterns, 12
- visual working memory, 11

Brain

- and attention, 9
- components, 24
- and computer co-processing, 175
- distributed cognition, 19–20
- evolution, 181–182
- and eye function, 5
- in eye movement sequence, 14
- and mental imagery, 151
- nested loop operation, 17–19
- neuron-based processing clock, 3
- object identification, 111–112
- pattern-testing loop, 40
- seeing and cognition, 2
- self-control, 171
- and sign languages, 130
- and stereoscopic depth, 94
- as thinking component, 166
- and visual field, 8

Brain pixels

- and detection field, 37–38
- and 2.5D space, 44
- form and function, 5, 7

- and information transfer, 10

Brainstorming, via scribbling, 154

Brightness, color-opponent channels, 72

British Sign Language (BSL), 130

Broca's area, and verbal working memory, 115

BSL, *see* British Sign Language (BSL)

Business process design, example, 98

C

Calatrava, Santiago, 158, 159

Card, Stuart, 175–176

Cartoon strips

- definition, 142
- and short-term prediction, 142
- as strong narratives, 145

Cash, Johnny, 124

Cast shadows, definition, 92

Change detection experiment, 1–2

Channels

- black-white channel, 68, 82, 84–85
- color-opponent channels, 68–74, 78–80, 84–85
- feature channels, 31, 33
- red-green channel, 68, 70, 71, 84–85
- in two-dimensional space, 52–53
- yellow-blue channel, 68, 70, 84–85

Children, and computer interaction, 103

Chromatic color, in color-opponent channels, 73

Cinematographers

- framing, 139–140
- shot transitions, 141–142

Clarity

- and color design, 85
- in graphic design, 35

A Clockwork Orange, 107

Coca Cola, images as symbols, 124

Cognitive affordances

- artificial interactive spaces, 102–103
- concept developments, 99
- definition, 99
- example, 100
- and space traversal, 104
- and where pathway, 100–102

Cognitive cost, and space traversal, 103–105

Cognitive economics

- design sketching, 158–160

- and visual learning, 177–179
- Cognitive process
 - brain and computer, 175
 - optimization, 174–177
 - and working memory, 115–116
- Cognitive resources
 - computer-based, 172
 - and seeing, 2
- Cognitive tasks
 - and graphic design, 14–15
 - imagery and desire, 126
- Cognitive thread, in visual narrative, 138–139
- Cognitive walk-through, design prototypes, 162–163
- Cohen, Jonathan, 116
- Colby, C. L., 14
- Color
 - and pop-out effects, 30, 31, 32
 - semantics, 84
 - on shaded surfaces, 83
- Color appearance, and color-opponent channels, 74
- Color blindness, in color-opponent channels, 73
- Color coding
 - basic considerations, 77
 - color sequences, 80–83
 - large and small areas, 77–78
- Color design
 - and clarity, 85
 - showing detail, 75–76
- Color-opponent channels
 - basic considerations, 68–69, 84–85
 - chromatic color, 73
 - color appearance, 74
 - color and attention, 73–74
 - color blindness, 73
 - color segmentation, 72–73
 - color solids, 74
 - contrast, 69–70
 - emphasis and highlighting, 78–80
 - luminance, lightness, brightness, 72
 - luminance nonlinearity, 72
 - luminance and saturation, 72
 - motion, 70
 - saturation, 72
 - shape-from-shading, 70–71
 - spatial detail, 70
 - stereoscopic depth, 70
 - unique hues, 70
- Color processing, basic machinery, 66–68
- Color segmentation, color-opponent channels, 72–73
- Color sequences, in color coding, 80–83
- Color solids, and color-opponent channels, 74
- Color vision
 - basic processing machinery, 66–68
 - channel properties, 69–72
 - opponent process theory, 68–69
 - overview, 65–66
- Commerce, visualization applications, 20
- Computer interfaces
 - and brain co-processing, 175
 - and children, 103
 - as cognitive tools, 172
 - design, 36
 - and space traversal, 104
 - thinking tools, 175
- Computer software
 - ACT-R, 181
 - design diagrams, 155
 - GeoZu4D, 173
 - graphical vs. written programming, 132
 - interface additions, 175
 - learning, 178
 - nested processes, 18–19
 - Photoshop, and visual novelty, 123
 - and visual novelty, 123
- Computer systems
 - parallel computing, 26
 - and visual working memory, 11
- Concept diagram, examples, 160
- Cone receptors
 - in color processing, 66–67
 - and color space, 74
 - in eye, 5
- Connor, Charles, 53
- Conscious awareness, and visual world, 3
- Consciousness experience, change detection
 - experiment, 2
- Consequences, and long-term memory, 117–118
- Consistency, in graphic design, 35
- Consolidation, in creative process, 148
- Constellation system, characteristics, 176–177

Constructive critique, in creative process, 148

Constructive perception

- in creative design, 158
- and visual imagination, 152

Contour extraction mechanism

- in pattern processing, 49
- in scribbles, 152–153

Contours

- in binding problem, 46–49
- generalized contour, 49–50
- and pattern for design, 58

Contrast

- in color-opponent channels, 69–70
- degree, 93
- in texture, 50
- and visual distinctness, 29

Copycat cells, definition, 137

Cornea, form and function, 7

Crames, Christy, 80

Creative design

- cognitive economics of sketching, 158–160
- perceptual critique, 160–162
- prototypes, 162–163

Creative process

- and mental imagery, 150
- scribble exercise, 153
- steps, 148

D

Data analysis

- and color coding, 81
- and three dimensions, 97

Deictic gestures, definition, 135–136

Dendrites, and binding, 47

Deneve, Sophie, 178

Depth cues

- components, 91–93
- definition, 89–90
- in 2.5D space, 45
- stereoscopic depth, 70, 100
- stereoscopic depth perception, 94–95
- structure from motion, 95
- and 2.5 design, 95–97
- types, 90

Depth of focus, definition, 93

Depth perception, and physical world, 89

Design process

- computer applications, 106
- early requirements, 156–157
- for information access, 103
- object identification, 121

Desire, and imagery, 125–126

Detail

- in color design, 75–76
- in color-opponent channels, 70
- via fovea, 5–6
- sensitivity to, 36
- and 2.5 design, 96
- and visual novelty, 123

Detection field

- and brain pixels, 37–38
- definition, 38
- scanning strategy, 38

Diagrams

- and abstract meaning, 155–156
- definition, 155
- examples, 155
- in visual designs, 131

Diexis, for word and image linking, 135–136

Digital camera, eye as, 5

Dimensions

- away dimension, 44–45
- sideways dimensions, 44–45, 89
- three-dimensional space, 97–99, 102, 110
- towards-away dimension, 89, 90
- 2.5 dimensions, 44–45
- up dimension, 89

Direction, in pattern perception, 57–58

Distributed cognition, in visual queries, 19–20

Drilling down

- definition, 176
- hover query, 176–177

Drive, and cognitive cost, 104

Duhamel, J. R., 14

E

Eames, Charles, 105

Eames, Ray, 105

Echoic loop, in verbal working memory, 115

Edge detector neurons

- and binding problem, 47
- function, 48
- Egocentric space, dimensions, 89
- Ekman, Paul, 137
- Emergence, into visual field, 36
- Emotion, in scenes and objects, 124–125
- Emphasis, in color-opponent channels, 78–80
- Enfants sauvage, definition, 130
- Engineering
 - design requirements, 156–157
 - and diagrams, 155
- Environment
 - scanning strategies, 38–39
 - and visual distinctness, 29
- Error-checking, in map planning, 17
- Establishing shot, in visual narrative, 142
- Experiments, change detection, 1–2
- Explicit memory, definition, 117
- Extension, in creative process, 148
- External consequences, and long-term memory, 117
- Externalization, in creative process, 148
- Eye
 - in brain model, 24
 - as digital cameras, 5
 - form and function, 7
 - muscle function, 7
 - as thinking component, 166
- Eye movements
 - and cognitive cost, 104
 - control loop, 39–40
 - and faces, 32
 - for information retrieval, 87–88
 - and mental imagery, 150, 151
 - nested loops, 18
 - and perception, 20–21
 - planning, 26–27
 - in seeing and cognition, 3
 - sequence, 13–14
 - and shot transitions, 141
 - and thinking in action, 121
 - tracking, 176
 - and 2.5D space, 45
 - and visual attention, 9
 - and where pathway, 102
- Eye muscles, function, 7

F

- Face recognition, and viewpoint, 108–109
- Faces, and pop-out effects, 32
- Facial expressions, and attention, 9
- Faigin, Gary, 138
- Feature analysis
 - in binding problem, 46–49
 - low-level machinery, 25, 168
 - and pop-out effects, 31
 - as thinking component, 167
- Feature channels
 - in design, 33
 - and pop-out effects, 31
- Feature detectors, and pattern perception, 56
- Feature processing, bottom-up processes, 10
- Features, and attention, 9
- Field of information, consciousness, 3
- Findlay, John, 14
- Fingers of instantiation (FINSTS)
 - and cartoon strips, 142–143
 - in visual narrative, 140–141
- FINSTS, *see* Fingers of instantiation (FINSTS)
- Fixation, and thinking in action, 120–121
- Flach, Tim, 123
- Flowchart, as visual language, 132–133
- Fly, and cognitive cost, 104
- fMRI, *see* Functional magnetic resonance imaging (fMRI)
- Foraging theory, definition, 176
- Fovea
 - brain pixels, 5
 - rapid eye movements, 7
 - visual detail, 5–6
- Framing, in visual narrative, 139–140
- Franklin, R. G., 144
- Fruit eaters, color vision, 65–66
- Functional magnetic resonance imaging (fMRI)
 - mental imagery studies, 150–151
 - visual skill studies, 163

G

- Ganis, Georgio, 150
- Generalized contour mechanism,
 - characteristics, 49–50
- Generalized views, and face recognition, 109

Geon theory, definition, 110–111
 GeoZu4D software, and whale movements, 173
 Gestalt, definition, 10
 Gestures
 deictic gestures, 135–136
 in presentations, 137
 Gibson, James, 99
 Gilchrist, Ian, 14
 Gist perception
 definition, 112
 embedded objects, 114
 scenes, 113
 and visual novelty, 122–123
 Goldberg, M. E., 14
The Graduate, 140
 Grain size, in texture, 50
 Grammar, in natural language, 132–133
 Graphical artifacts, in pattern queries, 60–62
 Graphic designs
 and active vision model, 172
 via attention and perception, 3
 and cognitive process optimization, 174–175
 colors, 33
 computer interfaces, 36
 consistency *vs.* clarity, 35
 and constructive seeing, 170
 and degree of contrast, 93
 difficulty in, 21
 feature channels, 33
 future research, 181–182
 goals, 14–15
 map example, 15–17
 multiscale structures, 40–41
 and pattern finding, 172–174
 pattern for, 58–59
 and semantic pattern mappings, 62
 software package learning, 178
 symbols, 34
 tunability, 34
 as visual and language hybrids, 134
 Grass eaters, color vision, 65

H

Hand movements, and stereoscopic depth perception, 94

Hard-to-discriminate textures, examples, 51
 Hawkins, Jeff, 11, 142
 Hayhoe, Mary, 13
 Highlighting, in color-opponent channels, 78–80
 Hillstrom, A. P., 36
 Hippocampus, and long-term memory, 117
 Homunculus fallacy, definition, 47
 Horizontal direction, in pattern perception, 57–58
 Hover query, in drilling down, 176–177
 Hubel, David, 25
 Hues, in color-opponent channels, 70
 Hyperlink, and cognitive cost, 105

I

Ideas, into diagrams, 155
 Imagery
 and desire, 125–126
 mental, *see* Mental imagery
 as symbols, 123–124
 word linking, 135–136
 Implicit memory, definition, 117
 Impression, awake world view, 1
 Inferotemporal cortex (IT)
 and information reception, 109
 and intermediate patterns, 53
 and mind control, 171
 and pattern learning, 55
 and pattern processing, 46
 and pattern recognition, 168
 and patterns, 114
 and scene perception, 113
 and visual working memory, 118–119
 and what channel, 108–109
 Information access
 and cognitive co-processing, 175
 and cognitive cost, 105–106
 and cognitive thread, 138
 in depth cues, 89–90
 efficient design, 103
 and eye movements, 87–88
 foraging theory, 176
 and long-term memory, 117
 and structure from motion, 95
 as thinking component, 168–169
 for third dimension, 97–99

- and visual area, 4, 109
- and working memory capacity, 126–127
- Information processing
 - and multimodal connections, 108
 - visual representations, 133
- Information scent, definition, 176
- Interactive graphic display, Tubeguru™, 17
- Interference, in two-dimensional space, 51
- Intermediate patterns, characteristics, 53–54
- Internal consequences, and long-term memory, 117
- IT, *see* Inferotemporal cortex (IT)
- J**
- Jiricna, Eva, 155
- Just-in-time strategy, in eye movement sequence, 14
- K**
- Kanwisher, Nancy, 49
- Katz, Steven, 139
- Kennedy, John M., 152
- Kitchens, affordances, 100
- Known objects, referencing, 93
- Kosslyn, Stephen M., 148, 150, 151
- Kourtzi, Zoe, 49
- L**
- Laeng, Bruno, 150
- Lakoff, George, 62
- Language-based thinking, *vs.* visual thinking, 130–131
- Language processing
 - and attention, 180
 - as thinking component, 169
- Lateral geniculate nucleus
 - in brain model, 24
 - and visual area, 1, 25
- Lateral interparietal area, function, 39
- Lateral intraparietal lobe (LIP), and visual space, 101
- Lateral occipital cortex (LOC)
 - and generalized contour, 49
 - in pattern processing, 46
- Lawson, Bryan, 158
- Learnability, in color coding, 77
- Learned symbols, in sign language, 131
- Lens, form and function, 7
- Levin, Daniel, 1–2
- Levy, Ellen, 119

- Lightness, color-opponent channels, 72
- Light receptors, in retina, 7
- Linear perspective, definition, 92
- Lines, and contour extraction mechanism, 152–153
- LIP, *see* Lateral intraparietal lobe (LIP)
- LOC, *see* Lateral occipital cortex (LOC)
- Logic, in natural language, 132–133
- Logical propositions, in mental imagery, 148
- London Underground map, as design example, 15–17
- Long-term memory
 - and consequences, 117–118
 - definition, 116
 - explicit and implicit, 117
 - and hippocampus, 117
 - and situated interactions, 116–117
 - as thinking component, 170
- Long-term priming, definition, 118
- Look-aheads, in eye movement sequence, 14
- Loops, in visual searches, 39–40
- Low-level feature analysis
 - machinery, 25–26
 - and pop-out effects, 31
- Luminance
 - color-opponent channels, 72
 - and detail, 75–76
 - and highlighting, 79
 - and shading, 83
- M**
- Mackinlay, Jock, 175
- Magritte, Rene, 123
- Maps
 - action, 101
 - cognitive plan, 16–17
 - color coding, 80–83
 - contour map reading, 171
 - graphic design implications, 15
 - and information retrieval, 88
 - London Underground map, 15–17
 - in parietal lobe, 101
 - pattern finding, 46
 - picture planes, 92
 - road map sample, 59
 - and thinking process, 180
 - water currents and temperature, 59

- Masseroni, Manfredo, 153
- Meaning
 - definition, 165
 - in scenes and objects, 124–125
- Mediaeval cityscape, scene perception, 113
- Medial intraparietal lobe (MIP), and visual space, 101
- Memory
 - implicit and explicit, 117
 - long-term memory, 116–118, 170
 - spatial, for scenes, 109
 - as thinking component, 170
 - working memory, 3, 9, 11, 88, 114–116, 118–121, 126–127, 158–159, 161, 177, 179
- Memory aids, as thinking component, 169
- Mental activity, in mental imagery, 149–150
- Mentalese, definition, 133
- Mental imagery
 - and eye movements, 150
 - as internalized active process, 152
 - as mental activities, 149–150
 - as normal seeing, 148–149
 - as symbolic and non-spatial, 148
- Meta-seeing
 - cognitive economics of sketching, 158–160
 - perceptual critique, 160–162
 - prototypes, 162–163
- Michelangelo, 147
- Michelson, Harold, 140
- Miller, Earl, 116
- MIPE, *see* Medial intraparietal lobe (MIP)
- Mirror neurons
 - discovery and function, 137
 - example, 138
- Models
 - active vision, 172
 - biased competition, 119
- Monroe, Marilyn, 108
- Motion
 - in color-opponent channels, 70
 - orienting response, 36
 - static details, 36
 - for visibility enhancement, 36
 - and visual pollution, 37
 - web design considerations, 36–37
- Motor working memory, examples, 120–121
- Mouse hover, and cognitive cost, 104
- Mouse select, and cognitive cost, 104–105
- Movements
 - and cognitive cost, 104
 - eye, *see* Eye movements
 - replay tool studies, 173–174
 - and stereoscopic depth perception, 94
- Move and scan loop, in visual searches, 39
- Movies, as strong narratives, 145
- Multimodal connections, in information processing, 108
- Multiscale structure design, visual searches, 40–41
- Munzner, Tamara, 176
- N**
- Naive physics of perception, definition, 102
- Narrative diagrams
 - advantages, 143–144
 - example, 143
 - main principles, 143
 - short-term predictions, 142–143
 - single-frame narratives, 144–145
- Natural language, grammar and
 - logic, 132–133
- Nested loops, brain operation, 17–19
- Neural pathways
 - activation and strengthening, 177–178
 - mirror neurons, 137–138
- Neuron-based processing clock, and eye
 - movement, 3
- Neurons
 - as Bayesian operators, 178
 - and binding problem, 47
 - and feature analysis, 25–26
 - function, 48
 - mirror neurons, 137, 138
- Node link diagrams, and pattern queries, 62
- Non-pictorial depth cues
 - definition, 90
 - structure from motion, 95
- Non-uniformity, in visual processing, 6
- Normal seeing, in mental imagery, 148–150
- Norman, Donald A., 20

Novelty

- and advertising, 122
- via computer software, 123
- search for, 122

O

Objects

- bottom-up processes, 10–11
- and contour extraction mechanism, 152–153
- ease of identification, 121
- in eye movement sequence, 14
- and gist perception, 114
- identification in inferotemporal cortex, 109
- known, referencing, 93
- meaning and emotion, 124–125
- and stereoscopic depth perception, 94
- structured objects, 110–112
- and thinking in action, 120–121
- and 2.5 design, 95
- from visual images, 119
- in visual and verbal working memory, 161
- in Visual Working Memory, 9
- and working memory, 126

Occlusion

- definition, 91
- and 2.5 design, 96

Office space, affordances, 100

Oliva, Aude, 112–113

On Intelligence, 11, 142

Optic chiasm, in brain model, 24

Optic nerve

- and attention, 9
- in brain model, 24
- form and function, 7
- and visual field, 8

Optic radiations, in brain model, 24

Optic tract, in brain model, 24

O'Regan, Kevin, 1

Orientation, in texture, 50

Orienting response, to motion, 36

Overlapping regions

- and pattern for design, 58
- in two-dimensional space, 52

P

Para-hippocampal area, in mental imagery, 151

Parallel computing

- definition, 26
- pre-attentive effects, 28

Parallel projection, definition, 92

PARC, *see* Xerox Palo Alto Research Center (PARC)

Parietal lobe, visual space maps, 101

Pasupathy, Anitha, 53

Pattern-finding circuits

- and design, 172–174
- as thinking components, 170
- tuning, 3

Pattern learning, and two-dimensional space, 54–55

Pattern perception

- and graphic design, 3
- horizontal and vertical direction, 57–58
- and pop-out effects, 29
- in scanning strategy, 38
- and spatial layout, 56–57

Pattern processing

- basic machinery, 46
- binding problem, 46–49
- and 2.5D space, 45
- elementary systems, 24
- generalized contour, 49

Patterns

- and attention, 9
- in bottom-up processes, 10, 12
- and cognitive process optimization, 174–175
- construction, 43
- for design, 58–59
- formation, 43
- and generalized views, 109
- intermediate patterns, 53–54
- and long-term memory, 117
- and object identification, 121
- and top-down processes, 13
- in two-dimensional space, 52–53
- in visual narrative, 139

Patterns of patterns, definition, 63

Pattern-testing loop

- nested loops, 18
- in visual searches, 40

Perception

- as active process, 20–21
- biomotion patterns, 54
- constructive perception, 152, 158
- depth, and physical world, 89
- future research, 181
- gist perception, 112–114, 122–123
- and graphic design, 3
- naive physics of perception, 102
- pattern perception, 3, 29, 38, 56–58
- scene perception, 112, 113, 124–125
- and shot transitions, 141
- structured objects, 110–111
- Perceptual critique, in preliminary design, 160
- Perceptual facilitation, definition, 118
- Perspective, definition, 92
- Photoshop, and visual novelty, 123
- Pictorial depth cues, definition, 90
- Picture plane, definition, 92
- Pirolli, Peter, 176
- Plato, 110
- Pointing
 - in presentations, 136–137
 - as thinking component, 169
- Poldrack, Russel A., 163
- Pop-out effects
 - attention, 28–29
 - basic channels, 41–42
 - environment, 29
 - examples, 27–28
 - feature channels, 31
 - patterns, 29–30
 - tunable properties, 29
 - and visual conjunctive search, 31
 - and visual learning, 32–33
- Potter, Mary, 119
- PowerPoint slides
 - function and design, 136–137
 - and luminance contrast, 79
- Pre-attentive effects, in pop out experiment, 28
- Prefrontal cortex, and working memory, 116
- Presentations, and pointing, 136–137
- Primary visual cortex
 - in brain model, 24

- in feature analysis, 25
- texture regions, 50

Priming

- definition, 118
- and top-down processes, 12

Problem-solving systems

- nested loops, 18
- Tubeguru™, 17

Prototype design

- definition, 160
- in meta-seeing, 162–163
- and sketching, 164

Proxies, as thinking component, 169

Puzzles, as novelties, 123

Pylyshyn, Zenon W., 141, 148

Q

Question and answer patterns, in visual narrative, 139

R

- Radial coordinates, in visual space, 44
- Random letters, visual queries, 24–25
- Rapid eye movements, and fovea, 7
- Rays, in 2.5D space, 44
- Real world experiments, change detection, 1–2
- Recognition, faces, and viewpoint, 108–109
- Red-green channel
 - chromatic contrast, 70
 - and color-opponent theory, 84–85
 - in opponent process theory, 68
 - shape-from-shading, 71
- Relationship, and pattern for design, 58–59
- Rensink, Ronald A., 114
- Repin, Ilya, 167
- Replay tool, early work, 173
- Retina
 - and binding problem, 47
 - and color-opponent theory, 84–85
 - and color space, 74
 - form and function, 7
 - and visual area, 1, 25
 - and visual field, 8
- Ribbons, and replay analysis tool, 174
- Ring of Fire*, 124

Rizzolati, Giacomo, 137

Rods, in color processing, 66

S

Saturation

in color-opponent channels, 72

and highlighting, 79

Scatter plots, visual query example, 33

Scene perception

examples, 113

meaning and emotion, 124–125

rapid characterization, 112

Science

design diagrams, 155

visualization applications, 20

Scribbles

as brainstorming, 154

contours and lines, 152–153

in creative process, 153

Seeing

and cognition, 2

with design prototypes, 162–163

in mental imagery, 148–150

process and apparatus, 5–7

as thinking component, 170

and visual novelty, 123

Selective tuning, in two-dimensional space, 51

Semantic pattern mappings, spatial metaphors,
62–63

Serial processing, and two-dimensional space, 55

Shaded surfaces, color, 83

Shading, definition, 93

Shape-from-shading, in color-opponent channels,
70–71

Sharpening, in color-opponent channels, 72

Short-term prediction, in narratives, 142–143

Short transitions, in visual narrative, 141–142

Sideways dimensions

in egocentric space, 89

in 2.5D space, 44–45

Sign languages

brain subsystems, 130

definition, 130

development, 130

grammar and logic, 132–133

history, 130

learned symbols, 131

as visual language, 129–130

Simons, Daniel, 1–2

Sistine Chapel, 147

Situated interactions, definition, 116–117

Size gradients, in perspective, 92

Sketching

by architects, 158

and cognitive economics, 158–160

and prototype design, 164

Social dynamics, during video conferencing, 89

Sound cues, during video conferencing, 89

Space

artificial interactive spaces, 102, 103

binding problem, 46–49

egocentric space dimensions, 89

generalized contour, 49–50

horizontal and vertical direction, 57–58

interference and selective tuning, 51

intermediate patterns, 53–54

maps in parietal lobe, 101

mental manipulation, 110

and non-spatial data, 97–99

office space affordances, 100

pattern for design, 58–59

and pattern learning, 54–55

pattern-processing machinery, 46

pattern queries with common graphical artifacts,
60–62

patterns, 43

patterns, channels, attention, 52–53

radial coordinates, 44

semantic pattern mappings, 62–64

serial processing, 55

spatial layout, 56–57

and structured objects, 111–112

structuring and patterns, 43

texture regions, 50–51

traversal *vs.* cognitive cost, 103–105

2.5D space, 44–45

uniform color example, 74

and where pathway, 102

- Spatial detail
 - in color-opponent channels, 70
 - and 2.5 design, 96
- Spatial ideas, as logical propositions, 148
- Spatial layout, and pattern perception, 56–57
- Spatial memory, for scenes, 109
- Spatial metaphors, and semantic pattern mappings, 62–63
- Spelke, Elizabeth, 102
- Spiral design, definition, 157
- Static detail, sensitivity to, 36
- Steinberg, Saul, 153
- Stereoscopic depth
 - and affordances, 100
 - in color-opponent channels, 70
 - definition, 94
 - precision, 94
 - and structure from motion, 95
- Stereo technology, history, 94
- Storyboards, in visual narrative, 140
- Street signs, attention competition, 35
- Structured objects
 - perception, 110–111
 - and two-dimensional patterns, 111–112
- Structure from motion, definition, 95
- Surfaces
 - in color-opponent channels, 73
 - shaded, color, 83
- Suwa, Masaki, 158
- Symbols
 - and cognitive process optimization, 174–175
 - design considerations, 34–35
 - images as, 123–124
 - learned, 131
- T**
- Tasks, and attention, 180
- Teodorescu, D. S., 150
- Text
 - emphasis, 79–80
 - and luminance, 76
 - and 2.5 design, 96
- Texture gradients, in perspective, 92
- Texture regions, primary factors, 50–51
- Thatcher Illusion, definition, 108
- Thinking in action, working memory example, 120–121
- Thomas, N. J. T., 149
- Thompson, W. L., 150
- Three-dimensional space
 - mental manipulation, 110
 - and non-spatial data, 97–99
 - and where pathway, 102
- Top-down processes
 - attention, 12
 - definition, 8, 9
 - patterns, 13
- Torralba, Antonio, 112, 114
- Towards-away dimension
 - in egocentric space, 89
 - information content, 90
- Train, and cognitive cost, 104
- Triesman, Anne, 27–29
- Tubeguru™, 17
- Tunable properties
 - definition, 29
 - design considerations, 34
 - faces, 32
- Tuning, in two-dimensional space, 51
- Turner, R. J. W., 144
- Tversky, Barbara, 143, 158
- Two-dimensional space
 - binding problem, 46–49
 - generalized contour, 49–50
 - horizontal and vertical direction, 57–58
 - interference and selective tuning, 51
 - intermediate patterns, 53–54
 - pattern for design, 58–59
 - and pattern learning, 54–55
 - pattern-processing machinery, 46
 - pattern queries with common graphical artifacts, 60–62
 - patterns, channels, attention, 52–53
 - semantic pattern mappings, 62–64
 - serial processing, 55
 - spatial layout, 56–57
 - and structured objects, 111–112
 - structuring and patterns, 43
 - texture regions, 50–51
- 2.5 design
 - affordances, 99–100

- definition, 95
- and depth cues, 106
- and three-dimensional information, 97–98

2.5 dimensions, components, 44–45

U

Uniform color space, example, 74

Unique hues, in color-opponent channels, 70

Unlearned patterns, and apprehendable chunk, 55

Up dimension, in egocentric space, 89

Up-down dimensions, in 2.5D space, 44–45

V

V1, *see* Visual area 1 (V1)

V2, *see* Visual area 2 (V2)

V3, *see* Visual area 3 (V3)

V4, *see* Visual area 4 (V4)

Verbal working memory

- function, 115
- thinking in action example, 120–121
- vs.* visual working memory, 127, 161

Vertical direction, in pattern perception, 57–58

Video conferencing, and social dynamics, 89

Video games, and naive physics of perception, 102

Viewpoint

- and face recognition, 108–109
- from patterns, 109

Vision, at visual field, 5–6

Visual area 1 (V1)

- and color opponent process theory, 68
- in feature analysis, 25
- function, 7
- and intermediate patterns, 53
- and mind control, 171
- and pattern learning, 54
- and pattern processing, 46
- and semantic pattern mapping, 63
- and texture, 51
- and visual conjunctive search, 31
- and visual working memory, 118
- where pathway, 100–101

Visual area 2 (V2)

- in feature analysis, 25
- and generalized contour, 49
- and intermediate patterns, 53

- and mind control, 171
- and pattern learning, 54
- and texture, 51
- and visual working memory, 118
- where pathway, 100–101

Visual area 3 (V3)

- definition, 26
- and generalized contour, 49

Visual area 4 (V4)

- complex compound shape identification, 168
- and information reception, 109
- and intermediate patterns, 53
- and long-term memory, 116
- and mind control, 171
- and pattern learning, 54–55
- and pattern perception, 57
- and visual working memory, 118

Visual concept, in creative process, 148

Visual conjunctive search, definition, 31

Visual cortex

- and feature distinctness, 30
- first stage processing, 167

Visual cues, during video conferencing, 89

Visual designs, and learned symbols, 131

Visual detail, via fovea, 5–6

Visual distinctness

- in color coding, 77
- feature properties, 30
- pop-out effects, 29
- in queries, 24

Visual field

- emergence into, 36
- in eye movement sequence, 14
- hole in, 8
- vision at, 5–6

Visualization, definition, 20

Visual languages, *vs.* written modes, 133–135

Visual learning

- and cognition economics, 177–179
- and pop-out effects, 32–33

Visual narrative

- cognitive thread, 138–139
- FINSTS and divided attention, 140–141
- framing, 139–140
- question and answer patterns, 139

- Visual narrative (*Contd.*)
 - shot transitions, 141–142
 - visual vs. written, 145
 - Visual pattern queries
 - and apprehendable chunk, 55
 - with common graphical artifacts, 60–62
 - multi-chunk queries, 56
 - Visual pollution, and motion, 37
 - Visual processing
 - and attention, 180
 - non-uniformity, 6
 - Visual queries
 - attention, 3
 - awake impressions, 1
 - bottom-up processes, 10–12
 - during casual tasks, 4
 - and color semantics, 84
 - conscious awareness, 3
 - designs as, 160–161
 - distinct objects, 24
 - distributed cognition, 19–20
 - and graphic design, 3–4, 14–17, 21
 - hover query, 176–177
 - as information shorthand, 21
 - nested loops, 17–19
 - perception, 8–10
 - random letters, 24–25
 - scatter plot example, 33
 - seeing and cognition, 2
 - seeing process and apparatus, 5–7
 - top-down processes, 12–14
 - Visual searches
 - basic pop-out properties, 41–42
 - in color coding, 77
 - detection field, 37–39
 - multiscale structure design, 40–41
 - process, 39–40
 - strategies and skills, 37
 - Visual signals, processing, 7
 - Visual skills, development, 163–164
 - Visual space
 - maps in parietal lobe, 101
 - patterns, 43
 - radial coordinates, 44
 - traversal vs. cognitive cost, 103–105
 - 2.5D space, 44–45
 - Visual task analysis, example, 157–158
 - Visual thinking components
 - brain, 166
 - constructive seeing, 170
 - eye, 166
 - first stage cortical processing, 167
 - hierarchy of skills, 171
 - information, 168–169
 - language processing, 169
 - long term memories, 170
 - memory aids, 169
 - mind control center, 171
 - pattern finding, 170
 - pointing, 169
 - where and what pathways, 168
 - Visual working memory
 - bottom-up processes, 11
 - and objects, 9
 - specifications, 177
 - vs. verbal working memory, 127, 161
 - and visual objects, 126
 - Visual world, while awake, 1
- ## W
- Walk, and cognitive cost, 104
 - Walk the Line*, 124
 - Web designers, motion considerations, 36–37
 - Web pages, and pattern queries, 61
 - Weisel, Torsten, 25
 - Whale movements, replay tool studies, 173–174
 - What pathways
 - definition, 26
 - in eye movement planning, 27
 - and inferotemporal cortex, 108–109
 - and intermediate patterns, 53
 - and pattern processing, 46
 - as thinking components, 168
 - and visual conjunctive search, 31
 - and visual working memory, 118
 - Where pathways
 - in brain, 100–101
 - definition, 26

- in eye movement planning, 27
- and eye movements, 102
- and scanning strategies, 39
- as thinking components, 168
- unconscious functions, 101–102

Wilhelmson, Robert B., 92

Williams, David, 66

Words, image linking, 135–136

Workbenches, affordances, 100

Working memory

- accessing, 118–120
- and attention, 3, 115–116, 179
- bottom-up processes, 11
- capacity, 126–127
- and cognitive process, 115–116
- definition, 114–115
- and design sketching, 158–159
- function, 115
- and information retrieval, 88
- motor examples, 120–121
- and objects, 9

- specifications, 177
- thinking in action example, 120–121
- verbal *vs.* visual, 127, 161
- and visual objects, 126

Written languages, *vs.* visual languages, 133–135

X

Xerox Palo Alto Research Center (PARC), 175

XPLANE, 98

Y

Yantis, S., 36

Yarbus, Alfred, 167

Yellow-blue channel

- chromatic contrast, 70
- and color-opponent theory, 84–85
- in opponent process theory, 68

Z

Zero, in data analysis, 81

Zooming, and cognitive cost, 104–105